

TOM M. APOSTOL

CALCULUS

VOLUME I

INTRODUCTION,
WITH VECTORS AND
ANALYTIC GEOMETRY



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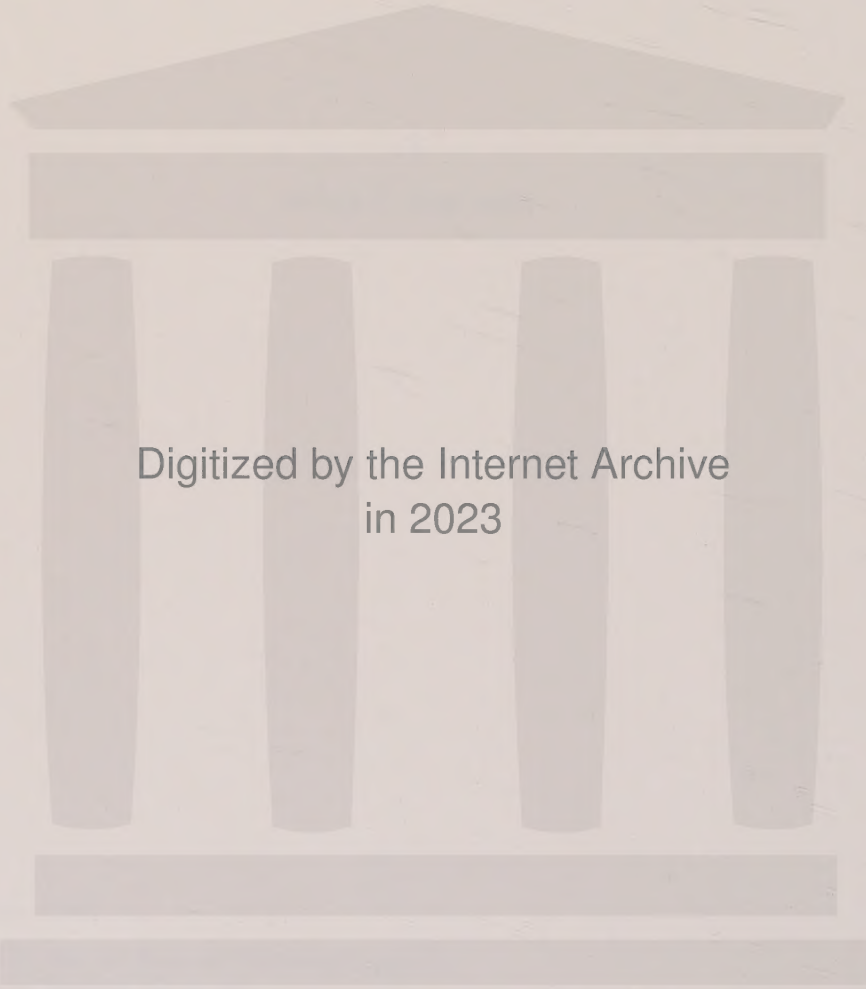
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To
Jane and Stephen



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PREFACE

There seems to be no general agreement as to what should constitute a first course in calculus and analytic geometry. Some people insist that the only way really to understand calculus is to start off with a thorough treatment of the real-number system and develop the subject step by step in a logical and rigorous fashion. Others argue that calculus is primarily a tool for engineers and physicists; they believe the course should stress applications of the calculus by appeal to intuition and by extensive drill on problems which develop manipulative skills. There is much that is sound in both these points of view. Calculus is a deductive science and a branch of pure mathematics. At the same time it is very important to remember that calculus has strong roots in physical problems and that it derives much of its power and beauty from the variety of its applications. It is possible to combine a strong theoretical development with sound training in technique; this book represents an attempt to strike a sensible balance between the two. While treating the calculus as a deductive science the book does not neglect applications to physical problems. Proofs of all the important theorems are presented as an essential part of the growth of mathematical ideas; the proofs are often preceded by a geometric or intuitive discussion to give the student some insight into why they take a particular form. Although these intuitive discussions will satisfy readers who are not interested in detailed proofs, the complete proofs are also included for those who prefer a more rigorous presentation.

The book is organized to accommodate a variety of backgrounds and interests. To avoid interrupting the main flow of ideas, some of the proofs appear in separate "starred" sections. Similarly, some chapters are accompanied by supplementary sections in which certain important topics related to calculus are discussed in detail. Some of these sections are also starred to indicate that they may be omitted or postponed without disrupting the continuity of the presentation. For example, one starred supplement contains a set of axioms for the real-number system; these axioms place the proofs on a solid mathematical foundation. Another starred supplement contains a discussion of mathematical induction. The extent to which such material is taught will depend on the student's background, his skill in algebra, and the depth of his interests.

The approach in this book has been suggested by the historical and philosophical

development of calculus and analytic geometry. For example, integration is treated before differentiation. Although to some this may seem unusual, it is historically correct and pedagogically sound. Moreover, it is the best way to make meaningful the true connection between the integral and the derivative.

The concept of the integral is defined first for step functions. Since the integral of a step function is merely a finite sum, integration theory in this case is extremely simple. As the student learns the properties of the integral for step functions he gains experience in the use of the summation notation and at the same time becomes familiar with the notation for integrals. This sets the stage so that the transition from step functions to more general functions seems easy and natural.

The book has been divided into two volumes. Volume I deals primarily with functions of one variable. The first two chapters develop the basic properties of the integral and the derivative, including their connection through the first and second fundamental theorems of the calculus. These concepts, in turn, are used in Chapter 3 to introduce the logarithm, the exponential, and the inverse trigonometric functions. Chapter 4 contains an early introduction to differential equations. This serves several purposes. First of all, it illustrates in a forceful way the usefulness of calculus in solving a variety of physical problems and, secondly, it gives the student an opportunity to review integration technique. Differential equations are returned to later in the book in many of the exercises and again in connection with the study of power series. A more extensive treatment of differential equations is carried out in the latter part of Volume II.

Surprisingly little analytic geometry is required to understand the rudiments of integration and differentiation; accordingly, in the first four chapters the geometry is introduced only as it is needed. A systematic treatment of analytic geometry begins in Chapter 5, where it is developed with the aid of vector algebra. Both vector algebra and calculus are used in Chapter 6 to extend the study of curves and surfaces.

Chapter 7 treats the mean-value theorem of differential calculus and Taylor's formula with remainder. The arrangement of the material gives the instructor considerable flexibility. He can discuss all the details or he can suggest that the better students read some of them on their own. Chapter 8 covers applications of the mean-value theorem to extremum problems and to indeterminate forms, and Chapter 9, which concludes Volume I, contains an introductory treatment of infinite sequences and series and a brief account of improper integrals.

The early chapters of Volume II develop both multiple integration and elementary probability theory as applications of the notion of set function. (All or part of the material on probability can be omitted for a shorter course.) Subsequent chapters contain a more thorough treatment of partial differentiation, including the elements of vector analysis, line integrals, and surface integrals. The latter part of Volume II contains more advanced topics in differential equations and an introduction to numerical analysis. Most of the chapters in this volume are independent of one another to allow flexibility in the choice of topics.

Throughout both volumes, an historical introduction precedes each important new concept, tracing its development from an early intuitive physical notion to its precise mathematical formulation. The student learns something of the struggles of the past and of the triumphs of the men who contributed most to the subject. Thus the student becomes an active participant in the evolution of ideas rather than a passive observer of results.

I acknowledge with pleasure my debt to many colleagues and friends whose advice

aided in the preparation of the manuscript. First of all, I am deeply indebted to the hundreds of Caltech freshmen and sophomores who provided the original incentive for this work. Their favorable response to the preliminary mimeographed edition reinforced my belief that the plan of the book is pedagogically sound. The plan itself evolved during the academic year 1957–58 after extensive discussion with many members of the Caltech mathematics staff, and the first edition of mimeographed notes was issued in 1958. Special thanks are due to Professor H. F. Bohnenblust who first convinced me of the value of introducing integration via step functions and who supported the spirit and approach used throughout the book. Professors A. Erdélyi and F. B. Fuller also made many constructive criticisms of the original notes and influenced the final organization of the material.

In transforming the notes to a book I received valuable help from Professors Kenneth Hoffman of the Massachusetts Institute of Technology, George Springer of the University of Kansas, and Herbert S. Zuckerman of the University of Washington, each of whom read the entire manuscript and made detailed and thoughtful comments that resulted in a number of improvements. My obligation to Professor Zuckerman extends more deeply than this, however. His influence on my thinking, ever since my undergraduate years, has been vital and continuing, and I owe him more than I know how to acknowledge.

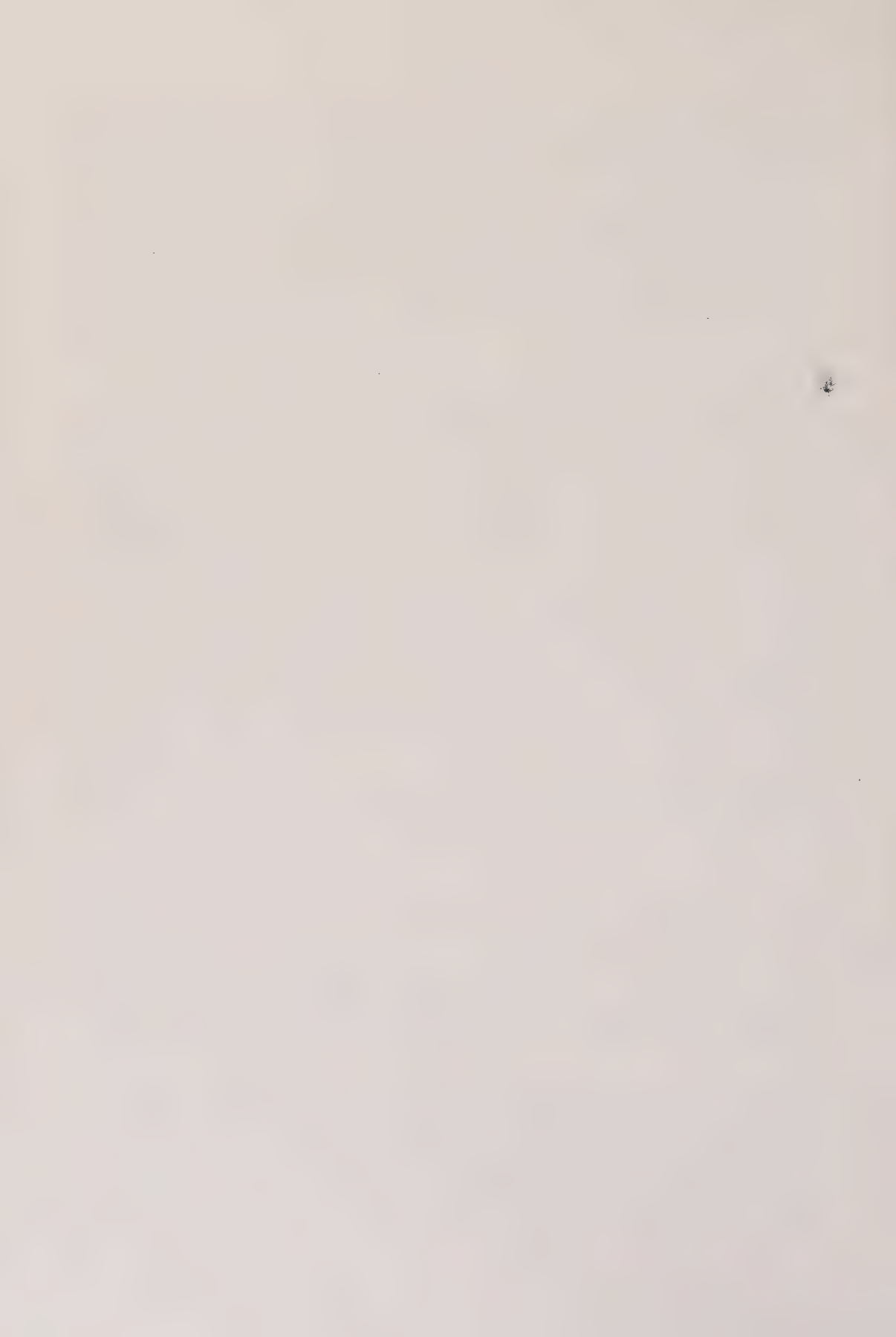
It has been a great pleasure to observe the imagination and skill of the Blaisdell Publishing Company in seeing this book through the press. I especially appreciate the partnership of effort between author and publisher that Mr. Warren Blaisdell has encouraged.

Thanks are due also to Jacquelin Sanborn for her competent typing and editing of the manuscript, to Lee Mazotti for her help with the illustrations, and to Roger Hill for his aid in preparing the answers to the exercises.

I owe the greatest debt of all to my wife for maintaining her cheerful disposition during the endless rereading of many passages that, thanks to her, are now more readable than they would have been otherwise.

TOM M. APOSTOL

California Institute of Technology
March 14, 1961



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1

INTEGRAL CALCULUS

Part I. The Method of Archimedes

1.1 Historical introduction

The remarkable progress that has been made in science and technology during the last century is due in large part to the development of mathematics. That branch of mathematics known as integral and differential calculus serves as a natural and powerful tool for attacking a variety of problems that arise in physics, astronomy, engineering, chemistry, geology, biology, and other fields including, rather recently, some of the social sciences.

To give the reader an idea of the many different types of problems that can be treated by the methods of calculus, we list here a few sample questions selected from the exercises that occur in later chapters of this book:

What is the shape of the surface of a liquid that is being rotated about a vertical axis? With what speed should a rocket be fired upward so that it never returns to earth? What is the radius of the smallest circular disk that can cover every isosceles triangle of a given perimeter L ? What volume of material is removed from a solid sphere of radius $2r$ by drilling a hole of radius r through the center? If a strain of bacteria grows at a rate proportional to the amount present and if the population doubles in one hour, by how much will it increase at the end of two hours? What is the mass of a nonhomogeneous sphere if its density at each point P is twice the distance of P from the center?

These examples, chosen from various fields, illustrate some of the technical questions that can be answered by more or less routine applications of calculus.

Calculus is more than a technical tool—it is a collection of fascinating and exciting ideas that have interested thinking men for centuries. These ideas have to do with *speed*, *area*, *volume*, *rate of growth*, *continuity*, *tangent line*, and other concepts from a variety of fields. Calculus forces us to stop and think carefully about the meanings of these concepts. Another remarkable feature of the subject is its unifying power. Most of these ideas can be formulated so that they revolve around two rather specialized problems of a geometric nature. We turn now to a brief description of these problems.

Consider a curve C which lies above a horizontal base line such as that shown in Figure 1.1. We assume this curve has the property that every vertical line intersects it once at

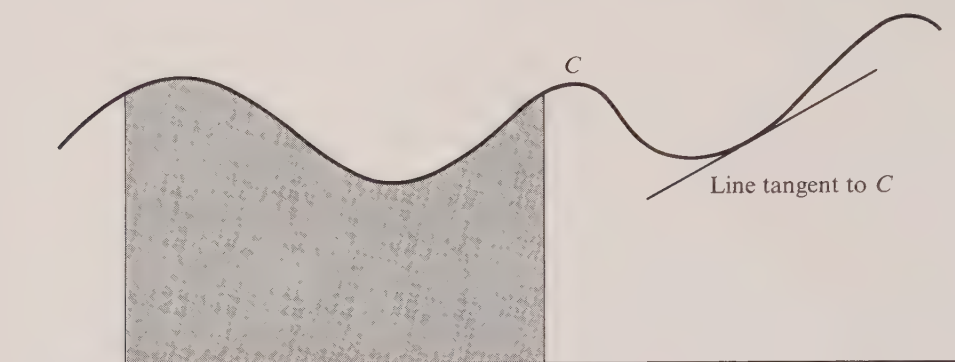


FIGURE 1.1

most. The shaded portion of the figure consists of those points which lie below the curve C , above the horizontal base, and between two parallel vertical segments joining C to the base. The first fundamental problem of calculus is this: *To assign a number which measures the area of this shaded region.*

Consider next a sloping line drawn tangent to the curve, as shown in Figure 1.1. The second fundamental problem may be stated as follows: *To assign a number which measures the steepness of this line.*

Basically, calculus has to do with the precise formulation and solution of these two special problems. It enables us to *define* the concepts of area and tangent line and to *calculate* the area of a given region or the steepness of a given tangent line. *Integral calculus* deals with the problem of area and will be discussed in this chapter. *Differential calculus* deals with the problem of tangents and will be introduced in Chapter 2.

Historically, the birth of integral calculus occurred more than 2000 years ago when the Greeks attempted to answer the question concerning area by devising a process which they called the *method of exhaustion*. The essential ideas of this method are really very simple and can be described briefly as follows: Given a region whose area is to be determined, we inscribe in it a polygonal region which approximates the given region and whose area we can easily compute. Then we choose another polygonal region which gives a better approximation and we continue the process taking polygons with more and more sides in an attempt to exhaust the given region. The method is illustrated for a semicircular region in Figure 1.2. It was used successfully by Archimedes (287–212 B.C.) to find exact formulas for the area of a circle and a few other special figures.

The development of the method of exhaustion beyond the point to which Archimedes carried it had to wait nearly eighteen centuries until the use of algebraic symbols and techniques became a standard part of mathematical learning. The elementary algebra that is familiar to most high-school students today was completely unknown in Archimedes' time and it would have been next to impossible to extend the method to any general class of regions without some convenient way of expressing rather lengthy calculations in a compact and simplified form.

A slow but revolutionary change in the development of mathematical notations began in the 16th century A.D. The cumbersome system of Roman numerals was gradually

displaced by the Hindu-Arabic characters used today, the symbols $+$ and $-$ were introduced for the first time, and the advantages of the decimal notation began to be recognized. During this same period, the brilliant successes of the Italian mathematicians Tartaglia, Cardano, and Ferrari in finding algebraic solutions of cubic and quartic equations stimulated a great deal of activity in mathematics and encouraged the growth and acceptance of a new and superior algebraic language. With the widespread introduction of well-chosen algebraic symbols, interest was revived in the ancient method of exhaustion and a large number of fragmentary results were discovered in the 16th century by such pioneers as Cavalieri, Toricelli, Roberval, Fermat, Pascal, and Wallis.

Gradually the method of exhaustion was transformed into the subject that came to be known as integral calculus, a new and powerful discipline that led to a large variety of applications, not only to geometrical problems concerned with areas and volumes but also to problems in other sciences. This method, which retained some of the original features of the method of exhaustion, received its biggest impetus in the 17th century, largely due to the efforts of Isaac Newton (1642–1727) and Gottfried Leibniz (1646–1716), and its development continued well into the 19th century before the subject was put on a firm mathematical basis by such men as Augustin-Louis Cauchy (1789–1857) and Bernhard Riemann (1826–1866). Further refinements and extensions of the theory are still being carried out in contemporary mathematics.

1.2 The method of exhaustion for the area under a parabola

Before we proceed to a systematic treatment of integral calculus, it will be instructive to apply the method of exhaustion directly to one of the special figures treated by Archimedes himself. The region in question is shown in Figure 1.3 and can be described as follows: If we choose an arbitrary point on the base of this figure and denote its distance from 0 by x , then the vertical distance from this point to the curve is x^2 . In particular, if the length of the base itself is b , the altitude of the figure is b^2 . The vertical distance from x to the curve is called the “ordinate” at x . The curve itself is an example of what is known as a *parabola*. The region bounded by it and the two line segments is called a *parabolic segment*.

This figure may be enclosed in a rectangle of base b and altitude b^2 , as shown in Figure 1.3. By examination of the figure it seems reasonable to assert that the area of the parabolic segment is less than half the area of the rectangle. Archimedes made the surprising discovery that the area of the parabolic segment is exactly *one-third* that of the rectangle; that is to say, $A = b^3/3$, where A denotes the area of the parabolic segment. We shall show presently how to arrive at this result.

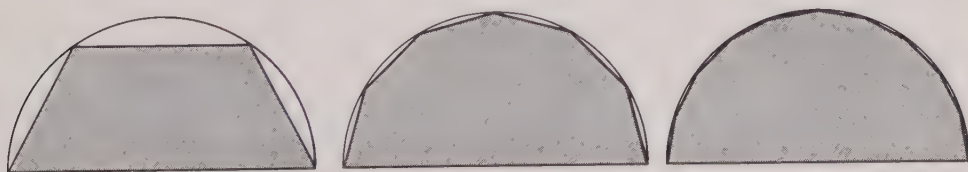


FIGURE 1.2 The method of exhaustion applied to a semicircular region.

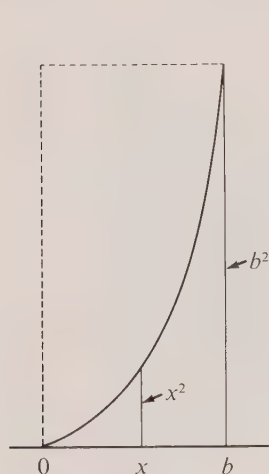


FIGURE 1.3 A parabolic segment.

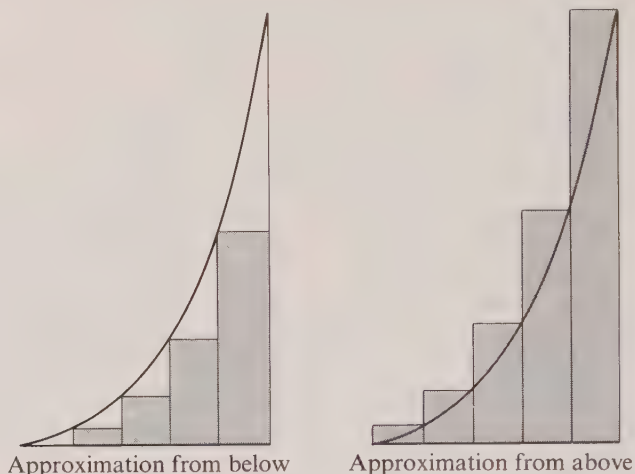


FIGURE 1.4

It is only fair to point out that the parabolic segment in Figure 1.3 is not shown exactly as Archimedes drew it and the details that follow are not exactly the same as those used by him. Nevertheless, the essential *ideas* are those of Archimedes: what is presented here can be thought of as the method of exhaustion in modern notation.

The method is simply this: We slice the figure into a number of strips and obtain two approximations to the region, one from below and one from above, by using two sets of rectangles† as illustrated in Figure 1.4. Archimedes would have concluded that the area of the parabolic segment is at least as large as the combined areas of the inner rectangles but certainly no larger than the combined areas of the outer rectangles.

If each strip is further subdivided to obtain a new approximation with a larger number of strips, the combined area of the inner rectangles *increases*, whereas the total area of the outer rectangles *decreases*. This is illustrated by comparing Figure 1.4 with Figure 1.5. The strips in Figure 1.5 were obtained by bisecting those in Figure 1.4. Archimedes realized that an approximation to the area within any desired degree of accuracy could be obtained by simply taking enough strips.

Let us carry out the actual computations that are required in this case. For the sake of simplicity, we subdivide the base into n equal parts, each of length b/n (see Figure 1.6). The points of subdivision correspond to the following values of x :

$$0, \frac{b}{n}, \frac{2b}{n}, \frac{3b}{n}, \dots, \frac{(n-1)b}{n}, \frac{nb}{n} = b.$$

A typical point of subdivision corresponds to $x = kb/n$, where k takes the successive values $k = 0, 1, 2, 3, \dots, n$. At each point kb/n we construct the outer rectangle of altitude $(kb/n)^2$ as illustrated in Figure 1.6(a). The area of this rectangle is the product of its base and altitude and is equal to

† We use rectangles rather than arbitrary polygons to simplify the computations.

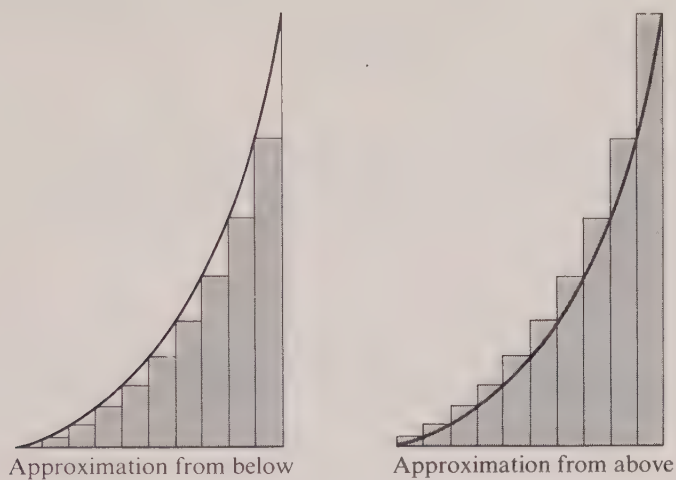


FIGURE 1.5

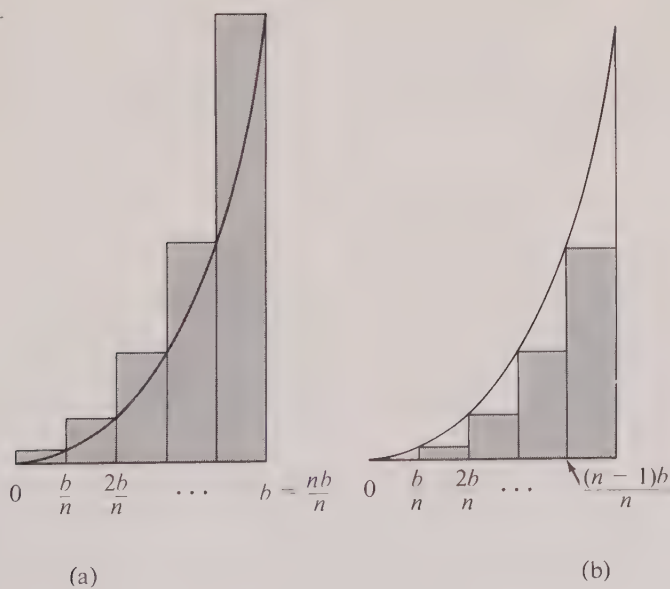


FIGURE 1.6

$$\left(\frac{b}{n}\right)\left(\frac{kb}{n}\right)^2 = \frac{b^3}{n^3} k^2.$$

Let us denote by S_n the sum of the areas of all the outer rectangles. Then, since the k th rectangle has area $(b^3/n^3)k^2$, we obtain the formula

$$(1.1) \quad S_n = \frac{b^3}{n^3} (1^2 + 2^2 + 3^2 + \cdots + n^2).$$

In the same way, by referring to Figure 1.6(b), we obtain a formula for the sum s_n of all the inner rectangles:

$$(1.2) \quad s_n = \frac{b^3}{n^3} [1^2 + 2^2 + 3^2 + \cdots + (n-1)^2].$$

This brings us to a very important stage in the calculation. Notice that the factor multiplying b^3/n^3 in Equation (1.1) is the sum of the squares of the first n integers:

$$1^2 + 2^2 + \cdots + n^2.$$

[The corresponding factor in Equation (1.2) is similar except that the sum has only $n-1$ terms.] For a large value of n the computation of this sum by direct addition of its terms is tedious and inconvenient. Fortunately there is an interesting identity which makes it possible to evaluate this sum in a simpler way, namely,

$$(1.3) \quad 1^2 + 2^2 + \cdots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}.$$

This identity is valid for every integer $n \geq 1$ and it may be proved by mathematical induction.† By subtracting n^2 from both sides of (1.3), we obtain the companion identity

$$(1.4) \quad 1^2 + 2^2 + \cdots + (n-1)^2 = \frac{n^3}{3} - \frac{n^2}{2} + \frac{n}{6}.$$

This also is valid for all $n \geq 1$ if we interpret the sum on the left to mean 0 when $n = 1$.

For our purposes we do not need the exact expressions given in the right-hand members of (1.3) and (1.4). All we need are the two *inequalities*

$$(1.5) \quad 1^2 + 2^2 + \cdots + (n-1)^2 < \frac{n^3}{3} < 1^2 + 2^2 + \cdots + n^2$$

which are valid for every integer $n \geq 1$. These inequalities may be deduced easily as consequences of (1.3) and (1.4) or they may be proved directly by induction. (A proof by induction is given in Section 1.17.)

If we multiply both inequalities in (1.5) by b^3/n^3 and make use of (1.1) and (1.2), we obtain

$$(1.6) \quad s_n < \frac{b^3}{3} < S_n$$

† A discussion of “proofs by induction” is given in Section 1.17.

for every n , and we note the appearance of the number $b^3/3$. The inequalities in (1.6) tell us that $b^3/3$ is a number which lies between s_n and S_n for every n . But now it is easy to prove that $b^3/3$ is the *only* number which has this property. In other words, we assert that if A is any number which satisfies the inequalities

$$(1.7) \quad s_n < A < S_n$$

for every positive integer n , then we must necessarily have $A = b^3/3$. It is because of this statement that Archimedes concluded that the area of the parabolic segment is $b^3/3$.

To prove that $A = b^3/3$ we use the inequalities in (1.5) once more. Adding n^2 to both sides of the leftmost inequality in (1.5), we obtain

$$1^2 + 2^2 + \cdots + n^2 < \frac{n^3}{3} + n^2.$$

Multiplying this by b^3/n^3 and using (1.1), we find

$$(1.8) \quad S_n < \frac{b^3}{3} + \frac{b^3}{n}.$$

Similarly, by subtracting n^2 from both sides of the rightmost inequality in (1.5) and multiplying by b^3/n^3 , we are led to the inequality

$$(1.9) \quad \frac{b^3}{3} - \frac{b^3}{n} < s_n.$$

Therefore, any number A satisfying (1.7) must also satisfy

$$(1.10) \quad \frac{b^3}{3} - \frac{b^3}{n} < A < \frac{b^3}{3} + \frac{b^3}{n}$$

for every integer $n \geq 1$. Now there are only three possibilities:

$$A > \frac{b^3}{3}, \quad A < \frac{b^3}{3}, \quad A = \frac{b^3}{3}.$$

If we show that each of the first two leads to a contradiction then we must have $A = b^3/3$ since, in the manner of Sherlock Holmes, this exhausts all the possibilities.

Suppose the inequality $A > b^3/3$ were true. From the second inequality in (1.10) we obtain

$$(1.11) \quad A - \frac{b^3}{3} < \frac{b^3}{n}$$

for every integer $n \geq 1$. Since $A - b^3/3$ is positive, we may divide both sides of (1.11) by $A - b^3/3$ and then multiply by n to obtain the equivalent statement

$$n < \frac{b^3}{A - b^3/3}$$

for every n . But this inequality is obviously false when $n > b^3/(A - b^3/3)$. Hence the inequality $A > b^3/3$ leads to a contradiction. By a similar argument we may show that the inequality $A < b^3/3$ also leads to a contradiction and therefore we must have $A = b^3/3$, as asserted.

1.3 Exercises

1. (a) Modify the region in Figure 1.3 by assuming that the ordinate at each x is $2x^2$ instead of x^2 . Draw the new figure. Check through the principal steps in the foregoing section and find what effect this has on the calculation of A . Do the same if the ordinate at each x is

- (b) $3x^2$,
- (c) $\frac{1}{4}x^2$,
- (d) $2x^2 + 1$,
- (e) $ax^2 + c$.

2. Modify the region in Figure 1.3 by assuming that the ordinate at each x is x^3 instead of x^2 . Draw the new figure.

(a) Use a construction similar to that illustrated in Figure 1.6 and show that the outer and inner sums S_n and s_n are given by

$$S_n = \frac{b^4}{n^4} (1^3 + 2^3 + \cdots + n^3), \quad s_n = \frac{b^4}{n^4} [1^3 + 2^3 + \cdots + (n-1)^3].$$

(b) Use the inequalities (which can be proved by mathematical induction; see Section 1.17)

$$(1.12) \quad 1^3 + 2^3 + \cdots + (n-1)^3 < \frac{n^4}{4} < 1^3 + 2^3 + \cdots + n^3$$

to show that $s_n < b^4/4 < S_n$ for every n , and prove that $b^4/4$ is the *only* number which lies between s_n and S_n for every n .

(c) What number takes the place of $b^4/4$ if the ordinate at each x is $ax^3 + c$?

3. The inequalities (1.5) and (1.12) are special cases of the more general inequalities

$$(1.13) \quad 1^k + 2^k + \cdots + (n-1)^k < \frac{n^{k+1}}{k+1} < 1^k + 2^k + \cdots + n^k$$

that are valid for every integer $n \geq 1$ and every integer $k \geq 1$. Assume the validity of (1.13) and generalize the results of Exercise 2.

1.4 A critical analysis of Archimedes' method

From calculations similar to those in Section 1.2 Archimedes concluded that the area of the parabolic segment in question is $b^3/3$. This fact was generally accepted as a mathematical theorem for nearly 2000 years before it was realized that one must reexamine the result from a more critical point of view. To understand why anyone would question the validity of Archimedes' conclusion it is necessary to know something about the important changes that have taken place in the recent history of mathematics.

Every branch of knowledge is a collection of ideas described by means of words and symbols and one cannot understand these ideas unless one knows the exact meanings of the words and symbols that are used. Certain branches of knowledge, known as *deductive systems*, are different from others in that a number of "undefined" concepts are chosen

in advance and all other concepts in the system are defined in terms of these. Certain statements about these undefined concepts are taken as *axioms* or *postulates* and other statements that can be deduced from these axioms are called *theorems*. The most familiar example of a deductive system is the Euclidean theory of elementary geometry that has been studied by well-educated men since the time of the ancient Greeks.

The spirit of early Greek mathematics, with its emphasis on the theoretical and postulational approach to geometry as presented in Euclid's *Elements*, dominated the thinking of mathematicians until the time of the Renaissance. A new and vigorous phase in the development of mathematics began with the advent of algebra in the 16th century and the next 300 years witnessed a flood of important discoveries. Conspicuously absent from this period was the logically precise reasoning of the deductive method with its use of axioms, definitions, and theorems. Instead, the pioneers in the 16th, 17th, and 18th centuries resorted to a curious blend of deductive reasoning combined with intuition, pure guesswork, and mysticism and it is not surprising to find that some of their work was later shown to be incorrect. However, a surprisingly large number of important discoveries emerged from this era and a great deal of the work has survived the test of history—a tribute to the unusual skill and ingenuity of these pioneers.

As the flood of new discoveries began to recede, a new and more critical period emerged. Little by little, mathematicians felt forced to return to the classical ideals of the deductive method in an attempt to put the new mathematics on a firm foundation. This phase of the development, which began early in the 19th century and has continued to the present day, has resulted in a degree of logical purity and abstraction that has surpassed all the traditions of Greek science. At the same time, it has brought about a clearer understanding of the foundations of not only calculus but of all of mathematics.

There are many ways to develop calculus as a deductive system. One possible approach is to take the real numbers as the undefined objects. Some of the rules governing the operations on real numbers may then be taken as axioms. One such set of axioms is listed in Supplement A, pp. 12–18. New concepts, such as *integral*, *limit*, *continuity*, *derivative*, must then be defined in terms of real numbers. Properties of these concepts are then deduced as theorems that follow from the axioms.

Looked at as part of the deductive system of calculus, Archimedes' result about the area of a parabolic segment cannot be accepted as a theorem until a satisfactory definition of area is given first. It is not clear whether Archimedes had ever formulated a precise definition of what he meant by area. He seems to have taken it for granted that every region has an area associated with it. On this assumption he then set out to calculate areas of particular regions. In his calculations he made use of certain facts about area that cannot be proved until we know what is *meant* by area. For instance, he assumed that if one region lies inside another, the area of the smaller region cannot exceed that of the larger region. Also, if a region is decomposed into two or more parts, the sum of the areas of the individual parts is equal to the area of the whole region. All these are properties we would like area to possess and we shall insist that any definition of area should be such that these properties may be derived as theorems. It is quite possible that Archimedes himself may have taken area to be an undefined concept and then used the properties we just mentioned as *axioms*.

Today we consider the work of Archimedes as being important not so much because it helps us to compute areas of particular figures, but rather because it suggests a reasonable way to *define* the concept of area for more or less *arbitrary* figures. As it turns out,

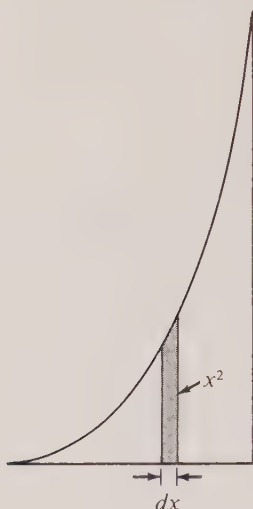


FIGURE 1.7 The area of a parabolic segment conceived of as a sum of infinitesimal rectangles.

the method of Archimedes suggests a way to define a much more general concept known as the *integral*. The integral, in turn, is used to define and compute not only area but also concepts such as arc length, volume, work, and others. When we try to find the work done by a nonconstant force that moves a particle from one point to another, the absence of an “intuitive picture” emphasizes the primary need to define what is meant by work.

If we look ahead and make use of the terminology of integral calculus, the result of the calculation carried out in Section 1.2 for the parabolic segment is often stated as follows:

“The integral of x^2 from 0 to b is $b^3/3$.”

It is written symbolically as

$$\int_0^b x^2 dx = \frac{b^3}{3}.$$

The symbol $\int$ (an elongated S) is called an *integral sign* and it was introduced by Leibniz in 1675. The process which produces the number $b^3/3$ is called *integration*. The numbers 0 and b which are attached to the integral sign are referred to as the *limits of integration*. The symbol $\int_0^b x^2 dx$ must be regarded as a whole. Its definition will treat it as such, just as the dictionary describes the word “lapidate” without reference to “lap,” “id,” or “ate.”

Leibniz’ symbol for the integral was readily accepted by many early mathematicians because they liked to think of integration as a kind of “summation process” which enabled them to add together infinitely many “infinitesimally small quantities.” For example, in the case of the parabolic segment, the area was conceived of as the “sum” of infinitely many infinitesimally small rectangles of height x^2 and base dx , as suggested in Figure 1.7. The

symbol† dx was used to represent a very small or infinitesimal change in x and the integral sign represented the process of summing together the areas of all these thin rectangles. This kind of thinking is very suggestive and often very helpful. Logically, however, it suffers from the defect that no one has been able to assign an exact meaning to the idea of an “infinitesimally small quantity.” Today we know how to introduce the integral in terms of the notion of real number without using mysterious and unexplained ideas like “infinitesimals.” This definition will be given in Part II of this chapter.

1.5 The approach to calculus to be used in this book

A thorough and complete treatment of either integral or differential calculus depends ultimately on a careful study of the real-number system. This study in itself, when carried out in full, is an interesting but somewhat lengthy program that requires a small volume for its complete exposition. The approach in this book is to begin with the real numbers as *undefined objects* and simply to list a number of fundamental properties of real numbers which we shall take as *axioms*. These axioms and some of the simplest theorems that can be deduced from them are discussed in a supplement which appears in the next few pages. Most of the properties of real numbers that are listed here are probably familiar to the reader from his study of elementary algebra. However, there are a few properties of real numbers that do not ordinarily come into consideration in elementary algebra but which play an important role in the calculus. These properties stem from the so-called *least-upper-bound axiom* (also known as the *continuity axiom*) which is dealt with here in some detail. The reader may wish to study the supplement before proceeding with the main body of the text, or he may postpone reading this material until later when he reaches those parts of the theory that make use of least-upper-bound properties. Material in the text that depends on the least-upper-bound axiom will be indicated clearly with specific references to the appropriate part of the supplement.

To develop calculus as a complete, formal mathematical theory it would be necessary to state, in addition to the axioms for the real-number system, a list of the various “methods of proof” which would be permitted for the purpose of deducing theorems from the axioms. Every statement in the theory would then have to be justified either as an “established law” (that is, an axiom, a definition, or a previously proved theorem) or as the result of applying one of the acceptable methods of proof to an established law. A program of this sort would be extremely long and tedious and would add very little to a beginner’s understanding of the subject. Fortunately, it is not necessary to proceed in this fashion in order to get a good understanding and a good working knowledge of calculus. In this book the subject is introduced in an informal way and ample use is made of geometric reasoning whenever it is convenient to do so. At the same time, the discussion proceeds in a manner that is consistent with modern standards of precision and clarity of thought. All the important theorems of the subject are explicitly stated and the essential details of their proofs are presented in full.

To avoid interrupting the principal flow of ideas, some of the proofs appear in separate starred sections. For the same reason, some of the chapters are accompanied by supplementary sections in which certain important topics related to calculus are dealt with in detail. Some of these are also starred to indicate that they may be omitted or postponed without disrupting the continuity of the presentation. The extent to which the starred

† The symbol “ dx ” is to be regarded as a single entity. It does *not* mean d times x .

sections are taken up or not will depend partly on the reader's background and skill in algebra and partly on the depth of his interests. A person who wants a thorough course in calculus, including theory as well as techniques, should read all the material. Those interested primarily in the basic ideas and techniques may omit the starred sections. Two groups of supplementary sections appear in the next few pages. The first group, called Supplement A, deals with the axioms for the real-number system. The second group, Supplement B, has to do with the concept of mathematical induction, and the summation notation. The reader should become familiar with the material in the unstarred sections in Supplement B before proceeding to the development of integral calculus in Section 1.25.

Supplements to Part I

Supplement A. A set of axioms for the real-number system

★1.6 Introduction

There are many ways to introduce the real-number system. One popular method is to begin with the positive integers 1, 2, 3, . . . and use them as building blocks to construct a more comprehensive system having the properties desired. Briefly, the idea of this method is to take the positive integers as undefined concepts, state some axioms concerning them, and then use the positive integers to build a larger system consisting of the positive *rational* numbers (quotients of positive integers). The positive rational numbers, in turn, may then be used as a basis for constructing the positive *irrational* numbers (real numbers like $\sqrt{2}$ and π that are not rational). The final step is the introduction of the negative real numbers and zero. The most difficult part of the whole process is the transition from the rational numbers to the irrational numbers.

Although the need for irrational numbers was apparent to the ancient Greeks from their study of geometry, satisfactory methods for constructing irrational numbers from rational numbers were not introduced until late in the 19th century. At that time, three different theories were outlined by Karl Weierstrass (1815–1897), Georg Cantor (1845–1918), and Richard Dedekind (1831–1916). In 1889, the Italian mathematician Guiseppe Peano (1858–1932) listed five axioms for the positive integers that could be used as the starting point of the whole construction. A detailed account of this construction, beginning with the Peano postulates and using the method of Dedekind to introduce irrational numbers, may be found in a book by E. Landau, *Foundations of Analysis* (New York, Chelsea Publishing Co., 1951).

The point of view we shall adopt here is nonconstructive. We shall start rather far out in the process, taking the real numbers themselves as undefined objects satisfying a number of properties that we use as axioms. That is to say, we shall assume there exist certain objects, called real numbers, which satisfy the 10 axioms listed in the next five sections. All the properties of real numbers that we shall use in this book are in this list or can be deduced from the axioms in the list. When the real numbers are defined by a constructive process, the properties we list as axioms must be proved as theorems.

In the axioms that appear below, lower-case letters $a, b, c, \dots, x, y, z$ represent arbi-

rary real numbers unless something is said to the contrary. The axioms fall in a natural way into three groups which we refer to as the *field axioms*, the *order axioms*, and the *least-upper-bound axiom* (also called the *axiom of continuity* or the *completeness axiom*).

*1.7 The field axioms

Along with the set of real numbers we assume the existence of two operations called *addition* and *multiplication* such that for every pair of real numbers x and y we can form the *sum* of x and y , which is another real number denoted by $x + y$, and the *product* of x and y , denoted by xy or by $x \cdot y$. The sum $x + y$ and the product xy are uniquely determined by x and y . We attach no special meanings to the symbols $+$ and $\cdot$ other than those contained in the axioms.

- AXIOM 1 *Commutative laws.* $x + y = y + x$, $xy = yx$.
- AXIOM 2 *Associative laws.* $x + (y + z) = (x + y) + z$, $x(yz) = (xy)z$.
- AXIOM 3 *Distributive law.* $x(y + z) = xy + xz$.
- AXIOM 4 *Existence of identity elements.* There exist two distinct real numbers, which we denote by 0 and 1, such that for every real x we have $0 + x = x + 0 = x$ and $1 \cdot x = x \cdot 1 = x$.
- AXIOM 5 *Existence of negatives.* For every real number x there is a real number y such that $x + y = y + x = 0$.
- AXIOM 6 *Existence of reciprocals.* For every real number $x \neq 0$ there is a real number y such that $xy = yx = 1$.

Note. The numbers 0 and 1 in Axioms 5 and 6 are those of Axiom 4.

From the above axioms we can deduce all the usual laws of arithmetic with which the reader is familiar from his study of elementary algebra. The most important of these laws are collected here as a list of theorems. In all these theorems the symbols a, b, c, d represent arbitrary real numbers.

- 1- 1 THEOREM. *Cancellation law for addition.* If $a + b = a + c$, then $b = c$. (In particular, this shows that the number 0 of Axiom 4 is unique.)
- 1- 2 THEOREM. *Possibility of subtraction.* Given a and b , there is exactly one x such that $a + x = b$. This x is denoted by $b - a$. In particular, $0 - a$ is written simply $-a$ and is called the negative of a .
- 1- 3 THEOREM. $b - a = b + (-a)$.
- 1- 4 THEOREM. $-(-a) = a$.
- 1- 5 THEOREM. $a(b - c) = ab - ac$.
- 1- 6 THEOREM. $0 \cdot a = a \cdot 0 = 0$.
- 1- 7 THEOREM. *Cancellation law for multiplication.* If $ab = ac$ and $a \neq 0$, then $b = c$. (In particular, this shows that the number 1 of Axiom 4 is unique.)
- 1- 8 THEOREM. *Possibility of division.* Given a and b with $a \neq 0$, there is exactly

one x such that $ax = b$. This x is denoted by b/a or $\frac{b}{a}$ and is called the quotient of b and a . In particular, $1/a$ is also written a^{-1} and is called the reciprocal of a .

- 1-9 THEOREM. If $a \neq 0$, $b/a = b \cdot a^{-1}$.
 1-10 THEOREM. If $a \neq 0$, $(a^{-1})^{-1} = a$.
 1-11 THEOREM. If $ab = 0$, then $a = 0$ or $b = 0$.
 1-12 THEOREM. $(-a)b = -(ab)$ and $(-a)(-b) = ab$.
 1-13 THEOREM. $(a/b)(c/d) = (ac)/(bd)$ if $b \neq 0$ and $d \neq 0$.
 1-14 THEOREM. $(a/b)/(c/d) = (ad)/(bc)$ if $b \neq 0$, $c \neq 0$, and $d \neq 0$.
 1-15 THEOREM. $(a/b) + (c/d) = (ad + bc)/(bd)$ if $b \neq 0$ and $d \neq 0$.

To illustrate how these statements may be obtained as consequences of the axioms we shall present proofs of Theorems 1-1 through 1-4. The reader may find it instructive to carry out proofs of the remaining theorems.

Proof of 1-1. Given $a + b = a + c$. By Axiom 5 we can choose y so that $y + a = 0$. Therefore $y + (a + b) = y + (a + c)$. Using the associative law, we obtain $(y + a) + b = (y + a) + c$ or $0 + b = 0 + c$. But by Axiom 4 we have $0 + b = b$ and $0 + c = c$, so that $b = c$. Notice that this theorem shows that there is only one real number having the property of 0 in Axiom 4. In fact, if 0 and $0'$ both have this property, then $0 + 0' = 0$ and $0 + 0 = 0$. Hence $0 + 0' = 0 + 0$ and, by the cancellation law, $0 = 0'$.

Proof of 1-2. Given a and b , choose y so that $a + y = 0$ and let $x = y + b$. Then $a + x = a + (y + b) = (a + y) + b = 0 + b = b$. Therefore there is at least one x such that $a + x = b$. But by Theorem 1-1 there is at most one such x . Hence there is *exactly* one.

Proof of 1-3. Let $x = b - a$ and let $y = b + (-a)$. We wish to prove that $x = y$. Now $x + a = b$ (by the definition of $b - a$) and $y + a = [b + (-a)] + a = b + [(-a) + a] = b + 0 = b$. Therefore $x + a = y + a$ and hence, by Theorem 1-1, $x = y$.

Proof of 1-4. We have $a + (-a) = 0$ by the definition of $-a$. But this equation tells us that a is the negative of $-a$. That is, $a = -(-a)$, as asserted.

The number 1 has been defined in Axiom 4. The number 2 is defined as $1 + 1$; 3 as $2 + 1$; 4 as $3 + 1$; and so on. The numbers obtained in this way are called the *positive integers*. Their negatives, $-1, -2, -3, \dots$, are called the *negative integers*. The word *integer* by itself can mean a positive or negative integer or 0 (zero).

Quotients of integers a/b (where $b \neq 0$) are called *rational* numbers. These include, in particular, all the integers. The reader should note that all six of the foregoing axioms are satisfied by the collection of rational numbers and that all, except Axiom 6 (which has to do with the existence of reciprocals), are satisfied by the collection of integers.

★1.8 Exercises

1. Prove Theorems 1–5 through 1–15 using Axioms 1 through 6 and Theorems 1–1 through 1–4 whenever it is convenient to do so.

In Exercises 2 through 10, prove the given statements or establish the given equations. You may use Axioms 1 through 6 and Theorems 1–1 through 1–15.

2. $-0 = 0$.
3. $1^{-1} = 1$.
4. Zero has no reciprocal.
5. $-(a + b) = -a - b$.
6. $-(a - b) = -a + b$.
7. $(a - b) + (b - c) = a - c$.
8. If $a \neq 0$ and $b \neq 0$, then $(ab)^{-1} = a^{-1}b^{-1}$.
9. $-(a/b) = (-a/b) = a/(-b)$ if $b \neq 0$.
10. $(a/b) - (c/d) = (ad - bc)/(bd)$ if $b \neq 0$ and $d \neq 0$.

★1.9 The order axioms

This group of axioms has to do with a concept which establishes an *ordering* among the real numbers. This ordering enables us to make statements about one real number being larger or smaller than another. We choose to introduce the order properties as a set of axioms about a new undefined concept called *positiveness* and then to define terms like *less than* and *greater than* in terms of positiveness.

We shall assume that among all the real numbers there exists a certain collection, which we call *positive* real numbers, that satisfies the following three order axioms:†

AXIOM 7. If x and y are positive, so are $x + y$ and xy .

AXIOM 8. For every real $x \neq 0$, either x is positive or $-x$ is positive, but not both.

AXIOM 9. The number 0 is not positive.

Now we can define the symbols $<$, $>$, $\leq$, and $\geq$, called, respectively, *less than*, *greater than*, *less than or equal to*, and *greater than or equal to*, as follows:

- $x < y$ means that $y - x$ is positive.
- $y > x$ means that $x < y$.
- $x \leq y$ means that either $x < y$ or $x = y$.
- $y \geq x$ means that $x \leq y$.

Thus, we have $x > 0$ if and only if x is positive. If $x < 0$, we say that x is *negative*; if $x \geq 0$, we say that x is *nonnegative*. A pair of simultaneous inequalities such as $x < y$, $y < z$ is usually written more briefly as $x < y < z$; similar interpretations are given to the compound inequalities $x \leq y < z$, $x < y \leq z$, and $x \leq y \leq z$.

Note. The order axioms are satisfied by the collection of rational numbers and also by the collection of integers.

† We attach no special meaning to the word *positive* other than that contained in these axioms. The positive integers, as defined at the end of Section 1.7, satisfy the properties of Axioms 7, 8, and 9, and hence they are examples of positive real numbers.

From the order axioms we can derive all the usual rules for calculating with inequalities. The most important of these are listed here as theorems.

- 1-16 THEOREM. For arbitrary real numbers a and b , exactly one of the three relations $a < b$, $b < a$, $a = b$ holds.
- 1-17 THEOREM. *Transitive law.* If $a < b$ and $b < c$, then $a < c$.
- 1-18 THEOREM. If $a < b$, then $a + c < b + c$.
- 1-19 THEOREM. If $a < b$ and $c > 0$, then $ac < bc$.
- 1-20 THEOREM. If $a \neq 0$, then $a^2 > 0$.
- 1-21 THEOREM. $1 > 0$.
- 1-22 THEOREM. If $a < b$ and $c < 0$, then $ac > bc$.
- 1-23 THEOREM. If $a < b$, then $-a > -b$. In particular, if $a < 0$ then $-a > 0$.
- 1-24 THEOREM. If $ab > 0$, then both a and b are positive or both are negative.
- 1-25 THEOREM. If $a < c$ and $b < d$, then $a + b < c + d$.

Again, we shall prove only a few of these theorems as samples to indicate how the proofs may be carried out. Proofs of the others are left as exercises for the reader.

Proof of 1-16. Let $x = b - a$. If $x = 0$, then $b - a = a - b = 0$ and hence, by Axiom 9, we cannot have $a > b$ or $b > a$. If $x \neq 0$, Axiom 8 tells us that either $x > 0$ or $x < 0$, but not both; that is, either $a < b$ or $b < a$ but not both. Therefore exactly one of the three relations $a = b$, $a < b$, $b < a$ holds.

Proof of 1-17. If $a < b$ and $b < c$, then $b - a > 0$ and $c - b > 0$. By Axiom 7 we may add to obtain $(b - a) + (c - b) > 0$. That is, $c - a > 0$ and hence $a < c$.

Proof of 1-18. Let $x = a + c$, $y = b + c$. Then $y - x = b - a$. But $b - a > 0$ since $a < b$. Hence $y - x > 0$ and this means that $x < y$.

Proof of 1-19. If $a < b$, then $b - a > 0$. If $c > 0$, then by Axiom 7 we may multiply c by $(b - a)$ to obtain $(b - a)c > 0$. But $(b - a)c = bc - ac$. Hence $bc - ac > 0$ and this means that $ac < bc$, as asserted.

Proof of 1-20. If $a > 0$, then $a \cdot a > 0$ by Axiom 7. If $a < 0$, then $-a > 0$ and hence $(-a) \cdot (-a) > 0$ by Axiom 7. In either case we have $a^2 > 0$.

Proof of 1-21. Apply Theorem 1-20 with $a = 1$.

★1.10 Exercises

1. Prove Theorems 1-22 through 1-25 using the earlier theorems and Axioms 1 through 9.

In Exercises 2 through 10, prove the given statements or establish the given inequalities. You may use Axioms 1 through 9 and Theorems 1-1 through 1-25.

2. There is no real number x such that $x^2 + 1 = 0$.

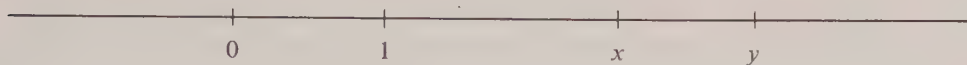


FIGURE 1.8 Real numbers represented geometrically on a line.

3. The sum of two negative numbers is negative.
4. If $a > 0$, then $1/a > 0$; if $a < 0$, then $1/a < 0$.
5. If $0 < a < b$, then $0 < b^{-1} < a^{-1}$.
6. If $a \leq b$ and $b \leq c$, then $a \leq c$.
7. If $a \leq b$ and $b \leq c$, and $a = c$, then $b = c$.
8. For all real a and b we have $a^2 + b^2 \geq 0$. If $ab \neq 0$, then $a^2 + b^2 > 0$.
9. There is no real number a such that $x \leq a$ for all real x .
10. If x has the property that $0 \leq x < h$ for every positive real number h , then $x = 0$.

★1.11 Geometric interpretation of real numbers

The reader is undoubtedly familiar with the geometric representation of real numbers by means of points on a straight line. A point is selected to represent 0 and another, to the right of 0, to represent 1, as illustrated in Figure 1.8. This choice determines the scale. If one adopts an appropriate set of axioms for Euclidean geometry, then each real number corresponds to exactly one point on this line and, conversely, each point on the line corresponds to one and only one real number. For this reason the line is often called the *real line* or the *real axis*, and it is customary to use the words *real number* and *point* interchangeably. Thus we often speak of the *point* x rather than the point corresponding to the real number x .

The ordering relation among the real numbers has a simple geometric interpretation. If $x < y$, the point x lies to the left of the point y , as shown in Figure 1.8. Positive numbers lie to the right of 0 and negative numbers to the left of 0. If $a < b$, a point x satisfies the inequalities $a < x < b$ if and only if x is *between* a and b .

This device for representing real numbers geometrically is a very worthwhile aid that helps us to discover and understand better certain properties of real numbers. However, the reader should realize that all properties of real numbers that are to be accepted as theorems must be deducible from the axioms without any reference to geometry. This does not mean that one should not make use of geometry in studying properties of real numbers. On the contrary, the geometry often suggests the method of proof of a particular theorem and sometimes a geometric argument is more illuminating than a purely *analytic* proof (one depending entirely on the axioms for the real numbers). In this book, geometric arguments are used to a large extent to help motivate or clarify a particular discussion. Nevertheless, the proofs of all the important theorems are presented in analytic form.

★1.12 The least-upper-bound axiom

The nine axioms listed above contain all the properties of real numbers usually discussed in elementary algebra. There is another axiom of fundamental importance in

calculus that is ordinarily not discussed in elementary algebra courses. This axiom (or something equivalent to it) is needed to establish the existence of irrational numbers.

Irrational numbers arise in elementary algebra when we try to solve certain quadratic equations. For example, it is desirable to have a real number x such that $x^2 = 2$. From the nine axioms above we cannot prove that such an x exists in the real-number system simply because these nine axioms are also satisfied by the rational numbers alone and there is no rational number x whose square is 2.† Axiom 10 allows us to introduce irrational numbers in the real-number system. We shall find also that it gives the real-number system a property of continuity that is especially important in the study of calculus.

Before describing Axiom 10, we shall introduce some special terminology. Let S denote any collection of real numbers. Such a collection is referred to more briefly as a *set*. The individual numbers in the collection are called *elements* or *members* of the set and they are said to *belong to* the set. If there is a real number b such that $x \leq b$ for every x in the collection S , then b is called an *upper bound* for the set S and we say that S is *bounded above*. We say *an* upper bound because if a set has one upper bound, say b , then every number greater than b is also an upper bound. Of course, a set need not have any upper bound at all. For example, the set of all positive real numbers has none. A set with no upper bound is said to be *unbounded above*.

The set N of all negative numbers is an example of a set which is bounded above. In fact, any positive real number b can be used as an upper bound for this set. The number 0 itself is also an upper bound for N . However, no number smaller than 0 can be an upper bound for N . We describe this by saying that 0 is the *least upper bound* of N .

In general, a number b is called a least upper bound of a set S of real numbers if b has the following two properties:

- (i) b is an upper bound for S , and
- (ii) no number $b' < b$ is an upper bound for S .

Because of property (ii), there cannot be more than one least upper bound for a given set S . Therefore, if a set has any least upper bound at all, it has *only* one, and we may speak of *the* least upper bound.

If S contains no numbers whatever (such a set is said to be *empty*), then every real number is an upper bound for S and in this case S has *no* least upper bound. The least-upper-bound axiom has to do with nonempty sets that are bounded above. (The term *nonempty* means that there is at least one real number in the set.)

AXIOM 10. If S is a nonempty set of real numbers that is bounded above, then there exists exactly one real number that is the least upper bound of S .

Definitions of the terms *lower bound*, *bounded below*, *unbounded below*, and *greatest lower bound* may be similarly formulated. Using Axiom 10, one can prove, as a theorem, that every nonempty set S that is bounded below has a greatest lower bound. In fact, let $-S$ denote the set of negatives of numbers in S . Then $-S$ is nonempty and bounded

† To prove this statement we may argue as follows: Suppose there exists a rational number whose square is 2, say $(a/b)^2 = 2$, where a and b are integers without a common factor. Then $a^2 = 2b^2$ and hence a^2 is even. Therefore a is even, say $a = 2c$. The equation $a^2 = 2b^2$ becomes $4c^2 = 2b^2$, or $2c^2 = b^2$, so b^2 , and hence b , is even. This contradicts the assumption that a and b have no common factor.

above and hence, by Axiom 10, $-S$ has a least upper bound, say b . It is easy to verify that $-b$ is the greatest lower bound of S .

To illustrate the power of the least-upper-bound axiom, we shall derive in the next two sections a number of rather important properties of the real-number system.

★1.13 The Archimedean property of the real-number system

1-26 THEOREM. The set S of positive integers $1, 2, 3, \dots$ is unbounded above.

Proof. Assume S is bounded above. We shall show that this leads to a contradiction. Since S is nonempty, Axiom 10 tells us that S has a least upper bound, say b . The number $b - 1$, being less than b , cannot be an upper bound for S . Hence there is at least one positive integer n such that $n > b - 1$. For this n we have $n + 1 > b$. Since $n + 1$ is in S , this contradicts the fact that b is an upper bound for S .

As corollaries of Theorem 1-26, we immediately obtain the following consequences:

1-27 THEOREM. For every real x there exists a positive integer n such that $n > x$.

Proof. If this were not so, x would be an upper bound for the set of positive integers, contradicting Theorem 1-26.

1-28 THEOREM. For every real $x > 0$ there exists a positive integer n such that $1/n < x$.

Proof. Apply Theorem 1-27 with x replaced by $1/x$.

1-29 THEOREM. If $x > 0$ and if y is an arbitrary real number, there exists a positive integer n such that $nx > y$.

Proof. Apply Theorem 1-27 with x replaced by y/x .

The property described in Theorem 1-29 is often called the *Archimedean property* of the real-number system. Geometrically it means that any line segment, no matter how long, may be covered by a finite number of line segments of a given positive length, no matter how small. In other words, a small ruler can measure arbitrarily large distances if it is repeatedly laid end to end. Archimedes, realizing that this was a fundamental property of the straight line, stated it explicitly as one of the axioms of geometry. In the 19th and 20th centuries, non-Archimedean geometries have been constructed in which this axiom is rejected.

★1.14 Exercises

1. If x and y are arbitrary real numbers, $x < y$, prove that there exists at least one real z satisfying $x < z < y$.

2. If x is an arbitrary real number, prove that there exist integers m and n such that $m < x < n$. [*Hint.* Use Theorem 1-27.]

3. If x is an arbitrary real number, prove that there is exactly one integer n which satisfies the inequalities

$$n \leq x < n + 1.$$

This n is called the *greatest integer in x* and is denoted by $[x]$. For example, $[5] = 5$, $[5/2] = 2$, $[-8/3] = -3$.

4. If x is an arbitrary real number, prove that there is exactly one integer n which satisfies $x \leq n < x + 1$.

5. If x and y are arbitrary real numbers, $x < y$, prove that there exists at least one rational number r satisfying $x < r < y$, and hence infinitely many. This property is often described by saying that the rational numbers are *dense* in the real-number system.

6. If x is rational, $x \neq 0$, and y irrational, prove that $x + y$, $x - y$, xy , x/y , and y/x are all irrational.

7. Is the sum or product of two irrational numbers always irrational?

8. If x and y are arbitrary real numbers, $x < y$, prove that there exists at least one irrational number z satisfying $x < z < y$, and hence infinitely many.

★1.15 Existence of square roots of nonnegative real numbers

It was pointed out earlier that the equation $x^2 = 2$ has no solutions among the rational numbers. With the help of Axiom 10 we can prove that the equation $x^2 = a$ has a solution among the *real* numbers if $a \geq 0$. Each such x is called a *square root* of a .

First, let us see what we can say about square roots without using Axiom 10. Negative numbers cannot have square roots because if $x^2 = a$, then a , being a square, must be nonnegative (by Theorem 1-20). Moreover, if $a = 0$, then $x = 0$ is the only square root (by Theorem 1-11). Suppose, then, that $a > 0$. If $x^2 = a$, then $x \neq 0$ and $(-x)^2 = a$, so both x and its negative are square roots. In other words, if a has a square root, then it has two square roots, one positive and one negative. Also, it has *at most two* because if $x^2 = a$ and $y^2 = a$, then $x^2 = y^2$ and $(x - y)(x + y) = 0$, and so, by Theorem 1-11, either $x = y$ or $x = -y$. Thus, if a has a square root, it has *exactly two*.

The existence of at least one square root will follow later from an important theorem in calculus known as the intermediate-value theorem for continuous functions, but it may be instructive to see how the existence of a square root can be proved directly from Axiom 10.

1-30 THEOREM. Every nonnegative real number a has a unique nonnegative square root.

Note. If $a \geq 0$, we denote its nonnegative square root by $a^{1/2}$ or by $\sqrt{a}$. If $a > 0$, the negative square root is $-a^{1/2}$ or $-\sqrt{a}$.

Proof. If $a = 0$, then 0 is the only square root. Assume, then, that $a > 0$. Let S be the set of all positive x such that $x^2 \leq a$. Since $(1 + a)^2 > a$, the number $1 + a$ is an upper bound for S . Also, S is nonempty because the number $a/(1 + a)$ is in S ; in fact, $a^2 \leq a(1 + a)^2$ and hence $a^2/(1 + a)^2 \leq a$. By Axiom 10, S has a least upper bound which we shall call b . Note that $b \geq a/(1 + a)$ so $b > 0$. There are only three possibilities: $b^2 > a$, $b^2 < a$, $b^2 = a$.

Suppose $b^2 > a$ and let $c = b - (b^2 - a)/(2b) = \frac{1}{2}(b + a/b)$. Then $0 < c < b$ and $c^2 = b^2 - (b^2 - a) + (b^2 - a)^2/(4b^2) = a + (b^2 - a)^2/(4b^2) > a$. Therefore $c^2 > x^2$ for each x in S and hence $c > x$ for each x in S . This means that c is an upper bound for S . Since $c < b$, we have a contradiction because b was the *least* upper bound for S . Therefore the inequality $b^2 > a$ is impossible.

Suppose $b^2 < a$. Since $b > 0$ we may choose a positive number c such that $c < b$ and such that $c < (a - b^2)/(3b)$. Then we have

$$(b + c)^2 = b^2 + c(2b + c) < b^2 + 3bc < b^2 + (a - b^2) = a.$$

Therefore $b + c$ is in S . Since $b + c > b$, this contradicts the fact that b is an upper bound for S . Therefore the inequality $b^2 < a$ is impossible and the only remaining alternative is $b^2 = a$.

The least-upper-bound axiom can also be used to show the existence of roots of higher order. For example, if n is a positive *odd* integer, then for each real x there is exactly one real y such that $y^n = x$. This y is called the *n th root* of x and is denoted by

$$(1.14) \quad y = x^{1/n} \quad \text{or} \quad y = \sqrt[n]{x}.$$

When n is *even*, the situation is slightly different. In this case, if x is negative, there is no real y such that $y^n = x$ because $y^n \geq 0$ for all real y . However, if x is positive, it can be shown that there is one and only one positive y such that $y^n = x$. This y is called the *positive n th root* of x and is denoted by the symbols in (1.14). Since n is even, $(-y)^n = y^n$ and hence each $x > 0$ has two real n th roots, y and $-y$. However, the symbols $x^{1/n}$ and $\sqrt[n]{x}$ are reserved for the *positive n th root*.

*1.16 Representation of real numbers by decimals

A real number of the form

$$(1.15) \quad r = a_0 + \frac{a_1}{10} + \frac{a_2}{10^2} + \cdots + \frac{a_n}{10^n},$$

where a_0 is a nonnegative integer and $a_1, a_2, \dots, a_n$ are integers satisfying $0 \leq a_i \leq 9$, is usually written more briefly as follows:

$$r = a_0.a_1a_2 \cdots a_n.$$

This is said to be a *finite decimal representation* of r . For example,

$$\frac{1}{2} = \frac{5}{10} = 0.5, \quad \frac{1}{50} = \frac{2}{10^2} = 0.02, \quad \frac{29}{4} = 7 + \frac{2}{10} + \frac{5}{10^2} = 7.25.$$

Real numbers like these are necessarily rational and, in fact, they all have the form $r = a/10^n$, where a is an integer. However, not all rational numbers can be expressed with finite decimal representations. For example, if $\frac{1}{3}$ could be so expressed, then we would have $\frac{1}{3} = a/10^n$ or $3a = 10^n$ for some integer a . But this is impossible since 3 is not a factor of any power of 10.

Nevertheless, we can approximate an arbitrary real number $x > 0$ to any desired degree of accuracy by a sum of the form (1.15) if we take n large enough. The reason for this may be seen by the following geometric argument: If x is not an integer, then x lies between two consecutive integers, say $a_0 < x < a_0 + 1$. The segment joining a_0 and $a_0 + 1$ may be subdivided into ten equal parts. If x is not one of the subdivision points,

then x must lie between two consecutive subdivision points. This gives us a pair of inequalities of the form

$$a_0 + \frac{a_1}{10} < x < a_0 + \frac{a_1 + 1}{10},$$

where a_1 is an integer ($0 \leq a_1 \leq 9$). Next we divide the segment joining $a_0 + a_1/10$ and $a_0 + (a_1 + 1)/10$ into ten equal parts (each of length 10^{-2}) and continue the process. If after a finite number of steps a subdivision point coincides with x , then x is a number of the form (1.15). Otherwise the process continues indefinitely and it generates an infinite set of integers $a_1, a_2, a_3, \dots$. In this case, we say that x has the infinite decimal representation

$$x = a_0.a_1a_2a_3 \dots$$

At the n th stage, x satisfies the inequalities

$$a_0 + \frac{a_1}{10} + \dots + \frac{a_n}{10^n} < x < a_0 + \frac{a_1}{10} + \dots + \frac{a_n + 1}{10^n}.$$

This gives us two approximations to x , one from above and one from below, by finite decimals that differ by 10^{-n} . Therefore we can achieve any desired degree of accuracy in our approximations by taking n large enough.

When $x = \frac{1}{3}$, it is easy to verify that $a_0 = 0$ and $a_n = 3$ for all $n \geq 1$ and hence the corresponding infinite decimal expansion is

$$\frac{1}{3} = 0.333 \dots$$

Every irrational number has an infinite decimal representation. For example, when $x = \sqrt{2}$ we may calculate by trial and error as many digits in the expansion as we wish. Thus, $\sqrt{2}$ lies between 1.4 and 1.5 because $(1.4)^2 < 2 < (1.5)^2$. Similarly, by squaring and comparing with 2, we find the following further approximations:

$$1.41 < \sqrt{2} < 1.42, \quad 1.414 < \sqrt{2} < 1.415, \quad 1.4142 < \sqrt{2} < 1.4143.$$

Note that the foregoing process generates a succession of intervals of lengths 10^{-1} , 10^{-2} , 10^{-3} , $\dots$, each contained in the preceding and each containing the point x . This is an example of what is known as a sequence of *nested intervals*, a concept that is sometimes used as a basis for constructing the irrational numbers from the rational numbers.

Since we shall do very little with decimals in this book, we shall not develop their properties in any further detail except to mention how decimal expansions may be defined analytically with the help of the least-upper-bound axiom.

If x is a given positive real number, let a_0 denote the largest integer $\leq x$. Having chosen a_0 , we let a_1 denote the largest integer such that

$$a_0 + \frac{a_1}{10} \leq x.$$

More generally, having chosen $a_0, a_1, \dots, a_{n-1}$, we let a_n denote the largest integer such that

$$(1.16) \quad a_0 + \frac{a_1}{10} + \frac{a_2}{10^2} + \cdots + \frac{a_n}{10^n} \leq x.$$

Let S denote the set of all numbers

$$(1.17) \quad a_0 + \frac{a_1}{10} + \frac{a_2}{10^2} + \cdots + \frac{a_n}{10^n}$$

obtained in this way for $n = 0, 1, 2, \dots$. Then S is nonempty and bounded above and it is easy to verify that x is actually the least upper bound of S . The integers $a_0, a_1, a_2, \dots$ so obtained may be used to define a decimal expansion of x if we write

$$x = a_0.a_1a_2a_3 \cdots$$

to mean that the n th digit a_n is the largest integer satisfying (1.16). For example, if $x = \frac{1}{8}$, we find $a_0 = 0, a_1 = 1, a_2 = 2, a_3 = 5$, and $a_n = 0$ for all $n \geq 4$. Therefore we may write

$$\frac{1}{8} = 0.125000 \cdots$$

If in (1.16) we replace the inequality sign $\leq$ by $<$, we obtain a slightly different definition of decimal expansions. The least upper bound of all numbers of the form (1.17) is again x although the integers $a_0, a_1, a_2, \dots$ need not be the same as those which satisfy (1.16). For example, if this second definition is applied to $x = \frac{1}{8}$, we find $a_0 = 0, a_1 = 1, a_2 = 2, a_3 = 4$, and $a_n = 9$ for all $n \geq 4$. This leads to the infinite decimal representation

$$\frac{1}{8} = 0.124999 \cdots$$

The fact that a real number might have two different decimal representations is merely a reflection of the fact that two different sets of real numbers can have the same least upper bound.

Supplement B. Mathematical induction, summation notation, and related topics

***1.17 Mathematical induction**

There is no *largest* integer because when we add 1 to an integer k we obtain $k + 1$, which is larger than k . Nevertheless, starting with the number 1, we can reach any positive integer whatever in a finite number of steps, passing successively from k to $k + 1$ at each step. This is the basis for a type of reasoning that mathematicians call *proof by induction*. We shall illustrate the use of this method by proving the pair of inequalities used in Section 1.2 in the computation of the area of a parabolic segment, namely,

$$(1.18) \quad 1^2 + 2^2 + \cdots + (n-1)^2 < \frac{n^3}{3} < 1^2 + 2^2 + \cdots + n^2.$$

Consider the leftmost inequality first and let us refer to this formula as $A(n)$ (an assertion involving n). It is easy to verify this assertion directly for the first few values of n . Thus, for example, when n takes the values 1, 2, and 3, the assertion becomes

$$A(1): 0 < \frac{1^3}{3}, \quad A(2): 1^2 < \frac{2^3}{3}, \quad A(3): 1^2 + 2^2 < \frac{3^3}{3},$$

provided we agree to interpret the sum on the left as 0 when $n = 1$.

Our object is to prove that $A(n)$ is true for every positive integer n . The procedure is as follows: Assume the assertion has been proved for a particular value of n , say for $n = k$. That is, assume we have proved

$$A(k): 1^2 + 2^2 + \cdots + (k-1)^2 < \frac{k^3}{3}$$

for a fixed $k \geq 1$. Now, *using this*, we shall deduce the corresponding result for $k+1$:

$$A(k+1): 1^2 + 2^2 + \cdots + k^2 < \frac{(k+1)^3}{3}.$$

Start with $A(k)$ (which is assumed to have been proven) and add k^2 to both sides. This gives the inequality

$$1^2 + 2^2 + \cdots + k^2 < \frac{k^3}{3} + k^2.$$

To obtain $A(k+1)$ as a consequence of this, it suffices to show that

$$\frac{k^3}{3} + k^2 < \frac{(k+1)^3}{3}.$$

But this follows at once from the equation

$$\frac{(k+1)^3}{3} = \frac{k^3 + 3k^2 + 3k + 1}{3} = \frac{k^3}{3} + k^2 + k + \frac{1}{3}.$$

Therefore we have shown that $A(k+1)$ follows from $A(k)$. Now, since $A(1)$ has been verified directly, we conclude that $A(2)$ is also true. Knowing that $A(2)$ is true, we conclude that $A(3)$ is true, and so on. Since every integer can be reached in this way, $A(n)$ is true for all positive integers n . This proves the leftmost inequality in (1.18).

The reader should make certain that he understands the *pattern* of the foregoing proof. First we proved the assertion $A(n)$ for $n = 1$. Next we showed that *if* the assertion is true for a particular integer *then* it is also true for the next integer. From this, we concluded that the assertion is true for all positive integers.

The idea of induction may be illustrated in many nonmathematical ways. For example, imagine a row of toy soldiers, numbered consecutively, that extends to the right without end, as suggested by the dots in Figure 1.9. Suppose they are so arranged that if any one of them falls, say the one labeled k , it will knock over the next one, labeled $k+1$. (See Figure 1.10.) Then anyone can visualize what would happen if soldier number 1 were

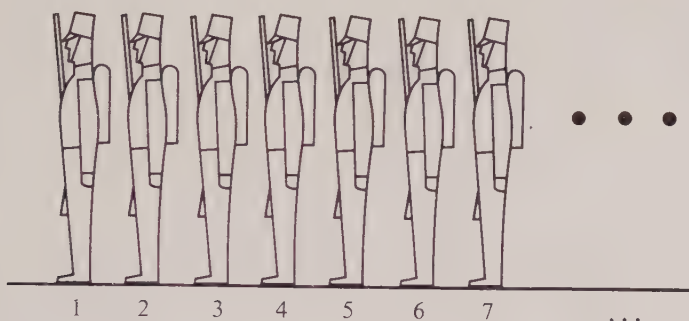


FIGURE 1.9



FIGURE 1.10

toppled backward. It is also clear that if a later soldier were knocked over first, say the one labeled n_1 , then all soldiers behind *him* would fall. This illustrates a slight generalization of the method of induction which can be described in the following way.

Method of proof by induction. Let $A(n)$ be an assertion involving an integer n . We conclude that $A(n)$ is true for every $n \geq n_1$ if we can perform the following two steps:

- (a) Prove that $A(n_1)$ is true.
- (b) Let k be an arbitrary but fixed integer $\geq n_1$. Assume that $A(k)$ is true and prove that $A(k + 1)$ is also true.

In actual practice n_1 is usually 1. The logical justification for this method of proof is the following theorem about real numbers.

1-31 THEOREM. *Principle of mathematical induction.* Let S be a set of real numbers which has the following two properties:

- (a) The number 1 is in the set S .
- (b) If an integer k is in S , then so is $k + 1$.

Then every positive integer is in the set S .

Proof. Since 1 is in S , we may apply property (b) with $n = 1$ to deduce that $1 + 1$ is in S . That is, 2 is in S . But now we may use (b) again with $n = 2$ to deduce that $2 + 1$ is in S . This shows that 3 is in S . Applying (b) once more with $n = 3$, we find that 4 is in S , and so on. If we recall the way we defined the positive integers (see p. 14), we find that S contains every positive integer, and this completes the proof.

It is important to realize that there may be numbers in S that are *not* positive integers. For example, the set of all rational numbers satisfies properties (a) and (b); so does the set of *all* real numbers. The theorem we have just proved tells us that the collection of positive integers is the *smallest* collection of real numbers which satisfies both properties (a) and (b).

Note. Whenever we carry out a proof of an assertion $A(n)$ for all $n \geq 1$ by mathematical induction, we are applying Theorem 1-31 to the collection S consisting of all the integers for which the as-

section $A(n)$ is true. If we want to prove that $A(n)$ is true only for all $n \geq n_1$, then we apply Theorem 1-31 to the collection of numbers n for which $A(n + n_1 - 1)$ is true.

There is another important statement about positive integers, called the *well-ordering principle*, that is also used as a basis for proofs by induction. It can be stated as follows:

1-32 THEOREM. *Well-ordering principle.* Let T be a nonempty† collection of positive integers. Then, among all the integers in T , there is one which is smallest.‡

The well-ordering principle is logically equivalent to the principle of induction. That is, it can be deduced from the principle of induction and, conversely, the principle of induction can be deduced from it. The equivalence is demonstrated in Section 1.24. We conclude this section with an example showing how the well-ordering principle can be used to prove theorems about positive integers.

Let $A(n)$ denote the following assertion:

$$A(n): 1^2 + 2^2 + \cdots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}.$$

Again, we note that $A(1)$ is true, since

$$1^2 = \frac{1}{3} + \frac{1}{2} + \frac{1}{6}.$$

Now there are only two possibilities. We have either

- (i) $A(n)$ is true for every positive integer n , or
- (ii) there is at least one positive integer n for which $A(n)$ is false.

We shall prove that alternative (ii) leads to a contradiction. Assume (ii) holds. Then by the well-ordering principle§ there must be a *smallest* positive integer, say k , for which $A(k)$ is false. This k must be greater than 1 because we have verified that $A(1)$ is true. Also, the assertion must be true for $k - 1$, since k was the smallest integer for which $A(k)$ is false; therefore we may write

$$A(k - 1): 1^2 + 2^2 + \cdots + (k - 1)^2 = \frac{(k - 1)^3}{3} + \frac{(k - 1)^2}{2} + \frac{k - 1}{6}.$$

Adding k^2 to both sides and simplifying the right-hand side, we find

$$1^2 + 2^2 + \cdots + k^2 = \frac{k^3}{3} + \frac{k^2}{2} + \frac{k}{6}.$$

But this equation states that $A(k)$ is true; therefore we have a contradiction because k is an integer for which $A(k)$ is false. In other words, statement (ii) leads to a contradiction. Therefore (i) holds and this proves that the identity in question is valid for all values of $n \geq 1$. A consequence of this identity is the rightmost inequality in (1.18).

† The term "nonempty" means that T contains at least one integer.

‡ It is important that the elements of T be positive integers. The theorem is not true for arbitrary collections of integers.

§ We apply the well-ordering principle to the collection T of all positive integers n for which $A(n)$ is false. Statement (ii) says that T is nonempty.

A proof like this which makes use of the well-ordering principle is also referred to as a proof by induction. Of course, the proof could also be put in the more usual form in which we verify $A(1)$ and then pass from $A(k)$ to $A(k + 1)$.

1.18 Exercises

1. Prove the following formulas by induction:

$$(a) \ 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

$$(b) \ 1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

$$(c) \ 1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2.$$

$$(d) \ 1^3 + 2^3 + \cdots + (n-1)^3 < \frac{n^4}{4} < 1^3 + 2^3 + \cdots + n^3.$$

2. Note that

$$\begin{aligned} 1 &= 1, \\ 1 - 4 &= -(1 + 2), \\ 1 - 4 + 9 &= 1 + 2 + 3, \\ 1 - 4 + 9 - 16 &= -(1 + 2 + 3 + 4). \end{aligned}$$

Guess the general law suggested and prove it by induction.

3. Note that

$$\begin{aligned} 1 + \frac{1}{2} &= 2 - \frac{1}{2}, \\ 1 + \frac{1}{2} + \frac{1}{4} &= 2 - \frac{1}{4}, \\ 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} &= 2 - \frac{1}{8}. \end{aligned}$$

Guess the general law suggested and prove it by induction.

4. Note that

$$\begin{aligned} 1 - \frac{1}{2} &= \frac{1}{2}, \\ (1 - \frac{1}{2})(1 - \frac{1}{3}) &= \frac{1}{3}, \\ (1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{4}) &= \frac{1}{4}. \end{aligned}$$

Guess the general law suggested and prove it by induction.

5. Guess a general law which simplifies the product

$$\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{9}\right)\left(1 - \frac{1}{16}\right) \cdots \left(1 - \frac{1}{n^2}\right)$$

and prove it by induction.

6. Let $A(n)$ denote the statement: $1 + 2 + \cdots + n = \frac{1}{8}(2n+1)^2$.

(a) Prove that if $A(k)$ is true for an integer k , then $A(k+1)$ is also true.

(b) Criticize the statement: "By induction it follows that $A(n)$ is true for all n ."

(c) Amend $A(n)$ by changing the equality to an inequality that is true for all positive integers n .

7. Given positive real numbers $a_1, a_2, a_3, \dots$, such that $a_n \leq ca_{n-1}$ for all $n \geq 2$, where c is a fixed positive number, use induction to prove that $a_n \leq a_1 c^{n-1}$ for all $n \geq 1$.

8. Prove the following statement by induction: If a line of unit length is given, then a line of length $\sqrt{n}$ can be constructed with straightedge and compass for each positive integer n .

9. Let b denote a fixed positive integer. Prove the following statement by induction: For every integer $n \geq 0$ there exist nonnegative integers q and r such that

$$n = qb + r, \quad 0 \leq r < b.$$

10. Let n and d denote integers. We say that d is a *divisor* of n if $n = cd$ for some integer c . An integer n is called a *prime* if $n > 1$ and if the only divisors of n are 1 and n . Prove, by induction, that every integer $n > 1$ is either a prime or a product of primes.

11. Describe the fallacy in the following “proof” by induction:

Statement. Given any collection of n blonde girls. If at least one of the girls has blue eyes, then all n of them have blue eyes.

“Proof.” The statement is obviously true when $n = 1$. The step from k to $k + 1$ can be illustrated by going from $n = 3$ to $n = 4$. Assume, therefore, that the statement is true when $n = 3$ and let G_1, G_2, G_3, G_4 be four blonde girls, at least one of which, say G_1 , has blue eyes. Taking G_1, G_2 , and G_3 together and using the fact that the statement is true when $n = 3$, we find that G_2 and G_3 also have blue eyes. Repeating the process with G_1, G_2 , and G_4 , we find that G_4 has blue eyes. Thus all four have blue eyes. A similar argument allows us to make the step from k to $k + 1$ in general.

Corollary. All blonde girls have blue eyes.

Proof. Since there exists at least one blonde girl with blue eyes, we can apply the foregoing result to the collection consisting of all blonde girls.

Note. This example is from G. Pólya, who suggests that the reader may want to test the validity of the statement by experiment.

1.19 The summation notation

In the calculations for the area of the parabolic segment we encountered the sum

$$(1.19) \quad 1^2 + 2^2 + 3^2 + \cdots + n^2.$$

Note that a typical term in this sum is of the form k^2 and we get all the terms by letting k run through the values $1, 2, 3, \dots, n$. There is a very useful and convenient notation which enables us to write sums like this in a more compact form. This is called the *summation notation* and it makes use of the Greek letter sigma, Σ . Using summation notation, we can write the sum in (1.19) as follows:

$$\sum_{k=1}^n k^2.$$

This symbol is read: “The sum of k^2 for k running from 1 to n .” The convention is that the numbers appearing under and above the sigma tell us the range of values taken by k . The letter k itself is referred to as the *index of summation*. Of course, it is not important that we use the letter k ; any other convenient symbol may take its place. For example, instead of $\sum_{k=1}^n k^2$ we could write $\sum_{i=1}^n i^2$, $\sum_{j=1}^n j^2$, $\sum_{m=1}^n m^2$, etc., all of which are considered as alternative notations for the same thing. The letters i, j, k, m , etc. that are used in this way are called *dummy indices*. It would not be a good idea to use the letter n for the dummy index in this particular example because n is already being used for the number of terms.

More generally, when we want to form the sum of several real numbers, say $a_1, a_2, \dots, a_n$, we denote such a sum by the symbol

$$(1.20) \quad a_1 + a_2 + \cdots + a_n$$

which, using summation notation, is also written more briefly as follows:

$$(1.21) \quad \sum_{k=1}^n a_k .$$

For example,

$$\begin{aligned} \sum_{k=1}^4 a_k &= a_1 + a_2 + a_3 + a_4 , \\ \sum_{i=1}^5 x_i &= x_1 + x_2 + x_3 + x_4 + x_5 . \end{aligned}$$

Sometimes it is convenient to begin summations from 0 or from some value of the index beyond 1. For example, we have

$$\begin{aligned} \sum_{i=0}^4 x_i &= x_0 + x_1 + x_2 + x_3 + x_4 , \\ \sum_{n=2}^5 n^3 &= 2^3 + 3^3 + 4^3 + 5^3 . \end{aligned}$$

Other uses of the summation notation are illustrated below:

$$\begin{aligned} \sum_{m=0}^4 x^{m+1} &= x + x^2 + x^3 + x^4 + x^5 , \\ \sum_{j=1}^6 2^{j-1} &= 1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 . \end{aligned}$$

To emphasize once more that the choice of dummy index is unimportant, we note that the last sum may also be written in each of the following forms:

$$\sum_{q=1}^6 2^{q-1} = \sum_{r=0}^5 2^r = \sum_{n=0}^5 2^{5-n} = \sum_{k=1}^6 2^{6-k} .$$

Note. From a strictly logical standpoint, the symbols in (1.20) and (1.21) do not appear among the axioms for the real-number system. In a more rigorous treatment one could define these new symbols in terms of the primitive undefined symbols of our system. This may be done by a process known as *definition by induction* which, like proof by induction, consists of two parts:

(a) We define

$$\sum_{k=1}^1 a_k = a_1 .$$

(b) Assuming that we have defined $\sum_{k=1}^n a_k$ for a fixed $n \geq 1$, we further define

$$\sum_{k=1}^{n+1} a_k = \left(\sum_{k=1}^n a_k \right) + a_{n+1} .$$

To illustrate, we may take $n = 1$ in (b) and use (a) to obtain

$$\sum_{k=1}^2 a_k = \sum_{k=1}^1 a_k + a_2 = a_1 + a_2.$$

Now, having defined $\sum_{k=1}^2 a_k$, we can use (b) again with $n = 2$ to obtain

$$\sum_{k=1}^3 a_k = \sum_{k=1}^2 a_k + a_3 = (a_1 + a_2) + a_3.$$

By the associative law for addition (Axiom 2), the sum $(a_1 + a_2) + a_3$ is the same as $a_1 + (a_2 + a_3)$ and therefore there is no danger of confusion if we drop the parentheses and simply write $a_1 + a_2 + a_3$ for $\sum_{k=1}^3 a_k$. Similarly,

$$\sum_{k=1}^4 a_k = \sum_{k=1}^3 a_k + a_4 = (a_1 + a_2 + a_3) + a_4.$$

In this case we can *prove* that the sum $(a_1 + a_2 + a_3) + a_4$ is the same as $(a_1 + a_2) + (a_3 + a_4)$ or $a_1 + (a_2 + a_3 + a_4)$ and therefore the parentheses can be dropped again without danger of ambiguity, and we agree to write

$$\sum_{k=1}^4 a_k = a_1 + a_2 + a_3 + a_4.$$

Continuing in this way, we find that (a) and (b) together give us a complete definition of the symbol in (1.21). The notation in (1.20) is considered to be merely an alternative way of writing (1.21). It is justified by a general associative law for addition which we shall not attempt to state or to prove here.

The reader should notice that *definition by induction* and *proof by induction* involve the same underlying idea. A definition by induction is also called a *recursive definition*.

1.20 Exercises

1. Find the numerical values of the following sums:

(a) $\sum_{k=1}^4 k,$

(d) $\sum_{n=1}^4 n^n,$

(b) $\sum_{n=2}^5 2^{n-2},$

(e) $\sum_{i=0}^5 (2i + 1),$

(c) $\sum_{r=0}^3 2^{2r+1},$

(f) $\sum_{k=1}^5 \frac{1}{k(k+1)}.$

2. Establish the following properties of the summation notation:

(a) $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$ (additive property).

(b) $\sum_{k=1}^n (ca_k) = c \sum_{k=1}^n a_k$ (homogeneous property).

(c) $\sum_{k=1}^n (a_k - a_{k-1}) = a_n - a_0$ (telescoping property).

Use the properties in Exercise 2 whenever possible to derive the formulas in Exercises 3 through 8.

3. $\sum_{k=1}^n 1 = n$. (This sum means $\sum_{k=1}^n a_k$, where each $a_k = 1$.) [Hint. Use Exercise 2(c) with $a_k = k$.]

4. $\sum_{k=1}^n (2k - 1) = n^2$. [Hint. $2k - 1 = k^2 - (k - 1)^2$.]

5. $\sum_{k=1}^n k = \frac{n^2}{2} + \frac{n}{2}$. [Hint. Use Exercises 3 and 4.]

6. $\sum_{k=1}^n k^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$. [Hint. $k^3 - (k - 1)^3 = 3k^2 - 3k + 1$.]

7. $\sum_{k=1}^n k^3 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}$.

8. (a) $\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}$ if $x \neq 1$. Note, x^0 is defined to be 1. [Hint. Apply Exercise 2 to $(1 - x) \sum_{k=0}^n x^k$.]

(b) What is the sum equal to when $x = 1$?

9. Prove, by induction, that the sum $\sum_{k=1}^{2n} (-1)^k (2k + 1)$ is proportional to n , and find the constant of proportionality.

10. (a) Give a reasonable definition of the symbol $\sum_{k=m}^{m+n} a_k$.

(b) Prove, by induction, that for $n \geq 1$ we have

$$\sum_{k=n+1}^{2n} \frac{1}{k} = \sum_{m=1}^{2n} \frac{(-1)^{m+1}}{m}.$$

1.21 Absolute values and the triangle inequality

Calculations with inequalities arise quite frequently in calculus. They are of particular importance in dealing with the notion of *absolute value*. If x is a real number, the absolute value of x is a nonnegative real number denoted by $|x|$ and defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x \leq 0. \end{cases}$$

When real numbers are represented geometrically on a real axis, the number $|x|$ is called the *distance* of x from 0. If $a > 0$ and if a point x lies between $-a$ and a , then $|x|$ is nearer to 0 than a is. The analytic statement of this fact is given by the following theorem.

1-33 THEOREM. If $a \geq 0$, then $|x| \leq a$ if and only if $-a \leq x \leq a$.

Proof. There are two statements to prove: first, that the inequality $|x| \leq a$ implies the two inequalities $-a \leq x \leq a$ and, conversely, that $-a \leq x \leq a$ implies $|x| \leq a$.

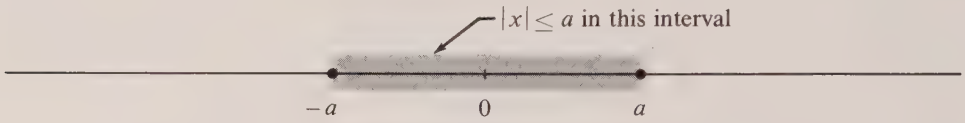


FIGURE 1.11 Geometrical significance of Theorem 1-33.

Suppose $|x| \leq a$. Then we also have $-a \leq -|x|$. But either $x = |x|$ or $x = -|x|$ and hence $-a \leq -|x| \leq x \leq |x| \leq a$. This proves the first statement.

To prove the converse, assume $-a \leq x \leq a$. Then if $x \geq 0$ we have $|x| = x \leq a$, whereas if $x \leq 0$ we have $|x| = -x \leq a$. In either case we have $|x| \leq a$, and this completes the proof.

Figure 1.11 illustrates the geometrical significance of this theorem.

As a consequence of Theorem 1-33 it is easy to derive an important inequality which states that the absolute value of a sum of two real numbers cannot exceed the sum of their absolute values.

1-34 THEOREM. For arbitrary real numbers x and y , we have

$$|x + y| \leq |x| + |y|.$$

Note. This property is called the *triangle inequality*, because when it is generalized to vectors it states that the length of any side of a triangle is less than or equal to the sum of the lengths of the other two sides.

Proof. Since either $x = |x|$ or $x = -|x|$, we have $-|x| \leq x \leq |x|$. Similarly, $-|y| \leq y \leq |y|$. Adding inequalities, we obtain

$$-(|x| + |y|) \leq x + y \leq |x| + |y|$$

and hence, by Theorem 1-33, we conclude that $|x + y| \leq |x| + |y|$.

If we take $x = a - c$ and $y = c - b$, then $x + y = a - b$ and the triangle inequality becomes

$$|a - b| \leq |a - c| + |b - c|.$$

This form of the triangle inequality is often used in practice.

Using mathematical induction, we may extend the triangle inequality as follows:

1-35 THEOREM. For arbitrary real numbers $a_1, a_2, \dots, a_n$, we have

$$\left| \sum_{k=1}^n a_k \right| \leq \sum_{k=1}^n |a_k|.$$

Proof. When $n = 1$ the inequality is trivial and when $n = 2$ it is the triangle inequality. Assume, then, that it is true for n real numbers. Then for $n + 1$ real numbers $a_1, a_2, \dots, a_{n+1}$, we have

$$\left| \sum_{k=1}^{n+1} a_k \right| = \left| \sum_{k=1}^n a_k + a_{n+1} \right| \leq \left| \sum_{k=1}^n a_k \right| + |a_{n+1}| \leq \sum_{k=1}^n |a_k| + |a_{n+1}| = \sum_{k=1}^{n+1} |a_k|.$$

Hence the theorem is true for $n + 1$ numbers if it is true for n . By induction, it is true for every positive integer n .

1.22 Exercises

1. Prove each of the following properties of absolute values.

- | | |
|--|---------------------------------------|
| (a) $ x = 0$ if and only if $x = 0$. | (f) $ xy = x y $. |
| (b) $ -x = x $. | (g) $ x/y = x / y $ if $y \neq 0$. |
| (c) $ x - y = y - x $. | (h) $ x - y \leq x + y $. |
| (d) $ x ^2 = x^2$. | (i) $ x - y \leq x - y $. |
| (e) $ x = \sqrt{x^2}$. | (j) $ x - y \leq x - y $. |

2. Prove, by induction, that $\sum_{k=1}^n a_k^2 \geq 0$ for arbitrary real numbers $a_1, \dots, a_n$. (This shows that a sum of squares cannot be negative.)

3. Each inequality (a_i), listed below, is equivalent to exactly one inequality (b_j). For example, $|x| < 3$ if and only if $-3 < x < 3$, and hence (a_1) is equivalent to (b_2). Determine all equivalent pairs.

- | | |
|--|--|
| (a ₁) $ x < 3$. | (b ₁) $4 < x < 6$. |
| (a ₂) $ x - 1 < 3$. | (b ₂) $-3 < x < 3$. |
| (a ₃) $ 3 - 2x \leq 1$. | (b ₃) $x > 3$ or $x < -1$. |
| (a ₄) $ 1 + 2x \leq 1$. | (b ₄) $x > 2$. |
| (a ₅) $ x - 1 > 2$. | (b ₅) $-2 \leq x < 4$. |
| (a ₆) $ x + 2 \geq 5$. | (b ₆) $-\sqrt{3} \leq x \leq -1$ or $1 \leq x \leq \sqrt{3}$. |
| (a ₇) $ 5 - x^{-1} < 1$. | (b ₇) $1 < x < 2$. |
| (a ₈) $ x - 5 < x + 1 $. | (b ₈) $x \leq -7$ or $x \geq 3$. |
| (a ₉) $ x^2 - 2 \leq 1$. | (b ₉) $\frac{1}{6} < x < \frac{1}{4}$. |
| (a ₁₀) $x < x^2 - 12 < 4x$. | (b ₁₀) $-1 \leq x \leq 0$. |

*1.23 Miscellaneous exercises involving induction

In this section we assemble a number of miscellaneous facts whose proofs are good exercises in the use of mathematical induction. Some of these exercises may serve as a basis for supplementary classroom discussion.

Factorials and binomial coefficients. The symbol $n!$ (read “ n factorial”) may be defined by induction as follows: $0! = 1$, $n! = (n - 1)n!$ if $n \geq 1$. Note that $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$.

If $0 \leq k \leq n$, the *binomial coefficient* $\binom{n}{k}$ is defined as follows:

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Note. Sometimes “ C_k ” is written for $\binom{n}{k}$. These numbers appear as coefficients in the binomial theorem. (See Exercise 4 below.)

1. Compute the values of the following binomial coefficients:

(a) $\binom{5}{3}$, (b) $\binom{7}{0}$, (c) $\binom{7}{1}$, (d) $\binom{7}{2}$, (e) $\binom{17}{14}$, (f) $\binom{0}{0}$.

2. (a) Show that $\binom{n}{k} = \binom{n}{n-k}$. (c) Find k , given that $\binom{14}{k} = \binom{14}{k-4}$.

(b) Find n , given that $\binom{n}{10} = \binom{n}{7}$. (d) Is there a k such that $\binom{12}{k} = \binom{12}{k-3}$?

3. Prove that $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$. This is called the *law of Pascal's triangle* and it provides a rapid way of computing binomial coefficients successively. Pascal's triangle is illustrated here for $n \leq 6$.

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 & & 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1
 \end{array}$$

4. Use induction to prove the binomial theorem:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Then use the theorem to derive the formulas

$$\sum_{k=0}^n \binom{n}{k} = 2^n \quad \text{and} \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

The product notation. The product of n real numbers $a_1, a_2, \dots, a_n$ is denoted by the symbol $\prod_{k=1}^n a_k$ which may be defined by induction. The symbol $a_1 a_2 \cdots a_n$ is an alternative notation for this product. Note that

$$n! = \prod_{k=1}^n k.$$

5. Give a definition by induction for the product $\prod_{k=1}^n a_k$.

Prove the following properties of products by induction:

6. $\prod_{k=1}^n (a_k b_k) = \left(\prod_{k=1}^n a_k \right) \left(\prod_{k=1}^n b_k \right)$ (multiplicative property).

An important special case is the relation $\prod_{k=1}^n (ca_k) = c^n \prod_{k=1}^n a_k$.

7. $\prod_{k=1}^n \frac{a_k}{a_{k-1}} = \frac{a_n}{a_0}$ if each $a_k \neq 0$ (telescoping property).

8. State the results of Exercises 4 and 5 of Section 1.18 using the product notation and derive them again as consequences of the telescoping property for products.

9. If $x \neq 1$, show that

$$\prod_{k=1}^n (1 + x^{2^{k-1}}) = \frac{1 - x^{2^n}}{1 - x}.$$

What is the value of the product when $x = 1$?

Some special inequalities.

10. If $a_k < b_k$ for each $k = 1, 2, \dots, n$, it is easy to prove by induction that $\sum_{k=1}^n a_k < \sum_{k=1}^n b_k$. Discuss the corresponding inequality for products:

$$\prod_{k=1}^n a_k < \prod_{k=1}^n b_k.$$

11. If $x > 1$, prove by induction that $x^n > x$ for every integer $n \geq 2$. If $0 < x < 1$, prove that $x^n < x$ for every integer $n \geq 2$.

12. Determine all positive integers n for which $2^n < n!$.

13. (a) Let p be a positive integer. Prove that

$$b^p - a^p = (b - a)(b^{p-1} + b^{p-2}a + b^{p-3}a^2 + \dots + ba^{p-2} + a^{p-1}).$$

[Hint. Write the right-hand member in summation notation and use the telescoping property.]

(b) Let p and n denote positive integers. Show that

$$n^p < \frac{(n+1)^{p+1} - n^{p+1}}{p+1} < (n+1)^p.$$

[Hint. Use part (a).]

(c) Use induction to prove that

$$\sum_{k=1}^{n-1} k^p < \frac{n^{p+1}}{p+1} < \sum_{k=1}^n k^p.$$

Part (b) will assist in making the inductive step from n to $n+1$.

14. The numbers 1, 2, 3, 5, 8, 13, 21, $\dots$, in which each term after the second is the sum of its two predecessors, are called *Fibonacci numbers*. They may be defined by induction as follows:

$$a_1 = 1, \quad a_2 = 2, \quad a_{n+1} = a_n + a_{n-1} \quad \text{if } n \geq 2.$$

Prove that

$$a_n < \left(\frac{1 + \sqrt{5}}{2} \right)^n$$

for every $n \geq 1$.

If $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are arbitrary real numbers, then we have

$$(1.22) \quad \left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right).$$

This is known as the *Cauchy-Schwarz inequality*, one of the most important inequalities in analysis. We shall make use of it in connection with our study of vector algebra in Chapter 5. A simple proof of (1.22) may be given as follows:

We have $\sum_{k=1}^n (a_k x + b_k)^2 \geq 0$ for every real x because a sum of squares can never be negative. This may be written in the form

$$(1.23) \quad Ax^2 + 2Bx + C \geq 0,$$

where

$$A = \sum_{k=1}^n a_k^2, \quad B = \sum_{k=1}^n a_k b_k, \quad C = \sum_{k=1}^n b_k^2.$$

We wish to prove that $B^2 \leq AC$. If $A = 0$, then each $a_k = 0$, so $B = 0$ and the result is trivial. If $A \neq 0$, we may complete the square and write

$$Ax^2 + 2Bx + C = A\left(x + \frac{B}{A}\right)^2 + \frac{AC - B^2}{A}.$$

The right side has its smallest value when $x = -B/A$. Putting $x = -B/A$ in (1.23), we obtain $B^2 \leq AC$. This proves (1.22).

15. Show that the equality sign holds in the Cauchy-Schwarz inequality if and only if there is a real number t such that $a_k = tb_k$ for every $k = 1, 2, \dots, n$ or such that $b_k = ta_k$ for every $k = 1, 2, \dots, n$.

16. Let $x_1, x_2, \dots, x_n$ be n positive real numbers. If p is a nonzero integer, the p th-power mean M_p of the n numbers is defined as follows:

$$M_p = \left(\frac{x_1^p + \dots + x_n^p}{n} \right)^{1/p}$$

The number M_1 is also called the *arithmetic mean*, M_2 the *root mean square*, and M_{-1} the *harmonic mean*. If $p > 0$, prove that $M_p < M_{2p}$ when $x_1, x_2, \dots, x_n$ are not all equal. [Hint. Apply the Cauchy-Schwarz inequality with $a_k = x_k^p$ and $b_k = 1$.]

17. Use the result of Exercise 16 to prove that

$$a^3 + b^3 + c^3 \geq \frac{16\sqrt{6}}{3}$$

if $a^2 + b^2 + c^2 = 8$ and $a > 0, b > 0, c > 0$.

18. Let $a_1, \dots, a_n$ be n positive real numbers whose product is equal to 1. Prove that $a_1 + \dots + a_n \geq n$ and that the equality sign holds only if every $a_k = 1$. [Hint. Consider two cases: (a) All $a_k = 1$; (b) not all $a_k = 1$. Use induction. In case (b) notice that if $a_1 a_2 \dots a_{k+1} = 1$, then at least one factor, say a_1 , exceeds 1 and at least one factor, say a_{k+1} , is less than 1. Let $b_1 = a_1 a_{k+1}$ and apply the induction hypothesis to the product $b_1 a_2 \dots a_k$.]

19. The *geometric mean* G of n positive real numbers $x_1, \dots, x_n$ is defined by the formula $G = (x_1 x_2 \dots x_n)^{1/n}$.

(a) Use the notation of Exercise 16 to prove that $G \leq M_1$ and that $G = M_1$ only when $x_1 = x_2 = \dots = x_n$. [Hint. Reduce this to Exercise 18.]

(b) Let p and q be integers, $q < 0 < p$. From part (a) deduce that $M_q < G < M_p$ when $x_1, x_2, \dots, x_n$ are not all equal.

20. Use the result of Exercise 19 to prove the following statement: If a , b , and c are positive real numbers such that $abc = 8$, then $a + b + c \geq 6$ and $ab + ac + bc \geq 12$.

21. Let $a_1, \dots, a_n$ be n real numbers, all having the same sign and all greater than -1 . Use induction to prove that

$$(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq 1 + a_1 + a_2 + \cdots + a_n.$$

In particular, when $a_1 = a_2 = \cdots = a_n = x$, where $x > -1$, this yields

$$(1.24) \quad (1 + x)^n \geq 1 + nx \quad (\text{Bernoulli's inequality}).$$

Show that when $n > 1$ the equality sign holds in (1.24) only for $x = 0$.

22. Derive the following as consequences of earlier exercises:

$$(a) \quad \left(1 + \frac{1}{n}\right)^n > 2 \quad \text{if } n > 1.$$

$$(b) \quad n! < \left(\frac{n+1}{2}\right)^n \quad \text{if } n > 1.$$

*1.24 Equivalence of the well-ordering principle and the principle of induction

In this section we show that the well-ordering principle (Theorem 1-32) and the principle of induction (Theorem 1-31) are logically equivalent.

Assume first that the well-ordering principle holds, and let S be a collection of real numbers having the following two properties:

(a) The integer 1 is in S .

(b) If an integer k is in S , then $k + 1$ is also in S .

Let T denote the collection of all positive integers *not* in S . We shall prove that T is empty. (This will show that S contains all positive integers.)

Suppose T is not empty. Then, by the well-ordering principle, there is a smallest positive integer in T which we shall call m . Now, m cannot be 1 because 1 is in S . Therefore $m \geq 2$. Also, the positive integer $m - 1$ cannot be in T because m is the smallest integer in T . Therefore $m - 1$ is in S . By property (b), m is also in S . But this is a contradiction because no integer can be in both S and T . Therefore the assumption that T is not empty leads to a contradiction. Hence T is empty and S contains all positive integers. This proves that the well-ordering principle implies the principle of induction.

Next we prove the converse statement. Assume the principle of induction holds, and let T be a nonempty collection of positive integers. We want to prove that T has a smallest member, that is, that there is a positive integer t_0 in T such that $t_0 \leq t$ for all t in T .

Suppose T has no smallest member. We shall show that this leads to a contradiction. The integer 1 cannot be in T (otherwise it would be the smallest member of T). Let S denote the collection of all positive integers n such that $n < t$ for all t in T . Now 1 is in S because $1 < t$ for all t in T . Next, let k be a positive integer in S . Then $k < t$ for all t in T . We shall prove that $k + 1$ is also in S . If this were not so, then for some t_1 in T we would have $t_1 \leq k + 1$. Since T has no smallest member there is an integer t_2 in T such that $t_2 < t_1$, and hence $t_2 < k + 1$. But this means that $t_2 \leq k$, contradicting the fact that $k < t$ for all t in T . Therefore $k + 1$ is in S . By the induction principle, S

contains all positive integers. Since T is nonempty there is a positive integer t in T . But this t must also be in S (since S contains all positive integers). It follows from the definition of S that $t < t$, which is a contradiction. Therefore, the assumption that T has no smallest member leads to a contradiction. It follows that T must have a smallest member and in turn this proves that the well-ordering principle is a consequence of the principle of induction.

Part II. The Concepts of Integral Calculus

In the remainder of this chapter we shall present the definition of the integral and some of its basic properties. To understand the definition one must have some acquaintance with the function concept, and the next few sections are devoted to an explanation of this and some related ideas.

1.25 The basic ideas of Cartesian geometry

As mentioned earlier, one of the uses of the integral is in the formulation of the concept of area. Ordinarily we do not talk about area by itself. Instead, we talk about the area of *something*. This means that we have certain objects (polygonal regions, circular regions, parabolic segments, etc.) whose areas we wish to measure. If we hope to arrive at a definition of area that will enable us to deal with many different kinds of objects, we must first find an effective way to describe these objects.

The most primitive way of doing this is by drawing figures, as was done by the ancient Greeks. A much better way was suggested by René Descartes (1596–1650) when he introduced the subject of analytic geometry in 1637. The backbone of Descartes' geometry (now known as *Cartesian geometry* or *analytic geometry*) is the idea of representing geometric points by *numbers*. The procedure for points in a plane is this:

Two perpendicular reference lines (called *coordinate axes*) are chosen, one horizontal (called the "*x*-axis"), the other vertical (the "*y*-axis"). Their point of intersection, denoted by 0, is called the *origin*. On the *x*-axis a convenient point is chosen to the right of 0 and its distance from 0 is called the *unit distance*. Vertical distances along the *y*-axis are measured with the same unit distance. Now, each point in the plane (sometimes called the *xy*-plane) is assigned a pair of numbers, called its *coordinates*. These coordinates represent the distances from the point to the axes. Figure 1.12 illustrates some examples. The point with coordinates (3, 2) lies three units to the right of the *y*-axis and two units above the *x*-axis. The number 3 is called the *x*-coordinate of the point, 2 its *y*-coordinate. Points to the left of the *y*-axis have a negative *x*-coordinate; those below the *x*-axis have a negative *y*-coordinate. The *x*-coordinate of a point is sometimes called its *abscissa* and the *y*-coordinate is called its *ordinate*.

When we write a pair of numbers such as (a, b) to represent a point we agree that the abscissa or *x*-coordinate, a , is written first. For this reason, the pair (a, b) is often referred to as an *ordered pair*. It is clear that two ordered pairs (a, b) and (c, d) represent the same

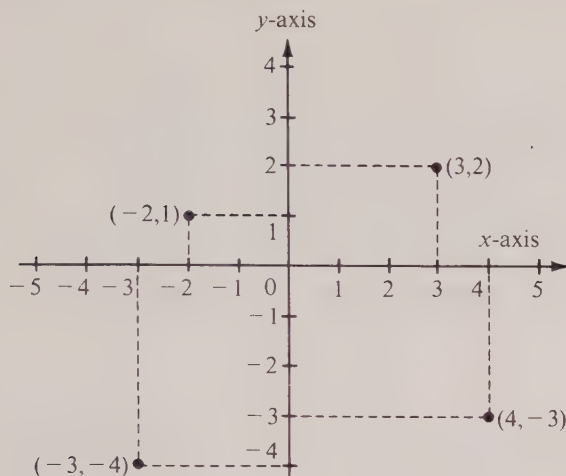
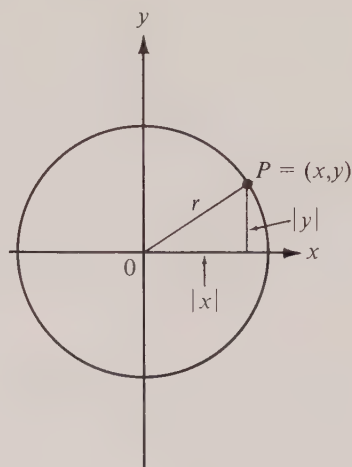


FIGURE 1.12

FIGURE 1.13 The circle represented by the Cartesian equation $x^2 + y^2 = r^2$.

point if and only if we have $a = c$ and $b = d$. Points (a, b) with both a and b positive are said to lie in the *first quadrant*; those with $a < 0$ and $b > 0$ are in the *second quadrant*; those with $a < 0$ and $b < 0$ are in the *third quadrant*; and those with $a > 0$ and $b < 0$ are in the *fourth quadrant*. Figure 1.12 shows one point in each quadrant.

The procedure for points in space is similar. We take three mutually perpendicular lines in space intersecting at a point (the origin). These lines determine three mutually perpendicular planes and each point in space can be completely described by specifying, with appropriate regard for signs, three numbers representing its distances from these planes. We shall discuss three-dimensional Cartesian geometry in more detail later on; for the present we confine our attention to plane analytic geometry.

A geometric figure, such as a curve in the plane, is a collection of points satisfying one or more special conditions. By translating these conditions into analytic expressions involving the coordinates x and y , we obtain one or more equations which characterize the curve in question. For example, suppose the curve is a circle of radius r with its center at the origin, as shown in Figure 1.13. Let P be an arbitrary point on this circle, and suppose P has coordinates (x, y) . Then the line segment OP is the hypotenuse of a right triangle whose legs have lengths $|x|$ and $|y|$ and hence, by the theorem of Pythagoras,

$$(1.25) \quad x^2 + y^2 = r^2.$$

This equation, called a *Cartesian equation* of the circle, is satisfied by all points (x, y) on the circle and by no others, so the equation completely characterizes the circle. This example illustrates how analytic geometry is used to reduce geometrical statements about points to analytical statements about real numbers.

Throughout their historical development, calculus and analytic geometry have been

intimately intertwined. New discoveries in one subject led to improvements in the other. The development of calculus and analytic geometry in this book is similar to the historical development, in that the two subjects are treated together. However, our primary purpose is to introduce the integral and differential calculus. Concepts from analytic geometry that are required for this purpose will be discussed as they are needed. Actually, only a few very elementary concepts of plane analytic geometry are required to understand the rudiments of calculus. A deeper study of analytic geometry is needed to extend the scope and applications of calculus, and this study will be carried out in Chapters 5 and 6 using vector methods as well as the methods of calculus. Until then, all that is required from analytic geometry is a little familiarity with drawing graphs of functions.

1.26 Functions and their graphs

Various fields of human endeavor have to do with relationships that exist between one collection of objects and another. Graphs, charts, curves, tables, formulas, and Gallup polls are familiar to everyone who reads the newspapers. These are merely artificial devices for describing special relationships in a quantitative fashion. Mathematicians refer to certain types of these relationships as *functions*.

Example 1. The force F necessary to stretch a steel spring a distance x beyond its natural length is proportional to x . That is, $F = cx$, where c is a number independent of x called the spring constant. This formula, discovered by Robert Hooke in the mid-17th century, is called *Hooke's law* and it is said to express the force as a function of the displacement.

Example 2. The volume of a cube is said to be a function of the lengths of its edges. If the edges have length x , the volume V is given by the formula $V = x^3$.

Example 3. A *prime* is any integer $n > 1$ that cannot be expressed in the form $n = ab$, where a and b are positive integers, both less than n . The first few primes are 2, 3, 5, 7, 11, 13, 17, 19. For a given real number $x > 0$ it is possible to count the number of primes less than x . This number is said to be a function of x even though no simple algebraic formula is known for computing it (without counting) when x is known.

The word "function" was introduced into mathematics by Leibniz, who used the term primarily to refer to certain kinds of mathematical formulas. It was later realized that Leibniz's idea of function was much too limited in its scope, and the meaning of the word has since undergone many stages of generalization. Today, the definition of function is essentially this: Given two collections of objects, say a collection X and another collection Y , a *function* is a correspondence which associates with each object in X one and only one object in Y . The collection X is called the *domain* of the function. Those objects in Y associated with the objects in X form another collection called the *range* of the function. (This may be all of Y , but it need not be.)

Letters of the English and Greek alphabets are often used to denote functions. The particular letters f , g , h , F , G , H , and ϕ are frequently used for this purpose. If f is a given function and if x is an object of its domain, the notation $f(x)$ is used to designate that object in the range which is correlated to x by the function f , and it is called the *value of f at x* or the *image of x under f* . The symbol $f(x)$ is read as " f at x " or " f of x ."

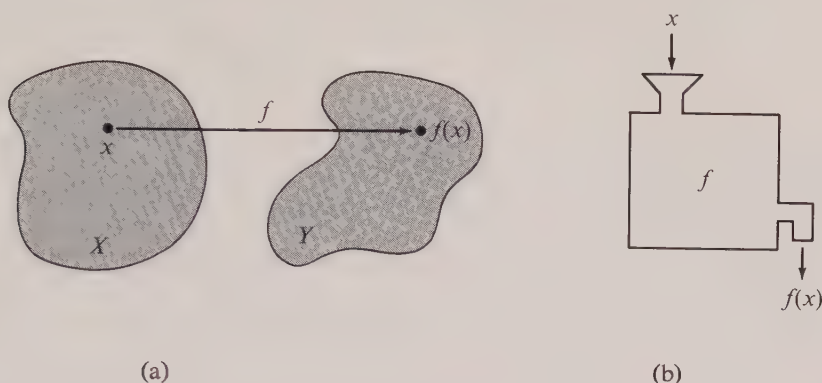


FIGURE 1.14 Schematic representations of the function idea.

Example 4. If f denotes the correspondence described in Example 3, then $f(x)$ denotes the number of primes less than x . For example, by actual counting we find the following function values:

$$f(1) = 0, \quad f(2) = 0, \quad f(3) = 1, \quad f(3\sqrt{2}) = 2, \quad f(13/2) = 3, \quad f(20) = 8.$$

Example 5. The concept of absolute value is an example of a function which assigns to each real number x the nonnegative number $|x|$. If we denote this function by ϕ , then we have $\phi(x) = |x|$ for all x . For example, $\phi(0) = 0$, $\phi(2) = 2$, $\phi(-3) = 3$. We list here some of the properties of absolute values expressed in function notation:

- | | |
|---|---------------------------------|
| (a) $\phi(-x) = \phi(x)$. | (d) $\phi[\phi(x)] = \phi(x)$. |
| (b) $\phi(x^2) = x^2$. | (e) $\phi(x) = \sqrt{x^2}$. |
| (c) $\phi(x + y) \leq \phi(x) + \phi(y)$ (the triangle inequality). | |

The function idea may be illustrated schematically in many ways. For example, in Figure 1.14(a) the collections X and Y are thought of as sets of points and an arrow is used to suggest a “pairing” of a typical point x in X with the image point $f(x)$ in Y . Another scheme is shown in Figure 1.14(b). Here the function f is imagined to be like a machine into which objects of the collection X are fed and objects of Y are produced. When an object x is fed into the machine, the output is the object $f(x)$.

In elementary calculus, we are primarily interested in functions for which the domain and range are collections of real numbers. Such functions are called *real-valued functions of a real variable* or, more briefly, *real functions*, and they may be illustrated geometrically by a graph in the xy -plane. We plot the domain X on the x -axis and above each point x in X we plot the point (x, y) , where $y = f(x)$. The totality of such points (x, y) is called the *graph* of the function.

In each of Examples 6 through 10 the domain X is the *real line*, that is, the collection of all real numbers.

Example 6. $f(x) = x$.

Here $x = y$ for each point (x, y) on the graph. In this case the graph is a straight line making equal angles with the coordinate axes (see Figure 1.15). This function is often called the *identity function*. The range of f is the collection of all real numbers.

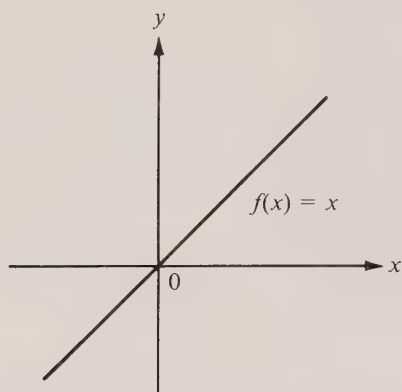


FIGURE 1.15 Graph of the identity function: $f(x) = x$.

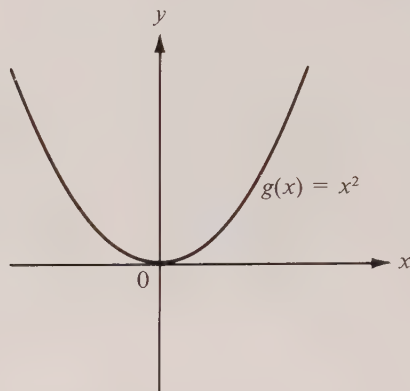


FIGURE 1.16 Graph of g when $g(x) = x^2$ (a parabola).

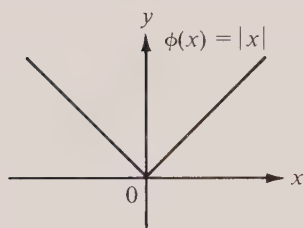


FIGURE 1.17 Absolute-value function, $\phi(x) = |x|$.

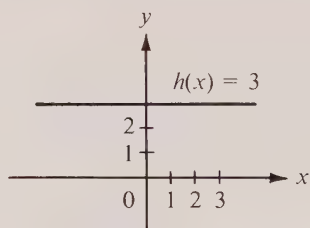


FIGURE 1.18 A constant function: $h(x) = 3$.

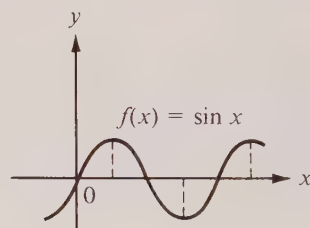


FIGURE 1.19 The sine function.

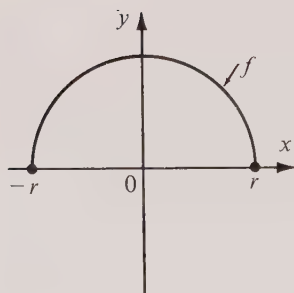


FIGURE 1.20 Graph of f , where $f(x) = \sqrt{r^2 - x^2}$, $|x| \leq r$.

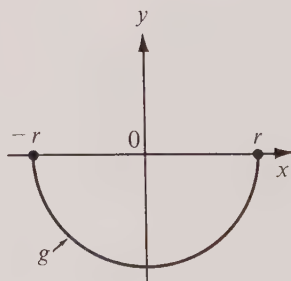


FIGURE 1.21 Graph of g , where $g(x) = -\sqrt{r^2 - x^2}$, $|x| \leq r$.

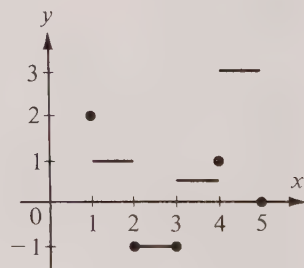


FIGURE 1.22 The graph of a step function.

Example 7. $g(x) = x^2$.

The graph of this function is a parabola, a portion of which is shown in Figure 1.16. The range of g consists of only nonnegative numbers and hence the graph never goes below the x -axis. Such a function is said to be *nonnegative*.

Example 8. $\phi(x) = |x|$.

The absolute-value function, described above in Example 5, is another nonnegative function. A portion of its graph is shown in Figure 1.17. For positive x we have $\phi(x) = x$ and the graph is the same as that of Example 6. For negative x we have $\phi(x) = -x$, and the graph may be obtained from that in Example 6 by reflection through the x -axis.

Example 9. $h(x) = 3$.

This is an example of a *constant function*, that is, one whose range consists of a single number. Its graph, shown in Figure 1.18, is a horizontal line cutting the y -axis at the point $(0, 3)$.

Example 10. $f(x) = \sin x$.

Figure 1.19 shows a portion of the graph of the *sine function*, one of the so-called *trigonometric functions*. Its range consists of all real numbers y satisfying $-1 \leq y \leq 1$. The trigonometric functions are described in further detail in Section 1.38.

The next two examples show that the domain of a real function need not be the entire real axis.

Example 11. Suppose we return to the Cartesian equation of a circle, $x^2 + y^2 = r^2$, and suppose we solve this equation for y in terms of x . There are two solutions given by†

$$y = \sqrt{r^2 - x^2} \quad \text{and} \quad y = -\sqrt{r^2 - x^2}.$$

There was a time when mathematicians would say that y is a *double-valued function* of x given by $y = \pm \sqrt{r^2 - x^2}$. However, the more modern point of view does not admit “double-valuedness” as a property of functions. The definition of function requires that for each x in the domain there corresponds one and only one y in the range. Geometrically, this means that vertical lines which intersect the graph do so at exactly one point. Therefore, to make this example fit the theory, we say that the two solutions for y define *two* functions, say f and g , where

$$f(x) = \sqrt{r^2 - x^2} \quad \text{and} \quad g(x) = -\sqrt{r^2 - x^2}$$

for each x satisfying $-r \leq x \leq r$. Each of these functions has for its domain the interval extending from $-r$ to r . If $|x| > r$, there is no real y such that $x^2 + y^2 = r^2$, and we say that the functions f and g are *not defined* for such x . Since $f(x)$ is the nonnegative square root of $r^2 - x^2$, the graph of f is the upper semicircle shown in Figure 1.20. The function values of g are ≤ 0 and hence the graph of g is the lower semicircle shown in Figure 1.21.

Example 12. Constant functions are the simplest kinds of real functions. There is an important class of functions, almost as simple as constant functions, which plays a fundamental role in the theory of the integral. This class is composed of so-called *step functions*, and these will be defined analytically in the next section. Geometrically, a step function is one whose graph consists of various horizontal line segments and may also contain certain isolated points. A typical example is shown in Figure 1.22. The domain of this particular step-function is the interval consisting of all x satisfying $1 \leq x \leq 5$. If we denote this function by s , its function values are defined as follows:

† We remind the reader that if $a > 0$ we use the symbol $\sqrt{a}$ to denote the *positive* square root of a . The negative square root is $-\sqrt{a}$.

$$\begin{aligned} s(1) &= 2, & s(x) &= 1 & \text{if } 1 < x < 2, & s(x) &= -1 & \text{if } 2 \leq x \leq 3, \\ s(x) &= \frac{1}{2} & \text{if } 3 < x < 4, & s(4) &= 1, & s(x) &= 3 & \text{if } 4 < x < 5, & s(5) &= 0. \end{aligned}$$

This particular function is not defined for $x < 1$ or for $x > 5$.

The next two examples are intended to give the reader some familiarity with the use of the functional notation. The reader should verify the correctness of each statement, and describe the set of values of x, y, t , etc., for which the given formulas are valid.

Example 13. If $f(x) = x^2$ for all real x , then we have:

$$\begin{aligned} (1) \quad f(-x) &= f(x), & (4) \quad f(2y) &= 4f(y), \\ (2) \quad f(y) - f(x) &= (y - x)(y + x), & (5) \quad f(t^2) &= [f(t)]^2, \\ (3) \quad f(x + h) - f(x) &= 2xh + h^2, & (6) \quad \sqrt{f(a)} &= |a|. \end{aligned}$$

Example 14. If $g(x) = \sqrt{4 - x^2}$ for $|x| \leq 2$, then we have:

$$\begin{aligned} (1) \quad g(-x) &= g(x), & (4) \quad g(a - 2) &= \sqrt{4a - a^2}, \\ (2) \quad g(2y) &= 2\sqrt{1 - y^2}, & (5) \quad g\left(\frac{s}{2}\right) &= \frac{1}{2}\sqrt{16 - s^2}, \\ (3) \quad g\left(\frac{1}{t}\right) &= \frac{\sqrt{4t^2 - 1}}{|t|}, & (6) \quad \frac{1}{2 + g(x)} &= \frac{2 - g(x)}{x^2}. \end{aligned}$$

1.27 Intervals and ordinate sets

In the theory of integration we are concerned primarily with real functions whose domains are intervals on the x -axis. Sometimes it is important to distinguish between intervals which include their endpoints and those which do not. This distinction is made by introducing the following definitions:

If $a < b$, we denote by $[a, b]$ the collection of all x satisfying the inequalities $a \leq x \leq b$ and refer to this collection as the *closed interval* from a to b . The corresponding *open interval*, written (a, b) , is the set of all x satisfying $a < x < b$. The closed interval $[a, b]$ includes the endpoints a and b , whereas the open interval does not. (See Figure 1.23.) The open interval (a, b) is also called the *interior* of $[a, b]$. Half-open intervals $(a, b]$ and $[a, b)$, which include just one endpoint, are defined by the inequalities $a < x \leq b$ and $a \leq x < b$, respectively.

Let f be a nonnegative function whose domain is a closed interval $[a, b]$. The portion of the plane between the graph of f and the x -axis is called the *ordinate set* of f . More

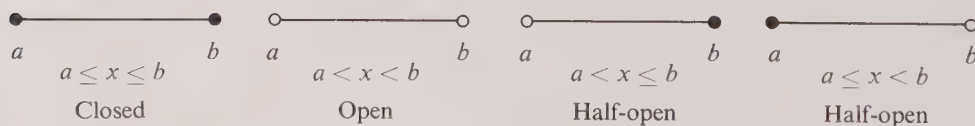


FIGURE 1.23 Examples of intervals.

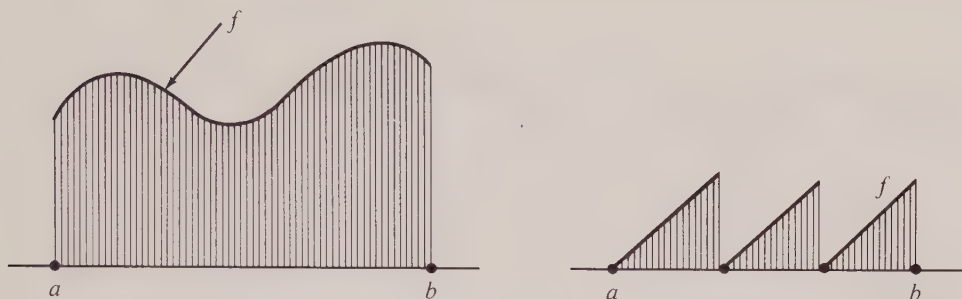


FIGURE 1.24 Examples of ordinate sets.

precisely, the ordinate set of f is the collection of all points (x, y) satisfying the inequalities

$$a \leq x \leq b, \quad 0 \leq y \leq f(x).$$

In each of the examples shown in Figure 1.24 the shaded portion represents the ordinate set of the corresponding function.

Ordinate sets are the geometric objects whose areas we want to define and compute by means of the integral calculus. We shall define the concept of integral first for step functions and then use the integral of a step function to formulate the definition of integral for more general functions. Integration theory for step functions is extremely simple and leads in a natural way to the corresponding theory for more general functions. To start this program, it is necessary to have an analytic definition of what we mean by a step function. This may be given most simply in terms of the concept of a *partition* to which we turn now.

1.28 Partitions and step functions

Suppose we decompose a given closed interval $[a, b]$ into n subintervals by inserting $n - 1$ points of subdivision, say $x_1, x_2, \dots, x_{n-1}$, subject only to the restriction

$$(1.26) \quad a < x_1 < x_2 < \dots < x_{n-1} < b.$$

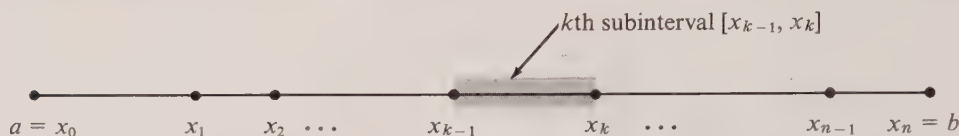
It is convenient to denote the point a itself by x_0 and the point b by x_n . A collection of points satisfying (1.26) is called a *partition* P of $[a, b]$, and we shall use the symbol

$$P = \{x_0, x_1, \dots, x_n\}$$

to designate this partition. The partition P determines n closed subintervals

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

A typical closed subinterval is $[x_{k-1}, x_k]$ and it is referred to as the k th closed subinterval of P ; an example is shown in Figure 1.25. The corresponding open interval (x_{k-1}, x_k) is called the k th open subinterval of P .

FIGURE 1.25 An example of a partition of $[a, b]$.

Now we are ready to formulate an analytic definition of a step function.

DEFINITION OF A STEP FUNCTION. A function s , whose domain is a closed interval $[a, b]$, is called a step function if there is a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ such that s is constant on each open subinterval of P . That is to say, for each $k = 1, 2, \dots, n$ there is a real number s_k such that

$$s(x) = s_k \quad \text{if } x_{k-1} < x < x_k.$$

Note. At each of the endpoints x_{k-1} and x_k the function must have some well-defined value, but this need not be the same as s_k .

Example. A familiar example of a step function is the “postage function,” whose graph is shown in Figure 1.26. Assume that the charge for first-class mail for parcels weighing up to 20 pounds is 4 cents for every ounce or fraction thereof. The graph shows the number of 4-cent stamps required for mail weighing up to 4 ounces. In this case the line segments on the graph are half-open intervals containing their right endpoints. The domain of the function is the interval $[0, 320]$.

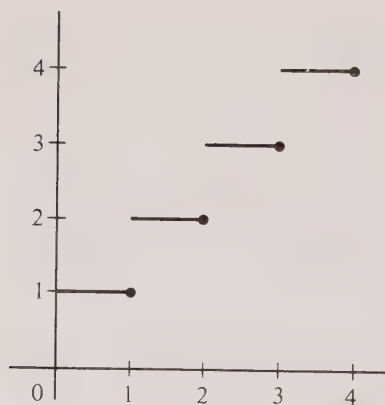
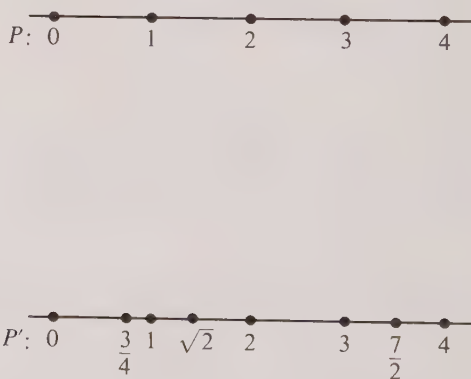


FIGURE 1.26 The postage function.

FIGURE 1.27 A partition P of $[0, 4]$ and a refinement P' .

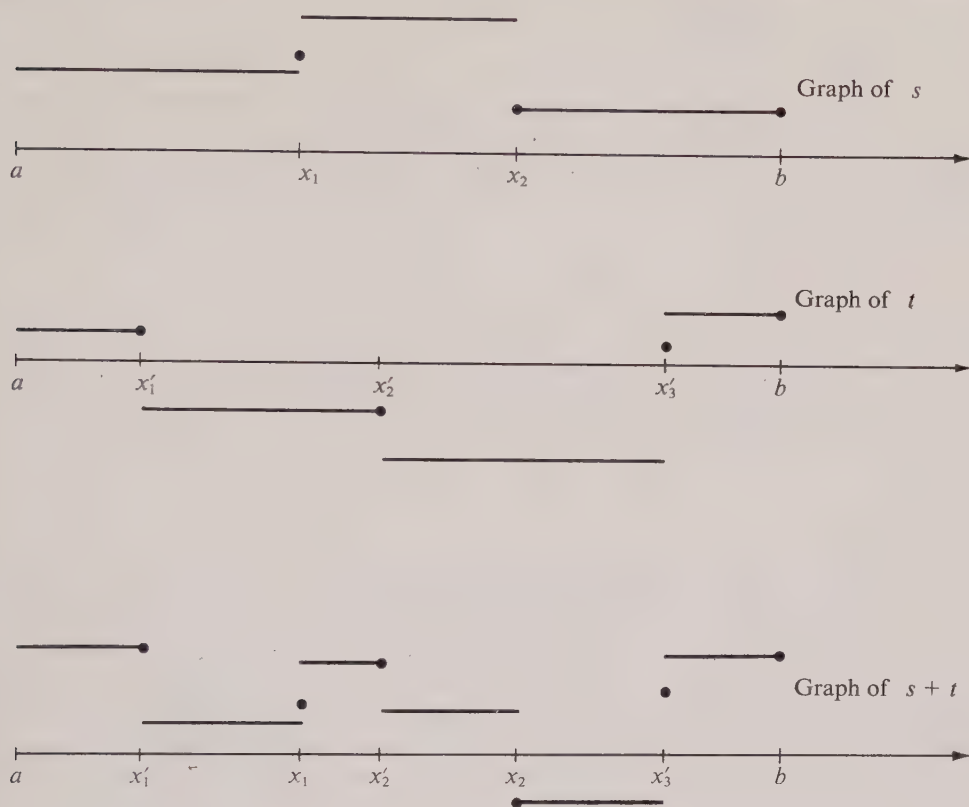


FIGURE 1.28 The sum of two step functions.

From a given partition P of $[a, b]$, we can always form a new partition P' by adjoining more subdivision points to those already in P . Such a partition P' is called a *refinement* of P and is said to be *finer than* P . For example, the collection $P = \{0, 1, 2, 3, 4\}$ is a partition of the interval $[0, 4]$. If we adjoin the points $3/4$, $\sqrt{2}$, and $7/2$, we obtain a new partition P' of $[0, 4]$, namely, $P' = \{0, 3/4, 1, \sqrt{2}, 2, 3, 7/2, 4\}$, which is a refinement of P . (See Figure 1.27.) If a step function is constant on the open subintervals of P , then it is also constant on the open subintervals of every refinement P' .

New step-functions may be formed from given step-functions by adding corresponding function values. For example, suppose s and t are step functions, both defined on the same interval $[a, b]$. Let P_1 and P_2 be partitions of $[a, b]$ such that s is constant on the open subintervals of P_1 and t is constant on the open subintervals of P_2 . Using s and t , we may define a new step-function u by the equation

$$u(x) = s(x) + t(x) \quad \text{if } a \leq x \leq b.$$

To show that u is actually a step function, we must exhibit a partition P such that u is constant on the open subintervals of P . For the new partition P we take all the points

of P_1 along with all the points of P_2 . This partition is called the *common refinement* of P_1 and P_2 and is denoted by the symbol $P_1 \cup P_2$ (read as " P_1 union P_2 "). Since both s and t are constant on the open subintervals of the common refinement $P_1 \cup P_2$, the same is true of u . The function u so obtained is called the *sum* of s and t and is denoted by $s + t$. An example is illustrated in Figure 1.28. The partition P_1 is $\{a, x_1, x_2, b\}$, the partition P_2 is $\{a, x'_1, x'_2, x'_3, b\}$, and the common refinement is $\{a, x'_1, x_1, x'_2, x_2, x'_3, b\}$.

Similarly, we can form a new step-function v from s and t by multiplying corresponding function values, thus:

$$v(x) = s(x)t(x) \quad \text{if } a \leq x \leq b.$$

This new step-function v is called the *product* of s and t and is denoted by $s \cdot t$. An important special case occurs when one of the factors, say t , is constant throughout $[a, b]$. If $t(x) = c$ for each x in $[a, b]$, then each function value $v(x)$ is obtained by multiplying the step function $s(x)$ by the constant c .

1.29 Exercises

Many examples of step functions may be constructed by use of the so-called *greatest-integer function*. Its value at each real x is the greatest integer less than or equal to x , and is denoted by $[x]$ (read as "square bracket x " or "the greatest integer in x "). Thus $[x]$ is that integer which satisfies the inequalities $[x] \leq x < [x] + 1$. For example, $[2.9] = 2$, $[-7/2] = -4$, $[5] = 5$, $[\sqrt{2}] = 1$, etc. Its graph over the interval $[-3, 3]$ is shown in Figure 1.29. Note that this function is constant on the half-open intervals of the form $[n, n + 1)$, where n is an integer. Therefore the horizontal line segments in Figure 1.29 contain their left endpoints but not their right endpoints.

If we put $f(x) = [2x]$, then f is a step function that is constant over half-open intervals of the form $[n/2, (n + 1)/2)$, where n is an integer. Its graph over $[-1, 3]$ is shown in Figure 1.30.

The following exercises are devoted to further properties of the greatest-integer function.

1. In each case, f is a function defined over the interval $[-3, 3]$ by the formula given. Draw the graph of f . If f is a step function, find a partition P of $[-3, 3]$ such that f is constant on the open subintervals of P .

- | | |
|----------------------------------|--|
| (a) $f(x) = x + [x]$. | (f) $f(x) = [x] + [x + \frac{1}{2}]$. |
| (b) $f(x) = x - [x]$. | (g) $f(x) = [x + \frac{1}{3}]$. |
| (c) $f(x) = [-x]$. | (h) $f(x) = [x + \frac{2}{3}]$. |
| (d) $f(x) = 2[x]$. | (i) $f(x) = [3x]$. |
| (e) $f(x) = [x + \frac{1}{2}]$. | (j) $f(x) = [x] + [x + \frac{1}{3}] + [x + \frac{2}{3}]$. |

[Hint for (f). Draw the graph of part (e) first and add ordinates corresponding with those in Figure 1.29.]

In Exercises 2 through 6 prove that the greatest-integer function has the properties indicated.

- $[x + n] = [x] + n$ for every integer n .
- $[-x] = \begin{cases} -[x] & \text{if } x \text{ is an integer,} \\ -[x] - 1 & \text{otherwise.} \end{cases}$
- $[x + y] = [x] + [y]$ or $[x] + [y] + 1$.
- $[2x] = [x] + [x + \frac{1}{2}]$.
- $[3x] = [x] + [x + \frac{1}{3}] + [x + \frac{2}{3}]$.

7. A point (x, y) in the plane is called a *lattice point* if both coordinates x and y are integers. Let f be a nonnegative function whose domain is the interval $[a, b]$, where a and b are integers, $a < b$. (An example is shown in Figure 1.31.) Let S denote the region in the xy -plane consisting

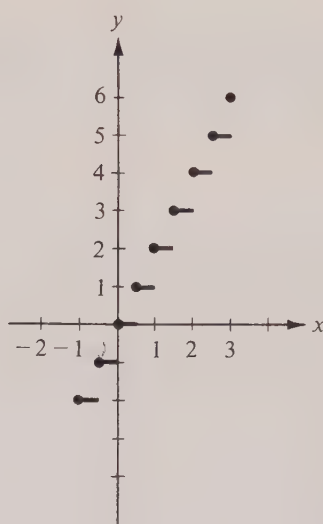
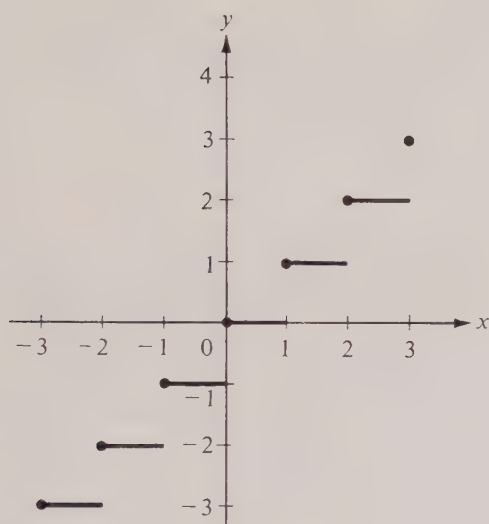
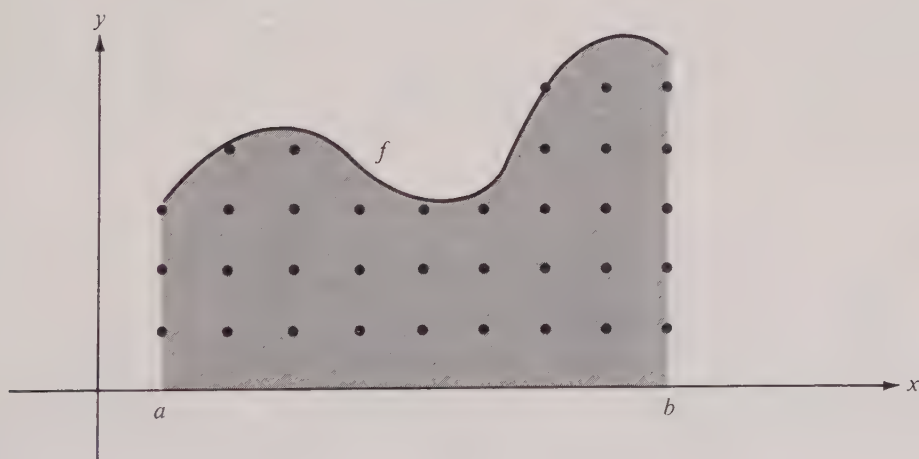
FIGURE 1.29 $f(x) = [x]$, for $-3 \leq x \leq 3$.FIGURE 1.30 $f(x) = [2x]$, for $-1 \leq x \leq 3$.

FIGURE 1.31 Lattice points in a plane region.

of all points (x, y) satisfying $a \leq x \leq b$ and $0 < y \leq f(x)$. Show that the number of lattice points in the region S is equal to the sum

$$\sum_{n=a}^b [f(n)].$$

8. If a and b are positive integers with no common factor, then we have the formula

$$\sum_{n=1}^{b-1} \left[\frac{na}{b} \right] = \frac{(a-1)(b-1)}{2}.$$

Note. When $b = 1$ the sum on the left is understood to be 0.

(a) Derive this result by a geometric argument, counting lattice points in a right triangle.

(b) Derive the result analytically as follows: By changing the index of summation, note that $\sum_{n=1}^{b-1} [na/b] = \sum_{n=1}^{b-1} [a(b-n)/b]$. Now apply Exercises 2 and 3 to the bracket on the right.

9. Prove that a triangle whose vertices are lattice points cannot be equilateral.

10. Let P be a polygon whose vertices are lattice points. The area of P is $I + \frac{1}{2}B - 1$, where I denotes the number of lattice points inside the polygon and B denotes the number on the boundary.

(a) Prove that this formula is correct for rectangles with sides parallel to the coordinate axes.

(b) Prove that the formula is correct for triangles.

(c) Try to construct a proof for general polygons.

1.30 The definition of the integral for step functions

In this section we shall introduce the definition of the integral for step functions. The definition will be constructed so that when a step function is nonnegative its integral is a number which agrees with our intuitive notion of what the area of its ordinate set "should be."

Let s be a step function defined on $[a, b]$ and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ such that s is constant on the open subintervals of P . Denote by s_k the constant value that s takes in the k th open subinterval, so that

$$s(x) = s_k \quad \text{if } x_{k-1} < x < x_k, \quad k = 1, 2, \dots, n.$$

DEFINITION OF THE INTEGRAL OF STEP FUNCTIONS. The integral of s from a to b , denoted by the symbol $\int_a^b s(x) dx$, is defined by the following formula:

$$(1.27) \quad \int_a^b s(x) dx = \sum_{k=1}^n s_k \cdot (x_k - x_{k-1}).$$

That is to say, to obtain the value of the integral, we multiply each constant value s_k by the length of the k th subinterval to form the product $s_k \cdot (x_k - x_{k-1})$, and then we add together all the products.

Note that the values of s at the subdivision points are immaterial since they do not appear on the right-hand side of (1.27). In particular, if s is constant on (a, b) , say $s(x) = c$ if $a < x < b$, then we have

$$\int_a^b s(x) dx = c \sum_{k=1}^n (x_k - x_{k-1}) = c(b - a),$$

regardless of the values $s(a)$ and $s(b)$.

Note also that if P is replaced by any finer partition P' , the sum on the right of (1.27) is not changed although some of the terms may be changed. For example, if we change from P to a finer partition P' by inserting exactly one new subdivision point, say the point x'_1 , where $x_0 < x'_1 < x_1$, then the first term on the right of (1.27) is replaced by the two terms

$$s_1 \cdot (x'_1 - x_0) + s_1 \cdot (x_1 - x'_1) = s_1 \cdot (x_1 - x_0),$$

and the rest of the terms are unchanged so the sum itself is unchanged. This means that the value of the integral is independent of the choice of P so long as s is constant on the open subintervals of P .

When the step function s is nonnegative we define the *area of its ordinate set* to be the value of the integral in (1.27). In particular, when s is constant on $[a, b]$ this definition implies that the area of a rectangle of height c and base $b - a$ is $c \cdot (b - a)$, the product of its base and height. Notice that, if s is constant on only one of the half-open intervals $(a, b]$ or $[a, b)$ or on the open interval (a, b) , the ordinate set of s is a rectangle with part or all of its vertical boundaries missing or perhaps with an extra vertical segment attached to one or both boundaries. The definition assigns the same area to all these regions. In the two examples shown in Figure 1.32, s has the constant value $5/4$ on the open interval $(1, 3)$. In Figure 1.32(a) we have $s(1) = s(3) = 0$, and in Figure 1.32(b) we have $s(1) = 3/2$, $s(3) = 3/4$. Both ordinate sets have the same area: $\int_1^3 s(x) dx = (5/4) \cdot 2 = 5/2$.

The ordinate set of a more general nonnegative step-function consists of a finite number of rectangles, one for each interval of constancy. The ordinate set may also contain or lack certain vertical line segments, depending upon how the function is defined at the subdivision points. The definition assigns to such an ordinate set an area equal to the sum of the areas of the individual rectangles which comprise it. The vertical segments

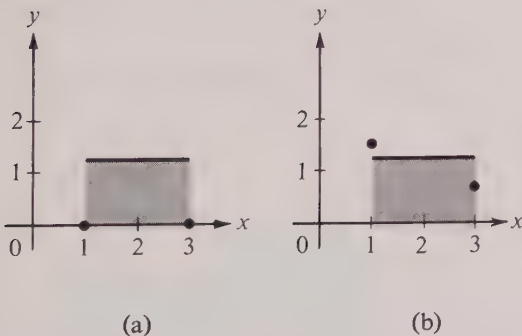


FIGURE 1.32 Changes in function values at two points do not alter area of ordinate set.

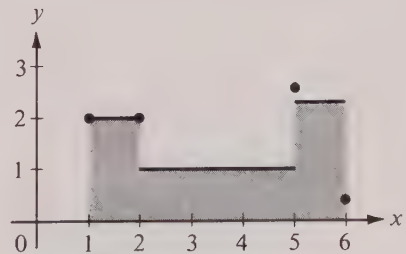


FIGURE 1.33 The ordinate set of a step function.

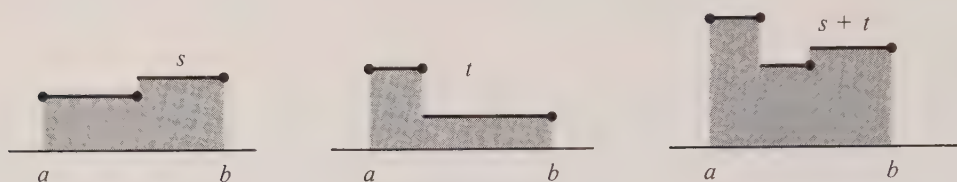


FIGURE 1.34 Illustrating the additive property of the integral.

make no contribution to the area and it is convenient to say that such segments have zero area. In the example shown in Figure 1.33 we have

$$s(x) = \begin{cases} 2 & \text{if } 1 \leq x \leq 2, \\ 1 & \text{if } 2 < x < 5, \\ 5/2 & \text{if } x = 5, \\ 7/4 & \text{if } 5 < x < 6, \\ 1/2 & \text{if } x = 6. \end{cases}$$

The area of the corresponding ordinate set is

$$\int_1^6 s(x) \, dx = 2 \cdot 1 + 1 \cdot 3 + \frac{7}{4} \cdot 1 = \frac{27}{4}.$$

1.31 Properties of the integral of a step function

In this section we describe a number of fundamental properties satisfied by the integral of a step function. Most of these properties seem obvious when they are interpreted geometrically and some of them may even seem trivial. All these properties carry over to integrals of more general functions and it will be a simple matter to prove them in the general case once we have them established for step functions. The properties are listed below as theorems, and in each case a geometric interpretation is given in terms of areas. Analytic proofs of the theorems are outlined in the next section.

The first property states that the integral of a sum of two step-functions is equal to the sum of the integrals. That is, we have

1-36 THEOREM.

$$\int_a^b [s(x) + t(x)] \, dx = \int_a^b s(x) \, dx + \int_a^b t(x) \, dx.$$

This is known as the *additive* property and it is illustrated in Figure 1.34.

The next property, illustrated in Figure 1.35, is called the *homogeneous* property. It states that if all the function values are multiplied by a constant c , then the integral is also multiplied by c .

1-37 THEOREM. *Homogeneous property.* For every real number c we have

$$\int_a^b c \cdot s(x) dx = c \int_a^b s(x) dx.$$

These two theorems can be combined into one formula known as the linearity property.

1-38 THEOREM. *Linearity property.* For every real c_1 and c_2 we have

$$\int_a^b [c_1 s(x) + c_2 t(x)] dx = c_1 \int_a^b s(x) dx + c_2 \int_a^b t(x) dx.$$

Next, we have a *comparison* theorem which tells us that if one step-function has larger values than another throughout $[a, b]$, its integral over this interval is also larger.

1-39 THEOREM. *Comparison theorem.* If $s(x) < t(x)$ for every x in $[a, b]$, then

$$\int_a^b s(x) dx < \int_a^b t(x) dx.$$

Interpreted geometrically, this theorem states that if one ordinate set lies inside another, the area of the smaller region cannot exceed that of the larger.

The foregoing properties all refer to step functions defined on a common interval. The integral has further important properties that relate integrals over different intervals. Among these we have

1-40 THEOREM. *Additivity with respect to the interval of integration.*

$$\int_a^c s(x) dx + \int_c^b s(x) dx = \int_a^b s(x) dx \quad \text{if } a < c < b.$$

This theorem reflects a desirable property of area, illustrated in Figure 1.36. If an ordinate set is decomposed into two ordinate sets, the sum of the areas of the two parts is equal to the area of the whole.

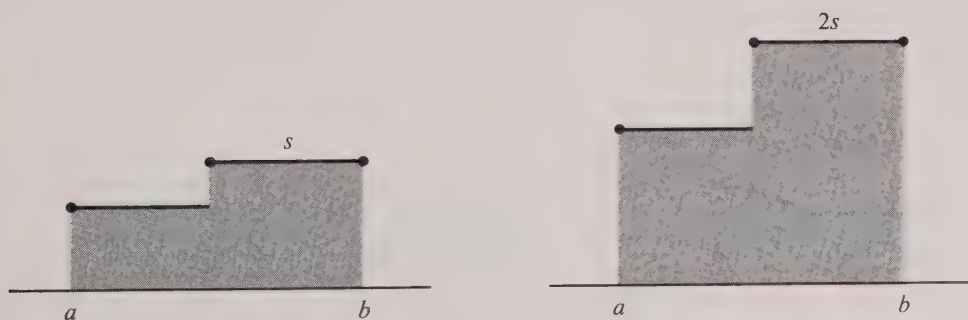


FIGURE 1.35 Illustrating the homogeneous property of the integral (with $c = 2$).

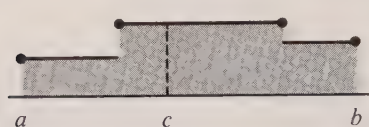


FIGURE 1.36 Additivity with respect to the interval of integration.

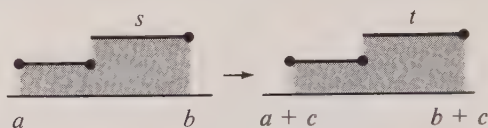


FIGURE 1.37 Illustrating invariance of the integral under translation: $t(x) = s(x - c)$.

The next theorem may be described as *invariance under translation*. If the ordinate set of a step function s is “shifted” by an amount c , the resulting ordinate set is that of another step function t related to s by the equation $t(x) = s(x - c)$. If s is defined on $[a, b]$, then t is defined on $[a + c, b + c]$, and their ordinate sets have the same area. This property is expressed analytically as follows:

1-41 THEOREM. *Invariance under translation.*

$$\int_a^b s(x) dx = \int_{a+c}^{b+c} s(x - c) dx \quad \text{for every real } c.$$

Its geometric meaning is illustrated in Figure 1.37 for $c > 0$. When $c < 0$ the ordinate set is shifted to the left.

The homogeneous property (Theorem 1-37) explains what happens to an integral under a change of scale on the y -axis. The following theorem deals with a change of scale on the x -axis. If s is a step function defined on an interval $[a, b]$ and if we distort the scale in the horizontal direction by multiplying all x -coordinates by a factor $c > 0$, then the new graph is that of another step function t defined on the interval $[ca, cb]$ and related to s by the equation

$$t(x) = s\left(\frac{x}{c}\right) \quad \text{if } ca \leq x \leq cb.$$

An example with $c = 2$ is shown in Figure 1.38 and it suggests that the distorted figure has an area twice that of the original figure. More generally, distortion by a positive factor c has the effect of multiplying the integral by c . Expressed analytically, this property assumes the following form:

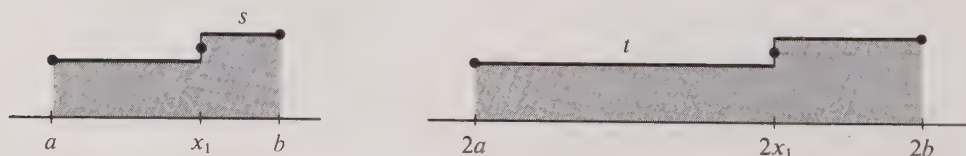


FIGURE 1.38 Change of scale on the x -axis: $t(x) = s(x/2)$.

1-42 THEOREM. *Expansion or contraction of the interval of integration.*

$$\int_{ca}^{cb} s\left(\frac{x}{c}\right) dx = c \int_a^b s(x) dx \quad \text{for every } c > 0.$$

Until now, when we have used the symbol $\int_a^b$ it has been understood that the lower limit a was less than the upper limit b . It is convenient to extend our ideas somewhat and consider integrals with a lower limit larger than the upper limit. This is done by defining

$$(1.28) \quad \int_b^a s(x) dx = - \int_a^b s(x) dx \quad \text{if } a < b.$$

We also define

$$\int_a^a s(x) dx = 0,$$

a definition that is suggested by putting $a = b$ in (1.28). These conventions allow us to conclude that Theorem 1-40 is valid not only when c is between a and b but for any arrangement of the points a, b, c . Theorem 1-40 is sometimes written in the form

$$\int_a^c s(x) dx + \int_c^b s(x) dx + \int_b^a s(x) dx = 0.$$

Similarly we can extend the range of validity of Theorem 1-42 and allow the constant c to be negative.

There is one remark that should be made concerning notation. The letter x that appears in the symbol $\int_a^b s(x) dx$ plays no essential role in the definition of the integral. Any other convenient symbol would serve equally well. The letters t, u, v, z are frequently used for this purpose and it is agreed that instead of $\int_a^b s(x) dx$ we may write $\int_a^b s(t) dt$, $\int_a^b s(u) du$, etc., all these being considered as alternative notations for the same thing. The symbols x, t, u , etc. that are used in this way are called "dummy variables." They are analogous to dummy indices used in the summation notation.

There is a tendency among some authors of calculus textbooks to omit the dummy variable and the d -symbol altogether and to write simply $\int_a^b s$ for the integral. One good reason for using this abbreviated symbol is that it suggests more strongly that the integral depends only on the *function* s and on the *interval* $[a, b]$. Also, certain formulas appear **simpler** in this notation. For example, the additive property becomes $\int_a^b (s + t) = \int_a^b s + \int_a^b t$. On the other hand, it becomes awkward to write formulas like Theorems 1-41 and 1-42 in the abbreviated notation. More important than this, we shall find later that the original Leibniz notation has certain practical advantages. The symbol dx , which appears to be rather superfluous at this stage, turns out to be an extremely useful computational device in connection with many routine calculations with integrals.

1.32 Exercises

1. Compute the value of each of the following integrals. You may use the theorems of the foregoing section whenever it is convenient to do so. The notation $[x]$ denotes the greatest integer $\leq x$.

$$(a) \int_{-1}^3 [x] dx,$$

$$(d) \int_{-1}^3 [2x] dx,$$

$$(b) \int_{-1}^3 [x + \frac{1}{2}] dx,$$

$$(e) \int_{-1}^3 ([x] + [x + \frac{1}{2}]) dx,$$

$$(c) \int_{-1}^3 2[x] dx,$$

$$(f) \int_{-1}^3 [-x] dx.$$

2. Compute the value of $\int_1^5 s(x) dx$, where s is the step function shown in Figure 1.22 on p. 42.

3. Show that

$$\int_a^b [x] dx + \int_a^b [-x] dx = a - b.$$

4. Give an example of a step function s , defined on the closed interval $[0, 5]$, which has the following properties:

$$\int_0^2 s(x) dx = 5 \quad \text{and} \quad \int_0^5 s(x) dx = 2.$$

Be sure to define $s(x)$ for every x in the closed interval $[0, 5]$.

5. Show that the translation property (Theorem 1-41) may be expressed in the equivalent form

$$\int_{a+c}^{b+c} f(x) dx = \int_a^b f(x+c) dx.$$

6. Show that the following property is equivalent to Theorem 1-42:

$$\int_{ca}^{cb} f(x) dx = c \int_a^b f(cx) dx.$$

Analytic proofs of the properties of the integral given in the foregoing section are requested in the following exercises. The proofs of Theorems 1-37 and 1-42 are worked out here as samples. Hints are given for the others.

Proof of Theorem 1-37: $\int_a^b c \cdot s(x) dx = c \int_a^b s(x) dx$ for every real c .

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ such that s is constant on the open subintervals of P . Assume $s(x) = s_k$ if $x_{k-1} < x < x_k$ ($k = 1, 2, \dots, n$). Then $c \cdot s(x) = c \cdot s_k$ if $x_{k-1} < x < x_k$, and hence by the definition of an integral we have

$$\int_a^b c \cdot s(x) dx = \sum_{k=1}^n c \cdot s_k \cdot (x_k - x_{k-1}) = c \sum_{k=1}^n s_k \cdot (x_k - x_{k-1}) = c \int_a^b s(x) dx.$$

Proof of Theorem 1-42: $\int_{ca}^{cb} s\left(\frac{x}{c}\right) dx = c \int_a^b s(x) dx$ if $c > 0$.

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of the interval $[a, b]$ such that s is constant on the open subintervals of P . Assume that $s(x) = s_k$ if $x_{k-1} < x < x_k$. Let $t(x) = s(x/c)$ if $ca \leq x \leq cb$. Then $t(x) = s_k$ if x lies in the open interval (cx_{k-1}, cx_k) ; hence $P' = \{cx_0, cx_1, \dots, cx_n\}$ is a partition of $[ca, cb]$ and t is constant on the open subintervals of P' . Therefore t is a step function whose integral is

$$\int_{ca}^{cb} t(x) dx = \sum_{k=1}^n s_k \cdot (cx_k - cx_{k-1}) = c \sum_{k=1}^n s_k \cdot (x_k - x_{k-1}) = c \int_a^b s(x) dx.$$

7. Prove Theorem 1-36 (the additive property). [*Hint.* Use the additive property for sums: $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$.]
8. Prove Theorem 1-38 (the linearity property). [*Hint.* Use the additive property and the homogeneous property.]
9. Prove Theorem 1-39 (the comparison theorem). [*Hint.* Use the corresponding property for sums: $\sum_{k=1}^n a_k < \sum_{k=1}^n b_k$ if $a_k < b_k$ for $k = 1, 2, \dots, n$.]
10. Prove Theorem 1-40 (additivity with respect to the interval). [*Hint.* If P_1 is a partition of $[a, c]$ and P_2 a partition of $[c, b]$, then the points of P_1 along with those of P_2 form a partition of $[a, b]$.]
11. Prove Theorem 1-41 (invariance under translation). [*Hint.* If $P = \{x_0, x_1, \dots, x_n\}$ is a partition of $[a, b]$ then $P' = \{x_0 + c, x_1 + c, \dots, x_n + c\}$ is a partition of $[a + c, b + c]$.]
12. If, instead of defining integrals of step functions by using formula (1.27), we used the definition

$$\int_a^b s(x) dx = \sum_{k=1}^n s_k^* \cdot (x_k - x_{k-1}),$$

a new and different theory of integration would result. Which of the following properties would remain valid in this new theory?

- (a) $\int_a^b s + \int_b^c s = \int_a^c s$.
- (b) $\int_a^b (s + t) = \int_a^b s + \int_a^b t$.
- (c) $\int_a^b c \cdot s = c \int_a^b s$.
- (d) $\int_{a+c}^{b+c} s(x) dx = \int_a^b s(x + c) dx$.
- (e) If $s(x) < t(x)$ for each x in $[a, b]$, then $\int_a^b s < \int_a^b t$.

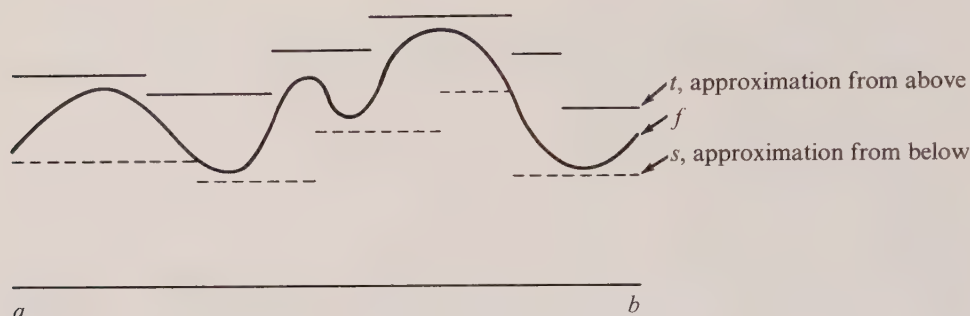
1.33 The integral of more general functions

The integral $\int_a^b s(x) dx$ has been defined when s is a step function. In this section we shall formulate a definition of $\int_a^b f(x) dx$ that will apply to more general functions f . The definition will be constructed so that the resulting integral has all the properties listed in Section 1.31.

The approach will be patterned somewhat after the method of Archimedes, which was explained above in Section 1.2. The idea is simply this: We begin by approximating the function f from below and from above by step functions, as suggested in Figure 1.39. By this we mean that we choose an arbitrary step function, say s , whose graph lies below that of f , and an arbitrary step function, say t , whose graph lies above that of f . Next, we consider the collection of all the numbers $\int_a^b s(x) dx$ and $\int_a^b t(x) dx$ obtained by choosing s and t in all possible ways. In general, we have

$$\int_a^b s(x) dx < \int_a^b t(x) dx$$

because of the comparison theorem. If the integral of f is to obey the comparison theorem, then it must be a number which falls between $\int_a^b s(x) dx$ and $\int_a^b t(x) dx$ for every pair of

FIGURE 1.39 Approximating a function f from above and below by step functions.

approximating functions s and t . If there is *only one* number which has this property, then it seems reasonable to define the integral of f to be this number.

There is only one thing that can cause trouble in this procedure, and it occurs in the very first step. Unfortunately, it is not possible to approximate *every* function from above and from below by step functions. For example, the function f given by the equations

$$f(x) = \frac{1}{x} \quad \text{if } x \neq 0, \quad f(0) = 0,$$

is defined for all real x , but on any interval $[a, b]$ containing the origin we cannot surround f by step functions. This is due to the fact that f has arbitrarily large values near the origin, or, as we say, f is *unbounded* in every neighborhood of the origin (see Figure 1.40). Therefore, we shall first restrict ourselves to those functions that are *bounded* on $[a, b]$, that is, to those functions f for which there exists a number $M > 0$ such that

$$(1.29) \quad -M \leq f(x) \leq M$$

for every x in $[a, b]$. Geometrically, the graph of such a function lies between the graphs

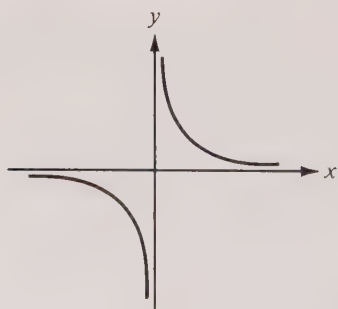


FIGURE 1.40 An unbounded function.

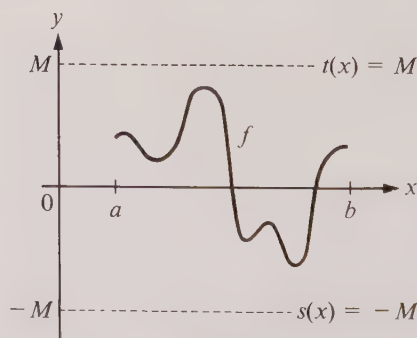


FIGURE 1.41 A bounded function.

of two constant step functions s and t having the values $-M$ and $+M$, respectively. (See Figure 1.41.) In a case like this, we say that f is bounded by M . The two inequalities in (1.29) can also be written as

$$|f(x)| \leq M.$$

With this point taken care of, we can proceed to carry out the plan described above and to formulate the definition of the integral.

DEFINITION OF THE INTEGRAL OF A BOUNDED FUNCTION. Let f be a function defined and bounded on $[a, b]$. Let s and t denote arbitrary step functions defined on $[a, b]$ such that

$$(1.30) \quad s(x) \leq f(x) \leq t(x)$$

for every x in $[a, b]$. If there is one and only one number I such that

$$(1.31) \quad \int_a^b s(x) dx \leq I \leq \int_a^b t(x) dx$$

for every pair of step functions s and t satisfying (1.30), then this number I is called the integral of f from a to b and it is denoted by the symbol $\int_a^b f(x) dx$. When such an I exists the function f is said to be integrable on $[a, b]$.

If $a < b$ we define $\int_b^a f(x) dx = - \int_a^b f(x) dx$, provided f is integrable on $[a, b]$. We also define $\int_a^a f(x) dx = 0$. If f is integrable on $[a, b]$, we say that the integral $\int_a^b f(x) dx$ exists. The function f is called the *integrand*, the numbers a and b are called the *limits of integration*, and the interval $[a, b]$ the *interval of integration*.

Note. Using the least-upper-bound axiom (Axiom 10 in Section 1.12), we can show that there is always *at least one* number I satisfying the inequalities in (1.31) for every pair of step functions s and t satisfying (1.30). (A proof of this fact is outlined in Section 1.49.) Therefore a bounded function f will fail to be integrable if and only if there is *more than one* number I satisfying (1.31). To show that this can actually happen, consider the example of a function f defined on the interval $[0, 1]$ as follows:

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational, } \dagger \\ 2 & \text{if } x \text{ is irrational.} \end{cases}$$

Now take any pair of step functions s and t satisfying (1.30). Each open subinterval (x_{k-1}, x_k) on which s is constant contains rational points and therefore we must have $s(x) \leq 1$ for each x in (x_{k-1}, x_k) . Similarly, each open subinterval on which t is constant contains irrational points and therefore $t(x) \geq 2$ on these intervals. By the comparison theorem we have the two inequalities

$$\int_0^1 s(x) dx \leq 1 \quad \text{and} \quad \int_0^1 t(x) dx \geq 2.$$

Therefore *any* number I which lies between 1 and 2 will automatically satisfy (1.31) for every s and t . Since there is more than one I satisfying (1.31), this particular function is *not integrable* on $[0, 1]$.

$\dagger$ A number x is called *rational* if it can be expressed as the quotient of two integers, say $x = p/q$. Otherwise, x is called *irrational*. Some of the properties of rational and irrational numbers are given in Sections 1.13–1.16. For this particular example, all we need to know is that every open interval contains both rational and irrational numbers.

In the next section we shall list a number of basic properties of the integral, which may be derived under the assumption that the integral exists. Later we shall discuss questions concerned with the problem of its existence.

1.34 The basic properties of the integral

From the definition of the previous section, it is possible to derive the following properties of the integral as theorems. Proofs are given in Section 1.48.

- 1-43 THEOREM. *Linearity with respect to the integrand.* If both f and g are integrable on $[a, b]$, so is $c_1f + c_2g$ for every pair of constants c_1 and c_2 . Furthermore, we have

$$\int_a^b [c_1f(x) + c_2g(x)] dx = c_1 \int_a^b f(x) dx + c_2 \int_a^b g(x) dx.$$

Note. By use of mathematical induction, the linearity property may be generalized as follows: If $f_1, \dots, f_n$ are integrable on $[a, b]$ then so is $c_1f_1 + \dots + c_nf_n$ for all real $c_1, \dots, c_n$, and we have

$$\int_a^b \sum_{k=1}^n c_k f_k(x) dx = \sum_{k=1}^n c_k \int_a^b f_k(x) dx.$$

- 1-44 THEOREM. *Additivity with respect to the interval of integration.* If two of the following three integrals exist, the third also exists, and we have

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

- 1-45 THEOREM. *Invariance under translation.* If f is integrable on $[a, b]$, then for every real c we have

$$\int_a^b f(x) dx = \int_{a+c}^{b+c} f(x - c) dx.$$

- 1-46 THEOREM. *Expansion or contraction of the interval of integration.* If f is integrable on $[a, b]$, then for every real $c \neq 0$ we have

$$\int_a^b f(x) dx = \frac{1}{c} \int_{ca}^{cb} f\left(\frac{x}{c}\right) dx.$$

Note. In both Theorems 1-45 and 1-46 the existence of one of the integrals implies the existence of the other.

- 1-47 THEOREM. *Comparison theorem.* If both f and g are integrable on $[a, b]$ and if $g(x) \leq f(x)$ for every x in $[a, b]$, then we have

$$\int_a^b g(x) dx \leq \int_a^b f(x) dx.$$

An important special case of Theorem 1-47 occurs when $g(x) = 0$ for every x . In this case the theorem states that, if $f(x) \geq 0$ everywhere on $[a, b]$, then $\int_a^b f(x) dx \geq 0$. In other words, a nonnegative function has a nonnegative integral. It can also be shown that, if we have the *strict* inequality $g(x) < f(x)$ for all x in $[a, b]$, then the same strict inequality holds for the integrals, but the proof is not easy to give at this stage.

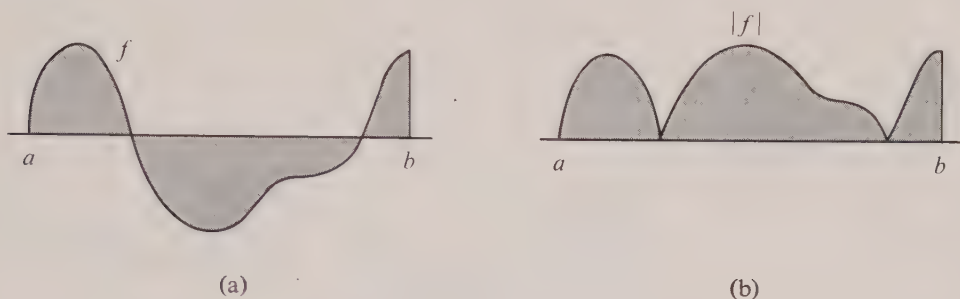


FIGURE 1.42 The absolute value of a function.

In Chapter 2 we shall discuss various methods for calculating the value of an integral without the necessity of using the definition in each case. These methods, however, are applicable to only a relatively small number of functions, and for most integrable functions the actual numerical value of the integral can only be estimated. This is usually done by approximating the integrand above and below by step functions or by other simple functions whose integrals can be evaluated exactly. Then the comparison theorem is used to obtain corresponding approximations for the integral of the function in question.

There is one “trivial” estimate that is often quite useful. If two constants m and M exist such that $m \leq f(x) \leq M$ for all x in $[a, b]$, then the definition of the integral immediately gives us the estimate

$$(1.32) \quad m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a).$$

In particular, if f is bounded by a positive constant M (as shown in Figure 1.41), this inequality holds with $m = -M$. In this case, by using absolute values, we can write the two inequalities in (1.32) as one, namely,

$$(1.33) \quad \left| \int_a^b f(x) \, dx \right| \leq M(b-a).$$

This is valid if $|f(x)| \leq M$ for all x in $[a, b]$.

Inequality (1.33) can be strengthened somewhat by introducing the *absolute value* of f . This is a new function, denoted by $|f|$, whose value at each x in the domain of f is $|f(x)|$. Its graph coincides with that of f at points where $f(x) \geq 0$. At points where $f(x) < 0$, the graph of $|f|$ may be obtained by reflecting the graph of f through the x -axis, as suggested in Figure 1.42. It can be shown that $|f|$ is integrable on an interval $[a, b]$ whenever f is integrable on this interval. Since we have $-|f(x)| \leq f(x) \leq |f(x)|$, the comparison theorem gives us the pair of inequalities

$$-\int_a^b |f(x)| \, dx \leq \int_a^b f(x) \, dx \leq \int_a^b |f(x)| \, dx.$$

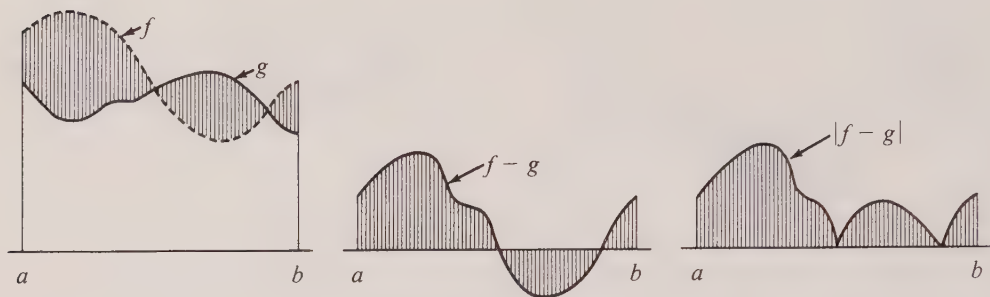


FIGURE 1.43

These are equivalent to the statement

$$(1.34) \quad \left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx .$$

1.35 The definition of area as an integral

In Section 1.30, the area of the ordinate set of a nonnegative step function was defined to be the integral of the function. In the same way, if f is nonnegative and integrable on $[a, b]$, we define the area of its ordinate set to be the integral of f from a to b . More generally, if f is integrable and takes on both positive and negative values, we define the area of that portion of the plane between the graph of f and the part of the x -axis from a to b to be the integral $\int_a^b |f(x)| \, dx$. In other words, the area of the shaded region in Figure 1.42(a) is, by definition, the area of the ordinate set in Figure 1.42(b).

If f and g are two functions, both of which are integrable on $[a, b]$, the area of the region between their graphs is defined to be the integral

$$(1.35) \quad \int_a^b |f(x) - g(x)| \, dx .$$

Note that $f - g$ is integrable because of the linearity property, hence $|f - g|$ is also integrable. Figure 1.43(a) shows the graphs of two functions f and g . The corresponding graphs of $f - g$ and $|f - g|$ are shown in Figure 1.43(b) and (c), respectively. The definition of area as given in (1.35) states that the area of the shaded region in Figure 1.43(a) is equal to that in Figure 1.43(c). To convince himself that this definition seems reasonable, the reader may imagine the shaded region in Figure 1.43(a) to be covered with thin pieces of wire as suggested by the vertical line segments shown between the two graphs. If we slide all these wires vertically until those endpoints touching the graph of g lie on the x -axis, we obtain the region shown in Figure 1.43(b). The portions of Figure 1.43(b) below the x -axis may be reflected through the x -axis to obtain the region shown in Figure 1.43(c). This region is the ordinate set of the nonnegative function $|f - g|$ and its area is given by the integral in (1.35).

A region of the type shown in Figure 1.43(a) has the property that every vertical line passing through a point in $[a, b]$ intersects the region in a line segment joining the graph of f to the graph of g . Let us refer to such a region as a region of Type I.

In Figure 1.44 is shown a region in which the roles of the x - and y -axes are interchanged. In this case the two curved portions of the boundary are the graphs of two functions F and G defined on an interval $[c, d]$ on the y -axis. The region consists of all points (x, y) for which $c \leq y \leq d$ and $F(y) \leq x \leq G(y)$. In this case horizontal lines intersect the region in line segments. The area of such a region, which we may call a region of Type II, is defined by the integral

$$\int_c^d |F(y) - G(y)| dy,$$

provided, of course, that the integral exists. Most of the regions that occur in practice are either of one of the two types just mentioned or else can be split into a finite number of pieces, each of which is of one of these two types. (An example is shown in Figure 1.45.) In a case like this the area of the whole region is defined to be the sum of the areas of the individual pieces. It can be shown that two different decompositions of the region always lead to the same sum of areas so the definition is unambiguous. (The proof of this fact is not trivial.)

Another property of area which also seems intuitively evident but whose proof is not entirely trivial is that *congruent regions have equal areas*. Congruence here is in the sense of elementary Euclidean geometry. Two sets S and T are said to be congruent if their points can be put in one-to-one correspondence in such a way that distances are preserved. That is to say, if two points p and q in S correspond to p' and q' in T , then the distance

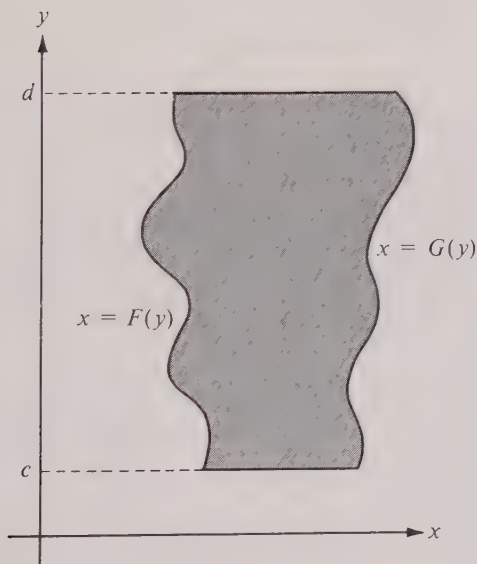


FIGURE 1.44 The area of the shaded region is $\int_c^d |F(y) - G(y)| dy$.

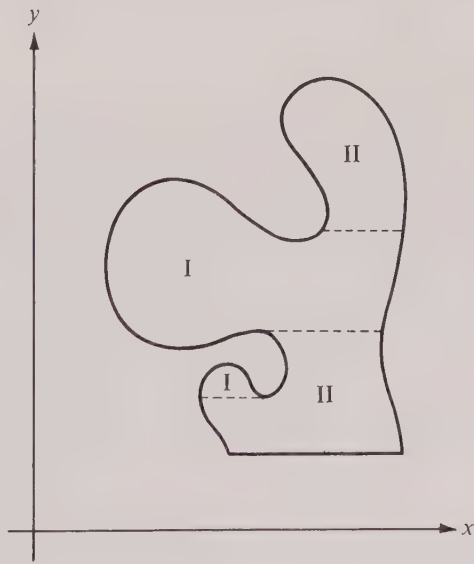


FIGURE 1.45 A region made up of a finite number of pieces of Types I and II.

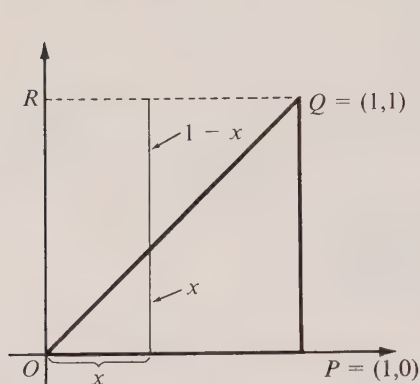


FIGURE 1.46 Area of triangle OPQ
 $= \int_0^1 x \, dx$.

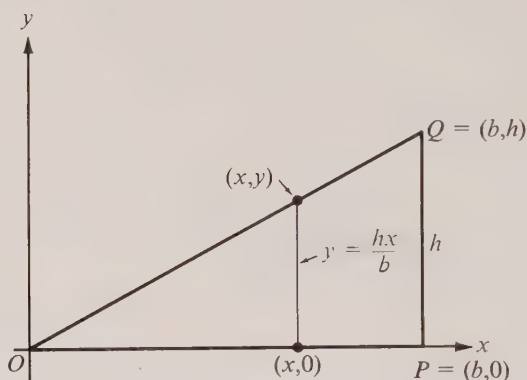


FIGURE 1.47 Area of triangle OPQ
 $= \int_0^b hx/b \, dx = bh \int_0^1 x \, dx$.

from p to q must be equal to the distance from p' to q' and this must be true for all choices of p and q . For example, suppose S is the region between the graphs of two integrable functions f and g , as shown in Figure 1.43(a), and suppose T is the region between the graphs of $f + c$ and $g + c$, where c is a constant. Then T differs from S by a vertical translation and, since $(f + c) - (g + c) = f - g$, the integral in (1.35) shows that S and T have the same area. When S and T are ordinate sets which differ only by a horizontal translation, the equality of their areas follows from Theorem 1-45.

1.36 Worked examples

The reader is undoubtedly familiar with various formulas for computing areas of certain simple geometric figures such as rectangles, triangles, and circles. We have already noted that our definition of area assigns to a rectangle of base b and altitude h the area bh , this being a consequence of the integral formula $\int_0^b h \, dx = bh$. In this section we shall derive a few more elementary integration formulas and use some of them to obtain further formulas for areas of special figures. In each case we shall assume that the integrals in question exist. Their existence will be established later as a consequence of a general theorem on integrability to be discussed in Section 1.41.

Example 1: The integral $\int_0^1 x \, dx$ and the area of an isosceles right triangle. Figure 1.46 shows the graph of the identity function $f(x) = x$ for $0 \leq x \leq 1$. Its ordinate set is a right triangle, of base 1 and altitude 1, whose area A , expressed as an integral, is

$$A = \int_0^1 x \, dx.$$

The figure suggests that $A = \frac{1}{2}$ because the area of triangle OPQ is half the area of the square $OPQR$. Assuming the integral exists, we shall prove that it has the value $\frac{1}{2}$ by using the general properties listed in Section 1.34.

The fact that triangles OPQ and ORQ have the same area may be expressed analytically as follows:

$$(1.36) \quad \int_0^1 x \, dx = \int_0^1 (1 - x) \, dx .$$

To prove (1.36) without recourse to a figure we use Theorem 1-46 with $c = -1$ and we write

$$\int_0^1 x \, dx = -\int_0^{-1} (-x) \, dx .$$

Next, we interchange the limits of integration and then use the translation property (Theorem 1-45) to obtain

$$-\int_0^{-1} (-x) \, dx = \int_{-1}^0 (-x) \, dx = \int_0^1 -(x - 1) \, dx = \int_0^1 (1 - x) \, dx .$$

This proves (1.36). Using the linearity property in the second integral of (1.36), we obtain

$$(1.37) \quad \int_0^1 (1 - x) \, dx = \int_0^1 1 \, dx - \int_0^1 x \, dx = 1 - \int_0^1 x \, dx .$$

Combining this with (1.36), we find $2 \int_0^1 x \, dx = 1$ and hence we have

$$(1.38) \quad \int_0^1 x \, dx = \frac{1}{2} .$$

Example 2: The integral $\int_0^b x \, dx$ and the area of an arbitrary right triangle. Figure 1.47 shows a right triangle of base b and altitude h . If we choose the coordinate axes as shown and select a point (x, y) on the hypotenuse, then by comparing similar triangles we find $y/x = h/b$ or $y = hx/b$. Therefore this triangle is the ordinate set of the function f defined by the equation

$$f(x) = \frac{hx}{b} \quad \text{if } 0 \leq x \leq b .$$

Its area A is the integral

$$(1.39) \quad A = \int_0^b \frac{hx}{b} \, dx = \frac{h}{b} \int_0^b x \, dx .$$

To evaluate the last integral we change the scale along the x -axis and use Theorem 1-46 (with $c = 1/b$) to obtain

$$(1.40) \quad \int_0^b x \, dx = b \int_0^1 (bx) \, dx = b^2 \int_0^1 x \, dx .$$

In Example 1 we found that $\int_0^1 x \, dx = \frac{1}{2}$. Substituting this in (1.40), we obtain

$$(1.41) \quad \int_0^b x \, dx = \frac{b^2}{2} .$$

Using this in (1.39), we find that $A = \frac{1}{2}bh$. That is to say, the area of a right triangle is one-half the product of its base and altitude if one side is along the x -axis. This is also true for a right triangle in any position because congruent figures have the same area. Once the result is known for right triangles it may be deduced also for an arbitrary triangle because a general triangle may be decomposed into two right triangles.

Example 3: The integral $\int_a^b x^p dx$, where p is a positive integer. In the foregoing example we evaluated the integral $\int_0^b x dx$ by expressing it in terms of the integral $\int_0^1 x dx$. In a similar way, using Theorem 1-46 with $c = 1/b$, we obtain

$$(1.42) \quad \int_0^b x^p dx = b \int_0^1 (bx)^p dx = b^{p+1} \int_0^1 x^p dx$$

for every positive integer p . Presently, we shall establish the formula

$$(1.43) \quad \int_0^1 x^p dx = \frac{1}{p+1}$$

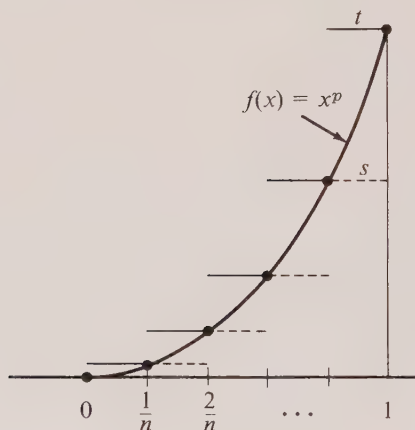


FIGURE 1.48 Graphs of $f(x) = x^p$ and two approximating step functions.

which, when combined with (1.42), gives us the integration formula

$$(1.44) \quad \int_0^b x^p dx = \frac{b^{p+1}}{p+1}.$$

The additive property $\int_a^b x^p dx = \int_0^b x^p dx - \int_0^a x^p dx$ then leads to the more general formula

$$(1.45) \quad \int_a^b x^p dx = \frac{b^{p+1} - a^{p+1}}{p+1}.$$

The special case of (1.44) in which $p = 2$ has already been mentioned in Section 1.4 in connection with the method of Archimedes for computing the area of a parabolic segment. A proof of (1.43) will now be given by a similar method. For this purpose we consider the special partition $P = \{x_0, x_1, \dots, x_n\}$ of the interval $[0, 1]$ in which each $x_k = k/n$. This partition decomposes

the interval $[0, 1]$ into n equal subintervals of length $1/n$. Next we define step functions s and t as follows:

$$\begin{aligned} s(x) &= x_{k-1}^p & \text{if } x_{k-1} \leq x < x_k, \\ t(x) &= x_k^p & \text{if } x_{k-1} < x \leq x_k, \\ s(1) &= 1, & t(0) = 0. \end{aligned}$$

The graphs of s and t are shown in Figure 1.48. Note that $s(x) \leq x^p \leq t(x)$ for all x in $[0, 1]$ and hence $\int_0^1 s(x) dx \leq \int_0^1 x^p dx \leq \int_0^1 t(x) dx$.

It is easy to compute the integrals of these step functions. For the integral of t we have

$$(1.46) \quad \int_0^1 t(x) dx = \sum_{k=1}^n x_k^p (x_k - x_{k-1}) = \sum_{k=1}^n \left(\frac{k}{n}\right)^p \frac{1}{n} = \frac{1}{n^{p+1}} \sum_{k=1}^n k^p.$$

Similarly, for the integral of s we find

$$(1.47) \quad \int_0^1 s(x) dx = \frac{1}{n^{p+1}} \sum_{k=1}^n (k-1)^p.$$

Now we make use of the following inequality, valid for every choice of the positive integers n and p :†

$$(1.48) \quad \sum_{k=1}^{n-1} k^p < \frac{n^{p+1}}{p+1} < \sum_{k=1}^n k^p.$$

Adding n^p to both sides of the leftmost inequality in (1.48), we find

$$\sum_{k=1}^n k^p < \frac{n^{p+1}}{p+1} + n^p.$$

Using (1.46), we obtain the inequality

$$\int_0^1 t(x) dx < \frac{1}{p+1} + \frac{1}{n}$$

which, in turn, implies

$$(1.49) \quad \int_0^1 x^p dx < \frac{1}{p+1} + \frac{1}{n}.$$

Since (1.49) is true for every n we must have‡

† An inductive proof of this inequality is outlined in Exercise 13 of Section 1.23.

‡ The only other alternative to (1.50) is $\int_0^1 x^p dx > 1/(p+1)$ and this contradicts (1.49) when n is chosen so large that

$$\frac{1}{n} < \int_0^1 x^p dx - \frac{1}{p+1}.$$

$$(1.50) \quad \int_0^1 x^p dx \leq \frac{1}{p+1}.$$

Similarly, by subtracting n^p from both sides of the rightmost inequality in (1.48), and using (1.47), we are led to the inequality

$$(1.51) \quad \int_0^1 x^p dx \geq \frac{1}{p+1}.$$

The two inequalities (1.50) and (1.51) together yield (1.43) and, as already mentioned, (1.43) implies (1.45). This argument gives us the values, if they exist, of the integrals in (1.43) and (1.45). As mentioned before, their existence will be established in Section 1.41.

Incidentally, formula (1.45) also holds when $p = 0$ if we define x^0 to be 1 for all real x . In this case the function being integrated is a constant step-function and the definition of the integral yields $b - a$ for the value of the integral. This is also the result of putting $p = 0$ in (1.45). The integral $\int_a^b 1 dx$ is usually written $\int_a^b dx$.

Example 4: Integration of polynomials. Using formula (1.45) and the linearity property, we can compute the integral of a constant times a power of x or of a finite sum of such terms. For example, to compute the integral $\int_1^3 (x^2 - 3x + 5) dx$ we find the integral of each term and then add the results. Thus we have

$$\begin{aligned} \int_1^3 (x^2 - 3x + 5) dx &= \int_1^3 x^2 dx - 3 \int_1^3 x dx + 5 \int_1^3 dx \\ &= \frac{3^3 - 1^3}{3} - 3 \frac{3^2 - 1^2}{2} + 5 \frac{3^1 - 1^1}{1} \\ &= \frac{26}{3} - 12 + 10 = \frac{20}{3}. \end{aligned}$$

The integral just computed is that of a *polynomial function*. By a polynomial we mean a function f defined for all real x by an equation of the form

$$f(x) = \sum_{k=0}^n c_k x^k.$$

The numbers $c_0, c_1, \dots, c_n$ are called the *coefficients* of the polynomial and the integer n is called its *degree* (if $c_n \neq 0$). To compute the integral of such a polynomial we integrate term by term, thus:

$$\int_a^b \sum_{k=0}^n c_k x^k dx = \sum_{k=0}^n c_k \int_a^b x^k dx = \sum_{k=0}^n c_k \frac{b^{k+1} - a^{k+1}}{k+1}.$$

Example 5: Area of a circle. We define the number π to be the area of a unit circle (that is, of a circle of radius 1). Using properties of the integral, we can prove that the area of a circle of radius r is πr^2 . In other words, the area of a circle of radius r is r^2 times the area of a unit circle.

Let us denote by $A(r)$ the area of a circle of radius r . To express $A(r)$ as an integral we use Equation (1.35) with

$$f(x) = \sqrt{r^2 - x^2} \quad \text{and} \quad g(x) = -\sqrt{r^2 - x^2}$$

for $-r \leq x \leq r$. (See Figure 1.49.) Since $g(x) = -f(x)$ we have $|f(x) - g(x)| = 2f(x)$, and the integral in (1.35) gives

$$(1.52) \quad A(r) = \int_{-r}^r 2f(x) dx = 2 \int_{-r}^r \sqrt{r^2 - x^2} dx.$$

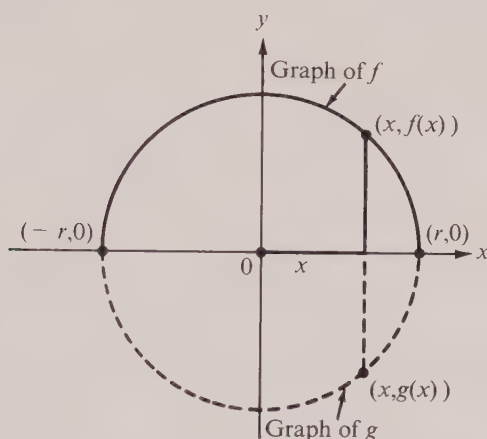


FIGURE 1.49 Area of circle $= \int_{-r}^r |f(x) - g(x)| dx = 2 \int_{-r}^r \sqrt{r^2 - x^2} dx$.

In particular, when $r = 1$ we have the formula

$$(1.53) \quad A(1) = 2 \int_{-1}^1 \sqrt{1 - x^2} dx = \pi.$$

If, now, we change the scale on the x -axis and use Theorem 1-46 with $c = 1/r$, we may write

$$\int_{-r}^r f(x) dx = r \int_{-1}^1 f(rx) dx = r \int_{-1}^1 \sqrt{r^2 - (rx)^2} dx = r^2 \int_{-1}^1 \sqrt{1 - x^2} dx.$$

Substituting this in (1.52) and using (1.53), we obtain the formula $A(r) = \pi r^2$.

The foregoing example illustrates a general property of area that has to do with a concept known as a *similarity transformation*. This concept may be described as follows: Suppose S is a given set of points in the plane and consider a new set of points obtained by multiplying the coordinates of each point of S by a constant factor $k > 0$. We denote this set by kS and say that it is *similar* to S . The process which produces kS from S is called a similarity transformation. Each point is moved along a straight line which passes through the origin to k times its original distance from the origin. If $k > 1$, the trans-

formation is also called a *stretching* or an *expansion* (from the origin) and if $0 < k < 1$ it is called a *shrinking* or a *contraction* (toward the origin).

For example, if S is the region bounded by a unit circle with center at the origin, then kS is a concentric circular region of radius k . In Example 5 we showed that for circular regions the area of kS is k^2 times the area of S . In a later chapter (Section 6.27) we shall prove that this property of area holds for more general regions as well. In the next example we prove that this property holds for circular sectors.

Example 6: The behavior of the area of a circular sector under a similarity transformation. Figure 1.50(a) shows a unit circle with center at the origin. The two radii OQ and OP determine a sector OQP , indicated by the shaded region, which we shall denote by S . The point Q has coordinates $(1, 0)$ and we denote the coordinates of P by (a, b) . The sector S is the ordinate set of a function f defined as follows:

$$f(x) = \begin{cases} \frac{b}{a}x & \text{if } 0 \leq x \leq a, \\ \sqrt{1-x^2} & \text{if } a \leq x \leq 1, \end{cases}$$

and therefore its area is given by the integral

$$\text{area of } S = \int_0^1 f(x) dx = \frac{b}{a} \int_0^a x dx + \int_a^1 \sqrt{1-x^2} dx.$$

If we apply a similarity transformation with a positive factor k , then kS is a new sector $OQ'P'$, where $Q' = (k, 0)$ and $P' = (ka, kb)$, as shown in Figure 1.50(b). The sector kS is the ordinate set of the function g defined by

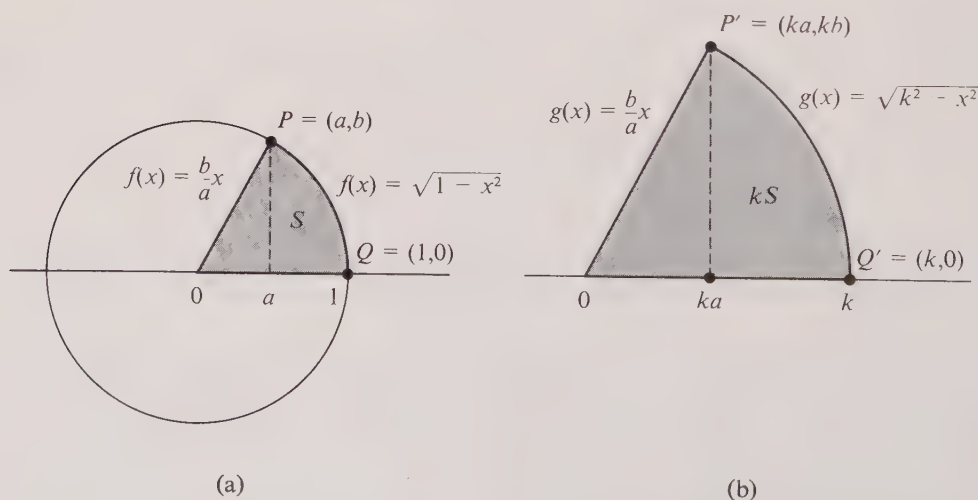


FIGURE 1.50 The area of kS is k^2 times that of S .

$$g(x) = \begin{cases} \frac{b}{a}x & \text{if } 0 \leq x \leq ka, \\ \sqrt{k^2 - x^2} & \text{if } ka \leq x \leq k, \end{cases}$$

and therefore its area is

$$\text{area of } kS = \int_0^k g(x) dx = \frac{b}{a} \int_0^{ka} x dx + \int_{ka}^k \sqrt{k^2 - x^2} dx.$$

If, now, we use arguments similar to those given in Examples 2 and 5 above, we find

$$\int_0^{ka} x dx = k^2 \int_0^a x dx \quad \text{and} \quad \int_{ka}^k \sqrt{k^2 - x^2} dx = k^2 \int_a^1 \sqrt{1 - x^2} dx,$$

and we see that the area of kS is k^2 times that of S , as asserted. In this example we used a sector for which the point P is in the first quadrant, but the same kind of argument shows that the result also holds if P is in the other quadrants.

1.37 Exercises

In Exercises 1 through 12 compute the values of the given integrals using formula (1.45) and general properties of the integral.

1. $\int_{-1}^2 x^2 dx.$
2. $\int_2^3 (3x^2 - 4x + 2) dx.$
3. $\int_0^{\frac{1}{2}} (8t^3 + 6t^2 - 2t + 5) dt.$
4. $\int_{-2}^4 (t - 1)(t - 2) dt.$
5. $\int_{-1}^0 (x + 1)^2 dx.$
6. $\int_0^{-1} (x + 1)^2 dx.$
7. $\int_0^3 (2x - 5)^3 dx.$
8. $\int_0^2 (x - 1)(2x - 1)(3x - 1) dx.$
9. $\int_0^2 |(x - 1)(2x - 1)(3x - 1)| dx.$
10. $\int_{-3}^3 (x^2 - 3)^3 dx.$
11. $\int_0^5 x^2(x - 5)^4 dx.$
12. $\int_{-2}^{-4} (x + 4)^{10} dx.$

In Exercises 13 through 18, find the area of the region between the graphs of f and g over the interval $[a, b]$ specified in each case. Draw a rough sketch of the two graphs and indicate (by shading) the region whose area is being computed. You may use Equation (1.45) and general properties of the integral.

- | | | | |
|-------------------------------|--------------------|-------------------------------|-------------------------------|
| 13. $f(x) = 4 - x^2,$ | $g(x) = 0,$ | $a = -2,$ | $b = 2.$ |
| 14. $f(x) = 4 - x^2,$ | $g(x) = 8 - 2x^2,$ | $a = -2,$ | $b = 2.$ |
| 15. $f(x) = x^3 + x^2,$ | $g(x) = x^3 + 1,$ | $a = -1,$ | $b = 1.$ |
| 16. $f(x) = x - x^2,$ | $g(x) = -x,$ | $a = 0,$ | $b = 2.$ |
| 17. $f(x) = x^2,$ | $g(x) = x + 1,$ | $a = \frac{1 - \sqrt{5}}{2},$ | $b = \frac{1 + \sqrt{5}}{2}.$ |
| 18. $f(x) = x(x + 1)(x - 1),$ | $g(x) = x,$ | $a = -\sqrt{2},$ | $b = \sqrt{2}.$ |

19. Find all values of c for which

$$(a) \int_0^c x(1 - x) dx = 0, \quad (b) \int_0^c |x(1 - x)| dx = 0.$$

In each case explain your result geometrically in terms of areas.

20. Compute each of the following integrals. You may use formula (1.45) and general properties of the integral. Draw the graph of f in each case.

$$(a) \int_0^2 f(x) dx, \quad \text{where } f(x) = \begin{cases} x^2 & \text{if } 0 \leq x \leq 1, \\ 2 - x & \text{if } 1 \leq x \leq 2. \end{cases}$$

$$(b) \int_0^1 f(x) dx, \quad \text{where } f(x) = \begin{cases} x & \text{if } 0 \leq x \leq c, \\ c \frac{1-x}{1-c} & \text{if } c \leq x \leq 1, \end{cases}$$

c being a fixed real number, $0 < c < 1$.

21. Derive the following property of the integral as a consequence of Theorems 1-45 and 1-46:

$$\int_a^b f(c-x) dx = \int_{c-b}^{c-a} f(x) dx.$$

22. Theorems 1-45 and 1-46 suggest a common generalization for the integral $\int_a^b f(Ax+B) dx$. Guess the formula suggested and prove it; use Theorems 1-45 and 1-46 in the proof. Discuss also the case $A = 0$.

23. Use Theorems 1-45 and 1-46 to derive the formula

$$\int_a^b f(x) dx = (b-a) \int_0^1 f[a+(b-a)x] dx.$$

24. Let f be a function whose domain contains $-x$ whenever it contains x . We say that f is an *even* function if $f(-x) = f(x)$ and an *odd* function if $f(-x) = -f(x)$ for all x in the domain of f . If f is integrable on $[0, b]$, prove that

$$(a) \int_{-b}^b f(x) dx = 2 \int_0^b f(x) dx \quad \text{if } f \text{ is even;}$$

$$(b) \int_{-b}^b f(x) dx = 0 \quad \text{if } f \text{ is odd.}$$

Interpret these results geometrically in terms of areas.

25. Show that the following are polynomials by converting them to the form $\sum_{k=0}^m a_k x^k$ for a suitable m . In each case n denotes a positive integer.

$$(a) (1+x)^{2n}, \quad (b) \frac{1-x^{n+1}}{1-x}, \quad x \neq 1.$$

$$(c) \prod_{k=0}^n (1+x^{2^k}). \quad [\text{Hint. Multiply by } 1-x.]$$

1.38 The trigonometric functions

In his study of elementary trigonometry the reader has gained some familiarity with the properties of the six trigonometric functions, sine, cosine, tangent, cotangent, secant, and cosecant; and their inverses, arc sine, arc cosine, arc tangent, etc. These are the principal mathematical tools used by surveyors and navigators who are required to calculate the sides and angles of triangles.

The trigonometric functions also arise in other branches of science, not so much because of their relation to the sides and angles of a triangle, but rather because of the analytical properties they possess as *functions*. All these functions have one important

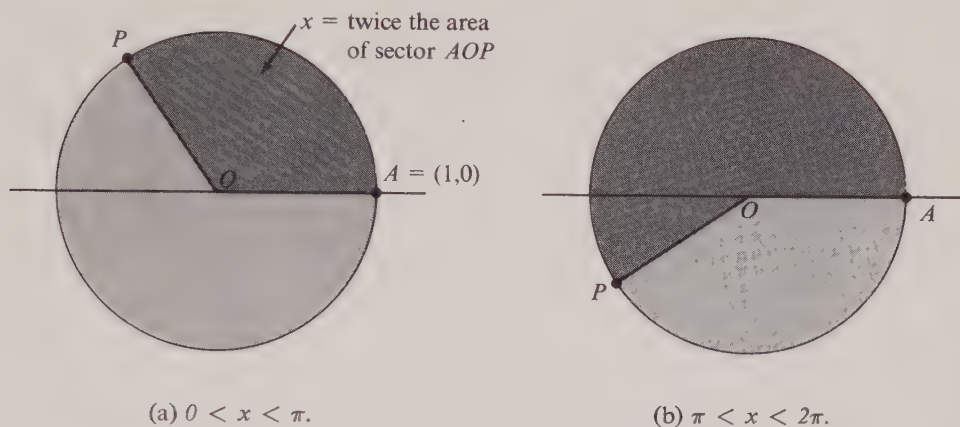


FIGURE 1.51 An angle $\angle AOP$ consisting of x radians.

feature in common, a property known as *periodicity*.† Many problems in physics and engineering deal with periodic phenomena (such as vibrations, planetary and wave motion) and the trigonometric functions form the basis for the mathematical analysis of such problems.

The definitions of the trigonometric functions as ordinarily given in elementary courses are based on certain geometric assumptions whose justification requires a great deal of careful analysis. We shall define these functions in a way that differs slightly from the usual procedure, the difference being not in the definition of the functions themselves but in the method used to measure angles. The procedure is the following:

Consider a circle of unit radius with center at the origin. Denote the point $(1, 0)$ by A and let P be any other point on the circumference of this circle. The two line segments OA and OP determine a geometric configuration called an *angle* which we denote by the symbol $\angle AOP$. (An example is shown in Figure 1.51.) We wish to assign to this angle a certain real number x which can be used as a measurement of its size. The most common way of doing this is to let x be the length of the circular arc AP , measured counterclockwise from A to P , and to say that $\angle AOP$ is an angle consisting of x radians. From a logical point of view this is unsatisfactory at the present stage because we have not yet given a *definition* of arc length. The notion of the length of a curve can be given a rigorous definition based on the properties of the real numbers alone, but it requires a type of analysis that is as complicated as the definition of area. Since the concept of area has already been defined, we prefer to use the area of the circular sector‡ AOP rather than the length of the arc AP as a measure of the size of $\angle AOP$.

† A function f is said to be *periodic* with period p if its domain contains $x + p$ whenever it contains x and if $f(x + p) = f(x)$ for every x in the domain of f . The smallest positive period (if one exists) is called the *fundamental period* of f . The sine, cosine, secant, and cosecant all have fundamental period 2π . The tangent and cotangent have fundamental period π .

‡ It is understood that the sector AOP is the smaller portion of the circle when P is above the real axis, and the larger portion when P is below the real axis. These are indicated by the shaded regions in Figures 1.51(a) and (b), respectively.

Later, when arc length is defined, we shall find that the number that expresses the length of the arc AP is exactly twice as large as the number that expresses the area of the sector AOP . Therefore, to get the same scale of measurement for angles by both methods we shall use twice the area of the sector AOP as a measure of the angle $\angle AOP$. However, to obtain a "dimensionless" measure of angles, that is, a measure independent of the unit of distance in our coordinate system, we can define the measure of $\angle AOP$ to be *twice the area of sector AOP divided by the square of the radius*. This ratio does not change if we expand or contract the circle, and therefore there is no loss in generality in restricting our considerations to a unit circle. The unit of measure so obtained is called the *radian*.

Let us denote the area of the unit circle by the symbol π . When $P = (-1, 0)$ the sector AOP is a semicircle of area $\frac{1}{2}\pi$ and it subtends an angle of π radians. We agree that the entire circle may be thought of as a sector consisting of 2π radians. If the point P is initially at $(1, 0)$ and if P moves once around the circle in a counterclockwise direction, the area of sector AOP increases from 0 to 2π and the area takes on every value in the interval $[0, 2\pi]$ exactly once. This property, which seems geometrically evident, can be proved analytically from the definition of area as an integral.

The next step is to define the sine and cosine of an angle. Actually, we prefer to speak of the sine and cosine of a *number* rather than of an *angle* so that the sine and cosine will give us *functions* to which we can apply the operations of calculus. We proceed as follows: Choose a number x satisfying $0 < x < 2\pi$ and let P be the point such that twice the area of sector AOP is equal to x . Let (a, b) denote the coordinates of P . The numbers a and b are completely determined by x . We define

$$\sin x = b \quad \text{and} \quad \cos x = a.$$

For example, when $x = \pi$ we have $P = (-1, 0)$ so $\sin \pi = 0$ and $\cos \pi = -1$. Similarly, when $x = \frac{1}{2}\pi$ we have $P = (0, 1)$ and hence $\sin \frac{1}{2}\pi = 1$ and $\cos \frac{1}{2}\pi = 0$. This procedure describes the sine and cosine as functions defined in the open interval $(0, 2\pi)$. We extend the definitions of the sine and cosine to the whole real axis by means of the following equations:

$$\sin 0 = 0, \quad \cos 0 = 1, \quad \sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x.$$

The other four trigonometric functions are now defined in terms of the sine and cosine by the usual formulas

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}.$$

These functions are defined for all real values of x except for certain isolated points where denominators may be zero. They all satisfy the periodicity property

$$f(x + 2\pi) = f(x).$$

The above definitions of the sine and cosine have a geometric flavor and should not be considered as analytic definitions. It is possible to formulate these concepts entirely in an analytic way without any reference whatever to geometry. There are several ways of doing this. One of these, which is essentially the analytic formulation of the definitions just given, makes use of integrals to define first one of the inverse functions, arc sine, arc cosine, or arc tangent; the entire theory of the trigonometric functions can be developed

from this starting point. We shall obtain these integrals in Chapter 3 when we treat the inverse trigonometric functions in detail. An entirely different procedure uses the theory of infinite series as a starting point. This will be discussed further in Chapter 9. All these methods are equivalent, in the sense that they all lead to the same functions.

The main properties of the sine and cosine that we shall make use of from time to time are listed here for easy reference.

$$\sin 0 = 0, \quad \sin \frac{\pi}{2} = 1, \quad \sin(-x) = -\sin x, \quad \sin(x + 2\pi) = \sin x,$$

$$\cos 0 = 1, \quad \cos \frac{\pi}{2} = 0, \quad \cos(-x) = \cos x, \quad \cos(x + 2\pi) = \cos x,$$

$$\sin^2 x + \cos^2 x = 1,$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y.$$

These are basic formulas from which all other identities can be derived. The list is not intended to be minimal. For example, by taking $y = -x$ in the formula for $\cos(x + y)$ we may deduce the identity $\sin^2 x + \cos^2 x = 1$. The following further identities are often used:

$$\sin 2x = 2 \sin x \cos x,$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x,$$

$$\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2},$$

$$\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}.$$

Extensive tables of values of the sine, cosine, tangent, and cotangent appear in most mathematical handbooks. The graphs of the six trigonometric functions are shown in Figure 1.52 as they appear over one complete period-interval. The rest of the curve in each case is obtained by appealing to periodicity. (The graphs are not drawn to scale.)

1.39 Exercises

Derive the identities in Exercises 1 through 6 from the properties of trigonometric functions listed in the foregoing section.

$$1. \sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x.$$

$$2. \sin(\pi + x) = -\sin x, \quad \cos(\pi + x) = -\cos x.$$

$$3. \sin 3x = 3 \sin x - 4 \sin^3 x, \quad \cos 3x = \cos x - 4 \sin^2 x \cos x.$$

$$4. \cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}, \quad \sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}.$$

Discuss when the $+$ or $-$ sign should be used.

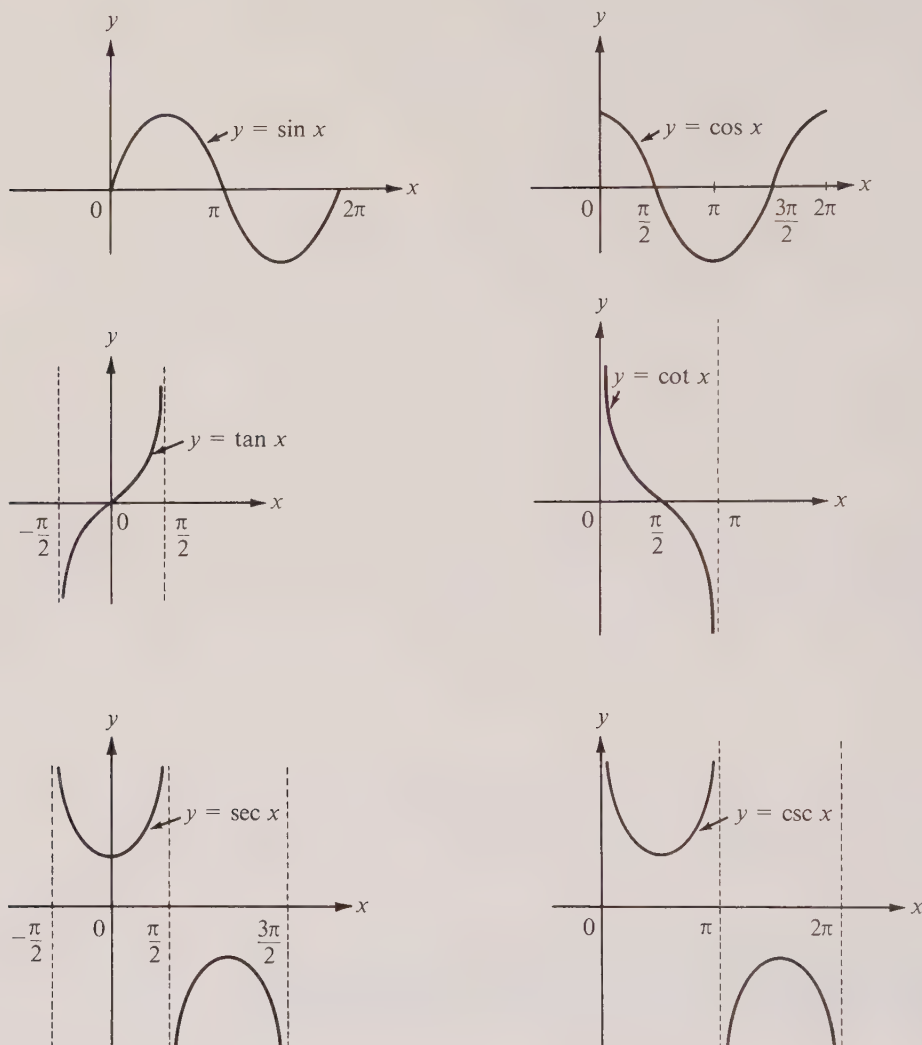


FIGURE 1.52 Graphs of the trigonometric functions.

5. $\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$.
6. Obtain similar formulas for $\tan(x + y)$, $\cot(x + y)$, and $\cot(x - y)$.
7. Prove that $\sin(x + \frac{1}{2}\pi) = \cos x$ and interpret geometrically.
8. Plot the graph of the function f given by $f(x) = C \sin(x + \alpha)$ when $C = 3$ and $\alpha = \frac{1}{3}\pi$.
9. Determine constants A and B such that

$$3 \sin\left(x + \frac{\pi}{3}\right) = A \sin x + B \cos x \quad (\text{for all } x).$$

10. Show that if C and α are given real numbers, there exist numbers A and B such that

$$(i) \quad C \sin(x + \alpha) = A \sin x + B \cos x \quad (\text{for all } x).$$

11. If A and B are given real numbers, show that there exist numbers C and α , with $C \geq 0$, such that (i) holds.

12. Determine C and α , with $C > 0$, such that

$$C \sin(x + \alpha) = -2 \sin x - 2 \cos x \quad (\text{for all } x).$$

13. If A and B are given real numbers, show that there exist numbers C and α such that

$$C \cos(x + \alpha) = A \sin x + B \cos x \quad (\text{for all } x).$$

Determine C and α if $A = B = 1$.

14. Plot the graphs of the functions f and g over the interval $0 \leq x \leq 2\pi$ if

$$f(x) = 2 \cos^2 \frac{x}{2}, \quad g(x) = \cos x.$$

Show that these two graphs are related by a simple geometrical construction.

15. (a) Show that $\sin n\pi = 0$ for every integer n and that these are the only values of x for which $\sin x = 0$.

(b) Find all real x such that $\cos x = 0$.

16. Find all real x such that

(a) $\sin x = 1$;

(b) $\cos x = 1$.

17. Find all real x such that

(a) $\sin x = -1$;

(b) $\cos x = -1$.

18. Find all real x such that

(a) $\sin x = \frac{1}{2}$;

(b) $\cos x = \frac{1}{2}$.

[Hint. If $\sin x = \frac{1}{2}$ then, by Exercise 3, $\sin 3x = 1$.]

19. Find all real x such that $\sin x = \cos x$.

20. Find all real x such that $\sin x - \cos x = 1$.

21. (a) Derive the identity

$$2 \sin \frac{x}{2} \cos kx = \sin(2k+1)\frac{x}{2} - \sin(2k-1)\frac{x}{2}.$$

(b) Use the identity in (a) and the telescoping property of finite sums to prove that, if $x \neq 2m\pi$ (m an integer), we have

$$\sum_{k=1}^n \cos kx = \frac{\sin \frac{1}{2}nx \cos \frac{1}{2}(n+1)x}{\sin \frac{1}{2}x}.$$

22. If $x \neq 2m\pi$ (m an integer), prove that

$$\sum_{k=1}^n \sin kx = \frac{\sin \frac{1}{2}nx \sin \frac{1}{2}(n+1)x}{\sin \frac{1}{2}x}.$$

23. If $x \neq m\pi$ (m an integer), prove that

$$(a) \quad \sum_{k=1}^n \cos(2k-1)x = \frac{\sin 2nx}{2 \sin x}, \quad (b) \quad \sum_{k=1}^n \sin(2k-1)x = \frac{\sin^2 nx}{\sin x}.$$

24. A circular arc AP of radius 1 subtending an angle of x radians, $0 < x < \frac{1}{2}\pi$, is shown in Figure 1.53. The coordinates of P are (a, b) , where $a = \cos x$ and $b = \sin x$.

(a) Show that the coordinates of a typical point (u, v) on the line segment AP satisfy

$$v = \frac{\sqrt{1-a^2}}{1-a} (1-u).$$

[Hint. From similar triangles, deduce that $v/b = (1-u)/(1-a)$.]

(b) If $0 \leq a \leq u \leq 1$, show that

$$\frac{\sqrt{1-a^2}}{1-a} (1-u) \leq \sqrt{1-u^2}.$$

Note. This proves that the line segment AP lies below the circular arc AP in the figure.

(c) Use (b), compare areas, and prove that

$$\sin x < x \quad \text{if } 0 < x < \frac{\pi}{2}.$$

(d) Deduce that

$$|\sin x| < |x| \quad \text{if } 0 < |x| < \frac{\pi}{2}.$$

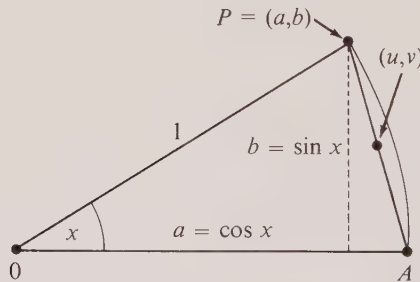
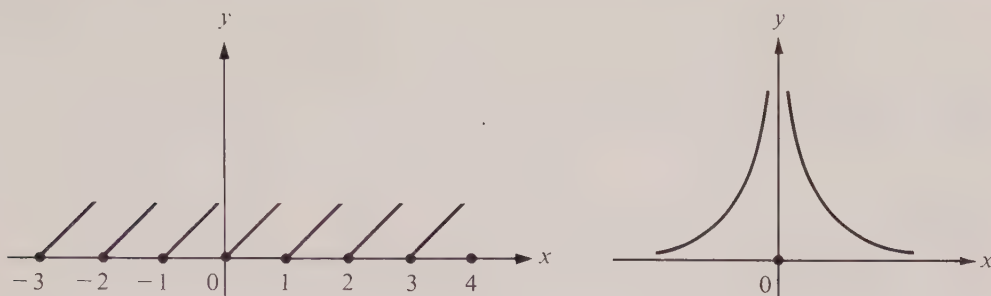


FIGURE 1.53

1.40 Informal remarks on continuity and the theory of integration

In Section 1.33 we gave an example of a function which is defined and bounded on an interval but which is *not* integrable. A natural question to ask at this point is: *What further restrictions must be placed on a bounded function to guarantee the existence of its integral?* This question comes under the heading “Theory of Integration,” and one answer to it lies in the concept of *continuity*, one of the most important and also one of the most fascinating ideas in all of mathematics. In Sections 2.7 and 2.8 we shall give a precise technical definition of continuity and discuss some of the basic theorems about continuous functions. In the meantime, to give the reader a feeling for the meaning of continuity without going into precise details, we shall briefly discuss the concept here in an informal and intuitive fashion.



(a) A jump discontinuity at each integer.

(b) An infinite discontinuity at 0.

FIGURE 1.54 Illustrating two kinds of discontinuities.

Roughly speaking, the situation is this: Suppose a function f has the value $f(x_0)$ at a certain point x_0 . Then f is said to be continuous at x_0 if at every nearby point x the function value $f(x)$ is close to $f(x_0)$. Another way of putting it is as follows: If we let x move toward x_0 , we want the corresponding function values $f(x)$ to become arbitrarily close to $f(x_0)$, regardless of the manner in which x approaches x_0 . We do *not* want sudden jumps in the values of a continuous function, as in the examples in Figure 1.54.

Figure 1.54(a) shows the graph of the function f defined by the equation $f(x) = x - [x]$, where $[x]$ denotes the greatest integer $\leq x$. At each integer we have what is known as a *jump discontinuity*. For example, $f(2) = 0$, but as x approaches 2 from the left, $f(x)$ approaches the value 1 which is not equal to $f(2)$. Therefore we have a discontinuity at 2. Note that $f(x)$ *does* approach $f(2)$ if we let x approach 2 *from the right*, but this by itself is not enough to establish continuity at 2. In a case like this the function is called *continuous from the right* at 2 and *discontinuous from the left* at 2. Continuity at a point requires both continuity from the left and from the right.

The example of a nonintegrable function described in Section 1.33 happens to be one which is *nowhere continuous*. At the other extreme are those functions that are *everywhere continuous* on a closed interval $[a, b]$. Polynomials are examples of such functions. An important theorem of integration theory states that, if a function is continuous on a closed interval, it is automatically integrable there. This theorem is proved in Section 2.32. Between these two extreme cases are those bounded functions that are continuous at some points and discontinuous at others. Step functions are in this category; they are integrable functions with a finite number of discontinuities. As it turns out, there exist examples of integrable functions with *infinitely many* discontinuities. Therefore to answer the question “Which bounded functions are integrable?” is by no means a trivial matter. A complete answer to this question lies in a concept known as *continuity almost everywhere*, a subject which is ordinarily discussed in detail in more advanced courses in analysis and which will not be dealt with in this book. However, a partial answer to this question can be given with very little effort by introducing an important class of functions known as *piecewise monotonic functions*. In the following section we define these functions and in Section 1.47 we prove that they are integrable. Fortunately, most of the functions that occur in practice are piecewise monotonic and therefore the results of the next section are quite comprehensive.

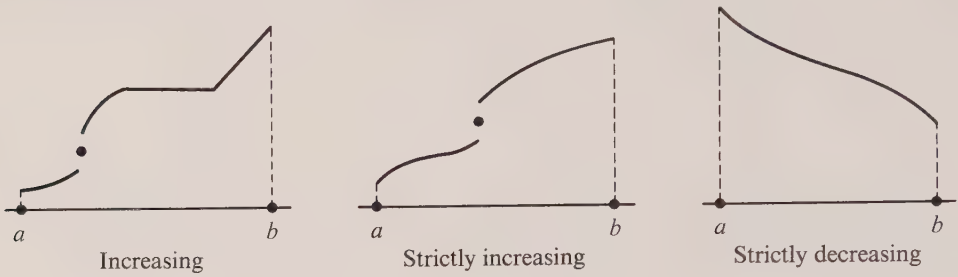


FIGURE 1.55 Monotonic functions.

1.41 Piecewise monotonic functions

A function f is said to be *increasing* on a set S if $f(x) \leq f(y)$ for every pair of points x and y in S with $x < y$. If the strict inequality $f(x) < f(y)$ holds for all $x < y$ in S , the function is said to be *strictly increasing* on S . Similarly, a function is called *decreasing* on S if $f(x) \geq f(y)$ for all $x < y$ in S . If $f(x) > f(y)$ for all $x < y$ in S , the function is called *strictly decreasing* on S . A function is called *monotonic* on S if it is increasing on S or if it is decreasing on S . The term *strictly monotonic* means that f is strictly increasing on S or strictly decreasing on S . Ordinarily the set S under consideration is either an open interval or a closed interval. Examples are shown in Figure 1.55.

A function f is said to be *piecewise monotonic* on an interval if its graph consists of a finite number of monotonic pieces. That is to say, f is piecewise monotonic on $[a, b]$ if there is a partition P of $[a, b]$ such that f is monotonic on each of the open subintervals of P . In particular, step functions are piecewise monotonic, as are all the examples shown in Figures 1.54, 1.55, and 1.56.

When p is a positive integer, we have the inequality

$$x^p < y^p \quad \text{if} \quad 0 \leq x < y,$$

which is easily proved by mathematical induction. This implies that the function f defined for all real x by the equation $f(x) = x^p$ is strictly increasing on the nonnegative real axis.

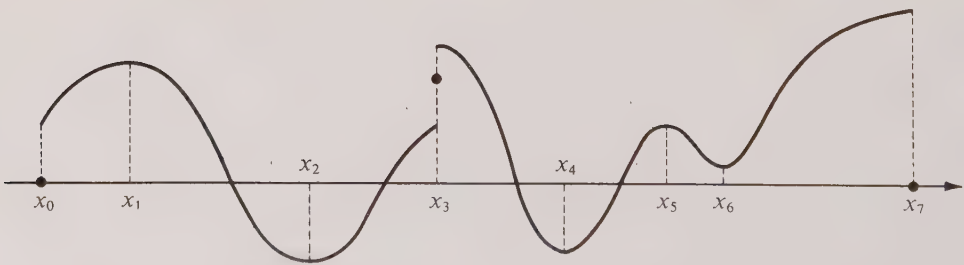


FIGURE 1.56 A piecewise monotonic function.

The same function is strictly monotonic on the negative real axis (it is decreasing if p is even and increasing if p is odd). Therefore f is piecewise monotonic on every finite interval.

The trigonometric functions are another source of piecewise monotonic functions and further examples will be encountered in Chapter 3 when we introduce the logarithmic and exponential functions. The importance of piecewise monotonic functions in integration theory is due to the following theorem:

1-48 THEOREM. Let f be defined and bounded on a closed interval $[a, b]$. If f is piecewise monotonic on $[a, b]$, then the integral $\int_a^b f(x) dx$ exists.

A simple proof of this theorem is given in Section 1.47. As a consequence of the theorem we deduce the existence of the integral $\int_0^1 x^p dx$ when p is a nonnegative integer. In Example 3 of Section 1.36 we derived the integration formula

$$(1.54) \quad \int_0^1 x^p dx = \frac{1}{p+1}$$

under the assumption that the integral exists, and then we used (1.54) to deduce the more general formula

$$(1.55) \quad \int_a^b x^p dx = \frac{b^{p+1} - a^{p+1}}{p+1}.$$

When we prove Theorem 1-48 in Section 1.47, the proof of (1.55) will be complete in every detail. From the linearity property (Theorem 1-43) and the existence of $\int_a^b x^p dx$ it follows that every polynomial is integrable on every finite interval.

The square-root function f , defined on the nonnegative real axis by the equation $f(x) = \sqrt{x}$, is strictly increasing. In fact, if $0 \leq x < y$ we have

$$\sqrt{y} - \sqrt{x} = \frac{y - x}{\sqrt{y} + \sqrt{x}}$$

and hence $\sqrt{y} - \sqrt{x} > 0$. From Theorem 1-48 it follows that the integral $\int_a^b \sqrt{x} dx$ exists whenever $0 \leq a < b$. The actual value of this integral may be obtained by substituting $p = \frac{1}{2}$ in formula (1.55), a fact that will be proved in Chapter 2 by a new method. Eventually we shall extend formula (1.55) to all real values of p except $p = -1$. Theorem 1-48 also establishes the existence of the integral $\int_{-r}^r \sqrt{r^2 - x^2} dx$ if $r > 0$ because the integrand is piecewise monotonic on $[-r, r]$. This is the integral encountered in Section 1.36 in the computation of the area of a circle of radius r .

The sine and cosine functions are piecewise monotonic (and hence integrable) on every finite interval $[a, b]$. This is apparent from their graphs and it is easy to prove analytically, using suitable identities. (See Exercise 14 in Section 1.43.) The integrals of the sine and cosine from a to b are given by the formulas

$$(1.56) \quad \int_a^b \sin x dx = \cos a - \cos b$$

and

$$(1.57) \quad \int_a^b \cos x \, dx = \sin b - \sin a.$$

These formulas may be proved by a method similar to the one we used in Section 1.36 for evaluating the integral $\int_0^1 x^p \, dx$. However, we shall not prove these formulas at this stage because they will be derived by an easier method in Chapter 2, with the help of differential calculus. (See Example 2 in Section 2.15.)

1.42 The integral as a function of the upper limit

Let f be a function which is integrable on $[a, b]$ and let us denote its integral as $\int_a^b f(t) \, dt$, where we have used the dummy variable t instead of x . It can be shown that f is integrable on every subinterval of $[a, b]$. In particular, if x is any point in $[a, b]$, the integral $\int_a^x f(t) \, dt$ exists. We want to keep a and f fixed and study this integral as a function of x . Let us denote the value of this integral by $A(x)$, so that

$$(1.58) \quad A(x) = \int_a^x f(t) \, dt \quad \text{if } a \leq x \leq b.$$

An equation like this enables us to construct a new function A from a given function f , the value of A at each point in $[a, b]$ being determined by Equation (1.58). The function A is sometimes referred to as an *indefinite integral* of f and it is said to be obtained from f by integration. We say *an* indefinite integral rather than *the* indefinite integral because A also depends on the lower limit a . Different values of a will lead to different functions A . If we use a different lower limit, say c , and define another indefinite integral F by the equation

$$F(x) = \int_c^x f(t) \, dt,$$

then the additive property tells us that

$$A(x) - F(x) = \int_a^x f(t) \, dt - \int_c^x f(t) \, dt = \int_a^c f(t) \, dt$$

and hence the difference $A(x) - F(x)$ is *independent* of x . Therefore any two indefinite integrals of the same function differ only by a constant (the constant depends on the choice of a and c).

When an indefinite integral of f is known, the value of an integral such as $\int_a^b f(t) \, dt$ may be evaluated by a simple subtraction. For example, if n is a nonnegative integer we have the formula

$$\int_0^x t^n \, dt = \frac{x^{n+1}}{n+1},$$

and the additive property implies that

$$\int_a^b t^n \, dt = \int_0^b t^n \, dt - \int_0^a t^n \, dt = \frac{b^{n+1} - a^{n+1}}{n+1}.$$

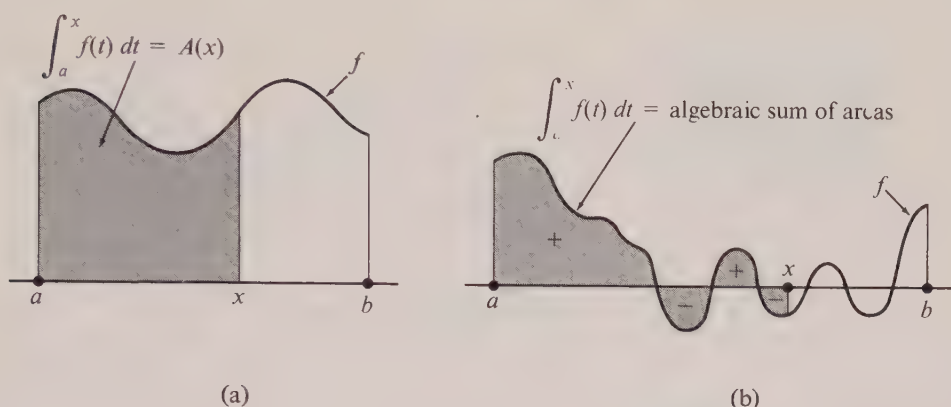


FIGURE 1.57 Indefinite integrals interpreted as area.

In general, if $F(x) = \int_c^x f(t) dt$, then we have

$$\int_a^b f(t) dt = \int_c^b f(t) dt - \int_c^a f(t) dt = F(b) - F(a).$$

A different choice of c merely changes $F(x)$ by a constant; this does not alter the difference $F(b) - F(a)$ because the constant cancels out in the subtraction. Many of the important functions that occur naturally in various branches of science arise exactly in this way, as indefinite integrals of other functions. For this reason a large part of calculus is devoted to the study of such functions.

There is, of course, a very simple geometric relationship between a function f and its indefinite integrals. An example is illustrated in Figure 1.57(a), where f is a positive function and the number $A(x)$ is equal to the area of the shaded region under the graph of f from a to x . If f assumes both positive and negative values, as in Figure 1.57(b), the integral $A(x)$ gives the sum of the areas of the regions above the x -axis minus the sum of the areas of the regions below the x -axis.

Sometimes it is instructive to compare the graph of f with that of A by drawing the graph of A alongside that of f . A few examples are illustrated in Figures 1.58 through 1.63. In each case the height of the curve at x in the right graph is equal to the area from 0 to x (the area of the shaded portion) of the left graph. In Figures 1.62 and 1.63, $A(x)$ is the algebraic sum of the areas taken with appropriate signs. The examples shown in the figures illustrate certain general properties of indefinite integrals. For instance, the indefinite integrals shown in Figures 1.58 and 1.60 are examples of increasing functions. That is to say, whenever $x < y$, we have

$$(1.59) \quad A(x) \leq A(y).$$

This happens because we are integrating a nonnegative function in each case. It is easy to prove that (1.59) always holds for the indefinite integral of any nonnegative function because we have

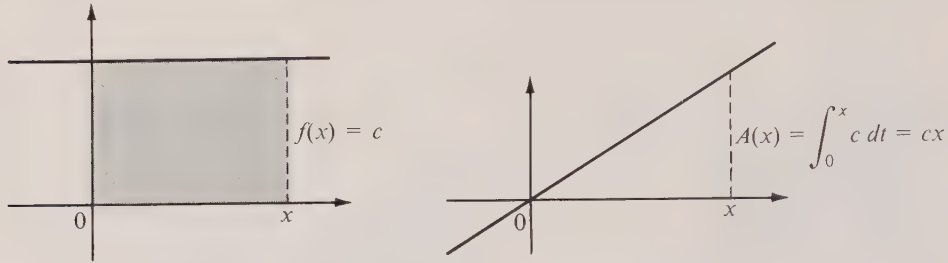


FIGURE 1.58

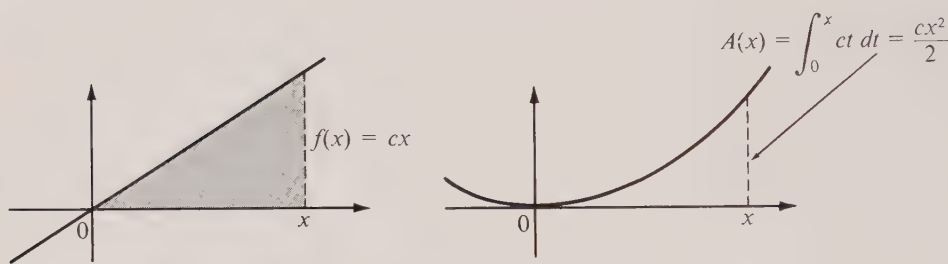


FIGURE 1.59

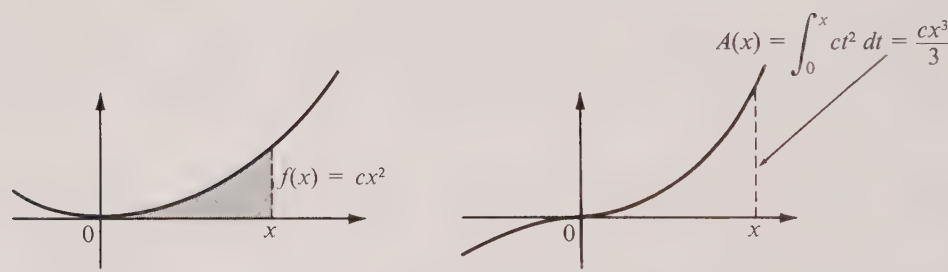


FIGURE 1.60

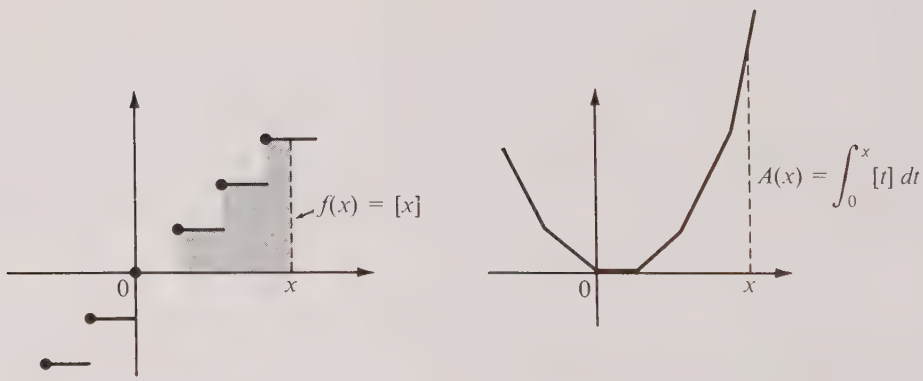


FIGURE 1.61

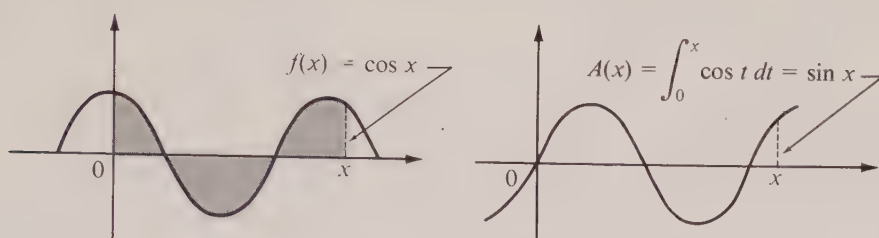


FIGURE 1.62

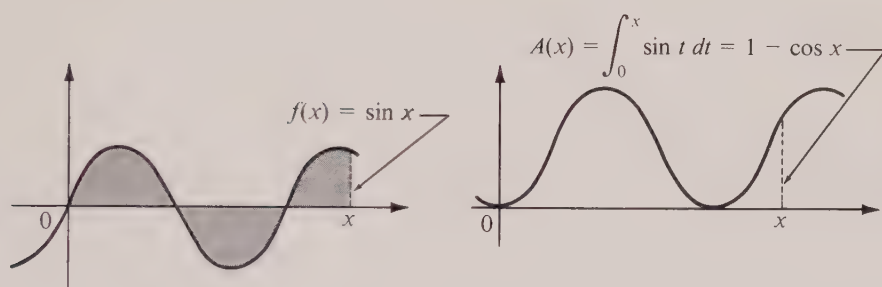


FIGURE 1.63

$$A(y) - A(x) = \int_a^y f(t) dt - \int_a^x f(t) dt = \int_x^y f(t) dt \geq 0,$$

the last inequality being a consequence of the comparison theorem. If f is a positive function, then every indefinite integral of f is a strictly increasing function.

Figure 1.61 contains a rather interesting example. The function f in this case is the greatest-integer function, introduced in Section 1.29, its function values being denoted by $[x]$. Observe that in this example the graph of f is made up of disconnected line segments. There are points on the graph of f where a small change in x produces a sudden jump in the value of the function. Note, however, that the corresponding indefinite integral does not exhibit this behavior. A small change in x produces only a small change in $A(x)$. That is why the graph of A is not disconnected. Again this illustrates a general property of indefinite integrals. It can be shown that the indefinite integral (whenever it exists) is always a *continuous function*. (See Exercise 19 in Section 2.21.)

1.43 Exercises

1. Evaluate the integrals in (a) through (d) in terms of x :

(a) $\int_0^x (1 + t + t^2) dt.$

(c) $\int_1^{1-x} (1 - 2t + 3t^2) dt.$

(b) $\int_{-1}^{2x} (1 + t + t^2) dt.$

(d) $\int_x^{x^2} (t^2 + 1)^2 dt.$

- (e) Find all real values of
- x
- such that

$$\int_0^x (t^3 - t) dt = \frac{1}{3} \int_{\sqrt{2}}^x (t - t^3) dt.$$

Draw a suitable figure and interpret this equation geometrically.

2. Let $f(x) = x - [x] - \frac{1}{2}$ if x is not an integer, and let $f(x) = 0$ if x is an integer. (As usual, $[x]$ denotes the greatest integer contained in x .) Define a new function P as follows:

$$P(x) = \int_0^x f(t) dt \quad \text{for every real } x.$$

- (a) Draw the graph of
- f
- over the interval
- $[-3, 3]$
- .

- (b) Prove that
- f
- is periodic with period 1; that is, prove that

$$f(x+1) = f(x) \quad \text{for every real } x.$$

- (c) Show that

$$P(x) = \frac{x^2}{2} - \frac{x}{2} \quad \text{if } 0 \leq x \leq 1,$$

and that P is periodic with period 1.

- (d) Express
- $P(x)$
- in terms of
- $[x]$
- .

- (e) Choose a constant
- c
- such that
- $\int_0^1 (P(t) + c) dt = 0$
- .

- (f) For the constant
- c
- of part (e), let
- $Q(x) = \int_0^x (P(t) + c) dt$
- . Show that

$$Q(x) = \frac{x^3}{6} - \frac{x^2}{4} + \frac{x}{12} \quad \text{if } 0 \leq x \leq 1,$$

and that Q is periodic with period 1.

Evaluate the integrals in Exercises 3 through 8. You may use the integration formulas (1.56) and (1.57) for the sine and cosine.

$$3. \int_0^{2\pi} (x + \sin x) dx.$$

$$6. \int_0^{\pi/2} \left(\sin 2x + \cos \frac{x}{2} \right) dx.$$

$$4. \int_0^{\pi/2} (x^2 + \cos x) dx.$$

$$7. \int_{-\pi}^{\pi} \left| \frac{1}{2} + \cos x \right| dx.$$

$$5. \int_0^{2\pi} |\sin x - \cos x| dx.$$

$$8. \int_0^{2\pi} \sin^2 \frac{x}{2} dx. \text{ [Hint, } \cos 2x = 1 - 2 \sin^2 x.]$$

Evaluate the integrals in Exercises 9, 10, and 11 in terms of x . You may use the integration formulas (1.56) and (1.57) for the sine and cosine.

$$9. \int_{-2}^x t^2(t^2 + 1) dt.$$

$$10. \int_x^{x^2} (t^2 + \sin 3t) dt.$$

$$11. \int_{-\pi}^x \left| \frac{1}{2} + \cos t \right| dt \quad \text{if } 0 \leq x \leq \pi.$$

12. Remember that $\sin^n t$ means $(\sin t)^n$. Given that the formula

$$\int_0^x \sin^n t dt = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int_0^x \sin^{n-2} t dt$$

is valid for $n = 2, 3, \dots$, derive the following integration formulas. You may use the integration formula (1.56) for the sine function.

$$(a) \int_0^x \sin^2 t \, dt = \frac{x}{2} - \frac{\sin 2x}{4}.$$

$$(b) \int_0^x \sin^3 t \, dt = \frac{x}{3} - \frac{1}{3}(2 + \sin^2 x) \cos x.$$

$$(c) \int_0^x \sin^4 t \, dt = \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}.$$

$$(d) \int_0^x \sin^5 t \, dt = \frac{8}{15} - \frac{\sin^4 x \cos x}{5} - \frac{4}{15} \sin^2 x \cos x - \frac{8}{15} \cos x.$$

13. Let f be increasing on an interval $[a, b]$. Let $A(x) = \int_a^x f(t) \, dt$ and prove that

$$A\left(\frac{x+y}{2}\right) \leq \frac{A(x) + A(y)}{2}.$$

Interpret this result geometrically. [Hint. If $x < y$, let $z = \frac{1}{2}(x+y)$. Then $\int_x^z f(t) \, dt \leq \int_z^y f(t) \, dt$.]

14. (a) Prove that the sine function is increasing on the interval $[0, \frac{1}{2}\pi]$. [Hint. Use the identity $\sin y - \sin x = 2 \sin \frac{1}{2}(y-x) \cos \frac{1}{2}(y+x)$.]

(b) Prove that the sine function is piecewise monotonic on every finite interval.

15. Use a method similar to that suggested in Exercise 14 to prove that the cosine function is piecewise monotonic on every interval.

16. (a) If a function f is periodic with period $p > 0$ and integrable on $[0, p]$, prove that $\int_0^p f(x) \, dx = \int_a^{a+p} f(x) \, dx$ for every real a .

(b) Use (a) and the translation property of the integral to show that

$$\int_0^\pi \sin^2 x \, dx = \int_0^\pi \cos^2 x \, dx = \frac{\pi}{2}.$$

[Hint. $\int_0^\pi (\sin^2 x + \cos^2 x) \, dx = \int_0^\pi dx = \pi$.]

17. From the integration formulas (1.56) and (1.57) it follows that

$$\int_0^{2\pi} \sin x \, dx = \int_0^{2\pi} \cos x \, dx = 0.$$

(a) Prove these formulas directly from general properties of the integral without using (1.56) and (1.57).

(b) From part (a) deduce that $\int_0^{2\pi} \sin nx \, dx = 0$ for all integers n , and that $\int_0^{2\pi} \cos nx \, dx = 0$ for all integers $n \neq 0$.

(c) Use part (b) and the identities for $\sin(x+y)$ and $\cos(x+y)$ to establish the following formulas (m and n denote integers, $m \neq n$):

$$\int_0^{2\pi} \sin nx \cos mx \, dx = \int_0^{2\pi} \sin nx \sin mx \, dx = \int_0^{2\pi} \cos nx \cos mx \, dx = 0,$$

$$\int_0^{2\pi} \sin^2 nx \, dx = \int_0^{2\pi} \cos^2 nx \, dx = \pi \quad \text{if } n \neq 0.$$

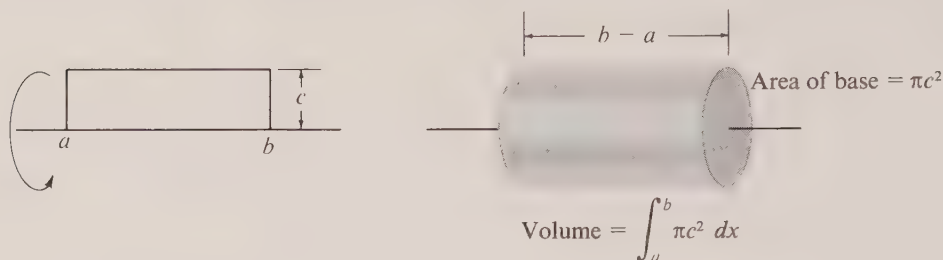


FIGURE 1.64 A cylinder generated by revolving a rectangle about its base.

1.44 Calculation by integration of volumes of solids of known cross-section

We have seen that the integral can be used to define and compute the areas of many geometric figures. The integral also provides an explanation of other geometrical ideas—for example, *volume*, *centroid*, and *arc length*—as well as certain topics that are of interest in physics and engineering, such as *work* and *moment of inertia*. The notion of volume for certain rather special figures (solids of known cross-section) is described in this section. More elaborate applications of the integral will be discussed in Volume II.

A rectangle with its base on the x -axis is shown in Figure 1.64; when this rectangle is rotated about the x -axis it sweeps out a circular cylinder as illustrated. Planes perpendicular to the x -axis intersect this solid in circular cross-sections, all of which have the same area. The volume of such a cylinder is defined to be the product of the cross-sectional area with the length. Thus for the example shown, the volume is $\pi c^2 (b - a)$. Notice that this is equal to the value of the integral $\int_a^b \pi c^2 dx$.

The graph of a nonnegative step-function s is shown in Figure 1.65. If the ordinate set of this function is revolved about the x -axis, another solid of revolution is swept out as illustrated in the figure. The solid consists of several adjoining cylinders; we *define* the volume of this solid to be the integral $\int_a^b \pi s^2(x) dx$, where $s^2(x)$ means $[s(x)]^2$. Note that $\pi s^2(x)$ denotes the area of the circular cross-section at x . Note also that this integral gives the sum of the volumes of the individual cylinders that make up the solid.

More generally, let f be integrable on $[a, b]$ and consider the solid of revolution obtained by rotating the ordinate set of $|f|$ about the x -axis. The volume of this solid is defined to be the integral

$$(1.60) \quad \int_a^b \pi f^2(x) dx,$$

where, in the integrand, $f^2(x)$ is obtained by squaring $f(x)$. This definition may be justified as follows:

Let s and t be nonnegative step-functions approximating $|f|$ from below and above, respectively, say $0 \leq s(x) \leq |f(x)| \leq t(x)$ for all x in $[a, b]$. Then the solid of revolution generated by $|f|$ lies between the corresponding solids generated by s and t . These latter solids are of the type shown in Figure 1.65 and their volumes are $\int_a^b \pi s^2(x) dx$ and $\int_a^b \pi t^2(x) dx$, respectively. Since we have $\pi s^2(x) \leq \pi f^2(x) \leq \pi t^2(x)$ for all x in $[a, b]$, the integral in (1.60) lies between the integrals $\int_a^b \pi s^2(x) dx$ and $\int_a^b \pi t^2(x) dx$ for all choices

of s and t . Since the integral in (1.60) is the *only* number with this property it seems natural to call it the volume of the solid in question.

Suppose now that two nonnegative functions f and g are integrable on a common interval $[a, b]$. When the region between their graphs is rotated about the x -axis it sweeps out a solid of revolution. The volume of this solid is defined to be the integral

$$\int_a^b \pi |f^2(x) - g^2(x)| dx.$$

The absolute-value sign is introduced so that the volume of a region between two solids of revolution may be obtained by subtracting the volume of the smaller solid from that of the larger.

Solids of revolution have the property that each cross-section made by a plane perpendicular to the axis of rotation is either a circular disk or an annulus (a region bounded by two concentric circles). Consider now a more general kind of solid whose cross sections, made by planes perpendicular to a given line, are not necessarily circular or annular regions but are regions with known areas. Specifically, let the x -axis be the given line and suppose $A(x)$ is the cross-sectional area at the point x , where $a \leq x \leq b$. The volume of such a solid is defined to be the integral

$$(1.61) \quad \int_a^b A(x) dx,$$

provided, of course, the integral exists. This definition of volume includes as special cases all the other definitions given in this section. In Volume II we shall present further justification for this definition. At that time we shall set forth a number of properties that we feel volume should have and then we shall prove that formula (1.61) is a consequence of these properties.

1.45 Exercises

1. Use integration to compute the volume of a right-circular cone generated by revolving the

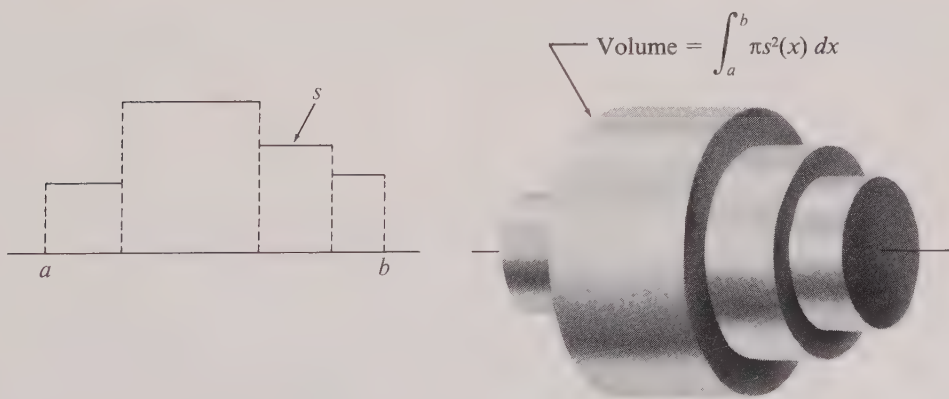


FIGURE 1.65 A solid generated by revolving the graph of a step function.

graph of the function f given by $f(x) = cx$ over the interval $0 \leq x \leq b$. (The result should be one-third the product of the area of the base and the length of the axis of the cone.)

2. A line segment joins two points $(0, a)$ and (h, b) in the xy -plane, where $0 < a < b$ and $h > 0$. When this segment is revolved about the x -axis it generates a *conical frustum*. Sketch the solid and compute its volume by integration.

In each of Exercises 3 through 6, compute the volume of the solid generated by revolving the ordinate set of each of the functions over the interval indicated. Sketch each of the solids. (You may use integration formulas given earlier in the text to evaluate the integrals.)

3. $f(x) = x^2$ if $0 \leq x \leq b$. 5. $f(x) = \sqrt{r^2 - x^2}$ if $-r \leq x \leq r$.
 4. $f(x) = 2 \cos x$ if $0 \leq x \leq \frac{\pi}{2}$. 6. $f(x) = \sin \frac{x}{2}$ if $0 \leq x \leq \pi$.

7. In each case sketch the solid obtained by rotating about the x -axis the region between the graphs of f and g and compute the volume of this solid if:

- (a) $f(x) = \sqrt{x}$, $g(x) = 1$, $0 \leq x \leq 1$;
 (b) $f(x) = \sqrt{r^2 - x^2}$, $g(x) = c$, $0 \leq c \leq r$, $0 \leq x \leq \sqrt{r^2 - c^2}$.

8. What volume of material is removed from a solid sphere of radius $2r$ by drilling a hole of radius r through the center?

9. Solid-geometry textbooks often discuss a proposition known as *Cavalieri's theorem*. It states that if two solids have equal altitudes, and if cross sections, made by planes parallel to the bases and at equal distances from them, are always in a given ratio, then the volumes of the solids are also in this ratio. Use (1.61) to prove this theorem.

10. The cross sections of a solid are squares perpendicular to the x -axis with their centers on the axis. If the square cut off at x has edge $2x^2$, find the volume of the solid between $x = 0$ and $x = a$. Make a sketch.

11. Find the volume of a solid whose cross section, made by a plane perpendicular to the x -axis, has the area $ax^2 + bx + c$ for each x in the interval $0 \leq x \leq h$. Express the volume in terms of the areas B_1 , M , and B_2 of the cross sections corresponding to $x = 0$, $x = h/2$, and $x = h$, respectively. The resulting formula is known as the *prismoid formula*.

12. Make a sketch of the region in the xy -plane consisting of all points (x, y) satisfying the simultaneous inequalities

$$0 \leq x \leq 2, \quad \frac{1}{4}x^2 \leq y \leq 1.$$

Compute the volume of the solid obtained by rotating this region about (a) the x -axis; (b) the y -axis; (c) the vertical line passing through $(2, 0)$; (d) the horizontal line passing through $(0, 1)$.

Supplement C. Proofs of the basic properties of the integral

This supplement contains analytic proofs of the basic properties of the integral listed in Section 1.34 and also a proof of the theorem that piecewise monotonic functions are integrable. It is convenient to present first a number of preliminary theorems concerning integrability of bounded functions. These will help to simplify the proofs of the theorems of Section 1.34.

***1.46 The Riemann condition**

Throughout this section, f denotes a function which is defined and bounded on a closed interval $[a, b]$ and, unless specified to the contrary, the letters s and t always denote step functions satisfying the inequalities

$$(1.62) \quad s(x) \leq f(x) \leq t(x)$$

for every x in $[a, b]$. The step functions s and t are said to approximate f from below and from above, respectively.

1-49 THEOREM. Assume f is integrable on $[a, b]$ and let $I = \int_a^b f(x) dx$. Let ϵ (the Greek letter epsilon) be a given positive number. Then there exists at least one pair of step functions s and t such that

$$(1.63) \quad I - \epsilon < \int_a^b s(x) dx \quad \text{and} \quad \int_a^b t(x) dx < I + \epsilon.$$

This theorem shows that the integral of f can be approximated both from below and from above within any prescribed degree of accuracy by integrals of step functions. The positive number ϵ denotes the required degree of accuracy.

Proof. The proof is by contradiction. Suppose the assertion of the theorem were not true. This would mean that, for the given ϵ and for *all* choices of s and t , we would have

$$(1.64) \quad \int_a^b s(x) dx \leq I - \epsilon \quad \text{or} \quad I + \epsilon \leq \int_a^b t(x) dx.$$

Since f is integrable we also have

$$\int_a^b s(x) dx \leq I \leq \int_a^b t(x) dx$$

for all s and t . If the first inequality in (1.64) holds, then there are two distinct numbers, I and $I - \epsilon$, lying between the integral of s and the integral of t for all s and t , contradicting the fact that f is integrable. If the second inequality in (1.64) holds, the same contradiction is obtained with the two numbers I and $I + \epsilon$. This completes the proof of Theorem 1-49.

The next theorem tells us that if f is integrable then we can approximate f from below and from above by step functions whose integrals differ by arbitrarily small amounts. More precisely, we have:

1-50 THEOREM. Assume f is integrable on $[a, b]$. Then for every $\epsilon > 0$ there exists at least one pair of step functions s and t satisfying (1.62) such that

$$(1.65) \quad 0 \leq \int_a^b t(x) dx - \int_a^b s(x) dx < \epsilon.$$

The statement in the conclusion of this theorem is known as the *Riemann condition* and the theorem shows that every integrable function must necessarily satisfy the Riemann

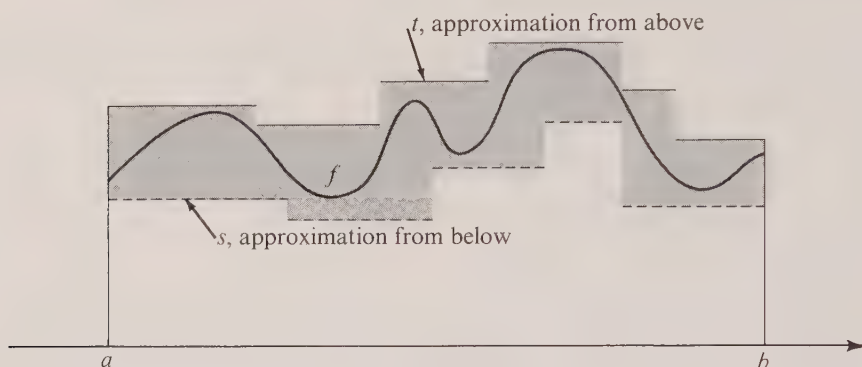


FIGURE 1.66 Illustrating the geometric meaning of the Riemann condition.
The area of the shaded region is $\int_a^b t(x) dx - \int_a^b s(x) dx$.

condition. The geometric meaning of the Riemann condition is illustrated in Figure 1.66. The area of the shaded region is equal to the difference $\int_a^b t(x) dx - \int_a^b s(x) dx$. The theorem states that if f is integrable and if a positive number ϵ is prescribed, then it is possible to choose step functions s and t so that this area does not exceed ϵ . It is important to realize that the choice of s and t will depend on the value of ϵ that is prescribed. But, *no matter how small* an ϵ is prescribed (provided only that ϵ is *positive*), there always corresponds at least one pair s and t satisfying (1.65).

Proof. Since the given ϵ is positive, $\frac{1}{2}\epsilon$ is also positive and we may apply Theorem 1-49 with ϵ replaced by $\frac{1}{2}\epsilon$ to find a pair of step functions s and t satisfying

$$(1.66) \quad I - \frac{\epsilon}{2} < \int_a^b s(x) dx \quad \text{and} \quad \int_a^b t(x) dx < I + \frac{\epsilon}{2}.$$

The first inequality in (1.66) is equivalent to $-\int_a^b s(x) dx < -I + \epsilon/2$. Adding this to the second inequality, we obtain

$$\int_a^b t(x) dx - \int_a^b s(x) dx < \epsilon.$$

This proves the rightmost inequality in (1.65). The leftmost inequality follows from the fact that $t(x) \geq s(x)$ for all x in $[a, b]$.

We have just proved that the Riemann condition is *necessary* for the integrability of f . The next theorem shows that the Riemann condition is also *sufficient*. That is, we have:

1-51 THEOREM. Suppose that for every $\epsilon > 0$ there exists at least one pair of approximating step-functions s and t such that (1.65) holds. Then f is integrable on $[a, b]$.

Proof. The proof is by contradiction. If f is not integrable then there are at least two numbers, say I_1 and I_2 , with $I_1 < I_2$, such that

$$\int_a^b s(x) dx \leq I_1 \leq \int_a^b t(x) dx \quad \text{and} \quad \int_a^b s(x) dx \leq I_2 \leq \int_a^b t(x) dx$$

for all s and t satisfying (1.62). Combining these inequalities, we find

$$0 < I_2 - I_1 \leq \int_a^b t(x) dx - \int_a^b s(x) dx.$$

This shows that the Riemann condition is violated when $\epsilon = I_2 - I_1$.

Theorem 1-51 provides a very useful criterion for testing the integrability of bounded functions because for each ϵ we need exhibit *only one pair* of step functions satisfying (1.65). The Riemann condition is one of the major tools for proving theorems about the existence of integrals. In the next section we shall find that it leads to an extremely simple proof of the integrability of monotonic functions. It is also used extensively in the proofs of the basic properties of the integral given below in Section 1.48.

*1.47 Proof that piecewise monotonic functions are integrable

We first prove that monotonic functions are integrable and then use the additive property of the integral to prove that piecewise monotonic functions are integrable.

1-52 THEOREM. Let f be a function which is defined and monotonic on a closed interval $[a, b]$. Then f is integrable on $[a, b]$.

Proof. We shall prove this theorem by using Riemann's condition (Theorem 1-51). Assume first that f is increasing. Let $\epsilon > 0$ be given. We shall exhibit a pair of step functions s and t such that $s(x) \leq f(x) \leq t(x)$ for all x in $[a, b]$ and such that

$$(1.67) \quad \int_a^b t(x) dx - \int_a^b s(x) dx < \epsilon.$$

For this purpose consider a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with n equal subdivisions and let s and t be step functions such that for each k we have $s(x) = f(x_{k-1})$ and $t(x) = f(x_k)$ whenever x is in the open subinterval (x_{k-1}, x_k) . At the subdivision points define s and t so as to preserve the relation $s(x) \leq f(x) \leq t(x)$. An example is shown in Figure 1.67. Geometrically, the difference $\int_a^b t(x) dx - \int_a^b s(x) dx$ is equal to the area of the shaded rectangles in Figure 1.67(a). By sliding these rectangles to the right so they have a common base as suggested by Figure 1.67(b), we see that they fill out a rectangle of base $(b - a)/n$ and altitude $f(b) - f(a)$ whose area is $(b - a) \times [f(b) - f(a)]/n$. Therefore the figure suggests that we have the formula

$$(1.68) \quad \int_a^b t(x) dx - \int_a^b s(x) dx = \frac{(b - a)[f(b) - f(a)]}{n}.$$

From (1.68) we can easily satisfy (1.67) by choosing n large enough. In fact, any n greater than $(b - a)[f(b) - f(a)]/\epsilon$ will do. From this it follows that f satisfies Riemann's condition and hence f is integrable on $[a, b]$.

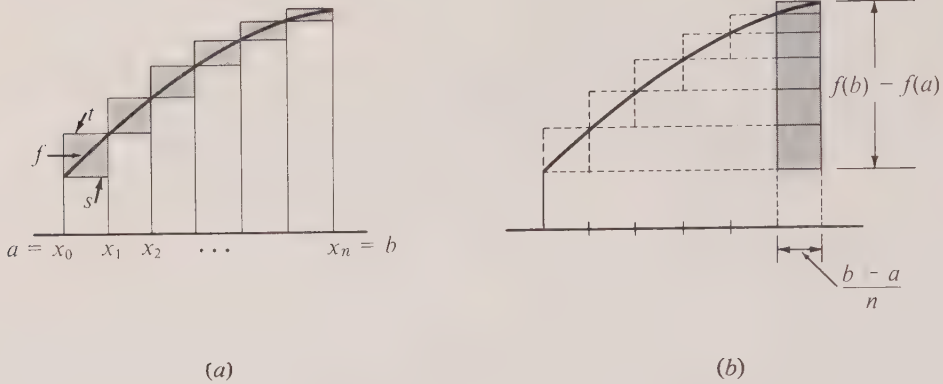


FIGURE 1.67 Proof of integrability of an increasing function.

To prove the formula in (1.68) analytically (without the aid of a figure) we note that

$$\begin{aligned}
 \int_a^b t(x) \, dx - \int_a^b s(x) \, dx &= \int_a^b [t(x) - s(x)] \, dx \\
 &= \sum_{k=1}^n [f(x_k) - f(x_{k-1})] (x_k - x_{k-1}) \\
 &= \frac{b-a}{n} \sum_{k=1}^n [f(x_k) - f(x_{k-1})] = \frac{(b-a)[f(b) - f(a)]}{n},
 \end{aligned}$$

where the last equation is a consequence of the telescoping property of finite sums.

In the foregoing proof we assumed that f was increasing on $[a, b]$. When f is decreasing we may apply the result just proved to the function $-f$.

Although Theorem 1-52 refers to functions which are monotonic on the *closed* interval $[a, b]$, a small modification of the proof shows that the conclusion also holds when f is monotonic on the *open* interval (a, b) , provided, of course, that f is bounded on the closed interval $[a, b]$. (This modification is left as an exercise for the reader.) Once we know this, it follows at once from the additive property of the integral that every function which is defined and bounded and piecewise monotonic on an interval $[a, b]$ is integrable on $[a, b]$. This completes the proof of Theorem 1-48.

★1.48 Proofs of Theorems 1-43 through 1-47

Proof of the linearity property (Theorem 1-43). We decompose the linearity property into two parts:

$$(A) \quad \int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx,$$

$$(B) \quad \int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx.$$

To prove (A), let $I = \int_a^b f(x) \, dx$ and $J = \int_a^b g(x) \, dx$. We wish to prove that $f + g$ is integrable and that $\int_a^b (f + g) = I + J$. To prove that $f + g$ is integrable we shall use the Riemann condition.

Let $\epsilon > 0$ be given. Using Theorem 1-49, we choose step functions s_1 and t_1 approximating f from below and from above such that

$$I - \frac{\epsilon}{4} < \int_a^b s_1(x) dx \quad \text{and} \quad \int_a^b t_1(x) dx < I + \frac{\epsilon}{4}.$$

Choose s_2 and t_2 similarly related to g and let $s = s_1 + s_2$, $t = t_1 + t_2$. Then $s(x) \leq f(x) + g(x) \leq t(x)$ for all x in $[a, b]$. By adding the inequalities on the integrals, we get

$$(1.69) \quad I + J - \frac{\epsilon}{2} < \int_a^b s(x) dx \leq \int_a^b t(x) dx < I + J + \frac{\epsilon}{2}.$$

These imply, first of all, that $0 \leq \int_a^b t(x) dx - \int_a^b s(x) dx < \epsilon$, which means the Riemann condition holds for $f + g$. Hence, by Theorem 1-51, $f + g$ is integrable. Let $K = \int_a^b [f(x) + g(x)] dx$. Since K lies between the integrals of s and t we may use (1.69) again to obtain

$$I + J - \frac{\epsilon}{2} < K < I + J + \frac{\epsilon}{2}.$$

But since ϵ is *arbitrary* it follows that $K = I + J$. (Make sure you understand the reason for this last statement.) This proves part (A) and the proof of part (B) is similar. The two parts together imply Theorem 1-43.

Proof of additivity with respect to the interval of integration (Theorem 1-44). Suppose that $a < b < c$, assume that the two integrals $\int_a^b f(x) dx$ and $\int_b^c f(x) dx$ exist, and denote their respective values by I and J . We wish to prove that the integral $\int_a^c f(x) dx$ exists and equals $I + J$. Again, we shall use the Riemann condition.

Let $\epsilon > 0$ be given, and choose step functions s_1 and t_1 which approximate f over $[a, b]$ such that

$$I - \frac{\epsilon}{4} < \int_a^b s_1(x) dx \quad \text{and} \quad \int_a^b t_1(x) dx < I + \frac{\epsilon}{4}.$$

Choose s_2 and t_2 similarly related to J . We can also assume that $s_1(b) = s_2(b)$ and $t_1(b) = t_2(b)$. Now define new step-functions s and t over the full interval $[a, c]$ as follows:

$$s(x) = \begin{cases} s_1(x) & \text{if } a \leq x \leq b, \\ s_2(x) & \text{if } b \leq x \leq c, \end{cases} \quad t(x) = \begin{cases} t_1(x) & \text{if } a \leq x \leq b, \\ t_2(x) & \text{if } b \leq x \leq c. \end{cases}$$

Then s and t satisfy $s(x) \leq f(x) \leq t(x)$ for all x in $[a, c]$ and, by adding the inequalities on the integrals, we have

$$I + J - \frac{\epsilon}{2} < \int_a^c s(x) dx \leq \int_a^c t(x) dx < I + J + \frac{\epsilon}{2}.$$

The rest of the proof is similar to that of Theorem 1-43. This proves Theorem 1-44 under the assumption that $\int_a^b f(x) dx$ and $\int_b^c f(x) dx$ exist. The proof is similar if, instead, we assume that one of these and $\int_a^c f(x) dx$ exist.

Proof of the translation property (Theorem 1-45). Let $I = \int_a^b f(x) dx$, and let g be the function defined on the interval $[a + c, b + c]$ by the equation $g(x) = f(x - c)$. We wish to prove that g is integrable and that $\int_{a+c}^{b+c} g(x) dx = I$. As in the foregoing proofs, we shall use the Riemann condition.

Let $\epsilon > 0$ be given and choose step functions s_1 and t_1 approximating f from below and from above such that

$$(1.70) \quad I - \epsilon < \int_a^b s_1(x) dx \leq \int_a^b t_1(x) dx < I + \epsilon.$$

Let $s(x) = s_1(x - c)$ and $t(x) = t_1(x - c)$. Then s and t are step functions satisfying $s(x) \leq g(x) \leq t(x)$ for all x in $[a + c, b + c]$. Because of the translation property for integrals of step functions (Theorem 1-41), we have $\int_a^b s_1(x) dx = \int_{a+c}^{b+c} s(x) dx$ and $\int_a^b t_1(x) dx = \int_{a+c}^{b+c} t(x) dx$ so (1.70) becomes

$$I - \epsilon < \int_{a+c}^{b+c} s(x) dx \leq \int_{a+c}^{b+c} t(x) dx < I + \epsilon.$$

These imply, first of all, that g satisfies the Riemann condition and, secondly, that the integral of g lies between $I - \epsilon$ and $I + \epsilon$. Since ϵ is arbitrary the integral of g must be equal to I , as asserted.

Proof of Theorem 1-46. The proof of this theorem is entirely analogous to that of Theorem 1-45 and therefore we shall outline only the principal steps. It suffices to treat the case $c > 0$. Let $I = \int_a^b f(x) dx$, and define g on the interval $[ac, bc]$ by the equation $g(x) = f(x/c)$. Let $\epsilon > 0$ be given and choose step functions s_1 and t_1 satisfying $s_1(x) \leq f(x) \leq t_1(x)$ such that (1.70) holds. Define s and t by the equations $s(x) = s_1(x/c)$ and $t(x) = t_1(x/c)$ and proceed as in the proof of Theorem 1-45, using Theorem 1-42 instead of 1-41.

Proof of the comparison theorem (Theorem 1-47). Let $I = \int_a^b f$ and $J = \int_a^b g$. We wish to prove that $I \leq J$. Let $\epsilon > 0$ be given and use Theorem 1-49 to choose step functions s and t such that

$$s(x) \leq f(x) \leq g(x) \leq t(x)$$

for all x on $[a, b]$ and such that

$$I - \epsilon < \int_a^b s(x) dx \quad \text{and} \quad \int_a^b t(x) dx < J + \epsilon.$$

Since $\int_a^b s \leq \int_a^b t$ we have $I - \epsilon < J + \epsilon$, or $I < J + 2\epsilon$. But since ϵ is arbitrary it follows that $I \leq J$, as asserted.

★1.49 Upper and lower integrals

We conclude this chapter with a few remarks related to the definition of the integral. Suppose s and t are step functions approximating f from below and from above. Let us denote by S the collection of all numbers $\int_a^b s(x) dx$ obtained by letting s vary over all step functions approximating f from below, and define T similarly as the collection of all numbers $\int_a^b t(x) dx$, where t is a step function approximating f from above. Let I

denote the least upper bound of the set S and let $\bar{I}$ denote the greatest lower bound of the set T . The number $\underline{I}$ is called the *lower integral* of f and $\bar{I}$ the *upper integral* of f over $[a, b]$. The upper and lower integrals of f *always exist*, even if f is not integrable. We shall prove presently that $\underline{I} = \bar{I}$ if and only if f is integrable. In general, all we can assert is that

$$(1.71) \quad \underline{I} \leq \bar{I}.$$

To prove this, we note that for every $\epsilon > 0$ there is at least one step function s such that $\underline{I} - \epsilon < \int_a^b s(x) dx$. (Otherwise $\underline{I} - \epsilon$ would be an upper bound of S smaller than the least upper bound.) Similarly, there is at least one step function t such that $\int_a^b t(x) dx < \bar{I} + \epsilon$. Since $\int_a^b s \leq \int_a^b t$, this implies $\underline{I} - \epsilon < \bar{I} + \epsilon$ or $\underline{I} < \bar{I} + 2\epsilon$. Since ϵ is arbitrary, this proves (1.71).

The inequality in (1.71) enables us to prove the following theorem:

1-53 THEOREM. Let f be defined and bounded on $[a, b]$ and let $\underline{I}$ and $\bar{I}$ denote its lower and upper integrals over $[a, b]$. Then f is integrable on $[a, b]$ if and only if $\underline{I} = \bar{I}$, in which case the common value of $\underline{I}$ and $\bar{I}$ is the integral of f .

Proof. Since f is bounded it has a lower integral $\underline{I}$ and an upper integral $\bar{I}$ and they satisfy the inequalities

$$(1.72) \quad \int_a^b s(x) dx \leq \underline{I} \leq \bar{I} \leq \int_a^b t(x) dx$$

for every pair of approximating step functions s and t . This proves that there is *at least one* number I which satisfies the inequalities

$$(1.73) \quad \int_a^b s(x) dx \leq I \leq \int_a^b t(x) dx$$

for all choices of s and t . If $\underline{I} < \bar{I}$, then (1.72) shows that there is more than one I satisfying (1.73) and in this case f is not integrable.

On the other hand, any I which satisfies (1.73) for all s and t also satisfies

$$(1.74) \quad \underline{I} \leq I \leq \bar{I},$$

by the very definition of the upper and lower integrals. Therefore, if the upper and lower integrals are equal, then (1.74) shows that every I which satisfies (1.73) is equal to the common value of the upper and lower integrals. In this case there is only one I which satisfies (1.73) for all s and t and therefore f is integrable. This completes the proof.

The upper and lower integrals of a bounded function are often denoted by the symbols

$$\bar{I} = \bar{\int}_a^b f \quad \text{and} \quad \underline{I} = \underline{\int}_a^b f.$$

Some of the properties of the integral are also shared by upper and lower integrals. For example, they satisfy the additive property with respect to the interval of integration. That is, we have

$$(1.75) \quad \bar{\int}_a^b f + \bar{\int}_b^c f = \bar{\int}_a^c f \quad \text{and} \quad \underline{\int}_a^b f + \underline{\int}_b^c f = \underline{\int}_a^c f.$$

The comparison theorem is also valid. That is, if $f(x) \leq g(x)$ for every x in $[a, b]$, we have

$$(1.76) \quad \bar{\int}_a^b f \leq \bar{\int}_a^b g \quad \text{and} \quad \underline{\int}_a^b f \leq \underline{\int}_a^b g.$$

The linearity property with respect to the integrand takes the form of *inequalities*, namely

$$(1.77) \quad \bar{\int}_a^b (f + g) \leq \bar{\int}_a^b f + \bar{\int}_a^b g \quad \text{and} \quad \underline{\int}_a^b (f + g) \geq \underline{\int}_a^b f + \underline{\int}_a^b g.$$

All these properties are easily proved from the definition of the upper and lower integrals. To indicate the ideas in the proofs, we shall prove the additive property (1.75) for lower integrals. The proofs of (1.76) and (1.77) are simpler.

Assume that $a < b < c$. Let s be a step function which approximates f from below on $[a, c]$. Using the additive property for integrals of step functions we have

$$\int_a^c s = \int_a^b s + \int_b^c s \leq \underline{\int}_a^b f + \underline{\int}_b^c f,$$

where the last inequality is a consequence of the definition of the lower integral. Since the number $\underline{\int}_a^b f + \underline{\int}_b^c f$ is an upper bound for $\int_a^c s$ for all s which approximate f from below, it cannot be less than their least upper bound, which is $\underline{\int}_a^c f$. This gives us the inequality

$$(1.78) \quad \underline{\int}_a^c f \leq \underline{\int}_a^b f + \underline{\int}_b^c f.$$

To prove the reverse inequality, let s_1 and s_2 be step functions approximating f from below on $[a, b]$ and $[b, c]$, respectively, and define a step function s as follows:

$$s(x) = \begin{cases} s_1(x) & \text{if } a \leq x < b, \\ s_2(x) & \text{if } b \leq x \leq c. \end{cases}$$

Then we have $\int_a^b s_1 + \int_b^c s_2 = \int_a^c s \leq \underline{\int}_a^c f$, and hence

$$\int_a^b s_1 \leq \underline{\int}_a^c f - \int_b^c s_2.$$

Now we keep s_2 fixed and let s_1 vary over all step functions approximating f from below on $[a, b]$. Since $\underline{\int}_a^c f - \int_b^c s_2$ is an upper bound for all the numbers $\int_a^b s_1$, it cannot be less than their least upper bound, which is $\underline{\int}_a^b f$. Therefore we have $\underline{\int}_a^b f \leq \underline{\int}_a^c f - \int_b^c s_2$, or

$$\int_b^c s_2 \leq \underline{\int}_a^c f - \underline{\int}_a^b f.$$

This, in turn, implies that $\underline{\int}_b^c f \leq \underline{\int}_a^c f - \underline{\int}_a^b f$, or $\underline{\int}_a^b f + \underline{\int}_b^c f \leq \underline{\int}_a^c f$. This last inequality, along with (1.78), proves (1.75) for lower integrals. The proof for upper integrals is entirely analogous.

2

DIFFERENTIAL CALCULUS

2.1 Historical introduction

Newton and Leibniz, quite independently of one another, were largely responsible for developing the ideas of integral calculus to the point where hitherto insurmountable problems could be solved by more or less routine methods. The successful accomplishments of these men were primarily due to the fact that they were able to fuse together the integral calculus with the second main branch of calculus, differential calculus.

The central idea of differential calculus is the notion of *derivative*. Like the integral, the derivative originated from a problem in geometry—the problem of finding the tangent line at a point of a curve. Unlike the integral, however, the derivative evolved very late in the history of mathematics. The concept was not formulated until early in the 17th century when the French mathematician, Pierre de Fermat, attempted to determine the maxima and minima of certain special functions.

Fermat's idea, basically very simple, can be understood if we refer to the curve in Figure 2.1. It is assumed that at each of its points this curve has a definite direction that can be described by a tangent line. Some of these tangents are indicated by broken lines in the figure. Fermat noticed that at certain points where the curve has a maximum or minimum, such as those shown in the figure with abscissae x_0 and x_1 , the tangent line must be horizontal. Thus the problem of locating such extreme values is seen to depend on the solution of another problem, that of locating the horizontal tangents.

This raises the more general question of determining the direction of the tangent line at an *arbitrary point* of the curve. It was the attempt to solve this general problem that led Fermat to discover some of the rudimentary ideas underlying the notion of derivative.

At first sight there seems to be no connection whatever between the problem of finding the area of a region lying under a curve and the problem of finding the tangent line at a point of a curve. The first person to realize that these two seemingly remote ideas are, in fact, rather intimately related appears to have been Newton's teacher, Isaac Barrow (1630–1677). However, Newton and Leibniz were the first to understand the real importance of this relation and they exploited it to the fullest, thus inaugurating an unprecedented era in the development of mathematics.

Although the derivative was originally formulated to study the problem of tangents, it was soon found that it also provides a way to calculate *velocity* and, more generally, the *rate of change* of a function. In the next section we shall consider a special problem

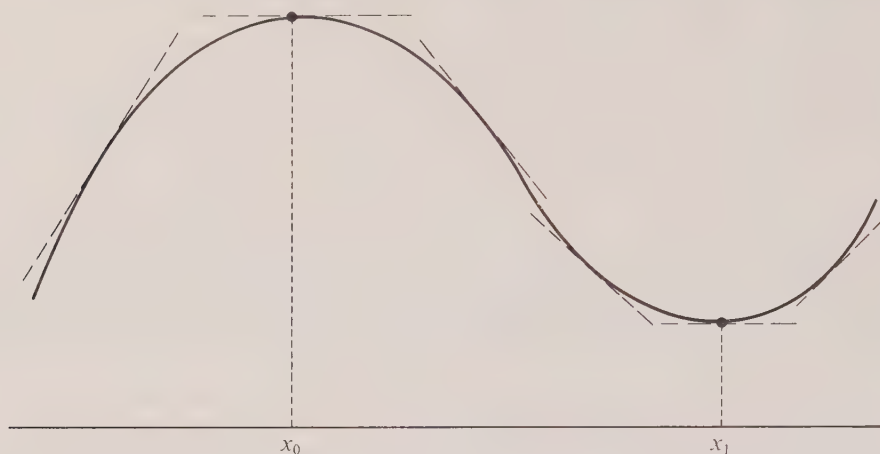


FIGURE 2.1 The curve has horizontal tangents above the points x_0 and x_1 .

involving the calculation of a velocity. The solution of this problem contains all the essential features of the derivative concept and may help to motivate the general definition of derivative which is given in Section 2.3.

2.2 A problem involving velocity

Suppose a projectile is fired straight up from the ground with initial velocity of 144 feet per second. Neglect friction, and assume the projectile is influenced only by gravity so that it moves up and back along a straight line. Let $f(t)$ denote the height in feet that the projectile attains t seconds after firing. If the force of gravity were not acting on it, the projectile would continue to move upward with a constant velocity, traveling a distance of 144 feet every second, and at time t we would have $f(t) = 144t$. In actual practice, gravity causes the projectile to slow down until its velocity decreases to zero and then it drops back to earth. Physical experiments suggest that as long as the projectile is aloft, its height $f(t)$ is given by the formula

$$(2.1) \quad f(t) = 144t - 16t^2.$$

The term $-16t^2$ is due to the influence of gravity. Note that $f(t) = 0$ when $t = 0$ and when $t = 9$. This means that the projectile returns to earth after 9 seconds and it is to be understood that formula (2.1) is valid only for $0 \leq t \leq 9$.

The problem we wish to consider is this: *To determine the velocity of the projectile at each instant of its motion.* Before we can understand this problem we must decide on what is *meant* by the velocity at each instant. To do this we introduce first the notion of *average velocity during a time interval*, say from time t to time $t + h$. This is defined to be the quotient

$$\frac{\text{change in distance during time interval}}{\text{length of time interval}} = \frac{f(t + h) - f(t)}{h}.$$

This quotient, called a *difference quotient*, is a number which may be calculated whenever both t and $t + h$ are in the interval $[0, 9]$. The number h may be positive or negative, but not zero. We shall keep t fixed and see what happens to the difference quotient as we take values of h with smaller and smaller absolute value.

For example, consider the instant $t = 2$. The distance traveled after 2 seconds is

$$f(2) = 288 - 64 = 224.$$

At time $t = 2 + h$ the distance covered is

$$f(2 + h) = 144(2 + h) - 16(2 + h)^2 = 224 + 80h - 16h^2.$$

Therefore the average velocity in the interval from $t = 2$ to $t = 2 + h$ is

$$\frac{f(2 + h) - f(2)}{h} = \frac{80h - 16h^2}{h} = 80 - 16h.$$

As we take values of h with smaller and smaller absolute value, this average velocity gets closer and closer to 80. For example, if $h = 0.1$ we get an average velocity of 78.4; when $h = 0.001$ we get 79.984; when $h = 0.00001$ we obtain the value 79.99984; and when $h = -0.00001$ we obtain 80.00016. The important thing is that we can make the average velocity as close to 80 as we please by taking $|h|$ sufficiently small. We describe this by saying that the average velocity *approaches 80 as a limit when h approaches zero*. It seems natural to call this limiting value the *instantaneous velocity* at time $t = 2$.

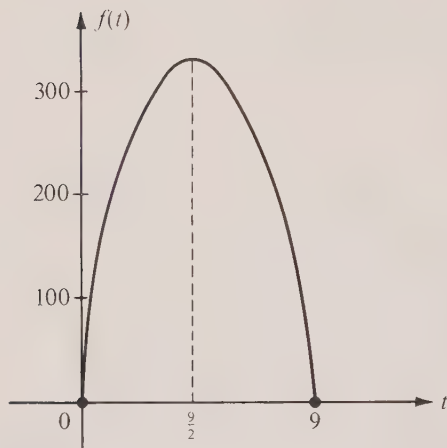
The same kind of calculation can be carried out for any other instant. The average velocity for an arbitrary time interval from t to $t + h$ is given by the quotient

$$\frac{f(t + h) - f(t)}{h} = \frac{[144(t + h) - 16(t + h)^2] - [144t - 16t^2]}{h} = 144 - 32t - 16h.$$

When h approaches zero the expression on the right approaches $144 - 32t$ as a limit and this limit is defined to be the *instantaneous velocity* at time t . If we denote the instantaneous velocity by $v(t)$ we may write

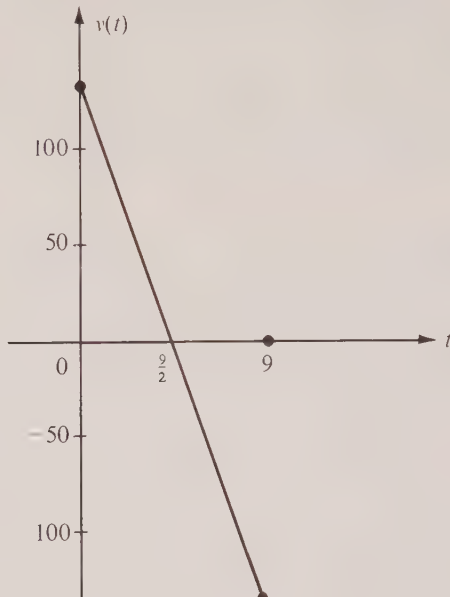
$$(2.2) \quad v(t) = 144 - 32t.$$

The formula in (2.1) for the distance $f(t)$ defines a function f which tells us how high the projectile is at each instant of its motion. We may refer to f as the *position function*. Its domain is the closed interval $[0, 9]$ and its graph is shown in Figure 2.2(a). [The scale on the vertical axis is distorted in both Figures 2.2(a) and (b).] The formula in (2.2) for the velocity $v(t)$ defines a new function v which tells us how fast the projectile is moving at each instant of its motion. This is called the *velocity function* and its graph is shown in Figure 2.2(b). As t increases from 0 to 9, $v(t)$ decreases steadily from $v(0) = 144$ to $v(9) = -144$. To find the time t for which $v(t) = 0$ we solve the equation $144 = 32t$ to obtain $t = 9/2$. Therefore, at the mid-point of the motion the influence of gravity just balances the upward forces, and the projectile is momentarily at rest. The height at this instant is $f(9/2) = 324$. When $t > 9/2$ the velocity is negative, indicating that the height is decreasing.



(a)

Graph of the position function:
 $f(t) = 144t - 16t^2$.



(b)

Graph of the velocity function:
 $v(t) = 144 - 32t$.

FIGURE 2.2

The process by which $v(t)$ is obtained from the difference quotient is called “taking the limit as h approaches zero” and it is denoted symbolically as follows:

$$(2.3) \quad v(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}.$$

This equation is used to define velocity not only for this particular example but, more generally, for any particle moving along a straight line, provided the position function f is such that the difference quotient tends to a definite limit as h approaches zero.

2.3 The derivative of a function

The example described in the foregoing section points the way to the introduction of the concept of derivative. We begin with a function f defined at least on some open interval (a, b) on the x -axis. Then we choose a fixed point x in this interval and introduce the difference quotient

$$\frac{f(x+h) - f(x)}{h},$$

where the number h , which may be positive or negative (but not zero), is such that $x+h$ also lies in (a, b) . The numerator of this quotient measures the change in the function

when x changes from x to $x + h$. The quotient itself is referred to as the *average rate of change* of f in the interval joining x to $x + h$.

Now we let h approach zero and see what happens to this quotient. If the quotient approaches some definite value as a limit (and if this limit is the same whether h approaches zero through positive values or through negative values), then this limit is called the *derivative* of f at x and is denoted by the symbol $f'(x)$ (read as “ f prime at x ”). Thus, the definition of $f'(x)$ is

$$(2.4) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h},$$

provided the limit exists. The number $f'(x)$ is also called the *rate of change* of f at x .

The equation in (2.4) is intended to convey the idea that the difference quotient can be made as close to $f'(x)$ as we please by taking $|h|$ sufficiently small. A more precise definition of this is given in Section 2.7.

By comparing (2.4) with (2.3) we see that the concept of instantaneous velocity is merely an example of the concept of derivative. The velocity $v(t)$ is equal to the derivative $f'(t)$, where f is the function which measures position. This is often described by saying that velocity is the rate of change of position with respect to time. In the example worked out in Section 2.2 the position function f is described by the equation

$$f(t) = 144t - 16t^2$$

and its derivative f' is a new function (velocity) given by

$$f'(t) = 144 - 32t.$$

In general, the limit process which produces $f'(x)$ from $f(x)$ gives us a way of obtaining a new function f' from a given function f . The process is called *differentiation* and f' is called the *first derivative* of f . If f' , in turn, is defined on an open interval we can try to compute its first derivative, denoted by f'' and called the *second derivative* of f . Similarly, the n th derivative of f , denoted by $f^{(n)}$, is defined to be the first derivative of $f^{(n-1)}$ ($n = 2, 3, \dots$).

In the case of rectilinear motion the first derivative of velocity (second derivative of position) is called *acceleration*. For example, to compute the acceleration in the example of Section 2.2 we can use Equation (2.2) to form the difference quotient

$$\frac{v(t + h) - v(t)}{h} = \frac{[144 - 32(t + h)] - [144 - 32t]}{h} = \frac{-32h}{h} = -32.$$

Since this quotient does not change as h approaches 0 we can regard it as *approaching* -32 (since it is near -32 when h is near 0). We conclude that the acceleration in this problem is constant and equal to -32 . This result tells us that the velocity is decreasing at the rate of 32 feet per second every second. In 9 seconds the total decrease in velocity is $9 \cdot 32 = 288$ feet per second. This agrees with the fact that during the 9 seconds of motion the velocity changes from $v(0) = 144$ to $v(9) = -144$.

2.4 Exercises

1. Assume that the height $f(t)$ of a projectile, t seconds after being fired directly upward from the ground with an initial velocity of v_0 feet per second, is given by the formula

$$f(t) = v_0 t - 16t^2.$$

(a) Use the method described in Section 2.2 to show that the average velocity of the projectile during a time interval from t to $t + h$ is $v_0 - 32t - 16h$ feet per second, and that the instantaneous velocity at time t is $v_0 - 32t$ feet per second.

(b) Compute (in terms of v_0) the time required for the velocity to drop to zero.

(c) What is the velocity on return to earth?

(d) What must the initial velocity be for the projectile to return to earth after 1 second? after 10 seconds? after T seconds?

(e) Show that the projectile moves with constant acceleration.

(f) Give an example of another formula for $f(t)$ which will lead to a constant acceleration of -20 feet per second per second.

2. Verify the entries in the following table of difference quotients and fill in the blank spaces. In each case, make a guess for the derivative $f'(x)$.

	$f(x)$	$\frac{f(x+h) - f(x)}{h}$	$f'(x)$
(a)	x	1	
(b)	x^2	$2x + h$	
(c)	x^3	$3x^2 + 3xh + h^2$	
(d)	x^4		
(e)	x^5		
(f)	$x^{-1} \quad (x \neq 0)$	$-\frac{1}{x(x+h)}$	
(g)	$x^{-2} \quad (x \neq 0)$	$-\frac{2}{x(x+h)^2} - \frac{h}{x^2(x+h)^2}$	
(h)	$x^{-3} \quad (x \neq 0)$		
(i)	$\sqrt{x} \quad (x > 0)$	$\frac{1}{\sqrt{x+h} + \sqrt{x}}$	
(j)	$\frac{1}{\sqrt{x}} \quad (x > 0)$	$\frac{-1}{x\sqrt{x+h} + (x+h)\sqrt{x}}$	
(k)	$x^{3/2} \quad (x \geq 0)$		
(l)	$x^{-3/2} \quad (x > 0)$		

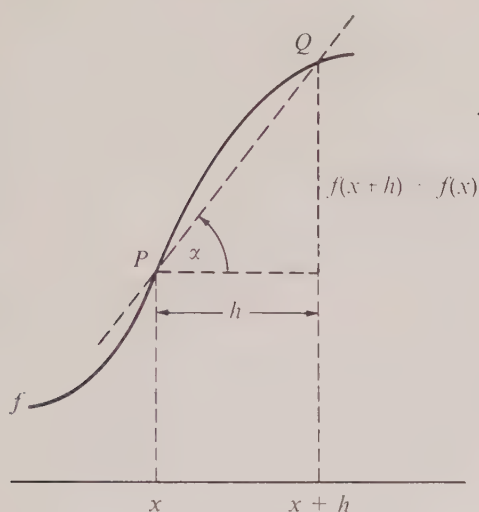


FIGURE 2.3 Geometric interpretation of the difference quotient as the tangent of an angle.

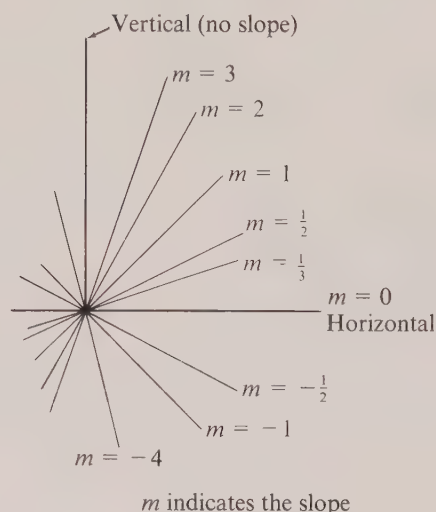


FIGURE 2.4 Lines of various slopes.

2.5 Geometric interpretation of the derivative as a slope

The procedure used to define the derivative has an interesting geometric interpretation which leads in a natural way to the idea of a tangent line to a curve. A portion of the graph of a function f is shown in Figure 2.3. Two of its points P and Q are shown with respective coordinates $(x, f(x))$ and $(x + h, f(x + h))$. Consider the right triangle with hypotenuse PQ ; its altitude, $f(x + h) - f(x)$, represents the difference of the ordinates of the two points Q and P . Therefore the difference quotient

$$\frac{f(x + h) - f(x)}{h}$$

represents the trigonometric tangent of the angle α that PQ makes with the horizontal. The real number $\tan \alpha$ is called the *slope* of the line through P and Q and it provides a way of measuring the “steepness” of this line. Some examples of lines of various slopes are shown in Figure 2.4. For a horizontal line, $\alpha = 0$ and the slope, $\tan \alpha$, is also 0. If α lies between 0 and $\frac{1}{2}\pi$, the line is rising as we move from left to right and the slope is represented by a positive number. If α lies between $\frac{1}{2}\pi$ and π , the line is falling as we move from left to right and the slope is negative. A line for which $\alpha = \frac{1}{4}\pi$ has slope 1. As α increases from 0 to $\frac{1}{2}\pi$, $\tan \alpha$ increases without bound and the corresponding lines of slope $\tan \alpha$ approach a vertical position. Since $\tan \frac{1}{2}\pi$ is not defined we say that *vertical lines have no slope*.

Suppose now that f has a derivative at x . This means that the difference quotient approaches a certain limit $f'(x)$ as h approaches 0. When this is interpreted geometrically it tells us that, as h gets nearer to 0, the point P remains fixed, Q moves along the curve

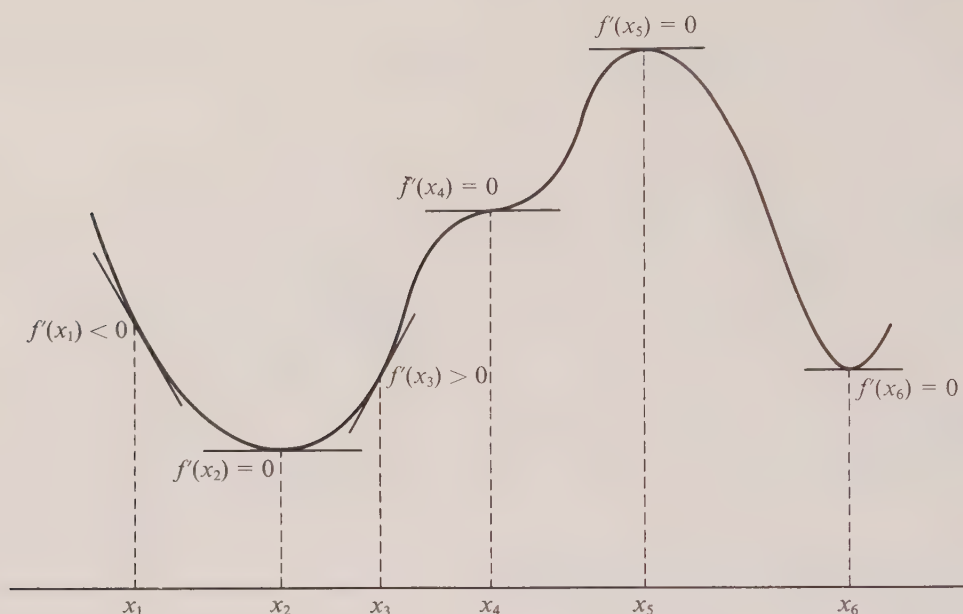


FIGURE 2.5 Geometric significance of the sign of the derivative.

toward P , and the line through PQ keeps changing its direction in such a way that the tangent of the angle α approaches the number $f'(x)$ as a limit. For this reason it seems natural to define the *slope of the curve* at P to be the number $f'(x)$. The line through P having this slope is called the *tangent line* at P .

Note. The concept of a line tangent to a circle (and to a few other special curves) was considered by the ancient Greeks. They defined a tangent line to a circle as a line having one of its points on the circle and all its other points outside the circle. From this definition, many properties of tangent lines to circles can be derived. For example, we can prove that the tangent at any point is perpendicular to the radius at that point. However, the Greek definition of tangent line is not easily extended to more general curves. The method described above, where the tangent line is defined in terms of a derivative, has proved to be far more satisfactory. Using this definition, we can prove that for a circle the tangent line has all the properties ascribed to it by the Greek geometers. Concepts such as perpendicularity and parallelism can be explained rather simply in analytic terms making use of slopes of lines. For example, from the trigonometric identity

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

it follows that two nonvertical lines with the same slope are parallel. Also, from the identity

$$\cot(\alpha - \beta) = \frac{1 + \tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$$

we find that two nonvertical lines with slopes having product -1 are perpendicular. These ideas will be developed systematically when we study analytic geometry in more detail in Chapter 5.

The algebraic sign of the derivative of a function gives us useful information about the behavior of its graph. For example, if x is a point in an open interval where the derivative is *positive*, then the graph is rising in the immediate vicinity of x as we move from left to right. This occurs above x_3 in Figure 2.5. A *negative* derivative in an interval means the graph is falling, as shown above x_1 , while a zero derivative at a point means a horizontal tangent line. At a maximum or minimum, such as those shown above x_2 , x_5 , and x_6 , the slope must be zero. Fermat was the first to notice that points like x_2 , x_5 , and x_6 , where f has a maximum or minimum, must occur among the roots of the equation $f'(x) = 0$. It is important to realize that $f'(x)$ may also be zero at points where there is no maximum or minimum, such as above the point x_4 . Note that this particular tangent line crosses the graph. This is an example of a situation not covered by the Greek definition of tangency.

The foregoing remarks concerning the significance of the algebraic sign of the derivative may seem quite obvious when we interpret them geometrically. Analytic proofs of these statements, based on general properties of derivatives, will be given in Chapter 8 (see also Example 3 of Section 2.15).

2.6 Some elementary formulas for derivatives

Example 1: Derivative of a constant function. Suppose f is a constant function, say $f(x) = c$ for all x . The graph of f is a horizontal line and hence has slope zero everywhere. Therefore the geometry suggests that $f'(x) = 0$ for all x . To prove this analytically we appeal directly to the definition of $f'(x)$. The difference quotient is

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h}.$$

Since this is equal to 0 when $h \neq 0$ we conclude that $f'(x) = 0$ for every x . In other words, a constant function has a zero derivative.

Example 2: Derivative of a linear function. Let f be a function whose graph is a straight line of slope m , and let us write b for $f(0)$. The number b is called the *y-intercept* of the line because it represents the y -coordinate of the point where the line crosses the y -axis. (See Figure 2.6.) If (x, y) is any point in the plane with $x \neq 0$, the slope of the line joining $(0, b)$ to (x, y) is $(y - b)/x$. Therefore (x, y) lies on the graph of f if and only if $(y - b)/x = m$, or, stated differently, if and only if

$$(2.5) \quad y = mx + b.$$

But if (x, y) is on the graph of f then we also have $y = f(x)$ and hence (2.5) becomes

$$f(x) = mx + b.$$

A function defined by a formula of this kind is called a *linear function* because its graph is a straight line. We shall prove now that the derivative of a linear function is a constant, this constant being equal to the slope of the line. If $h \neq 0$, we have

$$\frac{f(x+h) - f(x)}{h} = \frac{m(x+h) + b - (mx + b)}{h} = \frac{mh}{h} = m.$$

Since the difference quotient does not change when h approaches 0 we conclude that

$$f'(x) = m \quad \text{for every } x.$$

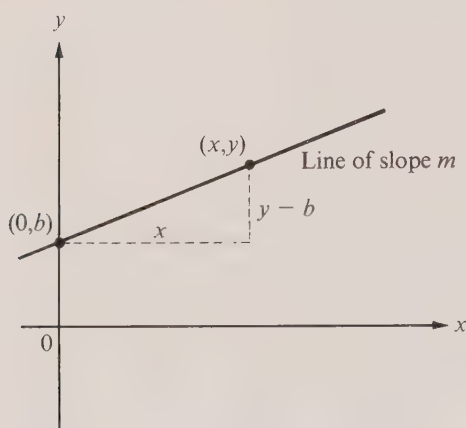


FIGURE 2.6 Graph of a linear function:
 $f(x) = mx + b$.

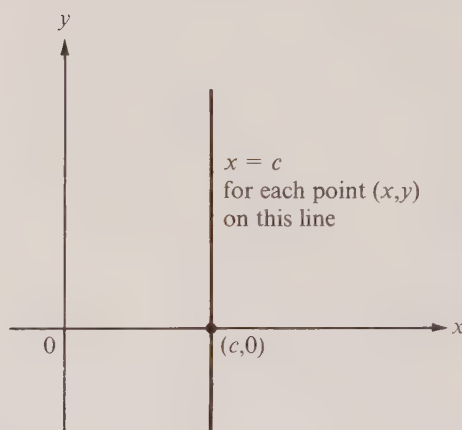


FIGURE 2.7 A vertical line with equation $x = c$.

We shall study straight lines in greater detail in Chapter 5 as part of the development of analytic geometry. At this point we wish to mention briefly (without proofs) some simple facts about straight lines with which the reader may or may not be familiar.

In the foregoing example we found that the coordinates of every point (x, y) on a line of slope m satisfy an equation of the form (2.5), and that no other points in the plane satisfy this relation. Equation (2.5) is called *an equation of the line* and it is also referred to as a linear equation. Every line which is not vertical has an equation of this form. If a line is vertical, all its points (x, y) have the same abscissa. If the common abscissa is c , the equation

$$(2.6) \quad x = c$$

is said to be an equation of the line. (See Figure 2.7.) All lines (including vertical lines) can be described by a more general equation of the form

$$(2.7) \quad Ax + By + C = 0,$$

where A , B , and C are constants determined by the inclination and position of the line. Conversely, every such equation in which A and B are not both zero describes a straight line. When $B \neq 0$ we can solve Equation (2.7) for y and transform it to the form (2.5). For example, the equation $2x - 3y + 6 = 0$ becomes $y = \frac{2}{3}x + 2$. The case $B = 0$ and $A \neq 0$ corresponds to a vertical line since we may solve for x and transform (2.7) to the form (2.6).

If we keep b fixed in Equation (2.5) and let m run through all real numbers, we obtain the collection of all possible nonvertical lines passing through the point $(0, b)$, as suggested by Figure 2.8. If we keep m fixed and let b run through all real numbers, we obtain the collection of all lines in the plane having the slope m . These lines are parallel to each other, as suggested by Figure 2.9.

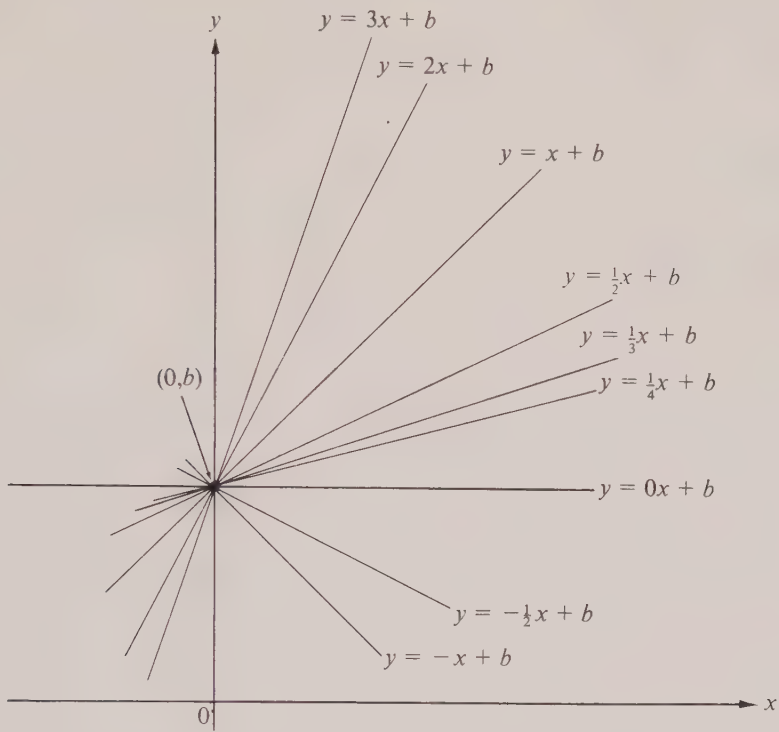
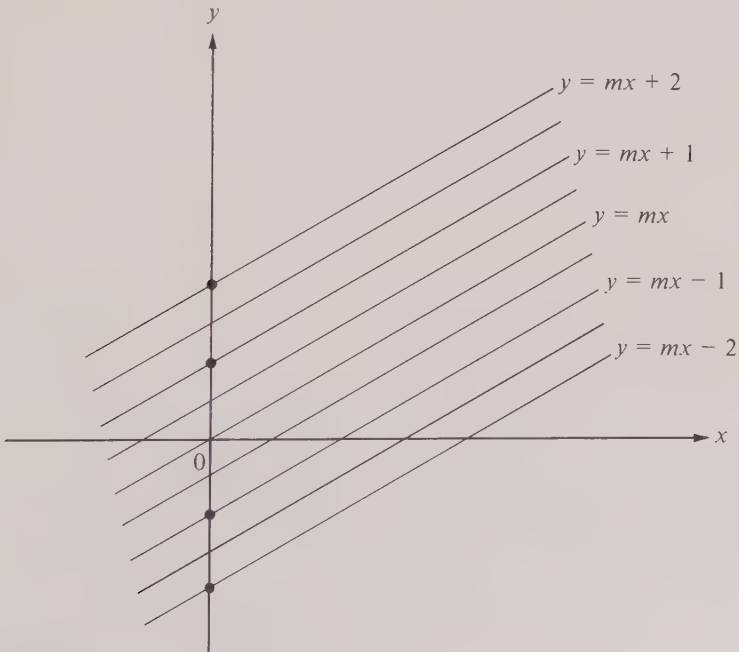
FIGURE 2.8 Lines of various slopes passing through $(0, b)$.

FIGURE 2.9 Lines having the same slope.

Example 3. Consider next the case $f(x) = x^n$, where n is a positive integer. The difference quotient becomes

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^n - x^n}{h}.$$

To study this quotient as h approaches 0 we can proceed in two ways, either by factoring the numerator as a difference of two n th powers or by using the binomial theorem to expand $(x+h)^n$. We shall carry out the details by the first method and leave the other method as an exercise for the reader. (See Exercise 12 in Section 2.9.)

From elementary algebra we have the identity†

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^k b^{n-1-k}.$$

If we take $a = x + h$ and $b = x$ and divide both sides by h , this identity becomes

$$(2.8) \quad \frac{(x+h)^n - x^n}{h} = \sum_{k=0}^{n-1} (x+h)^k x^{n-1-k}.$$

There are n terms in the sum. As h approaches 0, $(x+h)^k$ approaches x^k , the k th term approaches $x^k x^{n-1-k} = x^{n-1}$, and therefore the sum of all n terms approaches nx^{n-1} . From this it follows that we have

$$(2.9) \quad f'(x) = nx^{n-1} \quad \text{for every } x.$$

Differentiation, by its very definition, is a *limit process*. To develop various properties of derivatives we must constantly make use of a few basic rules for operating with limits. For example, in dealing with the sum appearing in (2.8) we computed the limit of each term and then added the results to find the limit of the sum. In so doing, we used the fact that the limit of a sum is the sum of the limits. Another basic rule states that the limit of a product is the product of the limits. This may be used to prove two further statements that were made in Example 3, namely, that $(x+h)^k$ approaches x^k and that $(x+h)^k x^{n-1-k}$ approaches x^{n-1} as h approaches 0. A third basic rule is that the limit of a quotient is the quotient of the limits. These “rules” are, in reality, *theorems* which can and must be proved from the axioms of the real-number system if calculus is to take its place as a *bona fide* deductive system.

The foregoing examples show that some familiarity with limit theorems is needed before derivative theory can be further developed in a logical fashion. Therefore we interrupt the development of differential calculus at this stage to introduce in the next two sections an exact definition of the limit concept and the basic theorems on limits. To avoid further delay in developing the differential calculus, the proofs of these theorems are postponed until Section 2.29. The proofs are not difficult, but our principal aim at present is to give the reader a feeling for what calculus is all about and how it can be used rather than to carry out detailed proofs of all its theorems. At the same time we want the reader to

† This identity is an immediate consequence of the telescoping property of finite sums. In fact, if we multiply each term of the sum by $(a-b)$ we find

$$(a-b) \sum_{k=0}^{n-1} a^k b^{n-1-k} = \sum_{k=0}^{n-1} (a^{k+1} b^{n-(k+1)} - a^k b^{n-k}) = a^n - b^n.$$

understand clearly the *over-all logical structure* of calculus. Thus, for example, he should realize that the techniques of differentiation which are being developed in this chapter consist of a chain of theorems that can be deduced from the properties of limits given in Section 2.8. A few properties of continuous functions are also needed to develop certain portions of derivative theory and it is convenient to present these now, together with the basic facts about limits.

2.7 The definitions of limit and continuity

In some of the earlier sections we have made informal remarks from time to time about limits and continuity. Now we are ready to formulate precise definitions of these concepts.

In the early development of calculus almost all functions that were dealt with were continuous and there was no real need at that time for a penetrating look into the exact meaning of continuity. It was not until late in the 18th century that discontinuous functions began appearing in connection with various kinds of physical problems. In particular, the work of J. B. J. Fourier (1758–1830) on the theory of heat forced mathematicians of the early 19th century to examine more carefully the exact meaning of such concepts as *function* and *continuity*. Although the meaning of the word “continuous” seems intuitively clear to most people, it is not obvious how a good definition of this idea should be formulated. One popular dictionary explains continuity as follows:

Continuity: Quality or state of being continuous.

Continuous: Having continuity of parts.

Trying to learn the meaning of continuity from these two statements alone is like trying to learn Chinese with only a Chinese dictionary. A satisfactory mathematical definition of continuity, expressed entirely in terms of properties of the real-number system, was first formulated in 1821 by the French mathematician, Augustin-Louis Cauchy (1789–1857). His definition, which is still used today, is most easily explained in terms of the limit concept to which we turn now.

The derivative has been described as the limit of a difference quotient. This is just a special case of a more general limit process which is indicated by the notation

$$(2.10) \quad \lim_{x \rightarrow x_0} f(x) = A$$

or by some similar notation. The function f is assumed to be defined in some interval containing the point x_0 although we do not insist that f be defined at the point x_0 itself. An equation like (2.10) is read: “The limit of $f(x)$, as x approaches x_0 , is equal to A ,” or “ $f(x)$ approaches A as x approaches x_0 .” It is also written without the limit symbol as follows:

$$(2.11) \quad f(x) \rightarrow A \quad \text{as } x \rightarrow x_0.$$

This symbolism is intended to convey the idea that we can make $f(x)$ as close to A as we please, provided we choose x sufficiently close to x_0 . To explain precisely what is meant by this, we introduce the absolute value $|f(x) - A|$, a number which measures the distance separating $f(x)$ from A , and also the absolute value $|x - x_0|$, which measures the distance separating x from x_0 . Roughly speaking, the meaning of the symbols in (2.10) and (2.11) is that we can make $|f(x) - A|$ *arbitrarily* small whenever $|x - x_0|$ is *sufficiently* small

(but not zero). Cauchy realized that he could express this idea solely in terms of real numbers. It is done by means of the following definition:

DEFINITION. The symbolism in (2.10) or in (2.11) means that for every number $\epsilon > 0$ there is another number $\delta > 0$ such that

$$(2.12) \quad |f(x) - A| < \epsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta.$$

The first thing to notice about this definition is that it involves two quantities ϵ and δ which do not appear in the limit symbols written in (2.10) or (2.11). The number ϵ is specified *first*; it tells us how small we are required to make $|f(x) - A|$. The whole essence of the definition is that, for *every* positive ϵ , *no matter how small*, some positive δ can be found to satisfy (2.12). The number δ tells us how close x should be to x_0 to guarantee that $f(x)$ will be within the given distance ϵ from A . In general, δ will depend on the choice of ϵ . A δ that works for one particular value of ϵ will also work for every larger ϵ although it might not be suitable for any smaller ϵ .

The next thing to notice is that the definition makes no assertion whatever about the behavior of f at the point x_0 itself. The inequality $0 < |x - x_0| < \delta$ refers only to points "near" x_0 but distinct from x_0 . It is not even necessary for the purposes of this definition that the function be *defined* at x_0 . Moreover, even if f is defined at x_0 , its value there need not be equal to the limit A . However, if it happens that f is defined at x_0 and if it also happens that $f(x_0) = A$, then we say the function f is *continuous* at x_0 . In other words, we have the following definition:

DEFINITION. A function f is said to be continuous at x_0 if

$$(a) \quad f \text{ is defined at } x_0,$$

and

$$(b) \quad \lim_{x \rightarrow x_0} f(x) = f(x_0).$$

In terms of ϵ and δ , this definition can be restated as follows:

A function f is continuous at x_0 if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$(2.13) \quad |f(x) - f(x_0)| < \epsilon \quad \text{whenever} \quad |x - x_0| < \delta.$$

The definitions of limit and continuity may be illustrated geometrically as in Figures 2.10 and 2.11. In Figure 2.10 the two horizontal lines with ordinates $A + \epsilon$ and $A - \epsilon$ determine a horizontal strip of height 2ϵ about the ordinate $y = A$. The number δ that corresponds to ϵ determines a vertical strip of width 2δ centered around the line $x = x_0$. These two strips intersect in a rectangle (shaded in the figure), and the definition asserts that the entire graph of f above the interval $(x_0 - \delta, x_0 + \delta)$ lies within this rectangle, except possibly for the point on the graph above x_0 itself. Figure 2.11 shows what happens in the case of a continuous function where the limiting value A is the same as $f(x_0)$.

Using the foregoing definition, we can easily prove that the identity function is continuous at every point. This is the function f defined for all x by the equation $f(x) = x$. To satisfy (2.13) we must show that for every given $\epsilon > 0$ there is a $\delta > 0$ such that

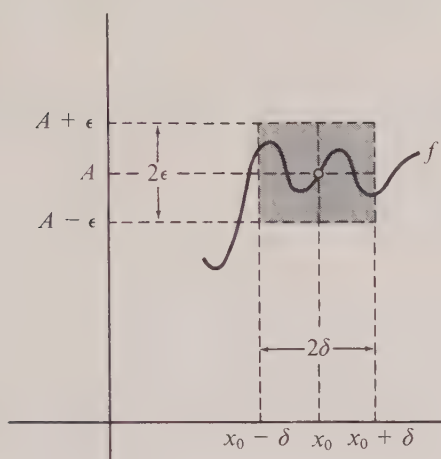


FIGURE 2.10 Here $\lim_{x \rightarrow x_0} f(x) = A$, but there is no assertion about f at x_0 .

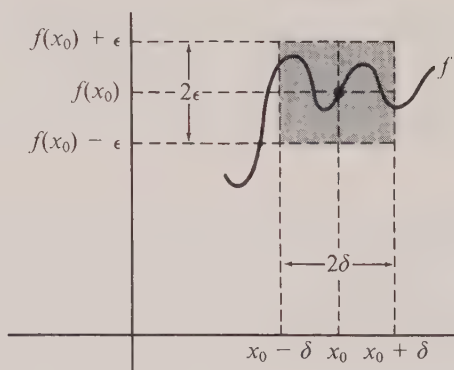


FIGURE 2.11 Here f is defined at x_0 and $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, so f is continuous at x_0 .

$|x - x_0| < \epsilon$ whenever $|x - x_0| < \delta$. The choice $\delta = \epsilon$ clearly suffices, so f is continuous at every point x_0 .

It is also easy to show that all constant functions are continuous everywhere. In fact, (2.13) is trivially satisfied for *any* choice of δ because $f(x) - f(x_0) = 0$ for all x when f is constant. Further examples of continuous functions will be constructed from these two simple examples by using the limit theorems to be described in the next section.

In dealing with limits as $x \rightarrow x_0$ we sometimes find it convenient to denote the difference $x - x_0$ by a new symbol, say h , and to let $h \rightarrow 0$. This simply amounts to a change in notation because, as may be easily verified, the following two equations are equivalent:

$$\lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{h \rightarrow 0} f(x_0 + h) = A.$$

In this notation, continuity at x_0 means that f is defined at x_0 and $\lim_{h \rightarrow 0} f(x_0 + h) = f(x_0)$. We also note that the two statements

$$\lim_{x \rightarrow x_0} f(x) = A \quad \text{and} \quad \lim_{x \rightarrow x_0} [f(x) - A] = 0$$

are equivalent. This follows at once from the definition of limit.

When we apply the idea of limit to define the derivative, we write

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}.$$

Here we think of x as fixed, and study the difference quotient as a function of h for small h . That is, we consider a function Q defined by the equation

$$Q(h) = \frac{f(x+h) - f(x)}{h}$$

for all $h \neq 0$ such that $x+h$ is in the domain of f . If $Q(h)$ has a limit as $h \rightarrow 0$, then this limit is called the derivative of f at x and it is denoted by $f'(x)$. The equation defining $f'(x)$ may also be written in the notation

$$f'(x) = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}$$

or in some similar notation.

2.8 The basic limit theorems. Examples of continuous functions

The basic rules for operating with limits are stated in the following theorem, whose proof is given in Section 2.29.

2- 1 THEOREM. Let f and g be functions such that

$$\lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{x \rightarrow x_0} g(x) = B.$$

Then we have

$$(i) \lim_{x \rightarrow x_0} [f(x) + g(x)] = A + B,$$

$$(ii) \lim_{x \rightarrow x_0} [f(x) - g(x)] = A - B,$$

$$(iii) \lim_{x \rightarrow x_0} [f(x) \cdot g(x)] = A \cdot B,$$

$$(iv) \lim_{x \rightarrow x_0} [f(x) / g(x)] = A / B \quad \text{if } B \neq 0.$$

Note. An important special case of (iii) occurs when f is constant, say $f(x) = A$ for all x . In this case, (iii) is written as $\lim_{x \rightarrow x_0} A \cdot g(x) = A \cdot B$.

In the theorem, statement (i) tells us that the limit of a sum is the sum of the limits, which can also be written as follows:

$$\lim_{x \rightarrow x_0} [f(x) + g(x)] = \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x).$$

Of course, similar remarks apply to the other three statements.

It is customary to denote by $f+g$, $f-g$, $f \cdot g$, and f/g the functions whose values at each x under consideration are

$$f(x) + g(x), \quad f(x) - g(x), \quad f(x) \cdot g(x), \quad \text{and} \quad f(x)/g(x),$$

respectively. These functions are called the *sum*, *difference*, *product*, and *quotient* of f and g . Of course, the quotient f/g is defined only at those points for which $g(x) \neq 0$. The

following corollary to Theorem 2-1 is stated in this terminology and notation and is concerned with continuous functions.

2- 2 THEOREM. Let f and g be continuous at a point x_0 . Then the sum $f + g$, the difference $f - g$, and the product $f \cdot g$ are also continuous at x_0 . The same is true of the quotient f/g if $g(x_0) \neq 0$.

Proof. Since f and g are continuous at x_0 we have $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ and $\lim_{x \rightarrow x_0} g(x) = g(x_0)$. Therefore we may apply the limit formulas in Theorem 2-1 with $A = f(x_0)$ and $B = g(x_0)$ to deduce Theorem 2-2.

We have already seen that the identity function and constant functions are continuous everywhere. Using these examples and Theorem 2-2, we may construct many more examples of continuous functions.

Example 1: Continuity of polynomials. If we take $f(x) = g(x) = x$, the result on continuity of products proves the continuity at each point for the function whose value at each x is x^2 . By mathematical induction, it follows that for every real c and every positive integer n the function f for which $f(x) = cx^n$ is continuous for all x . Since the sum of two continuous functions is itself continuous, by induction it follows that the same is true for the sum of any finite number of continuous functions. Therefore every polynomial is continuous at all points.

Example 2: Continuity of rational functions. The quotient of two polynomials is called a *rational function*. If r is a rational function, then we have

$$r(x) = \frac{p(x)}{q(x)},$$

where p and q are polynomials. The function r is defined for all real x for which $q(x) \neq 0$. Since quotients of continuous functions are continuous, we see that every rational function is continuous wherever it is defined. A simple example is $r(x) = 1/x$ if $x \neq 0$. This function is continuous everywhere except at $x = 0$, where it fails to be defined.

Example 3: Continuity of the sine and cosine. In Exercise 24 of Section 1.39 we outlined a proof of the inequality

$$(2.14) \quad |\sin x| < |x|,$$

valid for $0 < |x| < \frac{1}{2}\pi$. From this inequality it is easy to deduce that the sine and cosine are continuous everywhere. First we prove that the sine is continuous at 0. Since $\sin 0 = 0$, we must prove that $\lim_{x \rightarrow 0} \sin x = 0$. Returning to the definition of limit, we must show that for every $\epsilon > 0$ there is a $\delta > 0$ such that $|\sin x| < \epsilon$ whenever $|x| < \delta$. In view of (2.14) it suffices to choose $\delta = \epsilon$. This proves that the sine function is continuous at 0. In the same way we may show that $\lim_{x \rightarrow 0} \sin cx = 0$ for each real c , and hence the function f for which $f(x) = \sin cx$ is also continuous at 0.

Next we prove that the cosine is continuous at 0. For this purpose, we use the identity $\cos x = 1 - 2 \sin^2 \frac{1}{2}x$ to obtain

$$\lim_{x \rightarrow 0} \cos x = \lim_{x \rightarrow 0} \left(1 - 2 \sin^2 \frac{x}{2} \right) = 1 - 2 \left(\lim_{x \rightarrow 0} \sin \frac{x}{2} \right)^2 = 1.$$

Since $\cos 0 = 1$ this proves that the cosine is continuous at 0.

Continuity of the sine and cosine at 0 implies continuity at an arbitrary point x . To prove this we use the addition formulas for $\sin(x + h)$ and $\cos(x + h)$. For the sine, we have

$$\sin(x + h) = \sin x \cos h + \cos x \sin h.$$

We have just shown that $\cos h \rightarrow 1$ and $\sin h \rightarrow 0$ as $h \rightarrow 0$ so

$$\lim_{h \rightarrow 0} \sin(x+h) = \sin x \cdot 1 + \cos x \cdot 0 = \sin x.$$

This proves that the sine is continuous at every real x and the proof for the cosine is similar.

Example 4: Continuity of functions having derivatives. If a function f has a derivative at a point x , then it is also continuous at x . To prove this we use the identity

$$f(x+h) = f(x) + h \left(\frac{f(x+h) - f(x)}{h} \right)$$

which is valid for $h \neq 0$. If we let $h \rightarrow 0$, the difference quotient on the right approaches $f'(x)$ and, since this quotient is multiplied by a factor which tends to 0, the second term on the right approaches $0 \cdot f'(x) = 0$. This shows that $f(x+h) \rightarrow f(x)$ as $h \rightarrow 0$, and hence that f is continuous at x .

This example provides a new way of showing that functions are continuous. Every time we establish the existence of a derivative $f'(x)$ we also establish, at the same time, the continuity of f at x . It should be noted, however, that the converse is not true. Continuity at x does not necessarily mean that the derivative $f'(x)$ exists. For example, when $f(x) = |x|$ the point $x = 0$ is a point of continuity of f [since $f(x) \rightarrow 0$ as $x \rightarrow 0$] but there is no derivative at 0. (See Figure 2.12.) The difference quotient $[f(0+h) - f(0)]/h$ is equal to $|h|/h$. This has the value $+1$ if $h > 0$ and -1 if $h < 0$ and hence does not tend to a limit as $h \rightarrow 0$.

2.9 Exercises

In Exercises 1 through 10 find the limits and explain which limit theorems you are using in each case.

1. $\lim_{x \rightarrow 2} \frac{1}{x^2}.$

2. $\lim_{x \rightarrow 0} \frac{25x^3 + 2}{75x^7 - 2}.$

3. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}.$ [Hint. Reduce to lowest terms.]

4. $\lim_{x \rightarrow 1} \frac{2x^2 - 3x + 1}{x - 1}.$

5. $\lim_{h \rightarrow 0} \frac{(t+h)^2 - t^2}{h}.$

6. $\lim_{x \rightarrow 0} \frac{x^2 - a^2}{x^2 + 2ax + a^2}, \quad a \neq 0.$

7. $\lim_{a \rightarrow 0} \frac{x^2 - a^2}{x^2 + 2ax + a^2}, \quad x \neq 0.$

8. $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^2 + 2ax + a^2}, \quad a \neq 0.$

9. $\lim_{t \rightarrow 0} \tan t.$

10. $\lim_{t \rightarrow 0} (\sin 2t + t^2 \cos 5t).$

11. Prove that the tangent and cotangent functions are continuous at all points where they are defined.

12. Let $f(x) = x^n$, where n is a positive integer. Use the binomial theorem to expand $(x+h)^n$ and derive the formula

$$\frac{f(x+h) - f(x)}{h} = nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1}.$$

Express the sum on the right in summation notation. Let $h \rightarrow 0$ and deduce that $f'(x) = nx^{n-1}$. State which limit theorems you are using. (This result was derived in another way in Example 3 of Section 2.6.)

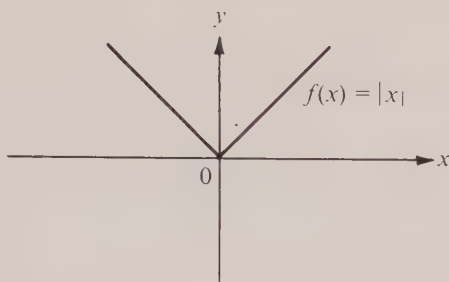


FIGURE 2.12 The function is continuous at 0 but $f'(0)$ does not exist.

13. If $x > 0$, let $f(x) = x^r$, where r is a positive rational number, say $r = m/n$ (where m and n are positive integers). Let

$$u = (x + h)^{1/n}, \quad v = x^{1/n}.$$

(a) If $m > 1$ and $n > 1$, show that

$$\frac{f(x+h) - f(x)}{h} = \frac{u^m - v^m}{u^n - v^n} = \frac{u^{m-1} + u^{m-2}v + \dots + uv^{m-2} + v^{m-1}}{u^{n-1} + u^{n-2}v + \dots + uv^{n-2} + v^{n-1}},$$

and obtain corresponding formulas for the difference quotient when $m = 1$ or $n = 1$.

(b) Prove that when $h \rightarrow 0$ the difference quotient in (a) approaches $(m/n)v^{m-n}$ and deduce from this that $f'(x) = rx^{r-1}$. State which limit theorems are used in this derivation. This exercise shows that the formula $f'(x) = nx^{n-1}$ for differentiating $f(x) = x^n$ when n is a positive integer is also valid for positive rational exponents. Eventually we shall extend the formula to all real exponents. Because of Example 4 in Section 2.8, this exercise also establishes the continuity of f for each $x > 0$.

Figure 2.13 shows the graph of the function f defined as follows:

$$f(x) = \frac{\sin x}{x} \quad \text{if } x \neq 0, \quad f(0) = 2.$$

The graph suggests the limit formula

$$(2.15) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

This is an important relation that will be proved later in Section 2.11. Notice that the denominator of the quotient $(\sin x)/x$ approaches 0 as $x \rightarrow 0$ and we cannot apply the quotient theorem on limits to deduce (2.15). The proof in Section 2.11 will be based on a method described below in Exercise 22. Since $f(0) = 2$, the function values $f(x)$ do not approach $f(0)$ as $x \rightarrow 0$, so this function has a discontinuity at 0. However, in this case the discontinuity can be “removed” by redefining the function at 0 to have the value 1 instead of 2. For this reason, a discontinuity of this type is called a *removable discontinuity*. Notice that jump discontinuities, such as those possessed by step functions, cannot be removed by merely changing the value of the function at one point.

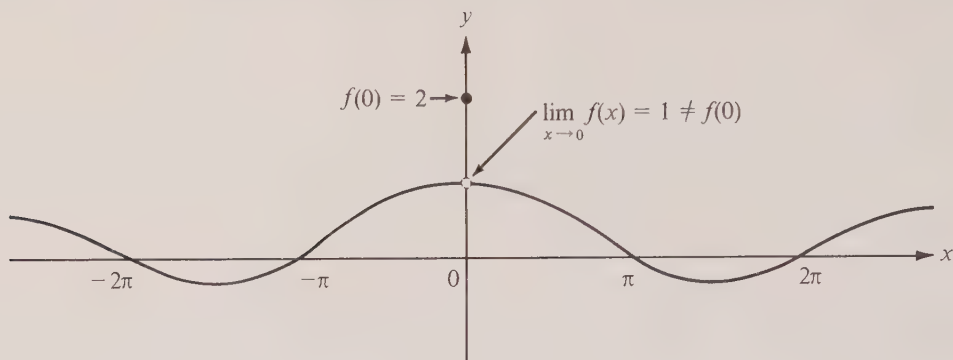


FIGURE 2.13 $f(x) = (\sin x)/x$ if $x \neq 0$, $f(0) = 2$. This function has a removable discontinuity at 0.

14. Let $f(x) = (\tan x)/x$ if $x \neq 0$. Sketch the graph of f over the half-open intervals $[-\frac{1}{4}\pi, 0)$ and $(0, \frac{1}{4}\pi]$. What happens to $f(x)$ as $x \rightarrow 0$? Can you define $f(0)$ so that f becomes continuous at 0? [Hint. Use (2.15) together with the formula $\tan x = (\sin x)/(\cos x)$.]

Use the limit formula (2.15) to establish the limit formulas in Exercises 15 through 20.

$$15. \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2. \left[\text{Hint. } \frac{\sin 2x}{x} = 2 \frac{\sin 2x}{2x} \right]$$

$$16. \lim_{x \rightarrow 0} \frac{\tan 2x}{\sin x} = 2.$$

$$17. \lim_{x \rightarrow 0} \frac{\sin 5x}{\sin x} = 5.$$

$$18. \lim_{x \rightarrow 0} \frac{\sin 5x - \sin 3x}{x} = 2.$$

$$19. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}. \left[\text{Hint. } 1 - \cos x = 2 \sin^2 \frac{x}{2} \right]$$

$$20. \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \cos a. \text{ [Equation (2.15) is a special case.]}$$

21. In the foregoing section we used the inequality $|\sin x| < |x|$ to prove that $\sin x \rightarrow 0$ as $x \rightarrow 0$. Use a similar argument to prove the following more general theorem: Suppose that $0 \leq f(x) \leq g(x)$ for all x in some interval with the possible exception of a single point x_0 . If $\lim_{x \rightarrow x_0} g(x) = 0$, then we also have $\lim_{x \rightarrow x_0} f(x) = 0$.

22. Prove the following theorem (a generalization of Exercise 21): Suppose F , G , and H are three functions satisfying the inequalities $F(x) \leq G(x) \leq H(x)$ for all x in some interval with the possible exception of a single point x_0 . Suppose also that

$$\lim_{x \rightarrow x_0} F(x) = \lim_{x \rightarrow x_0} H(x) = A.$$

Then we also have $\lim_{x \rightarrow x_0} G(x) = A$. In other words, if a function G is squeezed between two other functions which have equal limits as $x \rightarrow x_0$, then G also has this limit as $x \rightarrow x_0$. This property of limits is quite useful in practice because it is often possible to find approximating functions F and H that are easier to deal with than G . In particular, we shall use this result in Section 2.11 to prove the limit formula in Equation (2.15). [Hint. Show that for each x we have either $0 \leq |G(x) - A| \leq |H(x) - A|$ or $0 \leq |G(x) - A| \leq |F(x) - A|$.]

23. Figure 2.14 shows a portion of the graph of the function f defined as follows:

$$f(x) = \sin \frac{1}{x} \quad \text{if } x \neq 0.$$

For $x = 1/(n\pi)$, where n is an integer, we have $\sin(1/x) = \sin(n\pi) = 0$. Between two such points the function values rise to $+1$ and drop back to 0 or else drop to -1 and rise back to 0 . Therefore, between any such point and the origin, the curve has an infinite number of oscillations. This suggests that the function values do not approach any fixed value as $x \rightarrow 0$. Prove (analytically) that

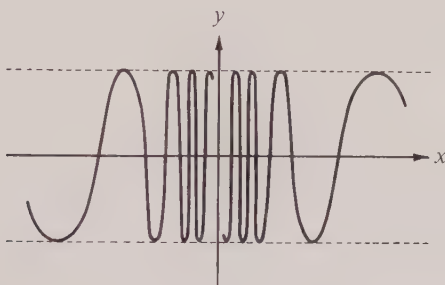


FIGURE 2.14 $f(x) = \sin(1/x)$ if $x \neq 0$. This function is discontinuous at 0 , no matter how $f(0)$ is defined.

there is no real number A such that $f(x) \rightarrow A$ as $x \rightarrow 0$. This shows that it is not possible to define $f(0)$ in such a way that f becomes continuous at 0 . [Hint. Assume such an A exists and obtain a contradiction.]

24. For $x \neq 0$, let $f(x) = [1/x]$, where $[t]$ denotes the greatest integer $\leq t$. Sketch the graph of f over the intervals $[-2, -\frac{1}{2}]$ and $[\frac{1}{2}, 2]$. What happens to $f(x)$ as $x \rightarrow 0$ through positive values? through negative values? Can you define $f(0)$ so that f becomes continuous at 0 ?

25. Same as Exercise 24, when $f(x) = (-1)^{[1/x]}$ for $x \neq 0$.

26. Same as Exercise 24, when $f(x) = x(-1)^{[1/x]}$ for $x \neq 0$.

2.10 The algebra of derivatives

We return now to the further development of differential calculus. Just as the limit theorems of Section 2.8 tell us how to compute limits of the sum, difference, product, and quotient of two functions, so the next theorem provides us with a corresponding set of rules for computing derivatives.

2- 3 THEOREM. Let f and g be two functions defined on a common interval. At each point where f and g have a derivative the same is true of the sum $f + g$, the difference $f - g$, the product $f \cdot g$, and the quotient f/g . (For f/g we need the extra proviso that g is not zero at the point in question.) The derivatives of these functions are given by the following formulas:

- (i) $(f + g)' = f' + g'$,
 (ii) $(f - g)' = f' - g'$,
 (iii) $(f \cdot g)' = f \cdot g' + g \cdot f'$,
 (iv) $\left(\frac{f}{g}\right)' = \frac{g \cdot f' - f \cdot g'}{g^2}$ at points x where $g(x) \neq 0$.

We shall prove this theorem in a moment but first we want to mention some of its consequences. A special case of (iii) occurs when one of the two functions is constant, say $g(x) = c$ for all x under consideration. In this case, (iii) becomes $(c \cdot f)' = c \cdot f'$. In other words, the derivative of a constant times f is the constant times the derivative of f . Combining this with the fact that the derivative of a sum is the sum of derivatives [property (i)], we find that for every pair of constants c_1 and c_2 we have

$$(c_1 f + c_2 g)' = c_1 f' + c_2 g' .$$

This is called the *linearity property* of the derivative and it is analogous to the linearity property of the integral. Using mathematical induction, we can extend the linearity property to arbitrary finite sums as follows:

$$\left(\sum_{i=1}^n c_i \cdot f_i\right)' = \sum_{i=1}^n c_i \cdot f_i' ,$$

where $c_1, \dots, c_n$ are constants and $f_1, \dots, f_n$ are functions with derivatives $f_1', \dots, f_n'$.

Every derivative formula can be written in two ways, either as an equality between two *functions* or as an equality involving *numbers*. The properties of Theorem 2-3, as written above, are equations involving functions. For example, property (i) states that the derivative of the function $f + g$ is the sum of the two functions f' and g' . When these functions are evaluated at a point x we obtain formulas involving numbers. Thus formula (i) implies

$$(f + g)'(x) = f'(x) + g'(x) .$$

Some people find it more convenient to express this as follows:

$$[f(x) + g(x)]' = f'(x) + g'(x) ,$$

where it is understood that $[f(x) + g(x)]'$ is merely an alternative way of writing $(f + g)'(x)$. Similarly, the product formula in (iii) is sometimes written as

$$[f(x) \cdot g(x)]' = f(x) \cdot g'(x) + f'(x) \cdot g(x) ,$$

where the left member really means $(f \cdot g)'(x)$, the derivative of $f \cdot g$ evaluated at the point x .

We proceed now to the proof of Theorem 2-3.

Proof of (i). Let x be a point where both derivatives $f'(x)$ and $g'(x)$ exist. The difference quotient for $f + g$ is

$$\frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} = \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}.$$

When $h \rightarrow 0$ the first quotient on the right approaches $f'(x)$, the second approaches $g'(x)$, and hence the sum approaches $f'(x) + g'(x)$. This proves (i), and the proof of (ii) is similar.

Proof of (iii). The difference quotient for the product $f \cdot g$ is

$$(2.16) \quad \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

To study this quotient as $h \rightarrow 0$ we add and subtract in the numerator a term which enables us to write (2.16) as a sum of two terms involving difference quotients of f and g . Adding and subtracting $g(x)f(x+h)$, we see that (2.16) becomes†

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = g(x) \frac{f(x+h) - f(x)}{h} + f(x+h) \frac{g(x+h) - g(x)}{h}.$$

When $h \rightarrow 0$ the first term on the right approaches $g(x)f'(x)$ and, since‡ $f(x+h) \rightarrow f(x)$, the second term approaches $f(x)g'(x)$. This proves (iii).

Proof of (iv). A special case of (iv) occurs when $f(x) = 1$ for all x . In this case $f'(x) = 0$ for all x and (iv) reduces to the formula

$$(2.17) \quad \left(\frac{1}{g}\right)' = -\frac{g'}{g^2}$$

provided $g(x) \neq 0$. We can deduce the general formula (iv) from this special case by writing f/g as a product and using (iii), since

$$\left(f \cdot \frac{1}{g}\right)' = \frac{1}{g} \cdot f' + f \cdot \left(\frac{1}{g}\right)' = \frac{f'}{g} - \frac{f \cdot g'}{g^2} = \frac{g \cdot f' - f \cdot g'}{g^2}.$$

Therefore it remains to prove (2.17). The difference quotient for $1/g$ is

$$(2.18) \quad \frac{[1/g(x+h)] - [1/g(x)]}{h} = -\frac{g(x+h) - g(x)}{h} \cdot \frac{1}{g(x)} \cdot \frac{1}{g(x+h)}.$$

When $h \rightarrow 0$ the first quotient on the right approaches $g'(x)$ and the third factor approaches $1/g(x)$. [This time the continuity of g at x is required since we are using the fact that $g(x+h) \rightarrow g(x)$ as $h \rightarrow 0$.] Hence the quotient in (2.18) approaches $-g'(x)/g(x)^2$, and this proves (2.17).

Note. In order to write (2.18) we need to know that $g(x+h) \neq 0$ for all sufficiently small h . This follows automatically from a general theorem on continuous functions which states that, if g is continuous at x and if $g(x) \neq 0$, then $g(x+h) \neq 0$ for all sufficiently small h . A proof of this theorem is given in Section 2.30. (See Theorem 2-9.)

† We could just as well have added and subtracted the term $f(x)g(x+h)$.

‡ In Section 2.8 it was shown that the existence of the derivative $f'(x)$ implies continuity of f at x .

2.11 Further examples of differentiation formulas

The rules developed in the foregoing section, when used with the examples worked out in Section 2.6, enable us to derive new examples of differentiation formulas.

Example 1: Polynomials. In Example 3 of Section 2.6 we showed that if $f(x) = x^n$, where n is a positive integer, then $f'(x) = nx^{n-1}$. The reader may find it amusing to rederive this result as a consequence of the special case $n = 1$, using mathematical induction in conjunction with the formula for differentiating a product.

Using this result along with the linearity property, we can differentiate any polynomial by computing the derivative of each term and adding the derivatives. Thus, if

$$f(x) = \sum_{k=0}^n c_k x^k,$$

then, by differentiating term by term, we obtain

$$f'(x) = \sum_{k=0}^n k c_k x^{k-1}.$$

Note that the derivative of a polynomial of degree n is a new polynomial of degree $n - 1$. For example, if $f(x) = 2x^3 + 5x^2 - 7x + 8$, then $f'(x) = 6x^2 + 10x - 7$.

Example 2: Rational functions. If r is the quotient of two polynomials, say $r(x) = p(x)/q(x)$, then the derivative $r'(x)$ may be computed by the quotient formula (iv) in Theorem 2-3. The derivative $r'(x)$ exists at every x for which the denominator $q(x) \neq 0$. Note that the function r' so defined is itself a rational function. In particular, when $r(x) = 1/x^m$, where m is a positive integer and $x \neq 0$, we find

$$(2.19) \quad r'(x) = \frac{x^m \cdot 0 - mx^{m-1}}{x^{2m}} = \frac{-m}{x^{m+1}}.$$

If this is written in the form $r'(x) = -mx^{-m-1}$, it provides an extension from positive exponents to negative exponents of formula (2.9) for differentiating n th powers.

Example 3: Differentiation of fractional powers. In Exercise 13 of Section 2.9 we outlined a method for proving that $f'(x) = rx^{r-1}$ if $f(x) = x^r$, where r is a positive rational number and $x > 0$. The quotient rule enables us to extend this result to negative rational exponents. The proof is identical to that of formula (2.19) except that the exponent m is assumed to be a rational number instead of an integer. The formula for differentiating x^r is also valid for negative x , provided r is such that x^r and x^{r-1} are defined. This occurs when $r = m/n$, where m and n are integers with no factor in common and the denominator n is odd and positive. If, in addition, $r > 1$, the formula is also valid for $x = 0$.

Example 4: The sine and cosine. Let s and c denote the sine and cosine functions, i.e., the functions defined by the equations

$$s(x) = \sin x, \quad c(x) = \cos x.$$

We shall show that their derivatives are given by the following formulas:

$$(2.20) \quad s'(x) = \cos x$$

and

$$(2.21) \quad c'(x) = -\sin x.$$

Expressed as equations between functions rather than numbers, (2.20) and (2.21) may be written as $\sin' = \cos$ and $\cos' = -\sin$, respectively. In other words, the derivative of the sine is the cosine and the derivative of the cosine is the negative of the sine.

Consider first the proof of (2.20). The difference quotient in question is

$$\frac{\sin(x+h) - \sin x}{h}.$$

To transform this into a form that makes it possible to calculate the limit as $h \rightarrow 0$ we use the trigonometric identity

$$\sin y - \sin x = 2 \sin \frac{y-x}{2} \cos \frac{y+x}{2}$$

with $y = x + h$. This leads to the formula

$$\frac{\sin(x+h) - \sin x}{h} = \frac{\sin(h/2)}{h/2} \cos\left(x + \frac{h}{2}\right).$$

As $h \rightarrow 0$ the factor $\cos(x + \frac{1}{2}h) \rightarrow \cos x$ because of the continuity of the cosine. Therefore, to complete the proof of (2.20), we need to know that

$$(2.22) \quad \frac{\sin(h/2)}{h/2} \rightarrow 1 \quad \text{as } h \rightarrow 0.$$

This follows from the limit formula

$$(2.23) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

which was encountered earlier in Equation (2.15) of Section 2.9. We shall prove (2.23) in a moment, thus completing the proof of (2.20).

The proof of (2.21) also depends on (2.23). In this case, we start with the identity

$$\cos y - \cos x = -2 \sin \frac{y-x}{2} \sin \frac{y+x}{2}$$

and take $y = x + h$. This leads to the formula

$$\frac{\cos(x+h) - \cos x}{h} = -\frac{\sin(h/2)}{h/2} \sin\left(x + \frac{h}{2}\right).$$

Continuity of the sine shows that $\sin(x + \frac{1}{2}h) \rightarrow \sin x$ as $h \rightarrow 0$; from (2.22), we obtain (2.21).

It remains to prove (2.23). A proof based on a geometric argument can be constructed with the aid of Figure 2.15, which shows a sector of a circle of radius 1 subtending an angle of x radians, where $0 < x < \frac{1}{2}\pi$. Because of the way we have defined angular measure (see Section 1.38), we have

$$\frac{x}{2} = \text{area of sector } OAP.$$

By comparing the area of sector OAP with that of triangles OQP and OAB , we obtain the inequalities

$$(2.24) \quad \frac{1}{2} \sin x \cos x < \frac{x}{2} < \frac{1}{2} \tan x.$$

Dividing by $\frac{1}{2} \sin x$ and taking reciprocals, we find

$$\cos x < \frac{\sin x}{x} < \frac{1}{\cos x}.$$

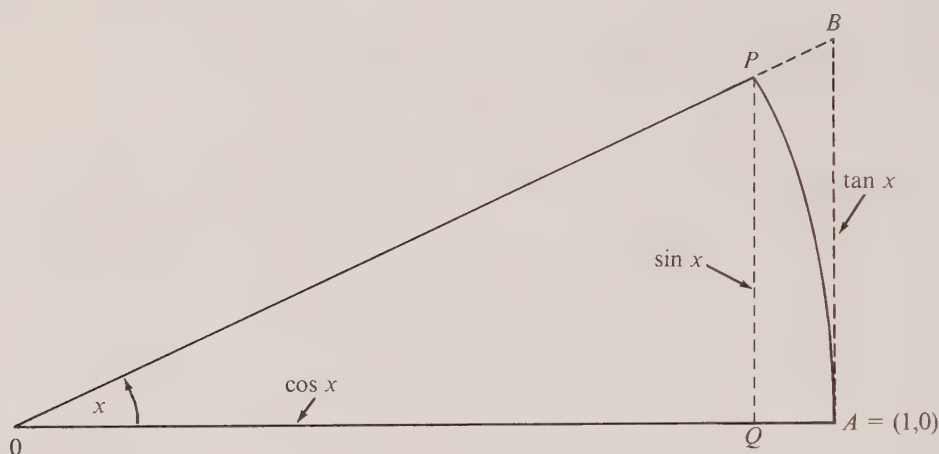


FIGURE 2.15 Area of triangle $OQP < \text{area of sector } OAP < \text{area of triangle } OAB$.

These inequalities have been established for $0 < x < \frac{1}{2}\pi$. They are also correct for $-\frac{1}{2}\pi < x < 0$, since $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, and hence they hold whenever $0 < |x| < \frac{1}{2}\pi$. Now, when $x \rightarrow 0$, $\cos x \rightarrow 1$, hence also $1/(\cos x) \rightarrow 1$. Therefore $(\sin x)/x$ is squeezed between two functions which approach 1 as $x \rightarrow 0$. From this we conclude that $(\sin x)/x \rightarrow 1$ as $x \rightarrow 0$. (See Exercise 22 in Section 2.9 for the rigorous justification of this last step.)

Note. The foregoing proof of (2.23) is not completely analytic because we deduced the inequalities in (2.24) by referring to a figure. The reader should not be overly concerned with this lack of rigor because we have not, as yet, given strictly analytic definitions of the sine and cosine functions nor have we given analytic proofs of the various trigonometric identities that we have used occasionally. In a later chapter, when we discuss infinite series, we shall indicate how the theory of the trigonometric functions can proceed in an entirely analytic fashion, without any reference to geometry. The standard identities and the limit relation in (2.23) are easily derived as consequences of the analytic definitions and the whole theory of trigonometric functions can be made to take its place as part of the deductive system of calculus.

2.12 Exercises

In Exercises 1 through 10, obtain a formula for $f'(x)$ if $f(x)$ is described as indicated.

1. $f(x) = x^2 + 3x + 2$.

6. $f(x) = \frac{x}{x-1}$, $x \neq 1$.

2. $f(x) = x^4 + \sin x$.

7. $f(x) = \frac{1}{2 + \cos x}$.

3. $f(x) = x^4 \sin x$.

8. $f(x) = \frac{x^2 + 3x + 2}{x^4 + x^2 + 1}$.

4. $f(x) = \frac{1}{x+1}$, $x \neq -1$.

9. $f(x) = \frac{2 - \sin x}{2 - \cos x}$.

5. $f(x) = \frac{1}{x^2 + 1} + x^5 \cos x$.

10. $f(x) = \frac{x \sin x}{1 + x^2}$.

11. If $f(x) = 2 + x - x^2$, determine $f'(0)$, $f'(\frac{1}{2})$, $f'(1)$, $f'(-10)$.

12. If $f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x$, find all x for which (a) $f'(x) = 0$; (b) $f'(x) = -2$; (c) $f'(x) = 10$.

In Exercises 13 through 20, obtain a formula for $f'(x)$.

13. $f(x) = (1 + nx^m)(1 + mx^n)$ (m and n are rational numbers, $x > 0$).

14. $f(x) = \frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3}, \quad x \neq 0.$

17. $f(x) = \frac{x}{(1-x)^2(1+x)^3}, \quad |x| \neq 1.$

15. $f(x) = \frac{2x}{1-x^2}, \quad |x| \neq 1.$

18. $f(x) = x + x^{1/2} + x^{1/3} + x^{1/4}, \quad x > 0.$

16. $f(x) = \frac{1+x-x^2}{1-x+x^2}.$

19. $f(x) = \frac{x}{1+\sqrt{x}} + \frac{\sqrt{x}}{1+x}, \quad x > 0.$

20. $f(x) = x^{-1} + x^{-1/2} + x^{-1/3} + x^{-1/4}, \quad x > 0.$

21. Let $f_1, \dots, f_n$ be n functions having derivatives $f'_1, \dots, f'_n$. Develop a rule for differentiating the product $g = f_1 \cdot \dots \cdot f_n$ and prove it by mathematical induction. Show that for those points x , where none of the function values $f_1(x), \dots, f_n(x)$ are zero, the result for $g'(x)$ can be written as follows:

$$\frac{g'(x)}{g(x)} = \frac{f'_1(x)}{f_1(x)} + \dots + \frac{f'_n(x)}{f_n(x)}.$$

22. Verify the entries in the following table of derivatives and fill in the blank spaces. It is understood that the formulas hold for those x for which $f(x)$ is defined.

	$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
(a)	$\tan x$	$\sec^2 x$	(g)	$\frac{\sin x}{x}$
(b)	$\cot x$		(h)	$\frac{1}{x + \sin x}$
(c)	$\sec x$	$\tan x \sec x$	(i)	$\frac{ax + b}{cx + d}$
(d)	$\csc x$		(j)	$\frac{\cos x}{2x^2 + 3}$
(e)	$\tan x \sec x$	$\sec x (1 + 2 \tan^2 x)$	(k)	$\frac{ax^2 + bx + c}{\sin x + \cos x}$
(f)	$x \tan x$			

23. What is the rate of change of the volume of a cube with respect to the length of each edge?

24. (a) The area of a circle of radius r is πr^2 and its circumference is $2\pi r$. (The formula for the circumference will be proved in a later chapter.) Show that the rate of change of the area with respect to the radius is equal to the circumference.

(b) The volume of a sphere of radius r is $\frac{4\pi r^3}{3}$ and its surface area is $4\pi r^2$. (The formula for the surface area will be proved in a later chapter.) Show that the rate of change of the volume with respect to the radius is equal to the surface area.

25. Consider the graph of the function f defined by the equation $f(x) = x^2 + ax + b$, where a and b are constants. (The curve is called a *parabola*.) Find values of a and b such that the line $y = 2x$ is tangent to this graph at the point $(2, 4)$.

26. Consider the graph of the function f defined by the equation $f(x) = x^2 + ax + b$, where a and b are constants.

(a) Find the slope of the chord joining the points on the graph for which $x = x_1$ and $x = x_2$.

(b) Find, in terms of x_1 and x_2 , all values of x for which the tangent line at $(x, f(x))$ has the same slope as the chord in part (a).

27. Show that the line $y = -x$ is tangent to the curve given by the equation $y = x^3 - 6x^2 + 8x$. Find the point of tangency. Does this tangent line intersect the curve anywhere else?

28. If $f(x) = (ax + b) \sin x + (cx + d) \cos x$, determine values of the constants a, b, c, d such that $f'(x) = x \cos x$.

29. If $g(x) = (ax^2 + bx + c) \sin x + (dx^2 + ex + f) \cos x$, determine values of the constants a, b, c, d, e, f such that $g'(x) = x^2 \sin x$.

30. Given the formula

$$1 + x + x^2 + \cdots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

(valid if $x \neq 1$), determine, by differentiation, formulas for the following sums:

(a) $1 + 2x + 3x^2 + \cdots + nx^{n-1}$,

(b) $1^2x + 2^2x^2 + 3^2x^3 + \cdots + n^2x^n$.

2.13 Other notations for derivatives

History has shown that notations have played an extremely important role in the development of mathematics. Some mathematical symbols, such as x^n or $n!$, are merely abbreviations that compress long statements or formulas into a short space. Others, like the integration symbol $\int_a^b f(x) dx$, not only remind us of the process being represented but also help us in carrying out computations.

Sometimes several different notations are used for the same idea, preference for one or another being dependent on the circumstances that surround the use of the symbols. This is especially true in differential calculus where many different notations are used for derivatives. The derivative of a function f has been denoted in our previous discussions by f' , a notation introduced by J. L. Lagrange (1736–1813) late in the 18th century. This emphasizes the fact that f' is a new function obtained from f by differentiation, its value at x being denoted by $f'(x)$. Each point (x, y) on the graph of f has its coordinates x and y related by the equation $y = f(x)$, and the symbol y' is also used to represent the derivative $f'(x)$. Similarly, $y'', \dots, y^{(n)}$ represent the higher derivatives $f''(x), \dots, f^{(n)}(x)$. For example, if $y = \sin x$, then $y' = \cos x$, $y'' = -\sin x$, etc. Lagrange's notation is not too far removed from that used by Newton who wrote $\dot{y}$ and $\ddot{y}$, instead of y' and y'' . Newton's dots are still used by some authors, especially to denote velocity and acceleration.

Another symbol was introduced in 1800 by L. Arbogast (1759–1803) who indicated the derivative of f by Df , a symbol that has widespread use today. The symbol D is called a *differentiation operator* and it helps to suggest that Df is a new function obtained from f by the operation of differentiation. Higher derivatives $f'', f''', \dots, f^{(n)}$ are written $D^2f, D^3f, \dots, D^n f$, respectively, the values of these derivatives at x being written $D^2f(x), D^3f(x), \dots, D^n f(x)$. Thus, we have $D \sin x = \cos x$ and $D^2 \sin x = D \cos x = -\sin x$. The rule for differentiating a sum of two functions becomes, in the D -notation, $D(f + g) = Df + Dg$. Evaluation of the derivatives at x leads to the formula $[D(f + g)](x) = Df(x) + Dg(x)$ which is also written in the form $D[f(x) + g(x)] = Df(x) + Dg(x)$. The reader may easily formulate the product and quotient rules in the D -notation.

Among the early pioneers of mathematical analysis, Leibniz, more than anyone else, understood the importance of well-chosen symbols. He experimented at great length and carried on extensive correspondence with other mathematicians, debating the merits or drawbacks of various notations. The tremendous impact that calculus has had on the development of modern mathematics is due in large part to its well-developed and highly suggestive symbols, many of them originated by Leibniz.

Leibniz developed a notation for derivatives quite different from those mentioned above. Using y for $f(x)$, he wrote the difference quotient

$$\frac{f(x + h) - f(x)}{h}$$

in the form

$$\frac{\Delta y}{\Delta x},$$

where Δx was written for h and Δy for $f(x + h) - f(x)$. The symbol Δ is called a *differencing operator*. For the limit of the difference quotient, that is, for the derivative $f'(x)$, Leibniz wrote dy/dx . In this notation, the definition of derivative becomes

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

Not only was Leibniz's notation different, but his way of thinking about derivatives was different. He thought of the limit dy/dx as a quotient of "infinitesimal" quantities dy and dx called "differentials," and he referred to the derivative dy/dx as a "differential quotient." Instead of using a limit process to define derivatives, Leibniz just passed from Δy and Δx to dy and dx without any more explanation than that Δy and Δx had become infinitesimals. Leibniz imagined infinitesimals as entirely new types of numbers which, although not zero, were smaller than every positive real number.

For a long time calculus was thought to be intrinsically difficult and somewhat mysterious because people could not grasp the idea of an infinitesimal. The work of Cauchy and others in the 19th century gradually led to the abandonment of infinitely small quantities as an essential part of mathematics. However, many people, especially those engaged in applied mathematics, have found it very helpful to try to think as Leibniz did in terms of infinitesimals. Quite often this kind of thinking leads quickly to results that can be proven correct by more conventional means.

Although some of Leibniz's ideas did not withstand the test of history, the same cannot be said of his notations.† The symbol dy/dx for the derivative has the obvious advantage that it summarizes the whole process of forming the difference quotient and passing to the limit. Later we shall find the further advantage that certain formulas become very easy to remember and to work with when derivatives are written as quotients of differentials.

The word "differential" which was introduced by Leibniz still appears in the current mathematical literature but it is now used for a concept that can be introduced in a logically satisfactory way. Differentials, as they are defined in Section 2.19, turn out to be useful in approximating complicated functions by simpler functions. Also, we shall find that they give us a formal symbolism that is helpful in the technique of integration.

† A full account of Leibniz's many contributions in this direction may be found in F. Cajori's *A History of Mathematical Notations*, vol. 2 (Chicago: The Open Court Publishing Co., 1928–29).

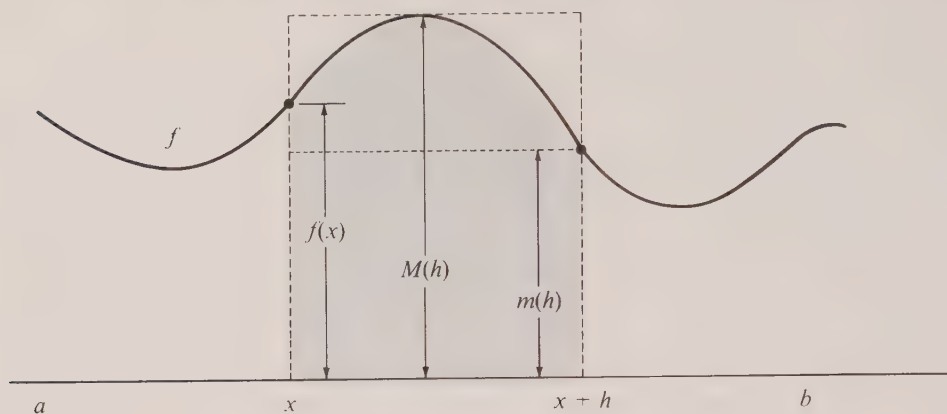


FIGURE 2.16 Geometric motivation for the first fundamental theorem of calculus.

2.14 The derivative of an indefinite integral. The first fundamental theorem of calculus

We come now to the remarkable connection that exists between integration and differentiation. The relationship between these two processes is somewhat analogous to that which holds between “squaring” and “taking the square root.” If we square a positive number and then take the positive square root of the result, we get the original number back again. Similarly, if we operate on a continuous function f by integration, we get a new function (an indefinite integral of f) which, when differentiated, leads back to the original function f . For example, if $f(x) = x^2$, then an indefinite integral A of f may be defined by the equation

$$A(x) = \int_c^x f(t) \, dt = \int_c^x t^2 \, dt = \frac{x^3}{3} - \frac{c^3}{3},$$

where c is a constant. Differentiating, we find $A'(x) = x^2 = f(x)$. This example illustrates a general result, called the first fundamental theorem of calculus, which may be stated as follows:

2- 4 THEOREM. *First fundamental theorem of calculus.* Let f be a function that is integrable on a closed interval $[a, b]$. Let c be such that $a \leq c \leq b$, and define a new function A as follows:

$$A(x) = \int_c^x f(t) \, dt \quad \text{if } a \leq x \leq b.$$

Then the derivative $A'(x)$ exists at each point x in the open interval (a, b) where f is continuous, and for such x we have

$$(2.25) \quad A'(x) = f(x).$$

First we give a geometric argument which suggests why the theorem ought to be true and then we give a rigorous analytic proof.

Geometric motivation. Figure 2.16 shows the graph of a positive function f over an interval $[a, b]$. In the figure, h is positive and the area of the shaded region is

$$\int_x^{x+h} f(t) dt = \int_c^{x+h} f(t) dt - \int_c^x f(t) dt = A(x+h) - A(x).$$

If we denote by $m(h)$ and $M(h)$, respectively, the smallest and largest values of $f(t)$ in the interval $[x, x+h]$, comparison of areas gives us the inequalities

$$hm(h) \leq A(x+h) - A(x) \leq hM(h).$$

From these we obtain

$$(2.26) \quad m(h) \leq \frac{A(x+h) - A(x)}{h} \leq M(h).$$

As $h \rightarrow 0$, both numbers $m(h)$ and $M(h)$ approach $f(x)$ as a limit because of the continuity of f at x . Therefore if we let $h \rightarrow 0$ in (2.26), we obtain (2.25).

To convert this argument into a rigorous proof we need two facts about continuous functions that we have not yet proved: first, that the function f has a largest value $M(h)$ and a smallest value $m(h)$ in the interval $[x, x+h]$ and, second, that both $m(h)$ and $M(h)$ approach $f(x)$ as $h \rightarrow 0$. Although these facts may seem geometrically evident when we examine a particular case, analytic proofs require a deeper investigation of the concept of continuity. This investigation will be carried out in a later chapter. By using a slightly different method, we can give a completely analytic proof of (2.25) which requires only the *definition* of continuity.

Analytic proof. Let x be a point of continuity of f , keep x fixed, and form the quotient

$$\frac{A(x+h) - A(x)}{h}.$$

To prove the theorem we must show that this quotient approaches the limit $f(x)$ as $h \rightarrow 0$. The numerator is

$$A(x+h) - A(x) = \int_c^{x+h} f(t) dt - \int_c^x f(t) dt = \int_x^{x+h} f(t) dt.$$

If we write $f(t) = f(x) + [f(t) - f(x)]$ in the last integral, we obtain

$$\begin{aligned} A(x+h) - A(x) &= \int_x^{x+h} f(x) dt + \int_x^{x+h} [f(t) - f(x)] dt \\ &= hf(x) + \int_x^{x+h} [f(t) - f(x)] dt, \end{aligned}$$

from which we find

$$(2.27) \quad \frac{A(x+h) - A(x)}{h} = f(x) + \frac{1}{h} \int_x^{x+h} [f(t) - f(x)] dt.$$

Therefore to complete the proof of (2.25) all we need to do is show that

$$(2.28) \quad \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} [f(t) - f(x)] dt = 0.$$

It is this part of the proof that makes use of the continuity of f at x .

Let us denote the second term on the right of (2.27) by $G(h)$. We are to prove that $G(h) \rightarrow 0$ as $h \rightarrow 0$. Using the definition of limit, we must show that for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$(2.29) \quad |G(h)| < \epsilon \quad \text{whenever} \quad 0 < |h| < \delta.$$

Continuity of f at x tells us that, if ϵ is given, there is a positive δ such that

$$(2.30) \quad |f(t) - f(x)| < \frac{1}{2}\epsilon$$

whenever

$$(2.31) \quad x - \delta < t < x + \delta.$$

If we choose h so that $0 < h < \delta$, then every t in the interval $[x, x + h]$ satisfies (2.31) and hence (2.30) holds for every such t . Using the property $|\int_x^{x+h} g(t) dt| \leq \int_x^{x+h} |g(t)| dt$ with $g(t) = f(t) - f(x)$, we see that the inequality in (2.30) leads to the relation

$$\left| \int_x^{x+h} [f(t) - f(x)] dt \right| \leq \int_x^{x+h} |f(t) - f(x)| dt \leq \int_x^{x+h} \frac{1}{2}\epsilon dt = \frac{1}{2}h\epsilon < h\epsilon.$$

If we divide by h , we see that (2.29) holds for $0 < h < \delta$. If $h < 0$, a similar argument proves that (2.29) holds whenever $0 < |h| < \delta$ and this completes the proof.

2.15 Primitive functions and the second fundamental theorem of calculus

If a function f is constant on an open interval (a, b) , then its derivative is zero everywhere on (a, b) . As noted earlier, the proof of this fact is an immediate consequence of the definition of derivative. The converse of this statement is also true. That is to say, if $f'(x) = 0$ for each x in (a, b) , then f is constant on (a, b) . This seems evident geometrically because if $f'(x) = 0$ at each x then the graph of f has a horizontal tangent at each of its points and this suggests that the graph itself must be a horizontal line. In spite of this strong geometric evidence, an analytic proof that f is constant is not easy to give with the materials developed thus far. A simple proof will be given in a later chapter as an application of the so-called mean-value theorem of differential calculus. In the meantime we shall want to use this property of derivatives rather extensively so we state it here as a formal theorem.

2- 5 THEOREM. If $f'(x) = 0$ for each x in an open interval (a, b) , then f is constant on (a, b) .

If f is the derivative of another function P , say

$$f(x) = P'(x)$$

for every x in some open interval, then P is called a *primitive* or an *antiderivative* of f . For example, the sine function is a primitive of the cosine function because $D \sin x = \cos x$. We speak of *a* primitive, rather than *the* primitive, because if P is a primitive of f then so is $P + k$ for every constant k . Conversely, any two primitives P and Q of the same function f can differ only by a constant because their difference $P - Q$ has the derivative

$$P'(x) - Q'(x) = f(x) - f(x) = 0$$

for every x in the open interval under consideration and hence, by Theorem 2-5, $P - Q$ must be constant on this interval.

The first fundamental theorem of calculus tells us that we can always construct a primitive of a continuous function by integration. When we combine this with the fact that two primitives of the same function can differ only by a constant, we obtain the second fundamental theorem of calculus:

2- 6 THEOREM. *Second fundamental theorem of calculus.* Let f be integrable on a closed interval $[a, b]$ and assume that f is continuous[†] on the open interval (a, b) . Let P be any primitive of f , so that we have $P'(x) = f(x)$ for each x in the open interval (a, b) . If $a < c < b$, then for each point x in (a, b) we have the following formula:

$$(2.32) \quad \int_c^x f(t) dt = P(x) - P(c).$$

Note. Since $f(t) = P'(t)$ we may write this formula as follows:

$$\int_c^x P'(t) dt = P(x) - P(c).$$

Expressed in this form, the second fundamental theorem tells us that, if we take the derivative of a function P and then integrate the result, we are led back to the original function P , plus a constant.

Proof. Let $A(x) = \int_c^x f(t) dt$ if $a < x < b$. We want to prove that

$$(2.33) \quad A(x) = P(x) - P(c).$$

Since f is continuous, the first fundamental theorem implies that A is also a primitive of f and, since two primitives of f can differ only by a constant, we must have $A(x) - P(x) = k$ for some constant k . When $x = c$ this formula states that $-P(c) = k$ since $A(c) = 0$. Therefore $A(x) - P(x) = -P(c)$, which is the same as (2.33). This completes the proof.

The real power of this theorem lies in the fact that it enables us to compute the value of an integral by a mere subtraction. The problem of evaluating an integral is thereby transferred to another problem—that of finding a primitive P of f . In actual practice the second problem is a great deal easier to deal with than the first. Every differentiation

[†] If f is continuous on $[a, b]$, then it can be shown that f is automatically integrable on $[a, b]$. This is an important theorem of integration theory that is proved in Section 2.32.

formula automatically gives us an example of a primitive of some function f and this, in turn, leads to an integration formula for this function.

If the hypotheses of Theorem 2-6 are satisfied in some open interval surrounding the closed interval $[a, b]$, then we may choose $c = a$ and $x = b$ in Equation (2.32) to write

$$(2.34) \quad \int_a^b f(x) dx = \int_a^b P'(x) dx = P(b) - P(a).$$

Sometimes the special symbol

$$P(x) \Big|_a^b$$

is used to designate the operation of evaluating $P(x)$ first for $x = b$ and then for $x = a$ and subtracting. Thus formula (2.34) may also be written as follows:

$$(2.35) \quad \int_a^b f(x) dx = P(x) \Big|_a^b = P(b) - P(a).$$

Note that $P(x) \Big|_a^b = [P(x) + C] \Big|_a^b$ if C is independent of x . When $P(x) = \int_c^x f(t) dt$ for some c in $[a, b]$, formula (2.35) becomes

$$\int_a^b f(x) dx = \int_c^x f(t) dt \Big|_a^b = P(b) - P(a).$$

From the differentiation formulas worked out thus far we can derive the following integration formulas as consequences of the second fundamental theorem:

Example 1: Integration of rational powers. The integration formula

$$(2.36) \quad \int_a^b x^n dx = \frac{b^{n+1} - a^{n+1}}{n+1} \quad (n = 0, 1, 2, \dots)$$

was proved in Section 1.36 directly from the definition of the integral. The result may be rederived and generalized to rational exponents by using the second fundamental theorem. First of all, we observe that the function P defined by the equation

$$(2.37) \quad P(x) = \frac{x^{n+1}}{n+1}$$

has the derivative $P'(x) = x^n$ if n is any nonnegative integer. Since this is valid for all real x , we may use (2.35) to write

$$\int_a^b x^n dx = \frac{x^{n+1}}{n+1} \Big|_a^b = \frac{b^{n+1} - a^{n+1}}{n+1}$$

for all intervals $[a, b]$. This formula, proved for all integers $n \geq 0$, also holds for all negative integers except $n = -1$, which is excluded because $n+1$ appears in the denominator. To prove (2.36) for negative n it suffices to show that (2.37) implies $P'(x) = x^n$ when n is negative and $\neq -1$, a fact which is easily verified by differentiating P as a rational function. Of course, when n is negative, neither $P(x)$ nor $P'(x)$ is defined for $x = 0$ and when we use (2.36) for negative n it is important to exclude those intervals $[a, b]$ that contain the point $x = 0$.

The results of Example 3 in Section 2.11 enable us to extend (2.36) to all *rational* exponents (except -1), provided the integrand is defined everywhere on the interval $[a, b]$ under consideration.

For example, if $0 < a < b$ and $n = -\frac{1}{2}$, we find

$$\int_a^b \frac{1}{\sqrt{x}} dx = \int_a^b x^{-1/2} dx = \left. \frac{x^{1/2}}{1/2} \right|_a^b = 2(\sqrt{b} - \sqrt{a}).$$

In the next chapter we shall define a general power function f such that $f(x) = x^c$ for every real exponent c . We shall find that this function has the derivative $f'(x) = cx^{c-1}$ and the primitive $P(x) = x^{c+1}/(c+1)$ if $c \neq -1$. This will enable us to extend (2.36) to all real exponents except -1 .

Note that we cannot get $P'(x) = 1/x$ by differentiation of any formula of the form $P(x) = x^n$. Nevertheless, there exists a function P whose derivative is $P'(x) = 1/x$. To exhibit such a function all we need to do is write a suitable indefinite integral; for example,

$$P(x) = \int_1^x \frac{1}{t} dt \quad \text{if } x > 0.$$

This integral exists because the integrand is monotonic. The function so defined is called the *logarithm* (more specifically, the *natural logarithm*). Its properties are developed systematically in Chapter 3.

Example 2: Integration of the sine and cosine. Since the derivative of the sine is the cosine and the derivative of the cosine is minus the sine, the second fundamental theorem also gives us the following formulas:

$$\begin{aligned} \int_a^b \cos x \, dx &= \sin x \Big|_a^b = \sin b - \sin a, \\ \int_a^b \sin x \, dx &= (-\cos x) \Big|_a^b = \cos a - \cos b. \end{aligned}$$

These formulas were stated earlier in Section 1.41 without proof.

Further examples of integration formulas can be obtained from Examples 1 and 2 by taking finite sums of terms of the form Ax^n , $B \sin x$, $C \cos x$, where A , B , C are constants.

Example 3. The formula $\int_c^x P'(t) \, dt = P(x) - P(c)$ often enables us to deduce properties of a function from properties of its derivative. For example, if we replace x by x_2 and c by x_1 in this formula, we find

$$P(x_2) - P(x_1) = \int_{x_1}^{x_2} P'(t) \, dt.$$

If $x_1 < x_2$ and $P'(t) \geq 0$ for all t in $[x_1, x_2]$, we deduce that $P(x_2) \geq P(x_1)$. This proves that a function with a continuous nonnegative derivative in an interval is monotonic increasing on this interval. If the derivative is positive in the interval, then P is strictly increasing.

2.16 Exercises

In each of Exercises 1 through 10, find a primitive of f ; that is, find a function P such that $P'(x) = f(x)$ and use the second fundamental theorem to evaluate $\int_a^b f(x) \, dx$.

1. $f(x) = 5x^3$.
2. $f(x) = 4x^4 - 12x$.
3. $f(x) = (x + 1)(x^3 - 2)$.
4. $f(x) = \frac{x^4 + x - 3}{x^3}$, $x \neq 0$.
5. $f(x) = (1 + \sqrt{x})^2$, $x > 0$.
6. $f(x) = \sqrt{2x} + \sqrt{\frac{1}{2}x}$, $x > 0$.
7. $f(x) = \frac{2x^2 - 6x + 7}{2\sqrt{x}}$, $x > 0$.
8. $f(x) = 2x^{1/3} - x^{-1/3}$, $x > 0$.
9. $f(x) = 3 \sin x + 2x^5$.
10. $f(x) = x^{4/3} - 5 \cos x$.
11. Prove that there is no polynomial f whose derivative is given by the formula $f'(x) = 1/x$.
12. Show that $\int_0^x |t| dt = \frac{1}{2}x|x|$ for all real x .
13. Show that

$$\int_0^x (t + |t|)^2 dt = \frac{2x^2}{3}(x + |x|) \quad \text{for all real } x.$$

14. Find a function f and a value of the constant c such that

$$\int_c^x f(t) dt = \cos x - \frac{1}{2} \quad \text{for all real } x.$$

15. Find a function f and a value of the constant c such that

$$\int_c^x tf(t) dt = \sin x - x \cos x - \frac{1}{2}x^2 \quad \text{for all real } x.$$

16. There is a function f , defined and continuous for all real x , which satisfies an equation of the form

$$\int_0^x f(t) dt = \int_x^1 t^2 f(t) dt + \frac{x^{16}}{8} + \frac{x^{18}}{9} + c,$$

where c is a constant. Find an explicit formula for $f(x)$ and find the value of the constant c .

17. In each case find a function f (with a continuous second derivative f'') which satisfies all the given conditions or else explain why such an example cannot exist. [Hint. Use Theorem 2-6.]

- (a) $f'''(x) > 0$ for every x , $f'(0) = 1$, $f'(2) = 0$.
- (b) $f'''(x) > 0$ for every x , $f'(0) = 1$, $f(x) \leq 100$ for every positive x .
- (c) $f'''(x) > 0$ for every x , $f'(0) = 1$, $f(x) \leq 100$ for every negative x .

2.17 Composite functions and the chain rule

Because of the second fundamental theorem of calculus, increased skill in differentiation carries with it a corresponding improvement in techniques of integration. With the differentiation formulas developed thus far, we can find derivatives of functions f for which $f(x)$ is a finite sum of products or quotients of constant multiples of $\sin x$, $\cos x$, and x^r (r rational). As yet, however, we have not learned to deal with something like $f(x) = \sin(x^2)$ without going back to the definition of derivative. The function f for which $f(x) = \sin(x^2)$ is an example of what is known as a *composite* function. In this section we shall present a new theorem, called the *chain rule*, that will enable us to differentiate composite functions and thereby increase substantially the number of functions that we can differentiate and integrate.

Given two functions u and v , we can create new functions by addition, subtraction, multiplication, and division, and we can compute the derivatives of these new combinations in terms of the derivatives of u and v . Another way to construct new functions

from two or more given ones is by means of an operation known as *composition*. The *composite* or *composition* of u and v (in that order) is defined to be the function f for which

$$f(x) = u[v(x)] \quad (\text{read as "}u \text{ of } v \text{ of } x").$$

That is, to evaluate f at x we first compute $v(x)$ and then evaluate u at the point $v(x)$. Of course, this presupposes that it makes sense to evaluate u at $v(x)$ and we must realize that f will be defined only at those points x for which $v(x)$ is in the domain of u .

For example, if $u(x) = \sqrt{x}$ and $v(x) = 1 - x^2$, then the composite f is given by $f(x) = \sqrt{1 - x^2}$. Note that v is defined for all real x , whereas u is defined only for $x \geq 0$. Therefore the composite f is defined only for those x satisfying $1 - x^2 \geq 0$.

Formally, $f(x)$ is obtained by substituting $v(x)$ for x in the expression $u(x)$. For this reason the function f is sometimes denoted by the symbol $f = u \circ v$ (read as " u of v "). Another notation that we shall use to denote composition is $f = u \cdot v$ (read as " u circle v "). This resembles the notation for the product $u \cdot v$. In fact, we shall see in a moment that the operation of composition has some of the properties possessed by multiplication.

The composite of three or more functions may be found by composing them two at a time. Thus, the function f given by

$$f(x) = \cos [\sin (x^2)]$$

is a composition, $f = u \circ (v \circ w)$, where

$$u(x) = \cos x, \quad v(x) = \sin x, \quad \text{and} \quad w(x) = x^2.$$

Notice that the same f can be obtained by composing u and v first and then composing $u \circ v$ with w , thus: $f = (u \circ v) \circ w$. This illustrates the *associative law* for composition which states that

$$(2.38) \quad u \circ (v \circ w) = (u \circ v) \circ w$$

for all functions u, v, w , provided it makes sense to form all the composites in question. The reader will find that the proof of (2.38) is a straightforward exercise.

It should be noted that the *commutative law*, $u \circ v = v \circ u$, does not always hold for composition. For example, if $u(x) = \sin x$ and $v(x) = x^2$, the composite $f = u \circ v$ is given by $f(x) = \sin x^2$ [which means $\sin (x^2)$], whereas the composition $g = v \circ u$ is given by $g(x) = \sin^2 x$ [which means $(\sin x)^2$].

We turn now to the *chain rule* which tells us how to express the derivative of the composite function $u \circ v$ in terms of the derivatives u' and v' .

2- 7 THEOREM. *Chain rule.* Let f be the composition of two functions u and v , say $f = u \circ v$. Suppose that both derivatives $v'(x)$ and $u'(y)$ exist, where $y = v(x)$. Then the derivative $f'(x)$ also exists and is given by the formula

$$(2.39) \quad f'(x) = u'(y) \cdot v'(x).$$

In other words, to compute the derivative of $u \circ v$ at x we compute first the derivative of u at the point y , where $y = v(x)$, and multiply this by $v'(x)$.

Before we discuss the proof of (2.39), we shall mention some alternative ways of expressing the chain rule and illustrate its use with some particular examples. If we write (2.39) entirely in terms of x , we obtain the formula

$$f'(x) = u'[v(x)] \cdot v'(x).$$

Expressed as an equation involving *functions* rather than numbers, the chain rule assumes the following form

$$(u \circ v)' = (u' \circ v) \cdot v'.$$

In the $u(v)$ -notation, let us write $u(v)'$ for the derivative of the composite function $u(v)$ and $u'(v)$ for the composition $u' \circ v$. Then the last formula becomes

$$u(v)' = u'(v) \cdot v'.$$

The chain rule is an excellent example to illustrate the usefulness of the Leibniz notation for derivatives. In fact, if we write (2.39) in the Leibniz notation, it assumes the appearance of a trivial algebraic identity. First we introduce new symbols, say

$$y = v(x) \quad \text{and} \quad z = u(y).$$

Then we write dy/dx for the derivative $v'(x)$, and dz/dy for $u'(y)$. The formation of the composite function is indicated by writing

$$z = u(y) = u[v(x)] = f(x),$$

and dz/dx is written for the derivative $f'(x)$. The chain rule, as expressed in Equation (2.39), now becomes

$$(2.40) \quad \frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$

The strong suggestive power of this formula is obvious. It is especially attractive to people who use calculus in physical problems. For example, suppose the foregoing symbol z represents a physical quantity measured in terms of other physical quantities x and y . The equation $z = f(x)$ tells us how to find z if x is given, and the equation $z = u(y)$ tells us how to find z if y is given. The relation between x and y is expressed by the equation $y = v(x)$. The chain rule, as expressed in (2.40), tells us that the rate of change of z with respect to x is equal to the product of the rate of change of z with respect to y and the rate of change of y with respect to x . The following example illustrates how the chain rule may be used in a special physical problem.

Example 1. Suppose a gas is pumped into a spherical balloon at a constant rate of 50 cubic centimeters per second. Assume that the gas pressure remains constant and that the balloon always has a spherical shape. How fast is the radius of the balloon increasing when the radius is 5 centimeters?

Solution. Let r denote the radius and V the volume of the balloon at time t . We are given dV/dt , the rate of change of volume with respect to time, and we want to determine dr/dt , the rate of change of the radius with respect to time, at the instant when $r = 5$. The chain rule provides the connection between the given data and the unknown. It states that

$$(2.41) \quad \frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt}.$$

To compute dV/dr we use the formula $V = 4\pi r^3/3$ which expresses the volume of the sphere in terms of its radius. Differentiation gives us $dV/dr = 4\pi r^2$ and hence (2.41) becomes

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Substituting $dV/dt = 50$ and $r = 5$, we obtain $dr/dt = 1/(2\pi)$. That is to say, the radius is increasing at a rate of $1/(2\pi)$ centimeters per second at the instant when $r = 5$.

The foregoing example is called a problem in *related rates*. Note that it was not necessary to express r as a function of t in order to determine the derivative dr/dt . It is this fact that makes the chain rule especially useful in related-rate problems.

The next two examples show how the chain rule may be used to obtain new differentiation formulas.

Example 2. Given $f(x) = \sin(x^2)$, compute $f'(x)$.

Solution. The function f is a composition, $f(x) = u[v(x)]$ where $v(x) = x^2$ and $u(x) = \sin x$. To use the chain rule we need to determine $u'[v(x)] = u'(x^2)$. Since $u'(x) = \cos x$, we have $u'(x^2) = \cos(x^2)$, and hence (2.39) gives us

$$f'(x) = \cos(x^2) \cdot v'(x) = \cos(x^2) \cdot 2x.$$

We may also solve the problem using the Leibniz notation. If we write $y = x^2$ and $z = f(x)$, then $z = \sin y$ and $dz/dx = f'(x)$. The chain rule yields

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = (\cos y)(2x) = \cos(x^2) \cdot 2x,$$

which agrees with the foregoing result for $f'(x)$.

Example 3. If $f(x) = [v(x)]^n$, where n is a positive integer, compute $f'(x)$ in terms of $v(x)$ and $v'(x)$.

Solution. The function f is a composition, $f(x) = u[v(x)]$, where $u(x) = x^n$. Since $u'(x) = nx^{n-1}$, we have $u'[v(x)] = n[v(x)]^{n-1}$ and the chain rule yields

$$f'(x) = n[v(x)]^{n-1}v'(x).$$

If we omit the reference to x and write this as an equality involving functions, we obtain the important formula

$$(v^n)' = nv^{n-1}v'$$

which tells us how to differentiate the n th power of v when v' exists. The formula is also valid for *rational* powers if v^n and v^{n-1} are defined. To solve the problem in the Leibniz notation we may write $y = v(x)$ and $z = f(x)$. Then $z = y^n$, $dz/dx = f'(x)$, and the chain rule gives us

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = ny^{n-1}v'(x) = n[v(x)]^{n-1}v'(x),$$

which agrees with the first solution.

Example 4. The equation $x^2 + y^2 = r^2$ represents a circle of radius r and center at the origin. If we solve this equation for y in terms of x , we obtain two solutions which serve to define two functions f and g given on the interval $[-r, r]$ by the formulas

$$f(x) = \sqrt{r^2 - x^2} \quad \text{and} \quad g(x) = -\sqrt{r^2 - x^2}.$$

(The graph of f is the upper semicircle and the graph of g the lower semicircle.) We may compute the derivatives of f and g by the chain rule. For f we use the result of Example 3 with $v(x) = r^2 - x^2$ and $n = \frac{1}{2}$ to obtain

$$(2.42) \quad f'(x) = \frac{1}{2}(r^2 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{r^2 - x^2}} = \frac{-x}{f(x)}$$

whenever $f(x) \neq 0$. The same method, applied to g , gives us

$$(2.43) \quad g'(x) = -\frac{-x}{\sqrt{r^2 - x^2}} = \frac{-x}{g(x)}$$

whenever $g(x) \neq 0$. Notice that if we let y stand for either $f(x)$ or $g(x)$, then both formulas (2.42) and (2.43) can be combined into one, namely,

$$(2.44) \quad y' = \frac{-x}{y} \quad \text{if } y \neq 0.$$

Another useful application of the chain rule has to do with a technique known as *implicit differentiation*. We shall explain the method and illustrate its advantages by rederiving the result of Example 4 in a simpler way.

Example 5: Implicit differentiation. Formula (2.44) may be derived directly from the equation $x^2 + y^2 = r^2$ without the necessity of solving for y . If we remember that y is a function of x [either $y = f(x)$ or $y = g(x)$], we may differentiate both sides of the equation $x^2 + y^2 = r^2$ to obtain

$$(2.45) \quad 2x + 2yy' = 0.$$

(The term $2yy'$ comes from differentiating y^2 as explained in Example 3.) When Equation (2.45) is solved for y' it yields (2.44).

The equation $x^2 + y^2 = r^2$ is said to define y *implicitly* as a function of x (it actually defines *two* functions), and the process by which (2.45) is obtained from this equation is called *implicit differentiation*. The end result is valid for either of the two functions f and g so defined. Notice that at a point (x, y) on the circle with $x \neq 0$ and $y \neq 0$ the tangent line has a slope $-x/y$, whereas the radius from the center to (x, y) has the slope y/x . The product of the two slopes is -1 so the tangent is perpendicular to the radius.

Proof of Theorem 2-7. We turn now to the proof of (2.39). We assume that v has a derivative at x and that u has a derivative at $v(x)$ and we wish to prove that f has a derivative at x given by the product $u'[v(x)] \cdot v'(x)$. The difference quotient for f is

$$(2.46) \quad \frac{f(x+h) - f(x)}{h} = \frac{u[v(x+h)] - u[v(x)]}{h}.$$

It is helpful at this stage to introduce some new notation. Let $y = v(x)$ and let $k = v(x+h) - v(x)$. (It is important to realize that k depends on h .) Then we have $v(x+h) = y+k$ and (2.46) becomes

$$(2.47) \quad \frac{f(x+h) - f(x)}{h} = \frac{u(y+k) - u(y)}{h}.$$

The right-hand side of (2.47) resembles the difference quotient whose limit defines $u'(y)$, except that h appears in the denominator instead of k . If $k \neq 0$ it is easy to complete the proof. We simply multiply numerator and denominator by k , and the right-hand side of (2.47) becomes

$$(2.48) \quad \frac{u(y+k) - u(y)}{k} \cdot \frac{k}{h} = \frac{u(y+k) - u(y)}{k} \cdot \frac{v(x+h) - v(x)}{h}.$$

When $h \rightarrow 0$ the last quotient on the right tends to $v'(x)$. Also, $k \rightarrow 0$ as $h \rightarrow 0$ because $k = v(x+h) - v(x)$ and v is continuous at x . Therefore the first quotient on the right of (2.48) approaches $u'(y)$ as $h \rightarrow 0$, and this leads at once to (2.39).

Although the foregoing argument seems to be the most natural way to proceed, it is not completely general. Since $k = v(x+h) - v(x)$, it may happen that $k = 0$ for infinitely many values of h as $h \rightarrow 0$, in which case the passage from (2.47) to (2.48) is not valid. To overcome this difficulty a slight modification of the proof is needed.

Let us return to Equation (2.47) and express the quotient on the right in a form that does not involve k in the denominator. For this purpose we introduce the difference between the derivative $u'(y)$ and the difference quotient whose limit is $u'(y)$. That is, we define a new function g as follows:

$$(2.49) \quad g(t) = \frac{u(y+t) - u(y)}{t} - u'(y) \quad \text{if } t \neq 0.$$

This equation defines $g(t)$ only if $t \neq 0$. Multiplying by t and rearranging terms, we may write (2.49) in the following form:

$$(2.50) \quad u(y+t) - u(y) = t[g(t) + u'(y)].$$

Although (2.50) has been derived under the hypothesis that $t \neq 0$, it also holds for $t = 0$ provided we assign some definite value to $g(0)$. The actual value we assign to $g(0)$ is not important for this proof but, since $g(t) \rightarrow 0$ as $t \rightarrow 0$, it seems natural to define $g(0)$ to be 0. If, now, we replace t in (2.50) by k , where $k = v(x+h) - v(x)$, and substitute the right-hand side of (2.50) in (2.47), we obtain

$$(2.51) \quad \frac{f(x+h) - f(x)}{h} = \frac{k}{h} [g(k) + u'(y)],$$

a formula that is valid even if $k = 0$. When $h \rightarrow 0$ the quotient $k/h \rightarrow v'(x)$ and $g(k) \rightarrow 0$ so the right-hand side of (2.51) approaches the limit $u'(y) \cdot v'(x)$. This completes the proof of the chain rule.

2.18 Exercises

Composite functions. In Exercises 1 through 12, the functions f and g are defined by the formulas given. Unless otherwise noted, the domains of f and g consist of all real numbers. Let $h(x) = f[g(x)]$ whenever $g(x)$ lies in the domain of f . In each case, describe the domain of h and give one or more formulas for determining $h(x)$.

- | | |
|---|------------------------------------|
| 1. $f(x) = x^2 - 2x$, | $g(x) = x + 1$. |
| 2. $f(x) = x + 1$, | $g(x) = x^2 - 2x$. |
| 3. $f(x) = \sqrt{x}$ if $x \geq 0$, | $g(x) = x^2$. |
| 4. $f(x) = \sqrt{x}$ if $x \geq 0$, | $g(x) = -x^2$. |
| 5. $f(x) = x^2$, | $g(x) = \sqrt{x}$ if $x \geq 0$. |
| 6. $f(x) = -x^2$, | $g(x) = \sqrt{x}$ if $x \geq 0$. |
| 7. $f(x) = \sin x$, | $g(x) = \sqrt{x}$ if $x \geq 0$. |
| 8. $f(x) = \sqrt{x}$ if $x \geq 0$, | $g(x) = \sin x$. |
| 9. $f(x) = x$, | $g(x) = x$. |
| 10. $f(x) = \cos x$, | $g(x) = \cos x$. |
| 11. $f(x) = \sqrt{x}$ if $x > 0$, | $g(x) = x + \sqrt{x}$ if $x > 0$. |
| 12. $f(x) = \sqrt{x + \sqrt{x}}$ if $x > 0$, | $g(x) = x + \sqrt{x}$ if $x > 0$. |
13. Let f and g be two functions defined as follows:

$$f(x) = \frac{x + |x|}{2} \quad \text{for all } x, \quad g(x) = \begin{cases} x & \text{for } x < 0, \\ x^2 & \text{for } x \geq 0. \end{cases}$$

Find a formula (or formulas) for computing

- | | |
|-----------------|-----------------|
| (a) $f[f(x)]$, | (c) $f[g(x)]$, |
| (b) $g[f(x)]$, | (d) $g[g(x)]$. |

14. Solve Exercise 13 when f and g are defined as follows:

$$f(x) = \begin{cases} 1 & \text{if } |x| \leq 1, \\ 0 & \text{if } |x| > 1, \end{cases} \quad g(x) = \begin{cases} 2 - x^2 & \text{if } |x| \leq 2, \\ 2 & \text{if } |x| > 2. \end{cases}$$

The chain rule. In Exercises 15 through 29, determine the derivative $f'(x)$.

15. $f(x) = \cos 2x - 2 \sin x$.
16. $f(x) = \sqrt{1 + x^2}$.
17. $f(x) = (2 - x^2) \cos x^2 + 2x \sin x^3$.
18. $f(x) = \sin(\cos^2 x) \cdot \cos(\sin^2 x)$.
19. $f(x) = \sin^n x \cdot \cos nx$.
20. $f(x) = \sin[\sin(\sin x)]$.
21. $f(x) = \tan \frac{x}{2} - \cot \frac{x}{2}$, $x \neq k\pi$, k an integer.
22. $f(x) = \frac{\sin^2 x}{\sin x^2}$, $x^2 \neq k\pi$, k an integer.
23. $f(x) = \sec^2 x + \csc^2 x$, $x \neq \frac{k\pi}{2}$, k an integer.
24. $f(x) = x\sqrt{1 + x^2}$.
25. $f(x) = \frac{x}{\sqrt{4 - x^2}}$, $|x| < 2$.
26. $f(x) = \left(\frac{1 + x^3}{1 - x^3} \right)^{1/3}$, $x^3 \neq 1$.
27. $f(x) = (1 + x)(2 + x^2)^{1/2}(3 + x^3)^{1/3}$, $x^3 \neq -3$.
28. $f(x) = \frac{1}{\sqrt{1 + x^2}(x + \sqrt{1 + x^2})}$.

29. $f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}$.

30. Determine the derivative $g'(x)$ in terms of $f'(x)$ if

(a) $g(x) = f(x^2)$.

(c) $g(x) = f[f(x)]$.

(b) $g(x) = f(\sin^2 x) + f(\cos^2 x)$.

(d) $g(x) = f\{f[f(x)]\}$.

31. Without attempting to evaluate the following indefinite integrals, find the derivative $f'(x)$ in each case if $f(x)$ is equal to

(a) $\int_0^x (1 + t^2)^{-3} dt$,

(b) $\int_0^{x^2} (1 + t^2)^{-3} dt$,

(c) $\int_{x^3}^{x^2} (1 + t^2)^{-3} dt$.

Related rates and implicit differentiation.

32. Each edge of a cube is expanding at the rate of 1 centimeter (cm) per second. How fast is the volume changing when the length of each edge is (a) 5 cm? (b) 10 cm? (c) x cm?

33. A reservoir has the shape of a right circular cone. The altitude is 10 feet, and the radius of the base is 4 feet. Water is poured into the reservoir at a constant rate of 5 cubic feet per minute. How fast is the water level rising when the depth of the water is 5 feet if (a) the vertex of the cone is up? (b) the vertex of the cone is down?

34. A baseball diamond is a 90-foot square. A ball is batted along the third-base line at a constant speed of 100 feet per second. How fast is its distance from first base changing when (a) it is halfway to third base? (b) it reaches third base?

35. A boat sails parallel to a straight beach at a constant speed of 12 miles per hour, staying 4 miles offshore. How fast is it approaching a lighthouse on the shoreline at the instant it is exactly 5 miles from the lighthouse?

36. A particle is constrained to move along a parabola whose equation is $y = x^2$. (a) At what point on the curve are the abscissa and the ordinate changing at the same rate? (b) Find this rate if the motion is such that at time t we have $x = \sin t$ and $y = \sin^2 t$.

37. The equation $x^3 + y^3 = 1$ defines y as one or more functions of x . (a) Assuming the derivative y' exists, and without attempting to solve for y , show that y' satisfies the equation $x^2 + y^2 y' = 0$. (b) Assuming the second derivative y'' exists, show that $y'' = -2xy^{-5}$ whenever $y \neq 0$.

38. If $0 < x < 5$, the equation $x^{1/2} + y^{1/2} = 5$ defines y as a function of x . Without solving for y , show that the derivative y' has a fixed sign. (You may assume the existence of y' .)

39. The equation $3x^2 + 4y^2 = 12$ defines y implicitly as two functions of x if $|x| \leq 4$. Assuming the second derivative y'' exists, show that it satisfies the equation $4y^3 y'' = -9$.

40. If $y = x^r$, where r is a rational number, say $r = m/n$, then $y^n = x^m$. Assuming the existence of the derivative y' , derive the formula $y' = rx^{r-1}$ using implicit differentiation and the corresponding formula for integer exponents.

2.19 Differentials

We mentioned earlier that Leibniz thought of a derivative as being a quotient of infinitesimal quantities, which he called differentials, and that this idea had to be discarded for lack of a satisfactory definition of the term "infinitesimal." In this section we shall give the modern definition of differentials and describe one way that they are used in practice. The basic idea underlying this discussion is the simple observation that a curve may be approximated by a collection of short line segments drawn tangent to the curve at each of its points, as suggested in Figure 2.17. To study the nature of this approximation quantitatively, we must investigate more closely the connection between a derivative and the difference quotient used to define it.



FIGURE 2.17 A curve approximated by a collection of short line segments tangent to it.

Suppose f has a derivative $f'(x)$ at a point x . This means that the difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

is nearly equal to $f'(x)$ when h is small or, what amounts to the same thing, the numerator of this quotient is approximately equal to the product $f'(x) \cdot h$. Suppose we write this as follows:

$$(2.52) \quad f(x+h) - f(x) \sim f'(x) \cdot h,$$

where $\sim$ can be read “is approximately equal to.” Figure 2.18 illustrates the geometric relationship between the difference $f(x+h) - f(x)$ and the product $f'(x) \cdot h$, and it suggests that these two quantities are nearly equal when h is small.

One way that an approximate formula like (2.52) is used in practice is as follows. Suppose the value of f and its derivative f' are known at a certain point, say at x , and suppose we want to find the value of f at a nearby point, say at $x+h$, using only the

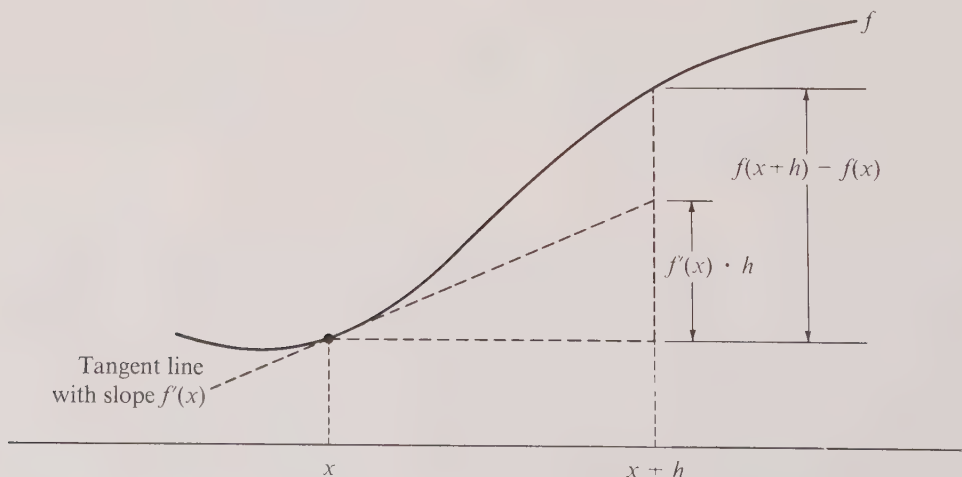


FIGURE 2.18 Illustrating the relationship between $f(x+h) - f(x)$ and $f'(x) \cdot h$.

information at our disposal about $f(x)$ and $f'(x)$. The approximation in (2.52) tells us that we have

$$(2.53) \quad f(x + h) \sim f(x) + f'(x) \cdot h.$$

In other words, to find an approximation to $f(x + h)$ we simply take the value of f at x and add to it the product $f'(x) \cdot h$. Geometrically this simply means that we are replacing the graph of f by its tangent line at the point $(x, f(x))$. It seems reasonable from Figure 2.18 that this line will lie very close to the curve in the immediate vicinity of this point.

Of course, an approximate formula such as (2.53) conveys no information whatever unless we also specify the *accuracy* of the approximation. The actual *error* made in approximating $f(x + h)$ by $f(x) + f'(x) \cdot h$ is defined to be the difference

$$(2.54) \quad \text{error} = f(x + h) - f(x) - f'(x) \cdot h.$$

We assert that this error is *small when compared with h* . By this we mean that the quotient

$$(2.55) \quad \frac{\text{error}}{h} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

In other words, the absolute value of the quotient, $|\text{error}/h|$, can be made less than any prescribed tolerance $\epsilon > 0$, by choosing h small enough. This follows immediately from the definition of $f'(x)$ because

$$\frac{\text{error}}{h} = \frac{f(x + h) - f(x) - f'(x) \cdot h}{h} = \frac{f(x + h) - f(x)}{h} - f'(x) \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Although the relation expressed in (2.55) provides some justification for us to expect the product $f'(x) \cdot h$ to be a good approximation to the difference $f(x + h) - f(x)$, it is not very helpful in actual practice because it does not tell us *how small* h should be to guarantee an error less than a prescribed tolerance. Moreover, in a specific problem, h is usually prescribed *first*, or else upper and lower bounds for h are given, and we want to know corresponding upper and lower bounds for the error associated with h . In general, this kind of information about the error requires that we know not only the numbers $f(x)$ and $f'(x)$, but also the second- or higher-order derivatives. We shall find in Chapter 7, when we study Taylor's formula, that the more knowledge we have about the higher derivatives the more we can say about the error as a function of h .

It may be instructive at this stage to look at an example which illustrates how the approximate formula in (2.52) may be used in practice. In this particular example we do not need an estimate for the error in (2.54) because we are able to compute the error explicitly as a function of h .

Example. Suppose a piece of laboratory equipment in the form of a cube comes from the manufacturer with the guarantee that each of its edges has length 5 centimeters with a tolerance of ± 0.001 cm. An experimenter wishes to know the tolerance limits that apply to the volume of this cube.

Analysis. The extreme values of the volume occur when all the edges are as large as possible or all are as small as possible. Therefore, let us assume that all three edges are equal, say each edge

has length $5 + h$ centimeters. The exact value of h is *unknown*, but the manufacturer guarantees that $|h| \leq 0.001$. The exact volume of this cube, expressed in terms of h , is

$$(5 + h)^3 = 5^3 + 3 \cdot 5^2h + 3 \cdot 5h^2 + h^3 = 125 + 75h + 15h^2 + h^3.$$

When we compare this with $5^3 = 125$, the difference is

$$(5 + h)^3 - 5^3 = 75h + 15h^2 + h^3.$$

Therefore, to obtain the tolerance limits that apply to the volume we must find the largest and smallest values taken on by the cubic polynomial $75h + 15h^2 + h^3$ in the interval $-0.001 \leq h \leq 0.001$. These extreme values occur at the endpoints of the interval in this case. Therefore, if we evaluate this cubic for $h = 0.001$ and $h = -0.001$, we find that the actual volume lies somewhere between

$$125 + 0.075015001 \quad \text{and} \quad 125 - 0.074985001.$$

Now let us compare this result with what we get by using the approximate formula in (2.52). Since we are interested only in an approximation, we make the added assumption of equal edges. With $f(x) = x^3$ and $x = 5$, the product $f'(x) \cdot h$ is $3x^2h = 75h$, and (2.52) states that

$$(5 + h)^3 - 5^3 \sim 75h.$$

The extreme values of $75h$ in the interval $-0.001 \leq h \leq 0.001$ occur at the endpoints (since $75h$ is a linear function of h) and they yield the tolerance limits ± 0.075 . Note that, in this particular interval, the linear term $75h$ is a very good approximation to the cubic $75h + 15h^2 + h^3$, the actual error being $15h^2 + h^3$.

This example is more or less typical of what happens in general. When we study Taylor's formula, we shall find that the error in (2.54) can be expressed in terms of h^2 and higher powers of h . The coefficients of these powers involve the higher derivatives $f''(x)$, $f'''(x)$, etc.

The product $f'(x) \cdot h$ which is used to approximate the difference $f(x + h) - f(x)$ is called the *differential of f at x with increment h* and will be denoted here by the symbol $df(x, h)$. Thus, by definition,

$$(2.56) \quad df(x, h) = f'(x) \cdot h.$$

We can interpret this equation as a means of constructing a new function df (the differential of f) from the given function f . This new function is an example of a *function of two variables*. To find its values we must be given both x and h . The number x must be a point where $f'(x)$ exists, whereas the increment h can be any real number whatever.

The difference $f(x + h) - f(x)$ is sometimes called the *increment of the function at x* , and it is denoted by $\Delta f(x, h)$. The approximate formula in (2.52) can now be written as follows:

$$\Delta f(x, h) \sim df(x, h),$$

the exact meaning of this being that

$$\frac{\Delta f(x, h) - df(x, h)}{h} \rightarrow 0 \quad \text{as} \quad h \rightarrow 0.$$

To fit this in with the Leibniz notation it is customary to denote the increment h by the symbol dx . Also, when y is written for $f(x)$, the symbol dy is used instead of $df(x, h)$.

In this notation, Equation (2.56) becomes

$$(2.57) \quad dy = f'(x) dx.$$

For example, if $y = \sin x$ then $dy = \cos x dx$ and if $y = x^n$ then $dy = nx^{n-1} dx$. From (2.57) it follows that $dy/dx = f'(x)$, whenever† the number $dx \neq 0$. Equation (2.57) may be written without reference to y as follows: $df(x) = f'(x) dx$. Thus we have $d(\sin x) = \cos x dx$ and $d(x^n) = nx^{n-1} dx$.

Sometimes the symbol d/dx is used in the same way as the differentiation operator D mentioned earlier. For example, the formula $D \sin x = \cos x$ is also written as

$$\frac{d}{dx} \sin x = \cos x \quad \text{or} \quad \frac{d(\sin x)}{dx} = \cos x,$$

both of which are read as “the derivative with respect to x of sine x is cosine x .” Again, it makes no difference whether we think of each of the left members of these equations as a quotient of differentials, $d(\sin x)$ divided by dx , or as a formula in which d/dx is used in the same way as the operator D .

Derivatives of higher order, $D^2, D^3, \dots, D^n$, are written with the symbols $d^2/dx^2, d^3/dx^3, \dots, d^n/dx^n$. For example, we have

$$\frac{d^2(\sin x)}{dx^2} = \frac{d}{dx} \frac{d(\sin x)}{dx} = \frac{d(\cos x)}{dx} = -\sin x, \quad \frac{d^3(\sin x)}{dx^3} = \frac{d}{dx} (-\sin x) = -\cos x.$$

2.20 Exercises

In Exercises 1, 2, and 3, let $\Delta f(x, h) = f(x + h) - f(x)$ and let $df(x, h) = f'(x) \cdot h$. Compute both the increment $\Delta f(x, h)$ and the differential $df(x, h)$ for the values of x and h specified.

1. $f(x) = x^3 - 2x + 1$.
(a) $x = 1, h = 1$; (b) $x = 1, h = 0.1$; (c) $x = 1, h = 0.01$.
2. $f(x) = \sqrt{x}$.
(a) $x = 4, h = -0.39$; (b) $x = 4, h = 0.04$.
3. $f(x) = 1/x$.
(a) $x = -1, h = -0.02$; (b) $x = 1, h = -0.02$.

In Exercises 4 through 7, α denotes a number whose value we want to estimate by means of differentials. This estimate is to be obtained as follows: First, choose a convenient function f and suitable values of x and h such that $\alpha = f(x + h)$. Then use the approximate formula $f(x + h) \sim f(x) + f'(x) \cdot h$ to estimate α . Try to choose f and x so that $f(x)$ is fairly easy to compute. You may test the accuracy of your approximations by comparing with the answers in the back of the book which give the value of α correct to four decimal places.

4. $\alpha = \sqrt[3]{1.02}$.
5. $\alpha = \sin 29^\circ$. [Hint. $x = \pi/6$; $h = -\pi/180$.]
6. $\alpha = \cos 151^\circ$.
7. $\alpha = (32.01)^{4/5}$.

† The symbol dy/dx was introduced earlier (in Section 2.13) as an alternative notation for the derivative $f'(x)$, without attaching a meaning to dy and dx separately. Now we have given individual meanings to dy and dx so that dy divided by dx is equal to the derivative $f'(x)$.

8. Derive the following approximate formula for computing square roots by using the differential of a suitable function as an approximation to its increment:

$$\sqrt{a^2 + b} \sim a + \frac{b}{2a},$$

where $a > 0$ and $|b|$ is "small" compared with a . Use this formula to obtain approximate values for (a) $\sqrt{3}$, (b) $\sqrt{34}$, (c) $\sqrt{120}$, (d) $\sqrt{1.45}$. [Four-place tables yield (a) 1.7321, (b) 5.8310, (c) 10.9545, (d) 1.2042.]

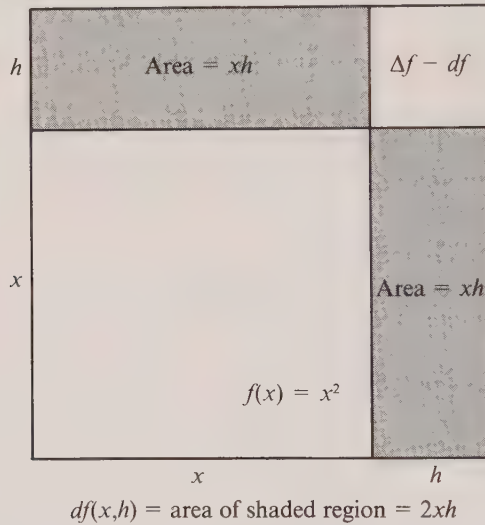


FIGURE 2.19 Exercise 10(a).

9. Same as Exercise 8 for n th roots:

$$\sqrt[n]{a^n + b} \sim a + \frac{b}{na^{n-1}}.$$

Use this to find approximate values for (a) $\sqrt[3]{29}$, (b) $\sqrt[4]{80}$, (c) $\sqrt[7]{100}$, (d) $\sqrt[10]{1000}$. [Four-place tables yield (a) 3.0723, (b) 2.9905, (c) 1.9309, (d) 1.9953.]

10. In this problem it is to be understood that the differential of a function is to be used as an approximation of its increment. Determine the approximate change in

- the area of a square due to a change h of each side.
- the volume of a cube due to a change h of each side.
- the area of a circle due to a change h of its radius.
- the area of a sphere due to a change h of its radius.
- the volume of a sphere due to a change h of its radius.
- the volume of a cylinder due to a change h of its radius.
- the volume of a cylinder due to a change h of its altitude.

In each case interpret this approximation geometrically. For example, in part (a) the differential is represented by the area of the shaded region in Figure 2.19.

★2.21 Miscellaneous exercises on differentiation and integration

1. Let f be a polynomial such that $f(0) = 1$ and let $g(x) = x^n f(x)$. Compute $g(0), g'(0), \dots, g^{(n)}(0)$.
2. Find a polynomial P of degree ≤ 5 such that $P(0) = 1, P(1) = 2, P'(0) = P''(0) = P'(1) = P''(1) = 0$.
3. If f is a polynomial of degree n with n distinct real roots $x_1, \dots, x_n$, show that $f'(x_k) \neq 0$ for $k = 1, 2, \dots, n$. [Hint. First show that $f(x) = (x - x_k)g_k(x)$, where $g_k(x_k) \neq 0$.]
4. Let $x_1, x_2, \dots, x_n$ be n distinct real numbers and let $y_1, y_2, \dots, y_n$ be n further numbers (not necessarily distinct). Define

$$A(x) = \prod_{i=1}^n (x - x_i) \quad \text{and} \quad A_k(x) = \prod_{\substack{i=1 \\ i \neq k}}^n (x - x_i).$$

In other words, $A_k(x)$ is the product of all factors $(x - x_i)$ except $(x - x_k)$. Note that $A_k(x)$ is a polynomial of degree $n - 1$ in x such that

$$A_k(x_i) = 0 \quad \text{if} \quad k \neq i.$$

- (a) Show that $A_k(x_k) = A'(x_k)$ for each $k = 1, 2, \dots, n$.
- (b) Let

$$f(x) = \sum_{k=1}^n \frac{y_k A_k(x)}{A'(x_k)}.$$

Show that f is a polynomial of degree $\leq n - 1$ such that $f(x_k) = y_k$ for each $k = 1, 2, \dots, n$. The formula for f is called *Lagrange's interpolation formula*. It tells us how to construct a polynomial of degree $\leq n - 1$ having assigned values $y_1, \dots, y_n$ at n given points $x_1, \dots, x_n$.

(c) Show that there is at most one polynomial having the properties in part (b). [Hint. A polynomial of degree m cannot have more than m roots.]

5. With the notation of Exercise 4, use Lagrange's interpolation formula to establish the following formulas. Part (c) of Exercise 4 will be of help.

$$(a) \sum_{k=1}^n A_k(x) = A'(x), \quad (b) \sum_{k=1}^n \frac{A_k(x)}{A'(x_k)} = 1, \quad (c) \sum_{k=1}^n \frac{x_k A_k(x)}{A'(x_k)} = x.$$

6. If $f(x) = \cos x$ and $g(x) = \sin x$, prove that

$$f^{(n)}(x) = \cos(x + \tfrac{1}{2}n\pi) \quad \text{and} \quad g^{(n)}(x) = \sin(x + \tfrac{1}{2}n\pi).$$

7. If $h(x) = f(x)g(x)$, prove that the n th derivative of h is given by the formula

$$h^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x),$$

where $\binom{n}{k}$ denotes the binomial coefficient. This is called *Leibniz's formula*.

8. If $f(x) = (1 - \sqrt{x})/(1 + \sqrt{x})$ for $x > 0$, find formulas for $Df(x)$, $D^2f(x)$, and $D^3f(x)$.

9. Given two functions f and g whose derivatives f' and g' satisfy the equations

$$(i) \quad f'(x) = g(x), \quad g'(x) = -f(x), \quad f(0) = 0, \quad g(0) = 1,$$

for every x in some open interval J containing 0. For example, these equations are satisfied when $f(x) = \sin x$ and $g(x) = \cos x$.

- (a) Prove that $f^2(x) + g^2(x) = 1$ for every x in J .
 (b) Let F and G be another pair of functions satisfying (i). Prove that $F(x) = f(x)$ and $G(x) = g(x)$ for every x in J . [Hint. Consider $h(x) = [F(x) - f(x)]^2 + [G(x) - g(x)]^2$.]
 (c) What more can you say about functions f and g satisfying (i)?
10. A function f is defined as follows:

$$f(x) = \begin{cases} x^2 & \text{if } x \leq c, \\ ax + b & \text{if } x > c, \end{cases} \quad (a, b, c \text{ constants}).$$

Find values of a and b (in terms of c) such that $f'(c)$ exists.

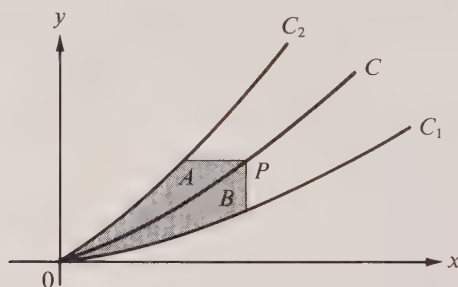


FIGURE 2.20 Exercise 16.

11. Solve Exercise 10 when f is defined as follows:

$$f(x) = \begin{cases} \frac{1}{|x|} & \text{if } |x| > c, \\ a + bx^2 & \text{if } |x| \leq c. \end{cases}$$

12. A function f , defined for all positive real numbers, satisfies the equation $f(x^2) = x^3$ for every $x > 0$. Determine $f'(4)$.
 13. A function g , defined for all positive real numbers, satisfies the following two conditions: $g(1) = 1$ and $g'(x^2) = x^3$ for all $x > 0$. Compute $g(4)$.
 14. Show that

$$\int_0^x \frac{\sin t}{t+1} dt \geq 0 \quad \text{for all } x \geq 0.$$

[Hint. Interpret the result geometrically and note that $\sin(t + \pi) = -\sin t$.]

15. Let $F(x) = \int_0^{g(x)} t^2 dt$, where g is a function having a derivative everywhere.

(a) Prove that $F'(x) = [g(x)]^2 g'(x)$.

(b) Generalize the result of part (a) if $F(x) = \int_0^{g(x)} f(t) dt$.

(c) Generalize further for integrals of the form $F(x) = \int_{h(x)}^{g(x)} f(t) dt$.

16. Let C_1 and C_2 be two curves passing through the origin as indicated in Figure 2.20. A curve C is said to "bisect in area" the region between C_1 and C_2 if, for each point P of C , the two shaded regions A and B shown in the figure have equal areas. Determine the upper curve C_2 , given that the bisecting curve C has the equation $y = x^2$ and that the lower curve C_1 has the equation $y = \frac{1}{2}x^2$.

17. A function f is defined for all x as follows:

$$f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

Let $Q(h) = f(h)/h$ if $h \neq 0$. (a) Prove that $Q(h) \rightarrow 0$ as $h \rightarrow 0$. (b) Prove that f has a derivative at 0, and compute $f'(0)$.

18. Suppose that, instead of the usual definition of the derivative $Df(x)$, we define a new kind of derivative, $D^*f(x)$, by the formula

$$D^*f(x) = \lim_{h \rightarrow 0} \frac{f^2(x+h) - f^2(x)}{h},$$

where $f^2(x)$ means $[f(x)]^2$.

(a) Derive formulas for computing the derivative D^* of a sum, difference, product, and quotient.

(b) Express $D^*f(x)$ in terms of $Df(x)$.

(c) For what functions does $D^*f = Df$?

19. If f is integrable on $[a, b]$ and $a \leq c \leq b$, let $A(x) = \int_c^x f(t) dt$. Prove that A is continuous at each point of $[a, b]$. In other words, prove that an indefinite integral of an integrable function is a continuous function. [Hint. Express $A(x+h) - A(x)$ as an integral and use the comparison theorem to obtain an inequality of the form $|A(x+h) - A(x)| \leq M|h|$, where M is a constant.]

20. A sequence of polynomials (called the *Bernoulli polynomials*) is defined by induction as follows:

$$P_0(x) = 1; \quad P'_n(x) = nP_{n-1}(x) \quad \text{and} \quad \int_0^1 P_n(x) dx = 0 \quad \text{if } n \geq 1.$$

(a) Determine explicit formulas for $P_1(x)$, $P_2(x)$, $\dots$, $P_5(x)$.

(b) Prove, by induction, that $P_n(x)$ is a polynomial in x of degree n , the term of highest degree being x^n .

(c) Prove that $P_n(0) = P_n(1)$ if $n \geq 2$.

(d) Prove that $P_n(x+1) - P_n(x) = nx^{n-1}$ if $n \geq 1$.

(e) Prove that for $n \geq 2$ we have

$$\sum_{r=1}^k r^n = \int_0^k P_n(x) dx = \frac{P_{n+1}(k) - P_{n+1}(0)}{n+1}.$$

(f) Prove that $P_n(1-x) = (-1)^n P_n(x)$ if $n \geq 1$.

(g) Prove that $P_{2n+1}(0) = 0$ and $P_{2n-1}(\frac{1}{2}) = 0$ if $n \geq 1$.

2.22 The Leibniz notation for primitives

Every differentiation formula can be rewritten as an equation involving differentials, as illustrated by the examples at the end of Section 2.19. Offhand there seems to be no particular virtue in writing $d(\sin x) = \cos x dx$ instead of $D \sin x = \cos x$. However, as we shall learn in the next few sections, the differential notation is a great help in developing skills for finding primitives.

If P is a primitive of f , then we may write $P'(x) = f(x)$ or, in terms of differentials, $dP(x) = f(x) dx$. If we know one primitive P of f , then any other primitive is given by $P + C$, where C is a suitable constant. Leibniz developed a symbol for primitives which combines the differential notation with the integral sign. He used the symbol $\int f(x) dx$ to denote a general primitive of f . In this notation, an equation like

$$(2.58) \quad \int f(x) \, dx = P(x) + C$$

is considered to be merely an alternative way of writing $P'(x) = f(x)$ or $dP(x) = f(x) \, dx$. For example, since $d(\sin x) = \cos x \, dx$ we may write

$$(2.59) \quad \int \cos x \, dx = \sin x + C.$$

The symbol C represents an arbitrary constant so each of Equations (2.58) and (2.59) is really a statement about a whole family of functions.

Despite similarity in appearance, the symbol $\int f(x) \, dx$ is conceptually distinct from the integration symbol $\int_a^b f(x) \, dx$. The symbols originate from two entirely different processes—differentiation and integration. Since, however, the two processes are related by the fundamental theorems of calculus, there are corresponding relationships between the two symbols.

The first fundamental theorem states that any indefinite integral of f is also a primitive of f . Therefore we may replace $P(x)$ in Equation (2.58) by $\int_{x_0}^x f(t) \, dt$ for some lower limit x_0 and write (2.58) as follows:

$$(2.60) \quad \int f(x) \, dx = \int_{x_0}^x f(t) \, dt + C.$$

This means that we can think of the symbol $\int f(x) \, dx$ as representing some indefinite integral of f , plus a constant.

The second fundamental theorem states that, for any primitive P of f and for any constant C , we have

$$\int_a^b f(x) \, dx = [P(x) + C] \Big|_a^b.$$

If we replace $P(x) + C$ by $\int f(x) \, dx$, this formula may be written in the form

$$(2.61) \quad \int_a^b f(x) \, dx = \int f(x) \, dx \Big|_a^b.$$

The two formulas in (2.60) and (2.61) may be thought of as symbolic expressions of the first and second fundamental theorems of calculus.

Because of long historical usage, many calculus textbooks refer to the symbol $\int f(x) \, dx$ as an “indefinite integral” rather than as a primitive or an antiderivative. This is justified, in part, by Equation (2.60), which tells us that the symbol $\int f(x) \, dx$ is, apart from an additive constant C , an indefinite integral of f . For the same reason, many handbooks of mathematical tables contain extensive lists of formulas labeled “tables of indefinite integrals” which, in reality, are tables of primitives. To distinguish the symbol $\int f(x) \, dx$ from $\int_a^b f(x) \, dx$, the latter is called a *definite* integral. Since the second fundamental theorem reduces the problem of integration to that of finding a primitive, the term “technique of integration” is used to refer to any systematic method for finding primitives. This terminology is widely used in the mathematical literature and it will be adopted also in this book. Thus, for example, when one is asked to “integrate” $\int f(x) \, dx$, it is to be understood that what is wanted is the most general primitive of f .

There are three principal techniques that are used to construct tables of indefinite

integrals and they should be learned by anyone who desires a good working knowledge of calculus. They are (1) *integration by substitution* (to be described in the next section), a method based on the chain rule; (2) *integration by parts*, a method based on the formula for differentiating a product (to be described in Section 2.25); and (3) *integration by partial fractions*, an algebraic technique which is discussed at the end of Chapter 3. These techniques not only explain how tables of indefinite integrals are constructed but also they tell us how certain formulas are converted to the basic forms listed in the tables.

Note. Leibniz's notation enables us to summarize the fundamental theorems of calculus in two very brief symbolic formulas. First of all, if we take the differential of both sides of the equation $\int f(x) dx = P(x) + C$, we obtain

$$d \int f(x) dx = d[P(x) + C] = P'(x) dx = f(x) dx .$$

If, now, we divide by dx , we find

$$(2.62) \quad \frac{d[\int f(x) dx]}{dx} = f(x) .$$

This can be interpreted to mean that integration of a function f , when followed by differentiation, leads back to the original function f . This, of course, is one interpretation of the first fundamental theorem of calculus.

To express the second fundamental theorem symbolically, we replace $f(x)$ by $P'(x)$ in the equation $\int f(x) dx = P(x) + C$ to obtain $\int P'(x) dx = P(x) + C$. Replacing $P'(x) dx$ by the differential $dP(x)$, we obtain

$$(2.63) \quad \int dP(x) = P(x) + C .$$

This formula can be interpreted as follows: If we operate on a function P by differentiation and then operate on the result by integration, we obtain P once again, plus a constant. This is one way in which the second fundamental theorem can be expressed.

2.23 Integration by substitution

If we know how to compute the derivative of $P(x)$, then the chain rule tells us that the derivative of the composition $h(x) = P[g(x)]$ is $h'(x) = P'[g(x)] g'(x)$. Suppose now that P is a primitive of another function f . Then $P'(x) = f(x)$, so $P'[g(x)] = f[g(x)]$, and hence $h'(x) = f[g(x)]g'(x)$. In other words, if we have

$$(2.64) \quad \int f(x) dx = P(x) + C ,$$

then we also have

$$(2.65) \quad \int f[g(x)]g'(x) dx = P[g(x)] + C .$$

For example, if $f(x) = \cos x$ and $g(x) = x^3$, then (2.65) becomes

$$(2.66) \quad \int \cos x^3 \cdot 3x^2 dx = \sin x^3 + C ,$$

a result that is easily verified directly because the derivative of $\sin x^3$ is $3x^2 \cos x^3$. [Here $\sin x^3$ means $\sin (x^3)$.]

The general formula in (2.65) can be derived from the special case in (2.64) by a simple mechanical process. Suppose we replace $g(x)$ everywhere in (2.65) by a new symbol u and replace $g'(x) dx$ by du . Then the general formula in (2.65) becomes

$$(2.67) \quad \int f(u) du = P(u) + C.$$

Notice that this has exactly the same form as (2.64) except that the symbol u appears everywhere instead of x . In other words, every integration formula such as (2.64) can be made to yield a more general integration formula if we simply substitute symbols. We replace x in (2.64) by a new symbol u to obtain (2.67) and then we think of u as representing a new function of x , say $u = g(x)$. Also, we think of the symbol du as representing the differential of this new function, $du = g'(x) dx$, and Equation (2.67) reduces to the more general formula which appears in (2.65).

For example, if we replace x by u in the formula $\int \cos x dx = \sin x + C$, we obtain

$$(2.68) \quad \int \cos u du = \sin u + C.$$

In this latter formula, u may be replaced by any function of x having a derivative, and a correct integration formula results, provided du is replaced by the differential of this function. Thus, if we let $u = x^3$, then $du = 3x^2 dx$ and (2.68) becomes (2.66).

When this mechanical process is used *in reverse*, it leads to the so-called method of *integration by substitution*. The object of the method is to transform an integral with a complicated integrand, such as $\int x^2 \cos x^3 dx$, into a more familiar integral, such as $\int \cos u du$. The method is applicable whenever the original integral can be written in the form $\int f[g(x)]g'(x) dx$, since the substitution $u = g(x)$, $du = g'(x) dx$ transforms this to $\int f(u) du$. If we can succeed in carrying out the integration indicated by $\int f(u) du$, we obtain a primitive, say $P(u)$, and then the original integral may be evaluated by replacing u by $g(x)$ in the formula for $P(u)$.

Success in this method depends on one's ability to determine at the outset which part of the integrand should be replaced by the symbol u , and this ability comes from a lot of experience in working out specific examples. The following specially selected examples illustrate how the method is carried out in actual practice.

Example 1. Integrate $\int x^3 \cos x^4 dx$.

Solution. Let us keep in mind that we are trying to write $x^3 \cos x^4$ in the form $f[g(x)]g'(x)$ with a suitable choice of f and g . Since $\cos x^4$ is a composition, this suggests that we take $f(x) = \cos x$ and $g(x) = x^4$ so that $\cos x^4$ becomes $f[g(x)]$. This choice of g gives $g'(x) = 4x^3$ and hence $f[g(x)]g'(x) = (\cos x^4)(4x^3)$. The extra factor 4 is easily taken care of by multiplying and dividing the integrand by 4. Thus we have

$$x^3 \cos x^4 = \frac{1}{4}(\cos x^4)(4x^3) = \frac{1}{4}f[g(x)]g'(x).$$

Now, we make the substitution $u = g(x) = x^4$, $du = g'(x) dx = 4x^3 dx$, and obtain

$$\int x^3 \cos x^4 dx = \frac{1}{4} \int f(u) du = \frac{1}{4} \int \cos u du = \frac{1}{4} \sin u + C.$$

Replacing u by x^4 in the end result, we obtain the formula

$$\int x^3 \cos x^4 dx = \frac{1}{4} \sin x^4 + C,$$

which can be verified directly by differentiation.

After a little practice one can perform some of the above steps mentally, and the entire calculation can be given more briefly as follows:

Let $u = x^4$. Then $du = 4x^3 dx$, and we obtain

$$\int x^3 \cos x^4 dx = \frac{1}{4} \int (\cos x^4) (4x^3 dx) = \frac{1}{4} \int \cos u du = \frac{1}{4} \sin u + C = \frac{1}{4} \sin x^4 + C.$$

Notice that the method works in this example because the factor x^3 has an exponent exactly one less than the power of x which appears in $\cos x^4$.

Example 2. Integrate $\int \cos^2 x \sin x dx$.

Solution. Let $u = \cos x$. Then $du = -\sin x dx$ and we get

$$\int \cos^2 x \sin x dx = - \int (\cos x)^2 (-\sin x dx) = - \int u^2 du = -\frac{u^3}{3} + C = -\frac{\cos^3 x}{3} + C.$$

Again, the final result is easily verified by differentiation.

Example 3. Integrate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

Solution. Let $u = \sqrt{x} = x^{1/2}$. Then $du = \frac{1}{2}x^{-1/2} dx$, or $dx/\sqrt{x} = 2 du$.

Hence

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = 2 \int \sin u du = -2 \cos u + C = -2 \cos \sqrt{x} + C.$$

Example 4. Integrate $\int \frac{x dx}{\sqrt{1+x^2}}$.

Solution. Let $u = 1 + x^2$. Then $du = 2x dx$ so $x dx = \frac{1}{2} du$, and we obtain

$$\int \frac{x dx}{\sqrt{1+x^2}} = \frac{1}{2} \int \frac{du}{\sqrt{u}} = \frac{1}{2} \int u^{-1/2} du = u^{1/2} + C = \sqrt{1+x^2} + C.$$

The method of substitution is, of course, also applicable to definite integrals. For example, to evaluate the definite integral $\int_0^{\pi/2} \cos^2 x \sin x dx$ we first determine the indefinite integral, as explained in Example 2, and then we use the second fundamental theorem to write

$$\int_0^{\pi/2} \cos^2 x \sin x dx = -\frac{1}{3} \cos^3 x \Big|_0^{\pi/2} = -\frac{1}{3} \left(\cos^3 \frac{\pi}{2} - \cos^3 0 \right) = \frac{1}{3}.$$

Sometimes it is desirable to apply the second fundamental theorem directly to the integral expressed in terms of u . This may be done by using new limits of integration. We shall illustrate how this is carried out in a particular example, and then we shall justify the process with a general theorem.

Example 5. Evaluate $\int_2^3 \frac{(x+1) dx}{\sqrt{x^2+2x+3}}$.

Solution. Let $u = x^2 + 2x + 3$. Then $du = (2x + 2) dx$, so

$$\frac{(x + 1) dx}{\sqrt{x^2 + 2x + 3}} = \frac{1}{2} \frac{du}{\sqrt{u}}.$$

Now we obtain new limits of integration by noting that $u = 11$ when $x = 2$, and that $u = 18$ when $x = 3$. Then we write

$$\int_2^3 \frac{(x + 1) dx}{\sqrt{x^2 + 2x + 3}} = \frac{1}{2} \int_{11}^{18} u^{-1/2} du = \sqrt{u} \Big|_{11}^{18} = \sqrt{18} - \sqrt{11}.$$

The same result is arrived at by expressing everything in terms of x , thus:

$$\int_2^3 \frac{(x + 1) dx}{\sqrt{x^2 + 2x + 3}} = \sqrt{x^2 + 2x + 3} \Big|_2^3 = \sqrt{18} - \sqrt{11}.$$

There is a general theorem which justifies the process used in Example 5. This theorem is concerned with two functions g and f related as follows: The function g is assumed to have a derivative g' which is continuous on an interval $[c, d]$. We denote by T the set of values taken on by g on the interval $[c, d]$ and we assume that f is continuous at each point† of T . This permits us to form the composition $f \circ g$ and prove that it is continuous‡ on $[c, d]$. Let $a = g(c)$, $b = g(d)$, and define two new functions F and G as follows:

$$F(x) = \int_a^x f(u) du \quad \text{if } x \text{ is in } T,$$

$$G(x) = \int_c^x f[g(t)]g'(t) dt \quad \text{if } x \text{ is in } [c, d].$$

The theorem in question asserts that $G(x) = F[g(x)]$; that is,

$$(2.69) \quad \int_c^x f[g(t)]g'(t) dt = \int_{g(c)}^{g(x)} f(u) du \quad \text{if } c \leq x \leq d.$$

The proof of the theorem is very simple. Since both F and G are indefinite integrals of continuous functions, they have derivatives given by

$$F'(x) = f(x), \quad G'(x) = f[g(x)]g'(x).$$

Now, let $H(x) = F[g(x)]$. Then by the chain rule we have

$$H'(x) = F'[g(x)]g'(x) = f[g(x)]g'(x) = G'(x),$$

hence $H(x) - G(x)$ is a constant. When $x = c$ we get $H(c) - G(c) = F[g(c)] - 0 = F(a) = 0$, so that $H(x) = G(x)$ for all x in $[c, d]$. This proves the theorem.

† If g is not a constant, the set T will be an *interval*. This follows from general properties of continuous functions which are proved in Section 2.30.

‡ A proof that $f \circ g$ is continuous is outlined in Exercise 5 in Section 2.31.

2.24 Exercises

In Exercises 1 through 20, evaluate the integrals by the method of substitution.

1. $\int \sqrt{2x+1} \, dx.$

11. $\int \frac{\sin x \, dx}{\sqrt{\cos^3 x}}.$

2. $\int x \sqrt{1+3x} \, dx.$

12. $\int_3^8 \frac{\sin \sqrt{x+1} \, dx}{\sqrt{x+1}}.$

3. $\int x^2 \sqrt{x+1} \, dx.$

13. $\int x^{n-1} \sin x^n \, dx, \quad n \neq 0.$

4. $\int_{-2/3}^{1/3} \frac{x \, dx}{\sqrt{2-3x}}.$

14. $\int \frac{x^5 \, dx}{\sqrt{1-x^6}}.$

5. $\int \frac{(x+1) \, dx}{(x^2+2x+2)^3}.$

15. $\int t(1+t)^{1/4} \, dt.$

6. $\int \sin^3 x \, dx.$

16. $\int (x^2+1)^{-3/2} \, dx.$

7. $\int z(z-1)^{1/3} \, dz.$

17. $\int x^2(8x^3+27)^{2/3} \, dx.$

8. $\int \frac{\cos x \, dx}{\sin^3 x}.$

18. $\int \frac{(\sin x + \cos x) \, dx}{(\sin x - \cos x)^{1/3}}.$

9. $\int_0^{\pi/4} \cos 2x \sqrt{4 - \sin 2x} \, dx.$

19. $\int \frac{x \, dx}{\sqrt{1+x^2} \sqrt{(1+x^2)^3}}.$

10. $\int \frac{\sin x \, dx}{(3 + \cos x)^2}.$

20. $\int \frac{(x^2+1-2x)^{1/5} \, dx}{1-x}.$

21. Deduce the formulas in Theorems 1-45 and 1-46 (p. 60) by the method of substitution.

22. Let

$$F(x, a) = \int_0^x \frac{t^p}{(t^2 + a^2)^q} \, dt,$$

where $a > 0$, and p and q are positive integers. Show that $F(x, a) = a^{p+1-2q} F(x/a, 1)$.

23. Show that

$$\int_x^1 \frac{dt}{1+t^2} = \int_1^{1/x} \frac{dt}{1+t^2} \quad \text{if } x > 0.$$

24. Show that

$$\int_0^1 x^m (1-x)^n \, dx = \int_0^1 x^n (1-x)^m \, dx$$

if m and n are positive integers.

25. Show that

$$\int_0^{\pi/2} \cos^m x \sin^n x \, dx = 2^{-m} \int_0^{\pi/2} \cos^m x \, dx$$

if m is a positive integer.

26. (a) Show that

$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

[Hint. $u = \pi - x$.]

(b) Use part (a) to deduce the formula

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \pi \int_0^1 \frac{dx}{1 + x^2}.$$

27. Show that $\int_0^1 (1 - x^2)^{n-1/2} dx = \int_0^{\pi/2} \cos^{2n} u du$ if n is a positive integer. [Hint. $x = \sin u$.]

The integral on the right can be evaluated by the method of integration by parts, to be discussed in the next section.

2.25 Integration by parts

We proved earlier in this chapter that the derivative of a product of two functions f and g is given by the formula

$$h'(x) = f(x)g'(x) + f'(x)g(x),$$

where $h(x) = f(x)g(x)$. When this is translated into the Leibniz notation for primitives it becomes $\int f(x)g'(x) dx + \int f'(x)g(x) dx = f(x)g(x) + C$, usually written as follows:

$$(2.70) \quad \int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx + C.$$

This equation, known as the formula for *integration by parts*, provides us with a new integration technique.

To evaluate an integral, say $\int k(x) dx$, using (2.70), we try to find two functions f and g such that $k(x)$ can be written in the form $f(x)g'(x)$. If we can do this, then (2.70) tells us that we have

$$\int k(x) dx = f(x)g(x) - \int g(x)f'(x) dx + C$$

and the difficulty has been transferred to the evaluation of $\int g(x)f'(x) dx$. If f and g are properly chosen, this last integral may be easier to evaluate than the original one. Sometimes two or more applications of (2.70) will lead to an integral that is easily evaluated or that may be found in a table. The examples worked out below have been chosen to illustrate the advantages of this method. For definite integrals, (2.70) leads to the formula

$$\int_a^b f(x)g'(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x) dx.$$

If we write $u = f(x)$ and $v = g(x)$, then we have $du = f'(x) dx$ and $dv = g'(x) dx$ and the formula for integration by parts assumes an abbreviated form that many people find easier to remember, namely,

$$(2.71) \quad \int u dv = uv - \int v du + C.$$

Example 1. Integrate $\int x \cos x \, dx$.

Solution. We choose $f(x) = x$ and $g'(x) = \cos x$. This means that we have $f'(x) = 1$ and $g(x) = \sin x$ so (2.70) becomes

$$(2.72) \quad \int x \cos x \, dx = x \sin x - \int \sin x \, dx + C = x \sin x + \cos x + C.$$

Note that in this case the second integral is one we have already calculated.

If we wish to carry out the same calculation in the abbreviated notation of (2.71), we write

$$\begin{aligned} u &= x, & dv &= \cos x \, dx, \\ du &= dx, & v &= \int \cos x \, dx = \sin x, \end{aligned}$$

$$\int x \cos x \, dx = uv - \int v \, du = x \sin x - \int \sin x \, dx + C = x \sin x + \cos x + C.$$

Had we chosen $u = \cos x$ and $dv = x \, dx$, we would have obtained $du = -\sin x \, dx$, $v = \frac{1}{2}x^2$, and (2.71) would have given us

$$\int x \cos x \, dx = \frac{1}{2}x^2 \cos x - \frac{1}{2} \int x^2(-\sin x) \, dx + C = \frac{1}{2}x^2 \cos x + \frac{1}{2} \int x^2 \sin x \, dx + C.$$

Since the last integral is one which we have not yet calculated, this choice of u and v is not as useful as the first choice. Notice, however, that we can solve this last equation for $\int x^2 \sin x \, dx$ and use (2.72) to obtain

$$(2.73) \quad \int x^2 \sin x \, dx = 2x \sin x + 2 \cos x - x^2 \cos x + C.$$

Example 2. Integrate $\int x^2 \cos x \, dx$.

Solution. Let $u = x^2$ and $dv = \cos x \, dx$. Then $du = 2x \, dx$ and $v = \int \cos x \, dx = \sin x$, so we have

$$(2.74) \quad \int x^2 \cos x \, dx = \int u \, dv = uv - \int v \, du + C = x^2 \sin x - 2 \int x \sin x \, dx + C.$$

The last integral can be evaluated by applying integration by parts once more. Since it is similar to Example 1, we simply state the result:

$$\int x \sin x \, dx = -x \cos x + \sin x + C.$$

Substituting in (2.74) and consolidating the two arbitrary constants into one, we obtain

$$\int x^2 \cos x \, dx = x^2 \sin x + 2x \cos x - 2 \sin x + C.$$

Example 3. The method sometimes fails because it leads back to the original integral. For example, let us try to integrate $\int x^{-1} \, dx$ by parts. If we let $u = x$ and $dv = x^{-2} \, dx$, then $\int x^{-1} \, dx = \int u \, dv$. For this choice of u and v we have $du = dx$ and $v = -x^{-1}$, so (2.71) gives us

$$(2.75) \quad \int x^{-1} \, dx = \int u \, dv = uv - \int v \, du + C = -1 + \int x^{-1} \, dx + C$$

and we are back where we started. Moreover, the situation does not improve if we try $u = x^n$ and $dv = x^{-n-1}$.

This example is often used to illustrate the importance of paying attention to the arbitrary constant C . If formula (2.75) is written without C , it leads to the equation $\int x^{-1} \, dx = -1 + \int x^{-1} \, dx$, which is sometimes used to give a fallacious proof that $0 = -1$.

2.26 Exercises

Use integration by parts to evaluate the integrals in Exercises 1 through 4.

$$1. \int x \sin x \, dx. \qquad 3. \int x^2 \sin x \, dx.$$

$$2. \int x^3 \cos x \, dx. \qquad 4. \int x^3 \sin x \, dx.$$

5. Use integration by parts to show that

$$\int \sqrt{1-x^2} \, dx = x \sqrt{1-x^2} - \int \sqrt{1-x^2} \, dx + \int \frac{dx}{\sqrt{1-x^2}} + C.$$

6. Evaluate the integral $\int_{-1}^3 t^3(4+t^3)^{-1/2} \, dt$, given that $\int_{-1}^3 (4+t^3)^{1/2} \, dt = 11.35$. Leave the answer in terms of $\sqrt{3}$ and $\sqrt{31}$.

7. Use integration by parts to show that

$$(a) \int_0^{\pi/2} \sin^2 x \, dx = \frac{1}{2} \int_0^{\pi/2} dx = \frac{\pi}{4}.$$

$$(b) \int_0^{\pi/2} \sin^4 x \, dx = \frac{3}{4} \int_0^{\pi/2} \sin^2 x \, dx = \frac{3\pi}{16}.$$

$$(c) \int_0^{\pi/2} \sin^6 x \, dx = \frac{5}{6} \int_0^{\pi/2} \sin^4 x \, dx = \frac{5\pi}{32}.$$

8. Let $J_n(x) = \int_0^x \sin^n t \, dt$, where n is an integer. Use integration by parts to derive the recursion formula

$$nJ_n(x) = (n-1)J_{n-2}(x) - \sin^{n-1} x \cos x \quad (n \geq 2).$$

9. Let $I_n = \int_0^{\pi/2} \sin^n t \, dt$. Use Exercise 8 to deduce the formula

$$I_{n+2} = \frac{n+1}{n+2} I_n.$$

10. Referring to Exercise 9, deduce the following relations:

$$I_{2n} = \frac{(2n-1)(2n-3)\cdots 3}{2n(2n-2)\cdots 2} \cdot \frac{\pi}{2}, \quad I_{2n+1} = \frac{2n(2n-2)\cdots 2}{(2n+1)(2n-1)\cdots 3}.$$

11. Referring to Exercise 10, show that

$$(a) \frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \cdot \frac{I_{2n}}{I_{2n+1}}.$$

$$(b) (n+1)I_n I_{n+1} = \frac{\pi}{2} \text{ for all integers } n \geq 1.$$

12. Let $K_n = \int_0^{\pi/2} \cos^n x \, dx$. Use integration by parts to find a formula relating K_n to K_{n-2} , and apply this formula to integrate

$$\int_0^{\pi/2} \cos^2 x \, dx \quad \text{and} \quad \int_0^{\pi/2} \cos^4 x \, dx.$$

13. Use integration by parts to derive the formula

$$(n-1) \int \tan^n x \, dx = \tan^{n-1} x - (n-1) \int \tan^{n-2} x \, dx + C.$$

Apply the formula to integrate $\int \tan^2 x \, dx$ and $\int \tan^4 x \, dx$.

14. (a) Use integration by parts to derive the formula

$$\int (a^2 - x^2)^n dx = \frac{x(a^2 - x^2)^n}{2n+1} + \frac{2a^2n}{2n+1} \int (a^2 - x^2)^{n-1} dx + C.$$

(b) Use part (a) to evaluate $\int_0^a (a^2 - x^2)^{3/2} dx$.

15. (a) If $I_n(x) = \int_0^x t^n(t^2 + a^2)^{-1/2} dt$, use integration by parts to show that

$$nI_n(x) = x^{n-1} \sqrt{x^2 + a^2} - (n-1)a^2 I_{n-2}(x) \quad \text{if } n \geq 2.$$

(b) Use part (a) to show that $\int_0^2 x^5(x^2 + 5)^{-1/2} dx = 168/5 - 40\sqrt{5}/3$.

In Exercises 16 through 20 use integration by parts to derive the given formulas.

$$16. \int x^n \sqrt{ax+b} dx = \frac{2}{a(2n+3)} \left(x^n(ax+b)^{3/2} - nb \int x^{n-1} \sqrt{ax+b} dx \right) + C \quad (n \neq -\frac{3}{2}).$$

$$17. \int \frac{x^m}{\sqrt{a+bx}} dx = \frac{2}{(2m+1)b} \left(x^m \sqrt{a+bx} - ma \int \frac{x^{m-1}}{\sqrt{a+bx}} dx \right) + C \quad (m \neq -\frac{1}{2}).$$

$$18. \int \frac{dx}{x^n \sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{(n-1)bx^{n-1}} - \frac{(2n-3)a}{(2n-2)b} \int \frac{dx}{x^{n-1} \sqrt{ax+b}} + C \quad (n \neq 1).$$

$$19. \int \frac{\cos^m x}{\sin^n x} dx = \frac{\cos^{m-1} x}{(m-n)\sin^{n-1} x} + \frac{m-1}{m-n} \int \frac{\cos^{m-2} x}{\sin^n x} dx + C \quad (m \neq n).$$

$$20. \int \frac{\cos^m x}{\sin^n x} dx = -\frac{\cos^{m+1} x}{(n-1)\sin^{n-1} x} - \frac{m-n+2}{n-1} \int \frac{\cos^{m-2} x}{\sin^{n-2} x} dx + C \quad (n \neq 1).$$

$$[\text{Hint. } \int \frac{\cos^m x}{\sin^n x} dx = \int \frac{\cos^m x (\sin^2 x + \cos^2 x)}{\sin^n x} dx = \int \frac{\cos^m x}{\sin^{n-2} x} dx + \int \frac{\cos^{m+2} x}{\sin^n x} dx.]$$

21. Show that

$$\int \frac{\sqrt{a+bx}}{x} dx = 2\sqrt{a+bx} + a \int \frac{dx}{x\sqrt{a+bx}} + C.$$

In Exercises 22 through 29, evaluate the given integrals. Try to simplify the calculations by using the method of substitution and/or integration by parts whenever possible.

$$22. \int_{-2}^1 x(x^2 - 1)^9 dx.$$

$$26. \int_1^2 x^{-2} \sin \frac{1}{x} dx.$$

$$23. \int_0^1 \frac{2x+3}{(6x+7)^3} dx.$$

$$27. \int \sin \sqrt[4]{x-1} dx.$$

$$24. \int x^4(1+x^5)^5 dx.$$

$$28. \int x \sin x^2 \cos x^2 dx.$$

$$25. \int_0^1 x^4(1-x)^{20} dx.$$

$$29. \int \sqrt{1+3\cos^2 x} \sin 2x dx.$$

30. Let $I_n = \int_0^1 (1-x^2)^n dx$ (n a positive integer). Show that

$$I_n = \frac{2n}{2n+1} I_{n-1}.$$

Use this formula to evaluate I_2 , I_3 , I_4 , and I_5 .

31. Let $F(m, n) = \int_0^x t^m(1+t)^n dt$, $m > 0$, $n > 0$. Show that

$$(m+1)F(m, n) + nF(m+1, n-1) = x^{m+1}(1+x)^n.$$

Use this to evaluate $F(10, 2)$.

32. Let $f(n) = \int_0^{\pi/4} \tan^n x dx$ where $n \geq 1$. Show that

$$(a) f(n+1) < f(n).$$

$$(b) f(n) + f(n-2) = \frac{1}{n-1} \quad \text{if } n > 2.$$

$$(c) \frac{1}{n+1} < 2f(n) < \frac{1}{n-1} \quad \text{if } n > 2.$$

33. Compute $f(0)$, given that $f(\pi) = 2$ and that $\int_0^\pi [f(x) + f''(x)] \sin x dx = 5$.

34. Let A denote the value of the integral

$$\int_0^\pi \frac{\cos x}{(x+2)^2} dx.$$

Compute

$$\int_0^{\pi/2} \frac{\sin x \cos x}{x+1} dx$$

in terms of A .

2.27 Partial derivatives

In Chapter 1, a function was defined to be a correspondence which associates with each object in a collection X one and only one object in another collection Y ; the collection X is referred to as the *domain* of the function. Up to now, we have dealt with functions having a domain consisting of points on the x -axis. Such functions are usually called *functions of one real variable*. It is not difficult to extend many of the ideas of calculus to functions of two or more real variables.

By a *real-valued function of two real variables* we mean one whose domain X is a collection of points in the xy -plane. If f denotes such a function, its value at a point (x, y) is a real number, written $f(x, y)$. It is easy to imagine how such a function might arise in a fictitious physical problem. For example, suppose a flat metal plate in the shape of a circular disk of radius 4 centimeters is placed on the xy -plane, with the center of the disk at the origin and with the disk heated in such a way that its temperature at each point (x, y) is $16 - x^2 - y^2$ degrees centigrade. If we denote the temperature at (x, y) by $f(x, y)$, then f is a function of two variables defined by the equation

$$(2.76) \quad f(x, y) = 16 - x^2 - y^2.$$

The domain of this function is the set of all points (x, y) whose distance from the origin does not exceed 4. The theorem of Pythagoras tells us that all points (x, y) at a distance r from the origin satisfy the equation

$$(2.77) \quad x^2 + y^2 = r^2.$$

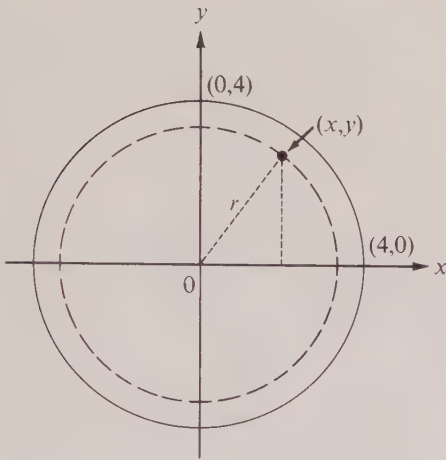


FIGURE 2.21 The temperature is constant on each circle with center at the origin.

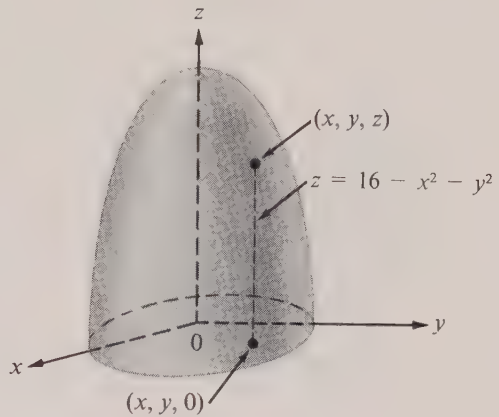


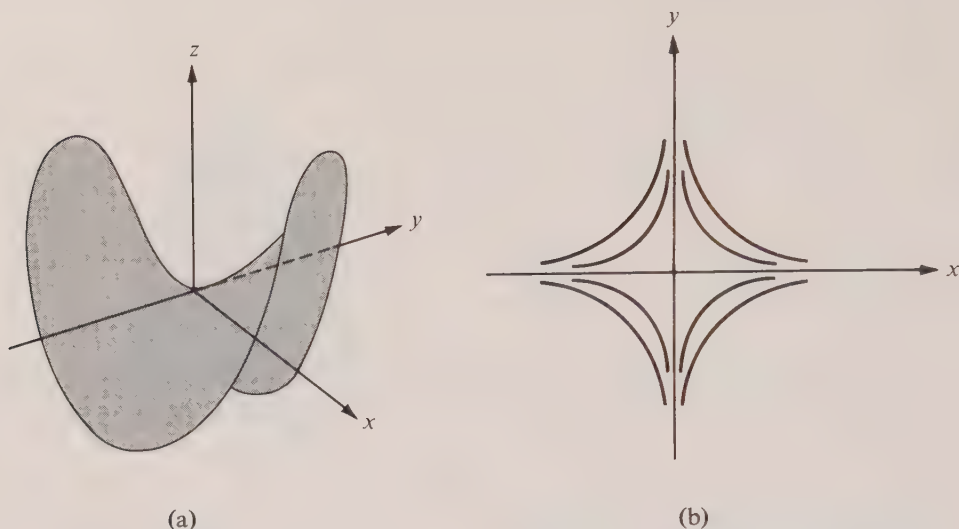
FIGURE 2.22 The surface represented by the equation $z = 16 - x^2 - y^2$.

Therefore the domain in this case consists of all points (x, y) which satisfy the inequality $x^2 + y^2 \leq 16$. Note that on the circle described by (2.77) the temperature is $f(x, y) = 16 - r^2$. That is, the function f is constant on each circle with center at the origin. (See Figure 2.21.)

There are two useful methods for obtaining a geometric picture of a function of two variables. One is by means of a *surface* in space. To construct this surface we introduce a third coordinate axis (called the z -axis); it passes through the origin and is perpendicular to the xy -plane. Above each point (x, y) we plot the point (x, y, z) whose z -coordinate is obtained from the equation $z = f(x, y)$.

The surface for the example described above is shown in Figure 2.22. If we placed a thermometer at a point (x, y) on the plate, the top of the mercury column would just touch the surface at the point (x, y, z) where $z = f(x, y)$ provided, of course, that unit distances on the z -axis are properly chosen.

A different kind of geometric picture of a function of two variables can be drawn entirely in the xy -plane. This is the method of *contour lines* that is used by map makers to represent a three-dimensional landscape by a two-dimensional drawing. We imagine that the surface described above has been cut by various horizontal planes (parallel to the xy -plane). They intersect the surface at those points (x, y, z) whose elevation z is constant. By projecting these points on the xy -plane we get a family of contour lines or *level curves*. Each level curve consists of those and only those points (x, y) whose coordinates satisfy the equation $f(x, y) = c$, where c is the constant elevation for that particular curve. In the example mentioned above, the level curves are concentric circles and they represent curves of constant temperature, or *isothermals*, as might be drawn on a weather map. Another example of a surface and its level curves is shown in Figure 2.23. The equation in this case is $z = xy$. The “saddle-shaped” surface is known as a *hyperbolic paraboloid*.



A surface whose equation is $z = xy$.

The corresponding level curves $xy = \text{constant}$.

FIGURE 2.23

Contour lines on topographic maps are often shown for every 100 feet of elevation. When they are close together the elevation is changing rapidly as we move from one contour to the next; this happens in the vicinity of a steep mountain. When the contour lines are far apart the elevation is changing slowly. We can get a general idea of the steepness of a landscape by considering the spacing of its level curves. However, to get precise information concerning the rate of change of the elevation we must describe the surface in terms of a function to which we can apply the ideas of differential calculus.

The rate at which the elevation is changing at a point (x_0, y_0) depends on the direction in which we move away from this point. For the sake of simplicity, we shall consider at this time just the two special directions parallel to the x - and y -axes. Suppose we examine a surface described by an equation of the form $z = f(x, y)$; let us cut this surface with a plane perpendicular to the y -axis, as suggested in Figure 2.24. Such a plane consists of all points (x, y, z) in space for which the y -coordinate is constant, say $y = y_0$. (The equation $y = y_0$ is called an equation of this plane.) The intersection of this plane with the surface is a plane curve, all points of which satisfy the equation $z = f(x, y_0)$. On this curve the elevation $f(x, y_0)$ is a function of x alone.

Suppose now we move from a point (x_0, y_0) to a point $(x_0 + h, y_0)$. The corresponding change in elevation is $f(x_0 + h, y_0) - f(x_0, y_0)$. This suggests that we form the difference quotient

$$(2.78) \quad \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

and let $h \rightarrow 0$. If this quotient approaches a definite limit as $h \rightarrow 0$, we call this limit the *partial derivative of f with respect to x at (x_0, y_0)* . There are various symbols that are used to denote partial derivatives, some of the most common ones being

$$\frac{\partial f(x_0, y_0)}{\partial x}, \quad f'_x(x_0, y_0), \quad f_x(x_0, y_0), \quad f_1(x_0, y_0), \quad D_1 f(x_0, y_0).$$

The subscript 1 in the last two notations refers to the fact that only the first coordinate is allowed to change when we form the difference quotient in (2.78). Thus we have

$$f_1(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}.$$

Similarly, we define the *partial derivative with respect to y* at (x_0, y_0) by the equation

$$f_2(x_0, y_0) = \lim_{k \rightarrow 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k},$$

alternative notations being

$$\frac{\partial f(x_0, y_0)}{\partial y}, \quad f'_y(x_0, y_0), \quad f_y(x_0, y_0), \quad D_2 f(x_0, y_0).$$

If we write $z = f(x, y)$, then $\partial z / \partial x$ and $\partial z / \partial y$ are also used to designate partial derivatives.

Partial differentiation is not a new concept. If we introduce another function g of one variable, defined by the equation

$$g(x) = f(x, y_0),$$

then the ordinary derivative $g'(x_0)$ is exactly the same as the partial derivative $f_1(x_0, y_0)$. Geometrically, the partial derivative $f_1(x, y_0)$ represents the slope of the tangent line at a typical point of the curve shown in Figure 2.24. In the same way, when x is constant, say $x = x_0$, the equation $z = f(x_0, y)$ describes the curve of intersection of the surface

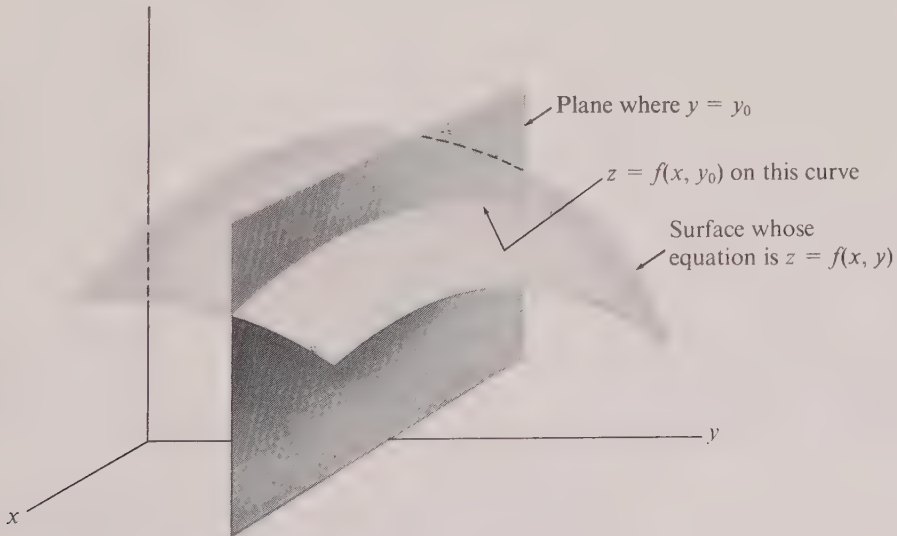


FIGURE 2.24 The curve of intersection of a surface $z = f(x, y)$ and a plane $y = y_0$.

with the plane whose equation is $x = x_0$. The partial derivative $f_2(x_0, y)$ gives the slope of the line tangent to this curve. From these remarks we see that, to compute the partial derivative of $f(x, y)$ with respect to x , we can treat y as though it were constant and use the ordinary rules of differential calculus. Thus, for example, if $f(x, y) = 16 - x^2 - y^2$, we get $f_1(x, y) = -2x$. Similarly, if we hold x fixed, we find $f_2(x, y) = -2y$.

Another example is the function given by

$$(2.79) \quad f(x, y) = x \sin y + y^2 \cos xy.$$

Its partial derivatives are

$$f_1(x, y) = \sin y - y^3 \sin xy, \quad f_2(x, y) = x \cos y - xy^2 \sin xy + 2y \cos xy.$$

Partial differentiation is a process which produces new functions $f_1 = \partial f / \partial x$ and $f_2 = \partial f / \partial y$ from a given function f . Since f_1 and f_2 are also functions of two variables we can consider *their* partial derivatives. These are called *second-order* partial derivatives of f , denoted as follows:

$$f_{1,1} = f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{1,2} = f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, \quad f_{2,1} = f_{yx} = \frac{\partial^2 f}{\partial x \partial y}, \quad f_{2,2} = f_{yy} = \frac{\partial^2 f}{\partial y^2}.$$

Notice that $f_{1,2}$ means $(f_1)_2$, the partial derivative of f_1 with respect to y . In the ∂ -notation we indicate the order of derivatives by writing

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right).$$

This does not always yield the same result as the other mixed partial derivative,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right).$$

However, equality of the two mixed partial derivatives does hold under certain conditions that are usually satisfied by most functions that occur in practice. We shall discuss these conditions further in Volume II.

Referring to the example in (2.76), we find that its second-order partial derivatives are given by the following formulas:

$$f_{1,1}(x, y) = -2, \quad f_{1,2}(x, y) = f_{2,1}(x, y) = 0, \quad f_{2,2}(x, y) = -2.$$

For the example in (2.79), we obtain

$$\begin{aligned} f_{1,1}(x, y) &= -y^4 \cos xy, \\ f_{1,2}(x, y) &= \cos y - xy^3 \cos xy - 3y^2 \sin xy, \\ f_{2,1}(x, y) &= \cos y - xy^3 \cos xy - y^2 \sin xy - 2y^2 \sin xy = f_{1,2}(x, y), \\ f_{2,2}(x, y) &= -x \sin y - x^2 y^2 \cos xy - 2xy \sin xy - 2xy \sin xy + 2 \cos xy \\ &= -x \sin y - x^2 y^2 \cos xy - 4xy \sin xy + 2 \cos xy. \end{aligned}$$

A more detailed study of partial derivatives will be undertaken in Volume II.

2.28 Exercises

In Exercises 1 through 8, compute all first- and second-order partial derivatives. In each case verify that the mixed partial derivatives $f_{1,2}(x, y)$ and $f_{2,1}(x, y)$ are equal.

1. $f(x, y) = x^4 + y^4 - 4x^2y^2$.

5. $f(x, y) = \sin(x^2y^3)$.

2. $f(x, y) = x \sin(x + y)$.

6. $f(x, y) = \sin[\cos(2x - 3y)]$.

3. $f(x, y) = xy + \frac{x}{y} \quad (y \neq 0)$.

7. $f(x, y) = \frac{x+y}{x-y} \quad (x \neq y)$.

4. $f(x, y) = \sqrt{x^2 + y^2}$.

8. $f(x, y) = \frac{x}{\sqrt{x^2 + y^2}} \quad (x, y) \neq (0, 0)$.

9. Show that $x(\partial z / \partial x) + y(\partial z / \partial y) = 2z$ if (a) $z = (x - 2y)^2$, (b) $z = (x^4 + y^4)^{1/2}$.

10. If $f(x, y) = xy/(x^2 + y^2)^2$ for $(x, y) \neq (0, 0)$, show that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$$

Supplement. Some basic theorems on limits and continuous functions

*2.29 Proofs of the fundamental limit theorems

The basic rules for operating with limits were described in Section 2.8 in a theorem which we restate here for ease of reference.

2-1 THEOREM. Let f and g be functions such that

$$\lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{x \rightarrow x_0} g(x) = B.$$

Then we have

- (i) $\lim_{x \rightarrow x_0} [f(x) + g(x)] = A + B,$
- (ii) $\lim_{x \rightarrow x_0} [f(x) - g(x)] = A - B,$
- (iii) $\lim_{x \rightarrow x_0} [f(x) \cdot g(x)] = A \cdot B,$
- (iv) $\lim_{x \rightarrow x_0} [f(x)/g(x)] = A/B \quad \text{if } B \neq 0.$

The principal algebraic tools used in the proof of this theorem are two properties of absolute values that have been mentioned earlier in Section 1.21. They are (1) the triangle inequality, which states that $|a + b| \leq |a| + |b|$ for all real a and b , and (2) the equation $|ab| = |a| |b|$, which states that the absolute value of a product is the product of absolute values.

Proofs of (i) and (ii). Since the two statements

$$\lim_{x \rightarrow x_0} f(x) = A \quad \text{and} \quad \lim_{x \rightarrow x_0} [f(x) - A] = 0$$

are entirely equivalent, and since we have

$$f(x) \pm g(x) - (A \pm B) = [f(x) - A] \pm [g(x) - B],$$

it suffices to prove parts (i) and (ii) of the theorem when the limits A and B are both zero.

Suppose, then, that $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow x_0$. We shall prove first that $f(x) + g(x) \rightarrow 0$ as $x \rightarrow x_0$. This means we must show that for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$(2.80) \quad |f(x) + g(x)| < \epsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta.$$

Let ϵ be given. Since $f(x) \rightarrow 0$ as $x \rightarrow x_0$, there is a $\delta_1 > 0$ such that

$$(2.81) \quad |f(x)| < \frac{\epsilon}{2} \quad \text{whenever} \quad 0 < |x - x_0| < \delta_1.$$

Similarly, since $g(x) \rightarrow 0$ as $x \rightarrow x_0$, there is a $\delta_2 > 0$ such that

$$(2.82) \quad |g(x)| < \frac{\epsilon}{2} \quad \text{whenever} \quad 0 < |x - x_0| < \delta_2.$$

If we let δ denote the smaller of the two numbers δ_1 and δ_2 , then both inequalities (2.81) and (2.82) are valid if $0 < |x - x_0| < \delta$ and hence, by the triangle inequality, we find

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This proves (2.80) which, in turn, proves (i). The proof of (ii) is entirely similar, except that in the last step we use the inequality $|f(x) - g(x)| \leq |f(x)| + |g(x)|$.

Proof of (iii). Suppose that we have proved part (iii) for the special case in which one of the limits is 0. Then the general case follows easily from this special case. In fact, all we need to do is write

$$f(x)g(x) - AB = f(x)[g(x) - B] + B[f(x) - A].$$

The special case implies that each term on the right approaches 0 as $x \rightarrow x_0$ and, by property (i), the sum of the two terms also approaches 0. Therefore, it remains to prove (iii) in the special case where one of the limits, say B , is 0.

Suppose, then, that $f(x) \rightarrow A$ and $g(x) \rightarrow 0$ as $x \rightarrow x_0$. We wish to prove that $f(x)g(x) \rightarrow 0$ as $x \rightarrow x_0$. To do this we must show that, if a positive ϵ is given, there is a $\delta > 0$ such that

$$(2.83) \quad |f(x)g(x)| < \epsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta.$$

Now we have

$$f(x)g(x) = [f(x) - A]g(x) + Ag(x)$$

so, by the triangle inequality and the property $|ab| = |a| |b|$, we may write

$$(2.84) \quad |f(x)g(x)| \leq |f(x) - A| |g(x)| + |A| |g(x)|.$$

The idea of the proof from here on is to show that both terms on the right of (2.84) can be made small if we take x sufficiently close to x_0 . This is easy to do because $g(x) \rightarrow 0$, and hence each term separately is small. To translate this idea into a rigorous proof we may argue as follows:

Since $f(x) \rightarrow A$ as $x \rightarrow x_0$, there is a δ_1 such that

$$(2.85) \quad |f(x) - A| < 1 \quad \text{whenever} \quad 0 < |x - x_0| < \delta_1.$$

For such x , inequality (2.84) gives us

$$(2.86) \quad |f(x)g(x)| \leq |g(x)| + |A| |g(x)| = (1 + |A|) |g(x)|.$$

Since $g(x) \rightarrow 0$ as $x \rightarrow x_0$, for every $\epsilon > 0$ there is a δ_2 such that

$$(2.87) \quad |g(x)| < \frac{\epsilon}{1 + |A|} \quad \text{whenever} \quad 0 < |x - x_0| < \delta_2.$$

Therefore, if we let δ be the smaller of the two numbers δ_1 and δ_2 , then both inequalities (2.86) and (2.87) are valid whenever $0 < |x - x_0| < \delta$, and for such x we deduce (2.83). This completes the proof of (iii).

Proof of (iv). Since the quotient $f(x)/g(x)$ is the product of $f(x)/B$ with $B/g(x)$, it suffices to prove that $B/g(x) \rightarrow 1$ as $x \rightarrow x_0$ and then appeal to (iii). Let $h(x) = g(x)/B$. Then $h(x) \rightarrow 1$ as $x \rightarrow x_0$ and we wish to prove that $1/h(x) \rightarrow 1$ as $x \rightarrow x_0$.

Let $\epsilon > 0$ be given. We must show that there is a $\delta > 0$ such that

$$(2.88) \quad \left| \frac{1}{h(x)} - 1 \right| < \epsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta.$$

The difference to be estimated may be written as follows:

$$(2.89) \quad \left| \frac{1}{h(x)} - 1 \right| = \frac{|h(x) - 1|}{|h(x)|}.$$

Since $h(x) \rightarrow 1$ as $x \rightarrow x_0$ we can choose a $\delta > 0$ such that both inequalities

$$(2.90) \quad |h(x) - 1| < \frac{\epsilon}{2} \quad \text{and} \quad |h(x) - 1| < \frac{1}{2}$$

are satisfied whenever $0 < |x - x_0| < \delta$. The second of these inequalities implies $h(x) > \frac{1}{2}$ so $1/|h(x)| = 1/h(x) < 2$ for such x . Using this in (2.89) along with the first inequality in (2.90), we obtain (2.88). This completes the proof of (iv).

Note. Theorem 2-1 is also valid if we restrict ourselves to "one-sided" limits. For example, if $f(x) \rightarrow A$ as $x \rightarrow x_0$ through values greater than x_0 , then we say that A is the *right-hand limit* of f at x_0 and we indicate this by writing

$$\lim_{x \rightarrow x_0+} f(x) = A.$$

In the ϵ, δ -terminology this means that for every $\epsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - A| < \epsilon$ whenever $0 < x - x_0 < \delta$. Also, we say that f is *continuous from the right* at x_0 if $f(x) \rightarrow f(x_0)$ as

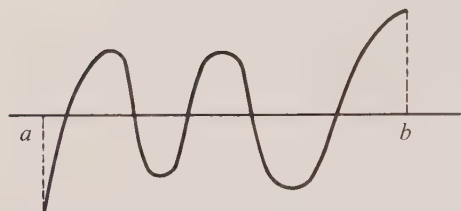
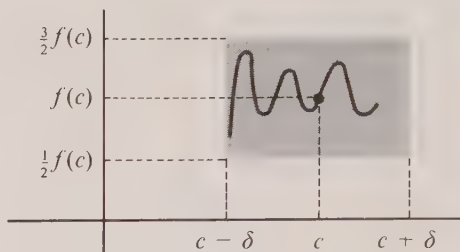


FIGURE 2.25 Illustrating Bolzano's theorem.

FIGURE 2.26 Here $f(x) > 0$ for x near c because $f(c) > 0$.

$x \rightarrow x_0+$. Left-hand limits, denoted by writing $x \rightarrow x_0-$, and left-hand continuity are defined in a similar fashion by restricting x to values less than x_0 . It is easy to check that the basic limit-theorems hold when the limits are all right-handed or all left-handed.

*2.30 Bolzano's theorem and the intermediate-value theorem

In this section we shall discuss certain special properties of continuous functions that are used quite frequently. Most of these properties appear obvious when interpreted geometrically; consequently many people are inclined to accept them as self-evident. However it is important to realize that these statements are no more self-evident than the definition of continuity itself and therefore they must be proved if they are to be used with any degree of generality.

Bernard Bolzano (1781–1848), a Catholic priest who made many important contributions to mathematics in the first half of the 19th century, was one of the first to recognize that many “obvious” statements about continuous functions require proof. His observations concerning continuity were published posthumously in 1850 in an important book, *Paradoxien des Unendlichen*. One of his results, now known as the *theorem of Bolzano*, is illustrated in Figure 2.25, where the graph of a continuous function f is shown. The graph lies below the x -axis at $x = a$ and above the axis at $x = b$. Bolzano's theorem asserts that the curve must cross the axis somewhere between a and b . This property may be stated formally as follows:

2– 8 THEOREM OF BOLZANO. Let f be continuous at each point of a closed interval $[a, b]$ and assume that $f(a)$ and $f(b)$ have opposite signs. Then there is at least one c in the open interval (a, b) such that $f(c) = 0$.

A proof of Bolzano's theorem can be made to depend on the following theorem which, in itself, is also a fundamental property† of continuous functions.

† The property expressed in Theorem 2–9 was used earlier in Section 2.10 in the proof of the formula for differentiating a quotient.

2-9 THEOREM. Let f be continuous at c and suppose that $f(c) \neq 0$. Then there is an open interval $(c - \delta, c + \delta)$ about c such that at each point x in this interval the numbers $f(x)$ and $f(c)$ have the same sign.

Proof. Suppose $f(c) > 0$. By continuity, for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$(2.91) \quad f(c) - \epsilon < f(x) < f(c) + \epsilon \quad \text{whenever} \quad c - \delta < x < c + \delta.$$

If we take the δ corresponding to $\epsilon = f(c)/2$ (this ϵ is *positive*), then (2.91) becomes

$$\frac{1}{2}f(c) < f(x) < \frac{3}{2}f(c) \quad \text{whenever} \quad c - \delta < x < c + \delta.$$

(See Figure 2.26.) Therefore $f(x) > 0$ in this interval and hence $f(x)$ and $f(c)$ have the same sign. If $f(c) < 0$, we take the δ corresponding to $\epsilon = -\frac{1}{2}f(c)$ and arrive at the same conclusion.

Proof of Bolzano's theorem. To be specific, assume $f(a) < 0$ and $f(b) > 0$, as shown in Figure 2.25. There may be many values of x between a and b for which $f(x) = 0$. Our problem is to find *one*. We shall do this by finding the largest x for which $f(x) = 0$. For this purpose we let S denote the set of all those points x in the interval $[a, b]$ for which $f(x) \leq 0$. There is at least one point in S because $f(a) < 0$. Therefore S is a nonempty set. Also, S is bounded above since all of S lies within $[a, b]$. Now, every nonempty set of real numbers which is bounded above has a *least upper bound* (this is exactly the statement of the least-upper-bound axiom for the real-number system). Call this least upper bound c . We shall prove that $f(c) = 0$.

There are only three possibilities: $f(c) > 0$, $f(c) < 0$, and $f(c) = 0$. If $f(c) > 0$, there is an interval $(c - \delta, c + \delta)$ about c such that $f(x)$ is positive if x is in this interval. Therefore no points of S can lie to the right of $c - \delta$, and hence $c - \delta$ is an upper bound for the set S . But $c - \delta < c$, and c is the *least* upper bound of S . Therefore the inequality $f(c) > 0$ is impossible. If $f(c) < 0$ there is an interval $(c - \delta, c + \delta)$ in which f is negative and hence $f(x) < 0$ for some $x > c$, contradicting the fact that c is an upper bound for S . Therefore $f(c) < 0$ is also impossible and the only remaining possibility is $f(c) = 0$. Also, $a < c < b$ because $f(a) < 0$ and $f(b) > 0$. This proves Bolzano's theorem.

An immediate consequence of Bolzano's theorem is the so-called *intermediate-value theorem* for continuous functions. (See Figure 2.27.)

2-10 THEOREM. Let f be continuous at each point of a closed interval $[a, b]$. Choose two arbitrary points $x_1 < x_2$ in $[a, b]$ such that $f(x_1) \neq f(x_2)$. Then f takes on every value between $f(x_1)$ and $f(x_2)$ somewhere in the interval $[x_1, x_2]$.

Proof. Suppose $f(x_1) < f(x_2)$ and let k be any value between $f(x_1)$ and $f(x_2)$. Let g be the function defined on $[x_1, x_2]$ as follows:

$$g(x) = f(x) - k.$$

Then g is continuous at each point of $[x_1, x_2]$, and we have

$$g(x_1) = f(x_1) - k < 0, \quad g(x_2) = f(x_2) - k > 0.$$

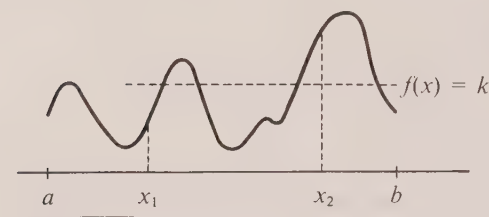


FIGURE 2.27 Illustrating the intermediate-value theorem.



FIGURE 2.28 An example for which Bolzano's theorem is not applicable.

Applying Bolzano's theorem to g we have $g(c) = 0$ for some c between x_1 and x_2 . But this means $f(c) = k$, and the proof is complete.

Note. In both Bolzano's theorem and the intermediate-value theorem it is assumed that f is continuous at each point of $[a, b]$, including the endpoints a and b . To understand why continuity at both endpoints is necessary, we refer to the curve in Figure 2.28. Here f is continuous everywhere in $[a, b]$ except at a . Although $f(a)$ is negative and $f(b)$ is positive, there is no x in $[a, b]$ for which $f(x) = 0$.

We conclude this section with an application of the intermediate-value theorem in which we prove that every positive real number has a positive n th root, a fact mentioned earlier in Section 1.15. We state this as a formal theorem.

2-11 THEOREM. If n is a positive integer and if $a > 0$, then there is exactly one positive b such that $b^n = a$.

Proof. Choose $c > 1$ such that $0 < a < c$, and consider the function f defined on the interval $[0, c]$ by the equation $f(x) = x^n$. This function is continuous on $[0, c]$ and at the endpoints we have $f(0) = 0$, $f(c) = c^n$. Since $0 < a < c < c^n$, the given number a lies between the function values $f(0)$ and $f(c)$. Therefore, by the intermediate-value theorem, we have $f(x) = a$ for some x in $[0, c]$, say for $x = b$. This proves the existence of at least one positive b such that $b^n = a$. There cannot be more than one such b because f is strictly increasing on $[0, c]$. This completes the proof.

★2.31 Exercises

1. Let f be a polynomial of degree n , say $f(x) = \sum_{k=0}^n c_k x^k$, such that the first and last coefficients c_0 and c_n have opposite signs. Prove that $f(x) = 0$ for at least one positive x .

2. A real number x_1 , such that $f(x_1) = 0$, is said to be a real root of the equation $f(x) = 0$. We say that a real root of an equation has been *isolated* if we exhibit an interval $[a, b]$ containing this root and no others. With the aid of Bolzano's theorem, isolate the real roots of each of the following equations (each has four real roots).

(a) $3x^4 - 2x^3 - 36x^2 + 36x - 8 = 0$.

(b) $2x^4 - 14x^2 + 14x - 1 = 0$.

(c) $x^4 + 4x^3 + x^2 - 6x + 2 = 0$.

3. If n is an odd positive integer and $a < 0$, prove that there is exactly one negative b such that $b^n = a$.

4. Let $f(x) = \tan x$. Although $f(\pi/4) = 1$ and $f(3\pi/4) = -1$, there is no x in the interval $[\pi/4, 3\pi/4]$ such that $f(x) = 0$. Explain why this does not contradict Bolzano's theorem.

5. If f is continuous at x_0 and g is continuous at $f(x_0)$, prove that the composite function $h = g \circ f$ is continuous at x_0 . [Hint. Let $y_0 = f(x_0)$. Given $\epsilon > 0$, find δ_1 so that $|y - y_0| < \delta_1$ implies $|g(y) - g(y_0)| < \epsilon$. Now, find δ so that $|x - x_0| < \delta$ implies $|f(x) - f(x_0)| < \delta_1$.]

6. Formulate an ϵ, δ -definition of right-hand continuity. Choose one of the basic limit-theorems and verify that its proof goes through for right-hand limits.

★2.32 Proof of the integrability of continuous functions

In this section we shall prove the following important theorem of integration theory:

2-12 THEOREM. Let f be a function defined and bounded on a closed interval $[a, b]$. If f is continuous[†] on $[a, b]$, then f is integrable on $[a, b]$.

Proof. We shall prove that the upper and lower integrals of f over $[a, b]$ are equal. For this purpose we define a function g on $[a, b]$ as follows:

$$g(x) = \bar{\int}_a^x f - \underline{\int}_a^x f.$$

Note that $g(a) = 0$. We wish to prove that $g(b) = 0$; we shall do this by showing that we have

$$(2.92) \quad g(b) \leq \epsilon(b - a)$$

for every $\epsilon > 0$.

The additive property of upper and lower integrals implies the following property of g :

$$\begin{aligned} g(y) - g(x) &= (\bar{\int}_a^y f - \underline{\int}_a^y f) - (\bar{\int}_a^x f - \underline{\int}_a^x f) = (\bar{\int}_a^y f - \bar{\int}_a^x f) - (\underline{\int}_a^y f - \underline{\int}_a^x f) \\ (2.93) \quad &= \bar{\int}_x^y f - \underline{\int}_x^y f. \end{aligned}$$

When $x < y$ this implies the inequalities

$$(2.94) \quad 0 \leq g(y) - g(x) \leq 2M(y - x),$$

where M is an upper bound for $|f|$ on $[a, b]$. From (2.94) it follows that g is continuous at each point in $[a, b]$.

Now let $\epsilon > 0$ be given and denote by S_ϵ the set of those points x in $[a, b]$ for which

$$(2.95) \quad g(x) \leq \epsilon(x - a).$$

This set is not empty because it contains a . Also, S_ϵ is bounded above by b . Therefore, by the least-upper-bound axiom, S_ϵ has a least upper bound, which we denote by c . We shall prove first that c is in S_ϵ and then that $c = b$. This will prove (2.92).

[†] Continuity of f on $[a, b]$ automatically implies boundedness of f on $[a, b]$. This follows from Theorem 8-1 of Section 8.1, which is proved in Section 8.8.

For this purpose we assume that c is *not* in S_ϵ and arrive at a contradiction. Since c is the least upper bound of S_ϵ , for every $h > 0$ there exists at least one point x of S_ϵ such that $c - h < x < c$. Also, since we are assuming that c is not in S_ϵ , we have $g(c) > \epsilon(c - a)$. Let $k = g(c) - \epsilon(c - a)$. Then $k > 0$ and, by the continuity of g at c , we can choose $h > 0$ such that

$$(2.96) \quad g(c) - g(x) < \frac{k}{2}$$

for all x in the interval $(c - h, c)$. If we choose such an x which also lies in S_ϵ then (2.96) gives us

$$\begin{aligned} g(c) &< g(x) + \frac{k}{2} \leq \epsilon(x - a) + \frac{k}{2} \leq \epsilon(c - a) + \frac{1}{2}g(c) - \frac{1}{2}\epsilon(c - a) \\ &= \frac{1}{2}g(c) + \frac{1}{2}\epsilon(c - a), \end{aligned}$$

from which we find $g(c) < \epsilon(c - a)$, contradicting the assumption that c is not in S_ϵ . This contradiction proves that c is in S_ϵ .

Next we prove that $c = b$. Suppose $c < b$. Then, because of the continuity of f at c , there is a $\delta > 0$ such that

$$(2.97) \quad f(c) - \frac{\epsilon}{2} < f(x) < f(c) + \frac{\epsilon}{2} \quad \text{whenever} \quad c - \delta < x < c + \delta.$$

Also, we can assume that $c + \delta \leq b$. Using (2.93) with $x = c$ and $y = c + \delta$, we have

$$(2.98) \quad g(c + \delta) - g(c) = \int_c^{c+\delta} f - \int_c^c f.$$

If we make use of (2.97) we find

$$\int_c^{c+\delta} f \leq \delta \left[f(c) + \frac{\epsilon}{2} \right] \quad \text{and} \quad \int_c^{c+\delta} f \geq \delta \left[f(c) - \frac{\epsilon}{2} \right],$$

and hence $\int_c^{c+\delta} f - \int_c^c f \leq \delta [f(c) + \frac{1}{2}\epsilon] - \delta [f(c) - \frac{1}{2}\epsilon] = \delta\epsilon$. Therefore (2.98) gives us the inequality

$$g(c + \delta) \leq g(c) + \delta\epsilon \leq \epsilon(c - a) + \delta\epsilon = \epsilon(c + \delta - a).$$

But this means that $c + \delta$ satisfies (2.95). Therefore $c + \delta$ is in S_ϵ , contradicting the fact that c is the least upper bound of S_ϵ . Since the inequality $c < b$ leads to a contradiction, we must have $c \geq b$. But $c \leq b$ and so $c = b$. As we have already noted, this implies that $g(b) = 0$ which, in turn, proves that f is integrable on $[a, b]$.

THE LOGARITHM, THE EXPONENTIAL, AND THE INVERSE TRIGONOMETRIC FUNCTIONS

3.1 Introduction

Whenever man focuses his attention on quantitative relationships, he is either studying the properties of a known function or trying to discover properties of an unknown function. The function concept is so broad and so general that it is not surprising to find an endless variety of functions occurring in nature. What *is* surprising is that a few rather special functions govern so many totally different kinds of natural phenomena. We shall study some of these functions in this chapter—first of all, the logarithm and its inverse (the exponential function) and secondly, the inverses of the trigonometric functions. Anyone who studies mathematics, either as an abstract discipline or as a tool for some other scientific field, will find that a good working knowledge of these functions and their properties is indispensable.

The reader probably has had occasion to work with logarithms to the base 10 in an elementary algebra or trigonometry course. The definition usually given in elementary algebra is this: If $x > 0$, the logarithm of x to the base 10, denoted by $\log_{10} x$, is that real number u such that $10^u = x$. If $x = 10^u$ and $y = 10^v$, the law of exponents yields $xy = 10^{u+v}$. In terms of logarithms, this becomes

$$(3.1) \qquad \log_{10} (xy) = \log_{10} x + \log_{10} y .$$

It is this fundamental property that makes logarithms particularly adaptable to computations involving multiplication. The number 10 is useful as a base because real numbers are commonly written in the decimal system, and certain important numbers like 0.01, 0.1, 1, 10, 100, 1000, . . . have for their logarithms the integers $-2, -1, 0, 1, 2, 3, \dots$, respectively.

It is not necessary to restrict ourselves to base 10. Any other positive base $b \neq 1$ would serve equally well. Thus

$$(3.2) \qquad u = \log_b x \qquad \text{means} \qquad x = b^u ,$$

and the fundamental property in (3.1) becomes

$$(3.3) \qquad \log_b (xy) = \log_b x + \log_b y .$$

If we examine the definition in (3.2) from a critical point of view, we find that it suffers from several logical gaps. First of all, to understand (3.2) we must know what is meant by b^u . This is easy to define when u is an *integer* or a *rational number* (the quotient of two integers), but it is not a trivial matter to define b^u when u is *irrational*. For example, how should we define $10^{\sqrt{2}}$? Even if we manage to obtain a satisfactory definition for b^u , there are further difficulties to overcome before we can use (3.2) as a good definition of logarithms. It must be shown that, for every $x > 0$, there actually *exists* a number u such that $x = b^u$. Also, the law of exponents, $b^u b^v = b^{u+v}$, must be established for all real exponents u and v in order to derive (3.3) from (3.2).

It is possible to overcome these difficulties and arrive at a satisfactory definition of logarithms by this method, but the process is long and tedious. Fortunately, however, the study of logarithms can proceed in an entirely different way which is much simpler and which illustrates the power and elegance of the methods of calculus. The idea is to introduce logarithms *first*, and then use logarithms to define b^u .

3.2 The definition of the natural logarithm

The logarithm is an example of a mathematical concept that can be defined in many different ways. When a mathematician tries to formulate a definition of a concept, such as the logarithm, he usually has in mind a number of properties he wants this concept to have. By examining these properties, he is often led to a simple formula or process that might serve as a definition from which all the desired properties spring forth as logical deductions. We shall illustrate how this procedure may be used to arrive at the definition of the logarithm which is given in this section.

One of the properties we want logarithms to have is that the logarithm of a product should be the sum of the logarithms of the individual factors. Let us consider this property by itself and see where it leads us. If we think of the logarithm as a function f , then we want this function to have the property expressed by the formula

$$(3.4) \quad f(xy) = f(x) + f(y)$$

whenever x , y , and xy are in the domain of f .

An equation like (3.4), which expresses a relationship between the values of a function at two or more points, is called a *functional equation*. Many mathematical problems can be reduced to solving a functional equation, a solution being any function which satisfies the equation. Ordinarily an equation of this sort has many different solutions and it is usually very difficult to find them all. It is easier to seek only those solutions which have some additional property such as continuity or differentiability. For the most part, these are the only solutions we are interested in anyway. We shall adopt this point of view and determine all differentiable solutions of (3.4). But first let us try to deduce what information we can from (3.4) alone, without any further restrictions on f .

One solution of (3.4) is the function that is zero everywhere on the real axis. In fact, this is the only solution of (3.4) that is defined for all real numbers. To prove this, let f be any function that satisfies (3.4). If 0 is in the domain of f , then we may put $y = 0$ in (3.4) to obtain $f(0) = f(x) + f(0)$ and this implies that $f(x) = 0$ for every x in the domain of f . In other words, if 0 is in the domain of f , then f must be identically zero. Therefore, a solution of (3.4) that is not identically zero cannot be defined at 0.

If f is a solution of (3.4) and if the domain of f includes 1, we may put $x = y = 1$ in (3.4) to obtain $f(1) = 2f(1)$, and this implies

$$f(1) = 0.$$

If both 1 and -1 are in the domain of f , we may take $x = -1$ and $y = -1$ to deduce that $f(1) = 2f(-1)$; hence $f(-1) = 0$. If now $x, -x, 1$, and -1 are in the domain of f , we may put $y = -1$ in (3.4) to deduce $f(-x) = f(-1) + f(x)$ and, since $f(-1) = 0$, we find

$$f(-x) = f(x).$$

In other words, any solution of (3.4) is necessarily an *even* function.

Suppose, now, we assume that f has a derivative $f'(x)$ at each $x \neq 0$. If we hold y fixed in (3.4) and differentiate with respect to x (using the chain rule on the left), we find

$$yf'(xy) = f'(x).$$

When $x = 1$ this gives us $yf'(y) = f'(1)$ and hence we have

$$f'(y) = \frac{f'(1)}{y} \quad \text{for each } y \neq 0.$$

From this equation we see that the derivative f' is monotonic and hence integrable on every closed interval not containing the origin. Also, f' is continuous on every such interval and we may apply the second fundamental theorem of calculus to write

$$f(x) - f(c) = \int_c^x f'(t) dt = f'(1) \int_c^x \frac{1}{t} dt.$$

If $x > 0$ this equation holds for any positive c , and if $x < 0$ it holds for any negative c . Since $f(1) = 0$, the choice $c = 1$ gives us

$$f(x) = f'(1) \int_1^x \frac{1}{t} dt \quad \text{if } x > 0.$$

If x is negative then $-x$ is positive and, since $f(x) = f(-x)$, we find

$$f(x) = f'(1) \int_1^{-x} \frac{1}{t} dt \quad \text{if } x < 0.$$

These two formulas for $f(x)$ may be combined into one formula that is valid for both positive and negative x , namely,

$$(3.5) \quad f(x) = f'(1) \int_1^{|x|} \frac{1}{t} dt \quad \text{if } x \neq 0.$$

Therefore we have shown that, if there is a solution of (3.4) which has a derivative at each point $x \neq 0$, then this solution must necessarily be given by the integral formula in

(3.5). If $f'(1) = 0$ then (3.5) implies that $f(x) = 0$ for all $x \neq 0$, and this solution agrees with the solution that is identically zero. Therefore if f is not identically zero we must have $f'(1) \neq 0$, in which case we can divide both sides of (3.5) by $f'(1)$ to obtain

$$(3.6) \quad g(x) = \int_1^{|x|} \frac{1}{t} dt \quad \text{if } x \neq 0,$$

where $g(x) = f(x)/f'(1)$. The function g is also a solution of (3.4), since cf is a solution whenever f is. This proves that if (3.4) has a solution that is not identically zero and if this solution has a derivative everywhere except at the origin, then the function g given by (3.6) is also a solution and *all* solutions may be obtained from this one by multiplying g by a suitable constant.

It should be emphasized that this argument does not prove that the function g in (3.6) actually *is* a solution because we derived (3.6) on the assumption that there is at least one solution that is not identically zero. Formula (3.6) suggests a way to construct such a solution. We simply operate in reverse. That is, we use the integral in (3.6) to define a function g and then we verify directly that this function actually satisfies (3.4). This suggests that we should define the logarithm to be the function g given by (3.6). If we did so, this function would have the property that $g(-x) = g(x)$ or, in other words, distinct numbers would have the same logarithm. For some of the things we want to do later, it is preferable to define the logarithm in such a way that no two distinct numbers have the same logarithm. This latter property may be achieved by defining the logarithm only for positive numbers. Therefore we use the following definition:

DEFINITION. If x is a positive real number, we define the natural logarithm of x , denoted temporarily by $L(x)$, to be the integral

$$(3.7) \quad L(x) = \int_1^x \frac{1}{t} dt.$$

When $x > 1$, $L(x)$ may be interpreted geometrically as the area of the shaded region shown in Figure 3.1.

The function L so defined, being an indefinite integral of a continuous function, has a derivative given by the formula

$$(3.8) \quad L'(x) = \frac{1}{x}$$

for each $x > 0$. We shall prove now that this function has the fundamental property

$$(3.9) \quad L(xy) = L(x) + L(y)$$

for all positive x and y . A proof can be given directly from the definition of $L(x)$ as an integral (see Exercise 2 of Section 3.5), but we prefer to base the proof on the derivative formula in (3.8). For this purpose we keep y fixed and introduce a new function K defined by the equation $K(x) = L(xy)$. Then, by the chain rule, we find $K'(x) = yL'(xy) = y/(xy) = 1/x = L'(x)$. Since K and L have the same derivative everywhere on the positive real axis, Theorem 2-5 tells us that they differ only by a constant. Therefore we can write

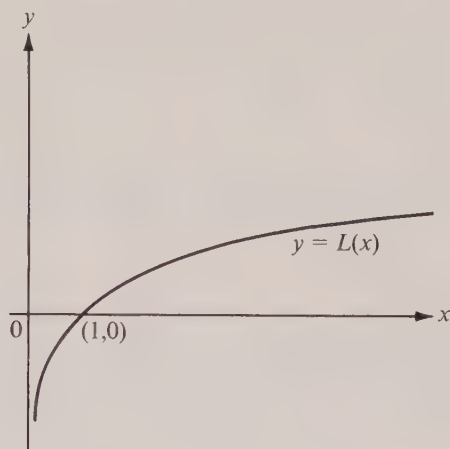
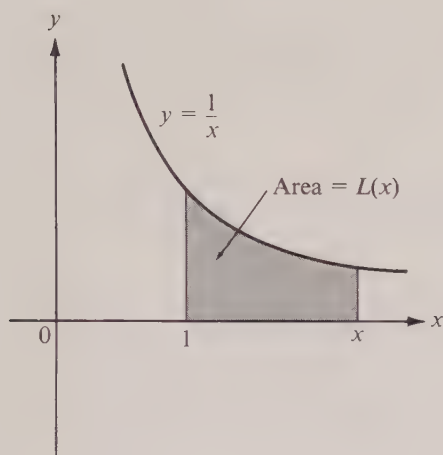


FIGURE 3.1 Interpretation of the logarithm as an area.

FIGURE 3.2 The graph of the natural logarithm.

$$(3.10) \quad K(x) = L(x) + c,$$

where c is a constant. If we put $x = 1$ in (3.10) and use the fact that $L(1) = 0$ [which follows at once from (3.7)], we find $c = K(1) = L(y)$. Using this in (3.10) and replacing $K(x)$ by $L(xy)$, we obtain (3.9).

3.3 The graph of the natural logarithm

The curve whose equation is $y = L(x)$ has the general shape shown in Figure 3.2. Many properties of this curve can be discovered without undue calculation simply by referring to the integral definition in (3.7) or to the fundamental properties expressed in (3.8) and (3.9).

From (3.7) it follows that the logarithm is a *strictly increasing* function; that is, we have

$$(3.11) \quad L(x_1) < L(x_2) \quad \text{whenever} \quad 0 < x_1 < x_2,$$

this being a general property of indefinite integrals of positive functions. In particular, (3.11) implies $L(x) > 0$ if $x > 1$ [since $L(1) = 0$]. Taking $y = 1/x$ in (3.9), we get

$$(3.12) \quad L(x) = -L\left(\frac{1}{x}\right),$$

so that $L(x) < 0$ if $0 < x < 1$. Therefore the curve lies above the x -axis if $x > 1$ and below the x -axis if $0 < x < 1$.

Since $L'(x) = 1/x$ the curve has slope 1 when $x = 1$. For $x > 1$ the slope gradually decreases toward zero as x increases indefinitely. For small values of x the slope is large and, moreover, it increases without bound as x decreases toward zero.

Since the curve tends to level off as x increases indefinitely, it might be suspected that the values of the function L have an upper bound. Actually, the function is *unbounded*; that is, for every positive number M (no matter how large) there exist values of x such that

$$(3.13) \quad L(x) > M.$$

To prove this we use (3.9). When $y = x$ we get $L(x^2) = 2L(x)$. Using (3.9) again with $y = x^2$, we obtain $L(x^3) = 3L(x)$. By induction we have the general formula

$$L(x^n) = nL(x)$$

for every integer $n \geq 1$. When $x = 2$ this becomes $L(2^n) = nL(2)$ and hence we have

$$(3.14) \quad L(2^n) > M \quad \text{when} \quad n > \frac{M}{L(2)}.$$

This proves the assertion in (3.13). Notice that for the same n we have

$$L\left(\frac{1}{2^n}\right) = -L(2^n) < -M$$

which shows that there also is no lower bound to the function values.

Finally we observe that the graph crosses every horizontal line exactly once. That is, given an *arbitrary* real number y (positive, negative, or zero), there is *one and only one* $x > 0$ such that

$$(3.15) \quad L(x) = y.$$

To prove this we can argue as follows: If $y > 0$, choose any integer $n > y/L(2)$. Then $L(2^n) > y$ because of (3.14). Now examine the function L on the closed interval $[1, 2^n]$. Its value at the left endpoint is $L(1) = 0$ and its value at the right endpoint is $L(2^n)$. Since $0 < y < L(2^n)$, the intermediate-value theorem for continuous functions (Theorem 2-10 in Section 2.30) guarantees the existence of at least one x such that $L(x) = y$. There cannot be another value x' such that $L(x') = y$ because this would mean $L(x) = L(x')$ for $x \neq x'$, thus contradicting the increasing property expressed in (3.11). Therefore the assertion in (3.15) has been proved for $y > 0$. The proof for negative y follows from this if we use the equation $L(1/x) = -L(x)$. In other words, we have proved the following theorem:

3-1 THEOREM. For every real number y there is exactly one positive real number x whose logarithm, $L(x)$, is equal to y .

In particular, there is exactly one number whose natural logarithm is equal to 1. This number, like π , occurs repeatedly in so many mathematical formulas that it was inevitable that a special symbol would be adopted for it. Leonard Euler (1707-1783) seems to have been the first to recognize the importance of this number and he modestly denoted it by e , a notation which soon became standard. Thus, by definition, e denotes that number such that

$$(3.16) \quad L(e) = 1.$$

When we study infinite series in Chapter 9, we shall obtain explicit formulas that enable us to calculate the decimal expansion of e to any desired degree of accuracy. Its value, correct to ten decimal places, is 2.7182818285. In Chapter 9 we also prove that e is irrational.

Natural logarithms are also called *Napierian logarithms*, in honor of their inventor, John Napier (1550–1617). It is common practice to use the symbols $\ln x$ or $\log x$ instead of $L(x)$ to denote the logarithm of x .

3.4 Logarithms referred to any positive base $b \neq 1$

The work of Section 3.2 tells us that the most general f which is differentiable on the positive real axis and which satisfies the functional equation $f(xy) = f(x) + f(y)$ is given by the formula

$$(3.17) \quad f(x) = c \log x,$$

where c is a constant. For each c we could call this $f(x)$ the logarithm of x associated with c although, of course, its value would not be necessarily the same as the natural logarithm of x . When $c = 0$, f is identically zero so this case is uninteresting. If $c \neq 0$, we may indicate in another way the dependence of f on c by introducing the concept of a *base* for logarithms.

From (3.17) we see that when $c \neq 0$ there exists a unique real number $b > 0$ such that $f(b) = 1$. This b is related to c by the equation $c \log b = 1$; hence $b \neq 1$, $c = 1/\log b$, and (3.17) becomes

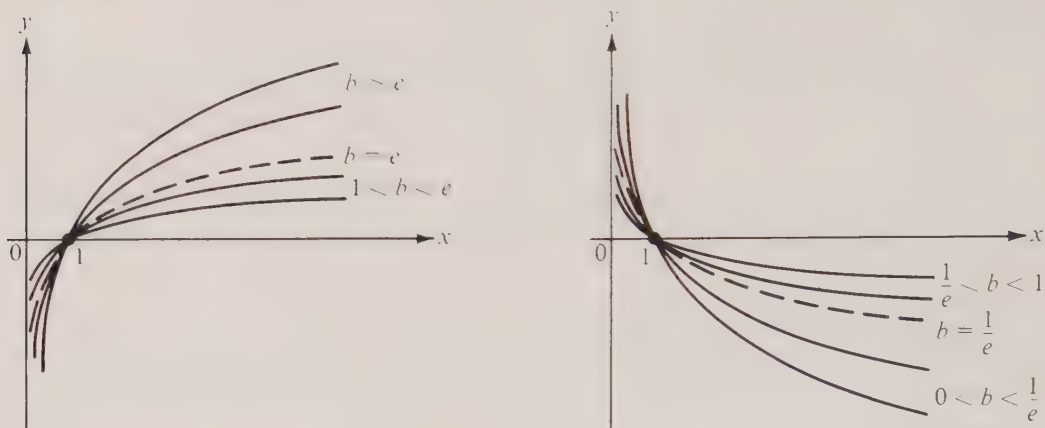
$$f(x) = \frac{\log x}{\log b}.$$

For this choice of c we say that $f(x)$ is *the logarithm of x to the base b* and we write $\log_b x$ for $f(x)$. In other words, by definition we have

$$(3.18) \quad \log_b x = \frac{\log x}{\log b} \quad \text{if } x > 0,$$

where the logarithms on the right are, of course, natural logarithms. Note that $\log_b b = 1$. Also, when $b = e$ we have $\log_e x = \log x$ so natural logarithms are those with base e . Since logarithms to base e are used so frequently in mathematics, the word logarithm almost invariably means *natural* logarithm. Later, in Section 3.8, we shall define b^u in such a way that the equation $b^u = x$ will mean exactly the same as the equation $u = \log_b x$.

Since logarithms to the base b are obtained from natural logarithms by multiplying by the constant $1/\log b$, the graph of the equation $y = \log_b x$ may be obtained from that of the equation $y = \log x$ by simply multiplying all ordinates by the same factor. When $b > 1$ this factor is positive and when $b < 1$ it is negative. Examples with $b > 1$ are shown in Figure 3.3(a). When $b < 1$ we note that $1/b > 1$ and $\log b = -\log(1/b)$, so the graph of $y = \log_b x$ may be obtained from that of $y = \log_{1/b} x$ by reflection through the x -axis. The examples in Figure 3.3(b) are related in this way to those in Figure 3.3(a).

(a) $b > 1$.(b) $0 < b < 1$.FIGURE 3.3 The graph of $y = \log_b x$ for various values of b .

3.5 Exercises

1. Derive the property $\log x = -\log (1/x)$ by making the substitution $u = 1/t$ in the integral which defines $\log x$.

2. Derive the property $\log (xy) = \log x + \log y$ by making an appropriate substitution in the integral which defines $\log (xy)$.

3. (a) Find an $x \neq 2$ such that

$$\frac{\log x}{x} = \frac{\log 2}{2}.$$

(b) Let $f(x) = \log [(1+x)/(1-x)]$ if $x > 0$. If a and b are given numbers, with $ab \neq -1$, find all x such that $f(x) = f(a) + f(b)$.

4. Show that

$$\log (1+x) = \int_0^x \frac{dt}{1+t} \quad \text{if } x > -1$$

and that

$$-\log (1-x) = \int_0^x \frac{dt}{1-t} \quad \text{if } x < 1.$$

5. (a) Let $f(t) = 1/t$ if $t > 0$. Choose suitable functions which approximate f from above and from below on the interval $[1, x]$, where $x > 1$, and derive the inequalities

$$1 - \frac{1}{x} < \log x < x - 1.$$

(b) Prove that these inequalities are also valid when $0 < x < 1$. (When $x = 1$ they become equalities.)

(c) Sketch graphs of the functions A and B defined by the equations

$$A(x) = x - 1, \quad B(x) = 1 - \frac{1}{x}$$

for $x > 0$, and interpret geometrically the inequalities in part (a).

6. Prove the limit relation

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

by the following two methods: (a) Using the definition of the derivative $L'(1)$. (b) Using the result of Exercise 5.

In Exercises 7 through 16, find the derivative $f'(x)$. The function f in each case is assumed to be defined for all real x for which the given formula for $f(x)$ is meaningful. [Hint. Use (3.8) and the chain rule.]

7. $f(x) = \log(1+x^2)$.

13. $f(x) = \sqrt{x+1} - \log(1+\sqrt{x+1})$.

8. $f(x) = \log \sqrt{1+x^2}$.

14. $f(x) = x \log(x + \sqrt{1+x^2}) - \sqrt{1+x^2}$.

9. $f(x) = \log \sqrt{4-x^2}$.

10. $f(x) = \log(\log x)$.

15. $f(x) = \frac{1}{2\sqrt{ab}} \log \frac{\sqrt{a} + x\sqrt{b}}{\sqrt{a} - x\sqrt{b}}$.

11. $f(x) = \log(x^2 \log x)$.

12. $f(x) = \frac{1}{4} \log \frac{x^2-1}{x^2+1}$.

16. $f(x) = x [\sin(\log x) - \cos(\log x)]$.

17. From the definition of the logarithm, derive the following inequalities, all valid for $x > 0$:

(a) $\log(1+x) < x$.

(b) $x - \frac{x^2}{2} < \log(1+x)$. [Hint. $1-t^2 < 1$, hence $1-t < \frac{1}{1+t}$ if $t > 0$.]

(c) $\log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3}$. [Hint. $1+t^3 > 1$, hence $1-t+t^2 > \frac{1}{1+t}$ if $t > 0$.]

(d) $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} < \log(1+x)$. [Hint. $1-t^4 < 1$ if $t > 0$.]

(e) Guess the generalization suggested and prove your result.

18. Derive the following inequalities for $\log(1-x)$:

(a) $x + \frac{x^2}{2} + \frac{x^3}{3} < -\log(1-x)$ if $0 < x < 1$.

[Hint. $1-t^3 < 1$ and hence $1+t+t^2 < \frac{1}{1-t}$ if $0 < t < 1$.]

(b) $-\log(1-x) \leq x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{2}$ if $0 < x \leq \frac{1}{2}$.

[Hint. $\frac{1}{1-t} = 1+t+t^2 + \frac{t^3}{1-t}$ if $t \neq 1$, and $\frac{t^3}{1-t} \leq 2t^3$ if $0 < t \leq \frac{1}{2}$.]

(c) Generalize the inequalities in parts (a) and (b).

19. Use Exercises 17 and 18 to deduce the following inequalities:

(a) $2\left(x + \frac{x^3}{3}\right) < \log \frac{1+x}{1-x}$ if $0 < x < 1$.

(b) $\log \frac{1+x}{1-x} < 2\left(x + \frac{x^3}{3} + \frac{x^4}{4}\right)$ if $0 < x \leq \frac{1}{2}$.

20. (a) Use Exercise 19 to show that

$$\frac{2}{3} + \frac{2}{81} < \log 2 < \frac{2}{3} + \frac{2}{81} + \frac{1}{162},$$

and hence that $0.691 < \log 2 < 0.698$.

(b) In a similar way, show that $1.083 < \log 3 < 1.115$.

(c) From (a) and (b) deduce inequalities for $\log 4$, $\log 6$, and $\log 9$.

21. Remember that the number e is defined by the equation $L(e) = 1$. Prove that $2 < e < 3$.

22. Let $P = \{a_0, a_1, a_2, \dots, a_n\}$ be any partition of the interval $[1, x]$, where $x > 1$.

(a) Integrate suitable step functions that are constant on the open subintervals of P to derive the following inequalities:

$$\sum_{k=1}^n \left(\frac{a_k - a_{k-1}}{a_k} \right) < \log x < \sum_{k=1}^n \left(\frac{a_k - a_{k-1}}{a_{k-1}} \right).$$

(b) Interpret the inequalities of part (a) geometrically in terms of areas.

(c) Specialize the partition to show that, for every integer $n > 1$,

$$\sum_{k=2}^n \frac{1}{k} < \log n < \sum_{k=1}^{n-1} \frac{1}{k}.$$

23. Prove the following formulas for changing from one logarithmic base to another:

$$(a) \log_b x = \log_b a \log_a x; \quad (b) \log_b x = \frac{\log_a x}{\log_a b}.$$

24. Given that $\log_e 10 = 2.302585$, correct to six decimal places, compute $\log_{10} e$ using one of the formulas in Exercise 23. How many correct decimal places can you be certain of in the result of your calculation?

Note. A table, correct to six decimal places, gives $\log_{10} e = 0.434294$.

3.6 The exponential function

By interchanging the roles of x and y in Theorem 3-1, we can state that for every real x there is one and only one y such that

$$(3.19) \quad L(y) = x.$$

This relation serves to define y as a function of x . If x is given, we denote by $E(x)$ that number y which satisfies (3.19). Thus the equation

$$(3.20) \quad y = E(x)$$

means exactly the same as (3.19). The new function E so defined is called the *exponential function* or the *antilogarithm*. Its graph, which is shown in Figure 3.4, is obtained from the graph of the logarithm by reflection through the line $y = x$. The domain of E is the whole real axis, and all the function values $E(x)$ are positive.

Note that (3.19) and (3.20) imply that

$$(3.21) \quad L[E(x)] = x$$

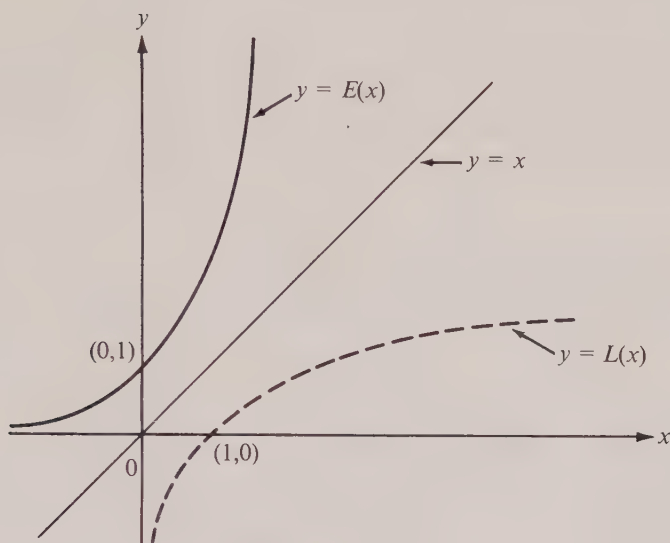


FIGURE 3.4 The graph of the exponential function is obtained from that of the logarithm by reflection through the line $y = x$.

for every x in the domain of E , and that

$$(3.22) \quad E[L(y)] = y$$

for every y in the domain of L . Two functions related in this way are called *inverses* of each other. A general discussion of inverse functions will be given in Section 3.14 and some of the things we do here with the logarithm and the exponential will serve as motivation for the more general discussion in Section 3.14.

Each property of the logarithm can be translated into a property of the exponential. For example, the two equations $L(1) = 0$ and $L(e) = 1$ become, respectively,

$$(3.23) \quad E(0) = 1 \quad \text{and} \quad E(1) = e.$$

Also, the functional equation for the logarithm,

$$(3.24) \quad L(xy) = L(x) + L(y),$$

can be transformed into a functional equation for the exponential, namely,

$$(3.25) \quad E(a)E(b) = E(a + b),$$

which is valid for all real a and b . To derive (3.25) from (3.24) we proceed as follows. Assume that a and b are given and let

$$x = E(a), \quad y = E(b), \quad c = L(xy).$$

Then

$$a = L(x), \quad b = L(y), \quad xy = E(c),$$

and the functional equation (3.24) becomes $c = a + b$. Therefore $E(c) = E(a + b)$ and since $E(c) = xy = E(a)E(b)$, this is exactly the same as (3.25).

Formula (3.25) has many interesting consequences. When $b = -a$, it yields

$$E(a)E(-a) = E(0) = 1,$$

and hence $E(-a) = 1/E(a)$ for every real a . Taking $b = a, b = 2a, \dots, b = na$ in (3.25), we obtain, successively, $E(2a) = E(a)^2$, $E(3a) = E(a)^3$, and, in general,

$$(3.26) \quad E(na) = E(a)^n$$

for every positive integer n . In particular, when $a = 1$ we obtain

$$E(n) = e^n,$$

whereas for $a = 1/n$ we obtain $E(1) = E(1/n)^n$ and hence

$$(3.27) \quad E\left(\frac{1}{n}\right) = e^{1/n}.$$

Therefore, if we put $a = 1/m$ in (3.26) and use (3.27), we find

$$E\left(\frac{n}{m}\right) = E\left(\frac{1}{m}\right)^n = e^{n/m}$$

for all positive integers m and n . In other words, we have

$$(3.28) \quad E(r) = e^r$$

for every positive rational number r . Since $E(-r) = 1/E(r) = e^{-r}$, Equation (3.28) also holds for all negative rational r .

3.7 The definition of e^x for arbitrary real x

In the foregoing section we *proved* that $e^x = E(x)$ when x is any *rational* number. Now we shall *define* e^x for irrational x by writing

$$(3.29) \quad e^x = E(x) \quad \text{for every real } x.$$

The main justification for this definition is that we can use it to prove that the law of exponents

$$(3.30) \quad e^a e^b = e^{a+b}$$

is valid for all real exponents a and b . When we use the definition in (3.29), the proof of (3.30) is a triviality because (3.30) is nothing but a restatement of (3.25).

The notation e^x for $E(x)$ is the one that is commonly used for the exponential. Occasionally $\exp(x)$ is written instead of e^x , especially when complicated formulas appear in the exponent. We shall continue to use $E(x)$ from time to time in this chapter, but later we shall switch to e^x .

We have defined the exponential function so that the two equations

$$y = e^x \quad \text{and} \quad x = \log y$$

mean exactly the same thing. In the next section we shall define more general powers so that the two equations $y = a^x$ and $x = \log_a y$ will be equivalent.

3.8 The definition of a^x for $a > 0$ and x real

Now that we have defined e^x for arbitrary real x , there is absolutely no difficulty in formulating a definition of a^x for every $a > 0$. One way to proceed is to let a^x denote that number y such that $\log_a y = x$. But this does not work for $a = 1$ since logarithms to the base 1 have not been defined. Another way is to define a^x by the formula

$$(3.31) \quad a^x = e^{x \log a}.$$

The second method is preferable because, first of all, it is meaningful for all positive a (including $a = 1$) and, secondly, it makes it easy to prove the following properties of exponentials:

$$\begin{aligned} \log a^x &= x \log a, & (ab)^x &= a^x b^x, \\ a^x a^y &= a^{x+y}, & (a^x)^y &= (a^y)^x = a^{xy}, \\ \text{If } a &\neq 1, \text{ then } y = a^x \text{ if and only if } x = \log_a y. \end{aligned}$$

The proofs of these properties are left as exercises for the reader.

Just as the graph of the exponential function was obtained from that of the logarithm by reflection through the line $y = x$, so the graph of $y = a^x$ can be obtained from that of $y = \log_a x$ by reflection through the same line; examples are shown in Figure 3.5. The curves in Figures 3.5(a) and (b) were obtained, respectively, by reflection of those in Figures 3.3(a) and (b). The graph corresponding to $a = 1$ is, of course, the horizontal line $y = 1$.

3.9 New examples of differentiation and integration formulas

One of the most remarkable properties of the exponential function is the fact that it equals its own derivative. That is to say, we have

$$(3.32) \quad E'(x) = E(x) \quad \text{for every } x.$$

This formula is very easy to prove if we take it for granted that the derivative $E'(x)$ actually *exists*. All we need do is form the composition of the logarithm L with E and then differentiate. If we let $g(x) = L[E(x)]$, then by (3.21) we have $g(x) = x$ for all x and hence $g'(x) = 1$. On the other hand, if we compute $g'(x)$ by the chain rule, we obtain

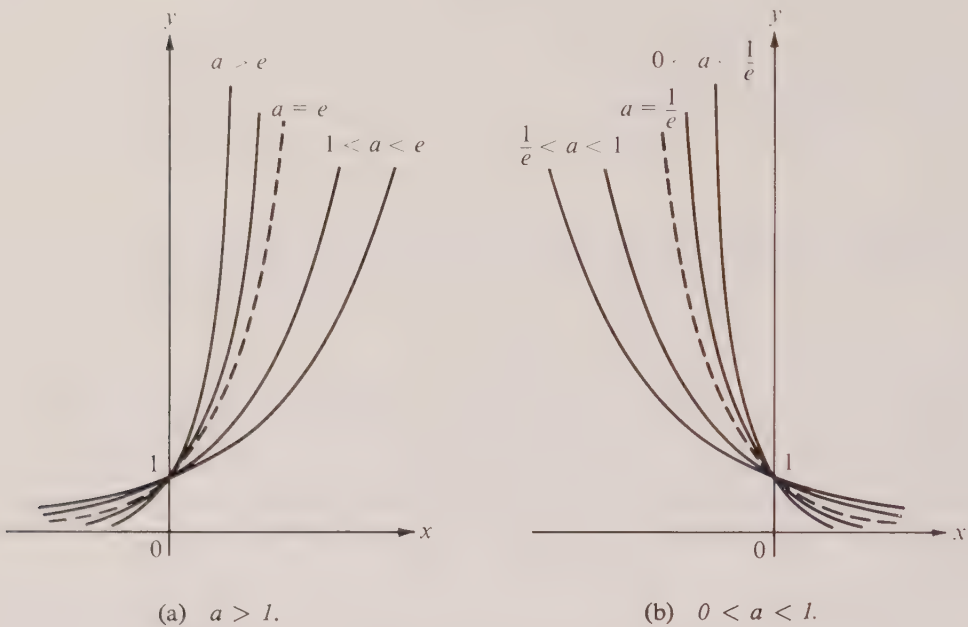


FIGURE 3.5 The graph of $y = a^x$ for various values of a .

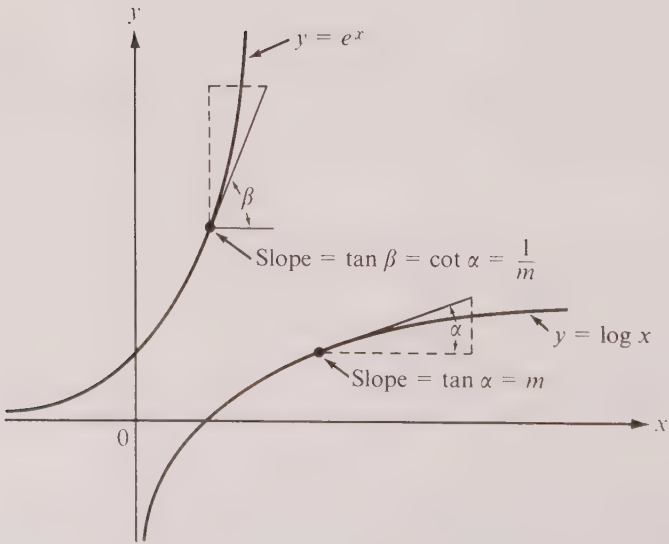


FIGURE 3.6 Geometric motivation for the derivative formula $E'(x) = E(x)$.

$$g'(x) = L'[E(x)] \cdot E'(x) = \frac{E'(x)}{E(x)}.$$

Since $g'(x) = 1$, this proves (3.32).

The actual existence of the derivative $E'(x)$ follows from a general theorem about derivatives of inverse functions given in Section 3.14 (see Theorem 3-3). For the present we shall examine Figure 3.6 and obtain strong geometric evidence both for the existence of $E'(x)$ and for the equation in (3.32).

Recall that the graph of the exponential function was obtained from that of the logarithm by reflection through the line $y = x$. Since the logarithmic curve has a tangent at each of its points, the reflected curve also has this property. Moreover, if the tangent line at a point of the logarithmic curve has slope m , then this tangent is reflected onto a tangent to the exponential curve with slope $1/m$, as suggested by the two triangles of Figure 3.6. This illustrates a general relationship which holds between any two functions that are inverses of each other. In this particular case we know that at a typical point (a, b) on the logarithmic curve the slope is $m = 1/a$. Therefore at the reflected point, (b, a) , the slope of the exponential curve is $1/m = a$. Since $a = e^b$, this means that the slope has the same value as the ordinate at each point on the exponential curve. This is exactly what (3.32) states.

If we use (3.32) along with the chain rule, we can obtain differentiation formulas for exponential functions with any positive base a . Suppose $f(x) = a^x$ if $x > 0$. By the definition of a^x we may write

$$f(x) = e^{x \log a} = E(x \log a);$$

hence, by the chain rule, we find

$$(3.33) \quad f'(x) = E'(x \log a) \cdot \log a = E(x \log a) \cdot \log a = a^x \log a.$$

In other words, differentiation of a^x simply multiplies a^x by the constant factor $\log a$, this factor being 1 when $a = e$.

Of course, these differentiation formulas automatically lead to corresponding integration formulas. For example, (3.32) yields the result

$$(3.34) \quad \int e^x dx = e^x + C,$$

whereas (3.33) gives us the more general formula

$$(3.35) \quad \int a^x dx = \frac{a^x}{\log a} + C \quad (a > 0, a \neq 1).$$

These may be generalized further by the method of substitution. We simply replace x everywhere in (3.34) and (3.35) by u to obtain

$$(3.36) \quad \int e^u du = e^u + C, \quad \int a^u du = \frac{a^u}{\log a} + C \quad (a > 0, a \neq 1),$$

where u now represents any function with a continuous derivative. If we write $u = f(x)$, then $du = f'(x) dx$ and the formulas in (3.36) become

$$(3.37) \quad \int e^{f(x)} f'(x) dx = e^{f(x)} + C, \quad \int a^{f(x)} f'(x) dx = \frac{a^{f(x)}}{\log a} + C,$$

the second of these being valid for $a > 0$, $a \neq 1$.

Similarly, we can write new integration formulas involving the logarithm. Since $D \log x = 1/x$ we have

$$\int \frac{1}{x} dx = \log x + C$$

and, more generally,

$$(3.38) \quad \int \frac{du}{u} = \log u + C \quad \text{or} \quad \int \frac{f'(x)}{f(x)} dx = \log f(x) + C.$$

Although the formulas in (3.37) hold for any function f that has a continuous derivative, some care must be exercised when using (3.38) because the logarithm is not defined for negative numbers. Therefore, the integration formulas in (3.38) are valid only if u , or $f(x)$, is positive.

Fortunately it is easy to extend the range of validity of these formulas to accommodate functions that are negative or positive (but *nonzero*). We simply introduce a new function L_0 defined for all real $x \neq 0$ by the equation

$$(3.39) \quad L_0(x) = \log |x| = \int_1^{|x|} \frac{1}{t} dt,$$

a definition suggested by Equation (3.6) of Section 3.2. The graph of L_0 is symmetric about the y -axis, as shown in Figure 3.7. The portion to the right of the y -axis is exactly the same as the logarithmic curve of Figure 3.2.

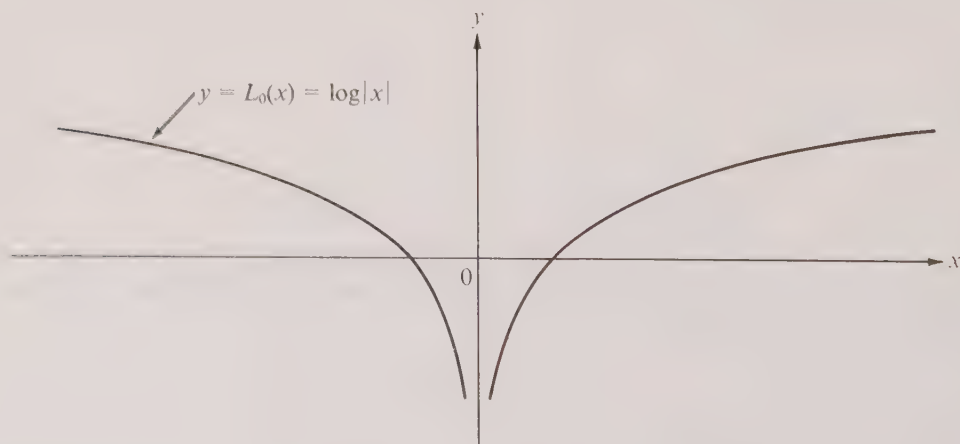


FIGURE 3.7 The graph of the function L_0 .

Since $\log |xy| = \log (|x| |y|) = \log |x| + \log |y|$, the function L_0 also satisfies the basic functional equation in (3.4). That is, we have

$$L_0(xy) = L_0(x) + L_0(y)$$

for all real x and y except 0. For $x > 0$ we have $L'_0(x) = 1/x$ since $L_0(x)$ is the same as $\log x$ for positive x . This derivative formula also holds for $x < 0$ because in this case $L_0(x) = L(-x)$, and hence $L'_0(x) = -L'(-x) = -1/(-x) = 1/x$. Therefore we have

$$(3.40) \quad L'_0(x) = \frac{1}{x} \quad \text{for all real } x \neq 0.$$

Hence, if we use L_0 instead of L in the foregoing integration formulas, we can extend their scope to include functions which assume negative values as well as positive values. For example, (3.38) can be generalized as follows:

$$(3.41) \quad \int \frac{du}{u} = \log |u| + C, \quad \int \frac{f'(x)}{f(x)} dx = \log |f(x)| + C.$$

Of course, when we use (3.41) along with the second fundamental theorem of calculus to evaluate a definite integral, we must avoid intervals that include points where u or $f(x)$ might be zero.

3.10 Worked examples

The integration formulas of the foregoing section, when used in conjunction with the method of substitution and the formula for integration by parts, enable us to deal with many integrals involving exponentials and logarithms. The following examples have been specially selected to illustrate some of the various techniques.

Example 1. Integrate $\int x^2 e^{x^3} dx$.

Solution. Let $u = x^3$. Then $du = 3x^2 dx$ and we obtain

$$\int x^2 e^{x^3} dx = \frac{1}{3} \int e^{x^3} (3x^2 dx) = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C = \frac{1}{3} e^{x^3} + C.$$

Example 2. Integrate $\int \frac{2^{\sqrt{x}}}{\sqrt{x}} dx$.

Solution. Let $u = \sqrt{x} = x^{1/2}$. Then $du = \frac{1}{2} x^{-1/2} dx = \frac{1}{2} dx/\sqrt{x}$. Hence

$$\int \frac{2^{\sqrt{x}}}{\sqrt{x}} dx = 2 \int 2^{\sqrt{x}} \left(\frac{1}{2} \frac{dx}{\sqrt{x}} \right) = 2 \int 2^u du = 2 \frac{2^u}{\log 2} + C = \frac{2^{1+\sqrt{x}}}{\log 2} + C.$$

Example 3. Integrate $\int \cos x e^{2 \sin x} dx$.

Solution. Let $u = 2 \sin x$. Then $du = 2 \cos x dx$ and hence

$$\int \cos x e^{2 \sin x} dx = \frac{1}{2} \int e^{2 \sin x} (2 \cos x dx) = \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C = \frac{1}{2} e^{2 \sin x} + C.$$

Example 4. Integrate $\int \frac{dx}{1 + e^x}$.

Solution. One way to treat this example is to rewrite the integrand as follows:

$$\frac{1}{1 + e^x} = \frac{e^{-x}}{e^{-x} + 1}.$$

Now put $u = e^{-x} + 1$. Then $du = -e^{-x} dx$ and we get

$$\int \frac{e^{-x}}{e^{-x} + 1} dx = - \int \frac{-e^{-x} dx}{e^{-x} + 1} = - \int \frac{du}{u} = -\log |u| + C = -\log (1 + e^{-x}) + C.$$

The result can be written in other ways if we manipulate the logarithm. For instance,

$$-\log (1 + e^{-x}) = \log \frac{1}{1 + e^{-x}} = \log \frac{e^x}{e^x + 1} = \log (e^x) - \log (e^x + 1) = x - \log (1 + e^x).$$

Another way to treat this same example is to write

$$\frac{1}{1 + e^x} = 1 - \frac{e^x}{1 + e^x}.$$

Then we have

$$\int \frac{dx}{1 + e^x} = x - \int \frac{e^x}{1 + e^x} dx = x - \int \frac{du}{u},$$

where $u = 1 + e^x$. Thus we find

$$\int \frac{dx}{1 + e^x} = x - \log (1 + e^x) + C,$$

which is one of the forms obtained above.

The next two examples illustrate the use of integration by parts.

Example 5. Integrate $\int \log |x| dx$.

Solution. Let $u = \log |x|$, $dv = dx$. Then $du = dx/x$, $v = x$, and we obtain

$$\int \log |x| dx = \int u dv = uv - \int v du = x \log |x| - \int x \frac{1}{x} dx = x \log |x| - x + C.$$

Example 6. Integrate $\int e^x \sin x dx$.

Solution. Let $u = e^x$, $dv = \sin x dx$. Then $du = e^x dx$, $v = -\cos x$, and we find

$$(3.42) \quad \int e^x \sin x dx = \int u dv = uv - \int v du = -e^x \cos x + \int e^x \cos x dx + C.$$

The integral $\int e^x \cos x dx$ is treated in the same way. We let $u = e^x$, $dv = \cos x dx$, $du = e^x dx$, $v = \sin x$, and we obtain

$$(3.43) \quad \int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx + C.$$

Substituting this in (3.42), we may solve for $\int e^x \sin x dx$ and consolidate the arbitrary constants to obtain

$$(3.44) \quad \int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x) + C.$$

Notice that we can use this in (3.43) to obtain also

$$(3.45) \quad \int e^x \cos x \, dx = \frac{e^x}{2} (\cos x + \sin x) + C.$$

The foregoing examples deal with integration formulas involving logarithms and exponentials. We shall describe now a technique known as *logarithmic differentiation* which is often a great help in computing derivatives. The method was developed in 1697 by Johann Bernoulli (1667–1748) and all it amounts to is a clever application of the chain rule.

Suppose we form the composition of L_0 with any differentiable function f ; say we let

$$(3.46) \quad g(x) = L_0[f(x)] = \log |f(x)|$$

for those x such that $f(x) \neq 0$. The chain rule, used in conjunction with (3.40), yields the formula†

$$(3.47) \quad g'(x) = L'_0[f(x)] \cdot f'(x) = \frac{f'(x)}{f(x)}.$$

If the derivative $g'(x)$ can be found in some other way, then we may use (3.47) to obtain $f'(x)$ by simply multiplying $g'(x)$ by $f(x)$. The process is useful in practice because in many cases $g'(x)$ is easier to compute than $f'(x)$ itself. In particular, this is true when f is a product or quotient of several simpler functions. The following example is typical.

Example 7. Compute $f'(x)$ if

$$f(x) = \frac{x^2 e^x \cos x}{(1 + x^3)^7} \quad (x \neq -1).$$

Solution. We take the logarithm of the absolute value of $f(x)$ and then we differentiate. Therefore we let

$$\begin{aligned} g(x) &= \log |f(x)| = \log x^2 + \log e^x + \log |\cos x| - \log |1 + x^3|^7 \\ &= 2 \log |x| + x + \log |\cos x| - 7 \log |1 + x^3|. \end{aligned}$$

Differentiation yields

$$g'(x) = \frac{f'(x)}{f(x)} = \frac{2}{x} + 1 - \frac{\sin x}{\cos x} - \frac{21x^2}{1 + x^3}.$$

Multiplying by $f(x)$, we obtain

$$f'(x) = \frac{2xe^x \cos x}{(1 + x^3)^7} + \frac{x^2 e^x \cos x}{(1 + x^3)^7} - \frac{x^2 e^x \sin x}{(1 + x^3)^7} - \frac{21x^4 e^x \cos x}{(1 + x^3)^8}.$$

3.11 Exercises

1. Find all real x such that

(a) $\log(x + 1) = \log(1 - x)$.

(b) $\log(x + 1) = 1 + \log(1 - x)$. [*Hint.* $1 = \log e$.]

(c) $\log(\sqrt{x} + \sqrt{x + 1}) = 1$.

† The quotient f'/f which appears on the right of (3.47) is called the *logarithmic derivative* of f .

2. Find constants a and b such that

$$e^x = b + \int_a^x e^t dt.$$

3. In Exercise 5 of Section 3.5 we outlined a proof of the inequalities $1 - 1/x < \log x < x - 1$, valid for $x > 0$, $x \neq 1$.

- (a) Using this, or by some other means, deduce the inequalities

$$e^x > 1 + x, \quad e^{-x} > 1 - x,$$

valid for all real $x \neq 0$. (When $x = 0$ these become equalities.)

Use part (a) and Exercise 2 to derive the following further inequalities, all valid for $x > 0$:

$$(b) \quad e^x > 1 + x + \frac{x^2}{2!}, \quad e^{-x} < 1 - x + \frac{x^2}{2!}.$$

$$(c) \quad e^x > 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}, \quad e^{-x} > 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!}.$$

- (d) Guess the generalization suggested and prove your result.

4. Let $f(x) = e^{cx}$, where c is a constant. Show that $f'(0) = c$, and use this to deduce the following limit relation:

$$\lim_{x \rightarrow 0} \frac{e^{cx} - 1}{x} = c.$$

5. Let f be a function defined everywhere on the real axis, with a derivative f' which satisfies the equation

$$f'(x) = cf(x) \quad \text{for every } x,$$

where c is a constant. Prove that there is a constant K such that

$$f(x) = Ke^{cx} \quad \text{for every } x.$$

[Hint. Let $g(x) = f(x)e^{-cx}$ and consider $g'(x)$.]

In Exercises 6 through 15, find the derivative $f'(x)$. In each case the function f is assumed to be defined for all real x for which the given formula for $f(x)$ is meaningful.

$$6. f(x) = e^{3x-1}.$$

$$11. f(x) = e^{\sin x}.$$

$$7. f(x) = e^{4x^2}.$$

$$12. f(x) = e^{\cos^2 x}.$$

$$8. f(x) = e^{-x^2}.$$

$$13. f(x) = e^{\log x}.$$

$$9. f(x) = e^{\sqrt{x}}.$$

$$14. f(x) = e^{e^x} \text{ (which means } e^{(e^x)}).$$

$$10. f(x) = e^{1/x}.$$

$$15. f(x) = e^{ee^x} \text{ (which means } e^{[e^{(e^x)}]}).$$

16. Let f be a function defined everywhere on the real axis. Suppose also that f satisfies the functional equation

$$(i) \quad f(x+y) = f(x)f(y) \quad \text{for all } x \text{ and } y.$$

(a) Using only the functional equation, prove that $f(0)$ is either 0 or 1. Also, prove that if $f(0) \neq 0$ then $f(x) \neq 0$ for all x .

Assume, in addition to (i), that $f'(x)$ exists for all x , and prove the following statements:

$$(b) \quad f'(x)f(y) = f'(y)f(x) \text{ for all } x \text{ and } y.$$

$$(c) \quad \text{There is a constant } c \text{ such that } f'(x) = cf(x) \text{ for all } x.$$

$$(d) \quad f(x) = e^{cx} \text{ if } f(0) \neq 0. \text{ [Hint. See Exercise 5.]}$$

17. Let $f(x) = x^r$, where $x > 0$ and r is any real number. The formula $f'(x) = rx^{r-1}$ was proved earlier for rational r .

(a) Show that this formula also holds for arbitrary real r . [Hint. Write $x^r = e^{r \log x}$.]

(b) Discuss under what conditions the result of part (a) applies for $x \leq 0$.

18. Use the definition $a^x = e^{x \log a}$ to derive the following properties of general exponentials:

(a) $\log a^x = x \log a$.

(b) $(ab)^x = a^x b^x$.

(c) $a^x a^y = a^{x+y}$.

(d) $(a^x)^y = (a^y)^x = a^{xy}$.

(e) If $a \neq 1$, then $y = a^x$ if and only if $x = \log_a y$.

19. Let $f(x) = \frac{1}{2}(a^x + a^{-x})$ if $a > 0$. Show that

$$f(x+y) + f(x-y) = 2f(x)f(y).$$

In Exercises 20 through 37, find the derivative $f'(x)$. In each case the function f is assumed to be defined for all real x for which the given formula for $f(x)$ is meaningful. Logarithmic differentiation may simplify the work in some cases.

20. $f(x) = 2^x$.

29. $f(x) = (\log x)^x$.

21. $f(x) = x^x$.

30. $f(x) = x^{\log x}$.

22. $f(x) = (1+x)(1+e^{x^2})$.

31. $f(x) = \log_x e$.

23. $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

32. $f(x) = \frac{(\log x)^x}{x^{\log x}}$.

24. $f(x) = 2^{x^2}$.

33. $f(x) = (\sin x)^{\cos x} + (\cos x)^{\sin x}$.

25. $f(x) = x^{a^a} + a^{x^a} + a^{a^x}$.

34. $f(x) = (x + \sqrt{1+x^2})^n$.

26. $f(x) = \log [\log (\log x)]$.

35. $f(x) = x^{1/x}$.

27. $f(x) = \log^+(e^x + \sqrt{1+e^{2x}})$.

36. $f(x) = \frac{x^2(3-x)^{1/3}}{(1-x)(3+x)^{2/3}}$.

28. $f(x) = x^{x^x}$.

37. $f(x) = \prod_{i=1}^n (x - a_i)^{b_i}$.

38. Derive the following inequalities. [Hint. Use Exercise 3.]

(a) $e^x < \frac{1}{1-x}$ if $x < 1$.

(b) $e^{-x} < \frac{1}{1+x}$ if $x > -1$.

(c) $e^{x/(1+x)} < 1+x$ if $x > -1$.

(d) $e^{xy/(x+y)} < \left(1 + \frac{x}{y}\right)^y$ if $x > 0$ and $y > 0$.

39. If n is a positive integer and if $x > 0$, show that

$$\left(1 + \frac{x}{n}\right)^n < e^x < \left(1 - \frac{x}{n}\right)^{-n}.$$

By choosing a suitable value of n , deduce that $2.5 < e < 2.99$.

40. Let $f(x, y) = x$ where $x > 0$. Show that

$$\frac{\partial f}{\partial x} = yx^{y-1} \quad \text{and} \quad \frac{\partial f}{\partial y} = x^y \log x.$$

Evaluate the indefinite integrals in Exercises 41 through 53.

41. $\int x e^{-x} dx.$

42. $\int x^3 e^{-x^2} dx.$

43. $\int x \log |x| dx.$

44. $\int x^2 \log |x| dx.$

45. $\int x^2 e^{-2x} dx.$

46. $\int e^{\sqrt{x}} dx.$

53. $\int \frac{dx}{\sqrt{x^2 + a^2}}.$ [Hint. Let $u = x + \sqrt{x^2 + a^2}$. Show that $du/u = dx/\sqrt{x^2 + a^2}$.]

47. $\int \frac{\log |x|}{x\sqrt{1 + \log |x|}} dx.$

48. $\int \tan x dx.$

49. $\int \frac{\cos^3 x}{\sin x} dx.$

50. $\int \frac{dx}{a + bx}.$

51. $\int x^5 e^{x^3} dx.$

52. $\int \frac{x^2}{\sqrt{4 + x^3}} dx.$

54. Let $A = \int e^{ax} \cos bx dx$ and $B = \int e^{ax} \sin bx dx$, where a and b are constants, not both zero. Use integration by parts to show that

$$aA - bB = e^{ax} \cos bx + C_1, \quad aB + bA = e^{ax} \sin bx + C_2,$$

where C_1 and C_2 are arbitrary constants. Solve for A and B to deduce the following integration formulas:

$$\int e^{ax} \cos bx dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C,$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C.$$

55. Show that

$$(a) \int_0^x e^{-t} t dt = e^{-x}(e^x - 1 - x),$$

$$(b) \int_0^x e^{-t} t^2 dt = 2!e^{-x}\left(e^x - 1 - x - \frac{x^2}{2!}\right).$$

$$(c) \int_0^x e^{-t} t^3 dt = 3!e^{-x}\left(e^x - 1 - x - \frac{x^2}{2!} - \frac{x^3}{3!}\right).$$

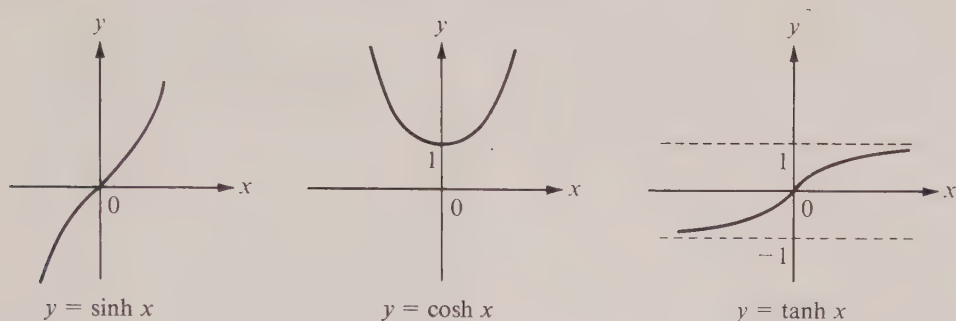
(d) Guess the generalization suggested and prove it by induction.

3.12 The hyperbolic functions

Certain combinations of exponential functions occur quite frequently in analysis and it is worth while to give these combinations special names and to study them as examples of new functions. These combinations, called the *hyperbolic sine* ($\sinh$), the *hyperbolic cosine* ($\cosh$), the *hyperbolic tangent* ($\tanh$), etc., are defined as follows:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

$$\operatorname{csch} x = \frac{1}{\sinh x}, \quad \operatorname{sech} x = \frac{1}{\cosh x}, \quad \operatorname{coth} x = \frac{1}{\tanh x}.$$

FIGURE 3.8 *Graphs of hyperbolic functions.*

The prefix “hyperbolic” is due to the fact that these functions are related geometrically to a hyperbola in much the same way as the trigonometric functions are related to a circle. This relation will be discussed in more detail in Chapter 6 when we study the hyperbola. The graphs of the $\sinh$, $\cosh$, and $\tanh$ are shown in Figure 3.8.

The hyperbolic functions possess many properties that resemble those of the trigonometric functions. Some of these are listed as exercises in the following section.

3.13 Exercises

Derive the properties of the hyperbolic functions listed in Exercises 1 through 15 and compare them, whenever possible, with the corresponding properties of the trigonometric functions.

1. $\cosh^2 x - \sinh^2 x = 1$.
2. $\sinh(-x) = -\sinh x$.
3. $\cosh(-x) = \cosh x$.
4. $\tanh(-x) = -\tanh x$.
5. $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$.
6. $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$.
7. $\sinh 2x = 2 \sinh x \cosh x$.
8. $\cosh 2x = \cosh^2 x + \sinh^2 x$.
9. $\cosh x + \sinh x = e^x$.
10. $\cosh x - \sinh x = e^{-x}$.
11. $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$ (n an integer).
12. $2 \sinh^2 \frac{1}{2}x = \cosh x - 1$.
13. $2 \cosh^2 \frac{1}{2}x = \cosh x + 1$.
14. $\tanh^2 x + \operatorname{sech}^2 x = 1$.
15. $\coth^2 x - \operatorname{csch}^2 x = 1$.
16. Find $\cosh x$ if $\sinh x = 4/3$.
17. Find $\sinh x$ if $\cosh x = 5/4$ and $x > 0$.
18. Find $\sinh x$ and $\cosh x$ if $\tanh x = 5/13$.
19. Find $\cosh(x + y)$ if $\sinh x = 4/3$ and $\sinh y = 3/4$.

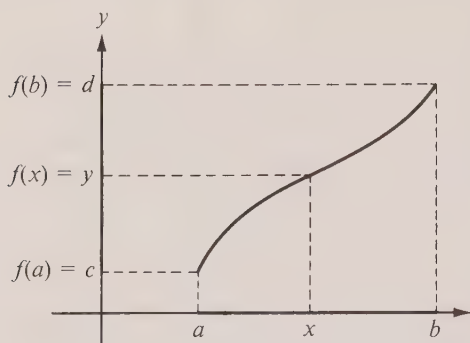


FIGURE 3.9 A continuous strictly increasing function.

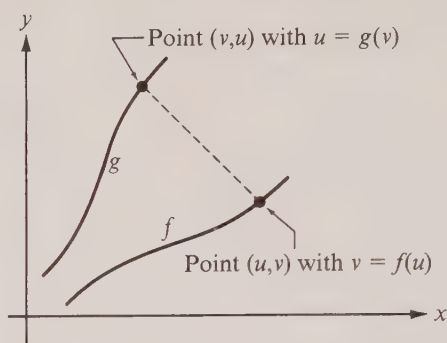


FIGURE 3.10 Illustrating the process of inversion.

20. Find $\tanh 2x$ if $\tanh x = 3/4$.

In Exercises 21 through 26, prove the differentiation formulas.

21. $D \sinh x = \cosh x$.

24. $D \coth x = -\operatorname{csch}^2 x$.

22. $D \cosh x = \sinh x$.

25. $D \operatorname{sech} x = -\operatorname{sech} x \tanh x$.

23. $D \tanh x = \operatorname{sech}^2 x$.

26. $D \operatorname{csch} x = -\operatorname{csch} x \coth x$.

3.14 Inverses of monotonic functions

The procedure employed in Section 3.6 to define the exponential function in terms of the logarithm illustrates an important method that is often used to construct new functions from given ones. The method is applicable to functions that are continuous and strictly monotonic on an interval. We shall describe the method only for strictly increasing functions, since the treatment for strictly decreasing functions is entirely analogous.

Let f be continuous and strictly increasing on $[a, b]$ and let $c = f(a)$, $d = f(b)$. (An example is shown in Figure 3.9.) The intermediate-value theorem for continuous functions (Theorem 2-10) tells us that, in the interval $[a, b]$, f takes on every value between c and d . Moreover, f cannot assume the same value twice because $f(x_1) < f(x_2)$ whenever $x_1 < x_2$. Therefore, for every y in $[c, d]$ there is exactly one y in $[c, d]$ such that

$$(3.48) \quad y = f(x),$$

and, conversely, for every y in $[c, d]$ there is exactly one x in $[a, b]$ related to y by Equation (3.48). We often describe this by saying that f establishes a one-to-one correspondence between the points of the two intervals $[a, b]$ and $[c, d]$. The correspondence which associates to each y in $[c, d]$ the uniquely determined x in $[a, b]$ satisfying (3.48) can be expressed as a functional relationship, $x = g(y)$. In other words, Equation (3.48) serves to define a new function g whose value at each point y is that number x such that $y = f(x)$. This new function is called the *inverse* of f . Its domain is the interval $[c, d]$. The process by which g is obtained from f is called *inversion*. Note that $g[f(x)] = x$ for all x in $[a, b]$ and $f[g(y)] = y$ for all y in $[c, d]$.

Many properties possessed by the function f are transmitted to the inverse g . Figure 3.10 illustrates the relationship between their graphs. One can be obtained from the other merely by reflection through the line $y = x$, because a point (u, v) lies on the graph of f if and only if the point (v, u) lies on the graph of g .

The properties of monotonicity and continuity possessed by f are transmitted to the inverse function g , as described by the following theorem.

- 3- 2 THEOREM. Assume f is strictly increasing and continuous on an interval $[a, b]$. Let $c = f(a)$ and $d = f(b)$ and let g be the function obtained from f by inversion. That is, for each y in $[c, d]$, let $g(y)$ be that x in $[a, b]$ such that $y = f(x)$. Then
- (a) g is strictly increasing on $[c, d]$;
 - (b) g is continuous on $[c, d]$.

The proof of this theorem is not difficult and it is outlined below in Exercises 13 and 14 of Section 3.15. The theorem has an important consequence which states that the process of inversion also transmits differentiability from a function to its inverse. In other words:

- 3- 3 THEOREM. Assume f and g are related as in Theorem 3-2. If the derivative $f'(x)$ exists and is nonzero at a point x in (a, b) , then the derivative $g'(y)$ also exists and is nonzero at the corresponding point y , where $y = f(x)$. Moreover, the two derivatives are reciprocals of each other; that is, we have

$$(3.49) \quad g'(y) = \frac{1}{f'(x)}.$$

Note. If we use the Leibniz notation and write y for $f(x)$, dy/dx for $f'(x)$, x for $g(y)$, and dx/dy for $g'(y)$, then Equation (3.49) becomes

$$\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)},$$

which has the appearance of a trivial algebraic identity.

Geometric motivation for the relation in (3.49) was given in Section 3.9 when we discussed the special case in which $f(x) = \log x$, $g(x) = e^x$. We turn now to the analytic proof of (3.49).

Proof. Assume x is a point in (a, b) where $f'(x)$ exists and is nonzero, and let $y = f(x)$. We shall show that the difference quotient

$$\frac{g(y+k) - g(y)}{k}$$

approaches the limit $1/f'(x)$ as $k \rightarrow 0$.

Let $h = g(y+k) - g(y)$. Since $x = g(y)$ this implies $h = g(y+k) - x$ or $x+h = g(y+k)$. Therefore $y+k = f(x+h)$ and hence $k = f(x+h) - f(x)$. Note that $h \neq 0$ if $k \neq 0$ because g is strictly increasing. Therefore, if $k \neq 0$, the difference quotient in question is

$$(3.50) \quad \frac{g(y+k) - g(y)}{k} = \frac{h}{f(x+h) - f(x)} = \frac{1}{[f(x+h) - f(x)]/h}.$$

As $k \rightarrow 0$ the difference $g(y+k) - g(y) \rightarrow 0$ because of the continuity of g at y [property (b) of Theorem 3-2]. This means that $h \rightarrow 0$ as $k \rightarrow 0$. But we know that the difference quotient in the denominator on the extreme right of (3.50) approaches $f'(x)$ as $h \rightarrow 0$ [since $f'(x)$ exists]. Therefore, when $k \rightarrow 0$ the quotient on the extreme left of (3.50) approaches the limit $1/f'(x)$. This proves Theorem 3-3.

It is clear that the method of inversion can also be used to define an inverse for every continuous function f which is strictly *decreasing* on $[a, b]$. On the other hand, if f is *not*

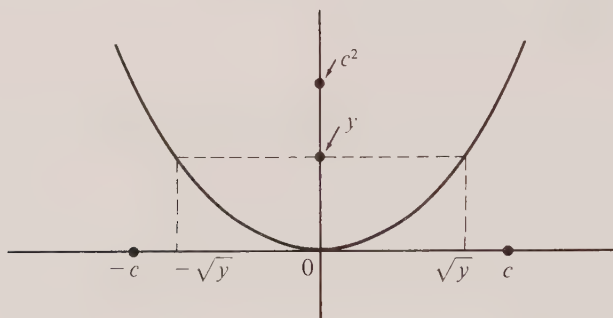


FIGURE 3.11 For each y in $[0, c^2]$ there are two values of x such that $y = x^2$.

monotonic on $[a, b]$, an inverse function is not uniquely determined by this process. For example, consider the case in which $f(x) = x^2$ on an interval of the form $[-c, c]$ on the x -axis. Each point x in this interval is carried by f into exactly one point y in the interval $[0, c^2]$, namely,

$$(3.51) \quad y = x^2.$$

But the correspondence between the points of these two intervals is *not* one-to-one. When we solve Equation (3.51) for x in terms of y we find that there are *two* values of x corresponding to each y in $(0, c^2]$, namely,

$$x = \sqrt{y} \quad \text{and} \quad x = -\sqrt{y}$$

(see Figure 3.11). As we have mentioned once before, there was a time when mathematicians would have said that the inverse g in this case is a *double-valued function* defined by

$$g(y) = \pm\sqrt{y}.$$

But since the more modern point of view does not admit double-valuedness as a property of functions, in a case like this we say that the process of inversion gives rise to *two* new functions, say g_1 and g_2 , where

$$(3.52) \quad g_1(y) = \sqrt{y} \quad \text{and} \quad g_2(y) = -\sqrt{y} \quad \text{for each } y \text{ in } [0, c^2].$$

To fit this in with the notion of inverse as explained above, we can look upon the equation $y = x^2$ as defining not *one* function f but *two* functions f_1 and f_2 , say, where

$$f_1(x) = x^2 \quad \text{if } 0 \leq x \leq c \quad \text{and} \quad f_2(x) = x^2 \quad \text{if } -c \leq x \leq 0.$$

These may be considered as *distinct* functions because they have different domains. Each function is monotonic on its domain and each has an inverse, the inverse of f_1 being g_1 and the inverse of f_2 being g_2 , where g_1 and g_2 are given by (3.52).

Note. In defining the logarithm in Section 3.2, we were careful to see that the resulting function was strictly monotonic. This made it possible to obtain the exponential function by inversion.

3.15 Exercises

In each of Exercises 1 through 9, show that f is strictly monotonic on the whole real axis. Let g denote the inverse of f . Describe the domain of g in each case. Write $y = f(x)$ and solve for x in terms of y ; thus find a formula (or formulas) for computing $g(y)$ for each y in the domain of g .

1. $f(x) = x + 1.$

5. $f(x) = e^{2x}.$

2. $f(x) = 2x + 5.$

6. $f(x) = e^{x^3}.$

3. $f(x) = 1 - x.$

7. $f(x) = \sinh x.$

4. $f(x) = x^3.$

8. $f(x) = \tanh x.$

9. $f(x) = \begin{cases} x & \text{if } x < 1, \\ x^2 & \text{if } 1 \leq x \leq 4, \\ 2^x & \text{if } x > 4. \end{cases}$

10. In each of Exercises 1 through 9 above, compute $f'(x)$ and $g'(y)$ whenever they exist and verify that they are reciprocals when $y = f(x)$.

11. Let $f(x) = \log |x|$ if $x < 0$. Show that f has an inverse, and denote this inverse by g . What is the domain of g ? Find a formula for computing $g(y)$ for each y in the domain of g . Sketch the graph of g .

12. Let $f(x) = \int_0^x (1 + t^3)^{-1/2} dt$ if $x \geq 0$. (Do not attempt to evaluate this integral.)

(a) Show that f is strictly increasing on the nonnegative real axis.

(b) Let g denote the inverse of f . Show that the second derivative of g is proportional to g^2 [that is, $g''(y) = cg^2(y)$ for each y in the domain of g] and find the constant of proportionality.

13. Let f be continuous and strictly increasing on an interval $[a, b]$. Let $c = f(a)$, $d = f(b)$, and let g be the inverse of f . Prove that g is strictly increasing on $[c, d]$. [*Hint.* If $y_1 < y_2$, let $x_1 = g(y_1)$, $x_2 = g(y_2)$. Then $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Now, use the monotonicity of f to prove that $x_1 < x_2$. Draw a graph.]

14. Let f and g be as in Exercise 13. Prove that g is continuous on the closed interval $[c, d]$.

Outline of proof. Choose a point y_0 in the open interval (c, d) . To prove g is continuous at y_0 we must show that for every $\epsilon > 0$ there is a $\delta > 0$ such that

(i) $g(y_0) - \epsilon < g(y) < g(y_0) + \epsilon \quad \text{whenever} \quad y_0 - \delta < y < y_0 + \delta.$

Let $x_0 = g(y_0)$ [so that $f(x_0) = y_0$]. Suppose ϵ is given. (There is no loss in generality if we consider only those ϵ small enough so that both $x_0 - \epsilon$ and $x_0 + \epsilon$ are in $[a, b]$.) Let δ be the smaller of the two numbers

$$f(x_0) - f(x_0 - \epsilon) \quad \text{and} \quad f(x_0 + \epsilon) - f(x_0).$$

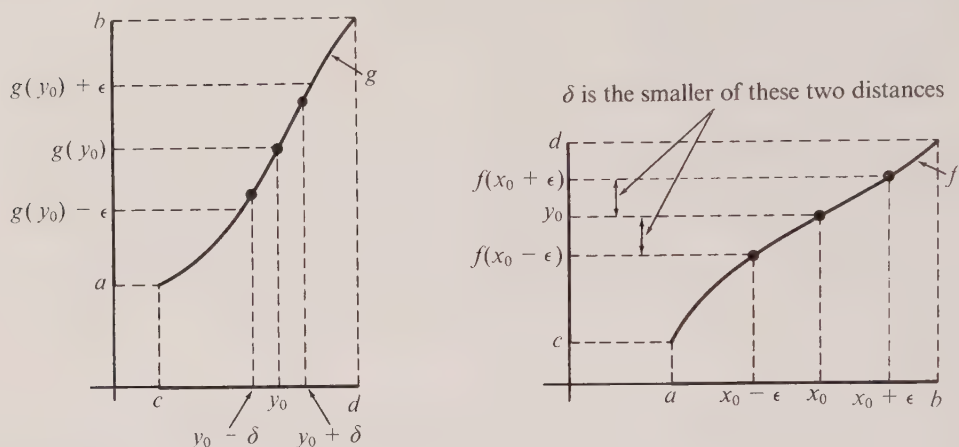


FIGURE 3.12 Proof of the continuity of the inverse function.

(See Figure 3.12.) It is easy to check that this δ works in (i). A slight modification of the argument proves continuity also at c and d .

Mean values. Let f be continuous and strictly monotonic on the positive real axis and let g denote the inverse of f . If $a_1 < a_2 < \cdots < a_n$ are n given positive real numbers, we define their *mean value* (or *average*) with respect to f to be the number M_f defined as follows:

$$M_f = g\left(\frac{1}{n} \sum_{i=1}^n f(a_i)\right).$$

In particular, when $f(x) = x^p$ for $p \neq 0$, M_f is called the p th *power mean*; when $f(x) = \log x$, M_f is called the *geometric mean*. (See also Exercises 16 and 19 of Section 1.23.) The exercises which follow deal with properties of mean values.

15. Show that $f(M_f) = (1/n) \sum_{i=1}^n f(a_i)$. In other words, the value of f at the average M_f is the arithmetic mean of the function values $f(a_1), \dots, f(a_n)$.

16. Show that $a_1 < M_f < a_n$. In other words, the average of $a_1, \dots, a_n$ lies between the largest and smallest of the a_i .

17. If $h(x) = af(x) + b$, where $a \neq 0$, show that $M_h = M_f$. This shows that different functions may lead to the same average. Interpret this theorem geometrically by comparing the graphs of h and f .

18. For each real p let F_p denote the function defined as follows:

$$F_p(x) = \int_1^x t^{p-1} dt \quad \text{if } x > 0.$$

(a) Show that F_p is strictly monotonic on the positive real axis. (b) Show that M_{F_p} is the p th power mean if $p \neq 0$ and the geometric mean if $p = 0$. [Hint. Use Exercise 17.]

3.16 Inverses of the trigonometric functions

The process of inversion may be applied to the trigonometric functions. Suppose we begin with the sine function. To determine a unique inverse we must consider the sine over some interval where it is monotonic. There are, of course, many such intervals, for example $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, $[\frac{1}{2}\pi, \frac{3}{2}\pi]$, $[-\frac{3}{2}\pi, -\frac{1}{2}\pi]$, etc., and it really does not matter which one of these we choose. It is customary to select $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ and define a new function f as follows:

$$f(x) = \sin x \quad \text{if} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}.$$

The function f so defined is strictly increasing and it assumes every value between -1 and $+1$ exactly once on the interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$. (See Figure 3.13.) Hence there is a uniquely determined function g defined on $[-1, 1]$ which assigns to each number y in $[-1, 1]$ that number x in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ for which $y = \sin x$. This function g is called the *inverse sine* or *arc sine* and its value at y is denoted by $\arcsin y$. Thus

$$u = \arcsin v \quad \text{means} \quad v = \sin u \quad \text{and} \quad -\frac{\pi}{2} \leq u \leq \frac{\pi}{2}.$$

The graph of the arc sine is shown in Figure 3.14. Note that the arc sine is not defined outside the interval $[-1, 1]$.

The derivative of the arc sine can be obtained from formula (3.49) of Section 3.14. In this case we have $f'(x) = \cos x$ and this is nonzero in the open interval $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$. Therefore formula (3.49) yields

$$g'(y) = \frac{1}{f'(x)} = \frac{1}{\cos x} = \frac{1}{\sqrt{1 - \sin^2 x}} = \frac{1}{\sqrt{1 - y^2}} \quad \text{if} \quad -1 < y < 1.$$

With a change in notation we can write this result as follows:

$$(3.53) \quad D \arcsin x = \frac{1}{\sqrt{1 - x^2}} \quad \text{if} \quad -1 < x < 1.$$

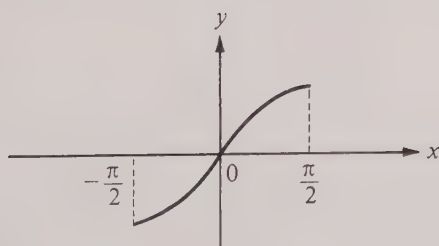


FIGURE 3.13 $y = \sin x$.

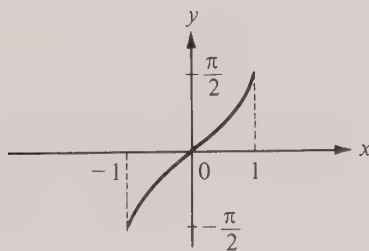
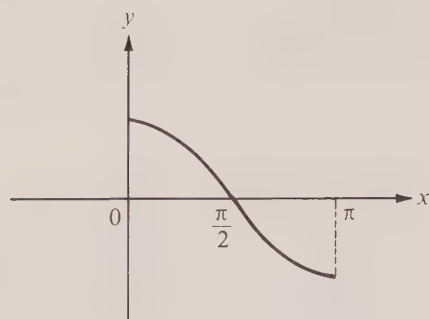
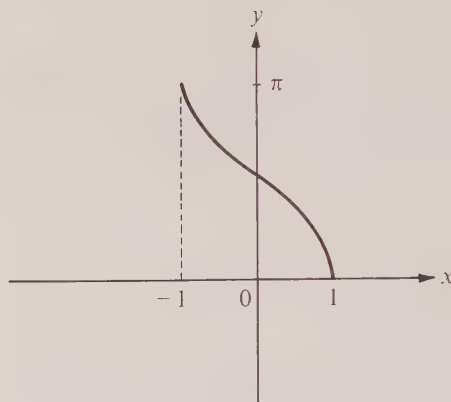


FIGURE 3.14 $y = \arcsin x$.

FIGURE 3.15 $y = \cos x$.FIGURE 3.16 $y = \arccos x$.

Of course, this now gives us a new integration formula,[†] namely

$$(3.54) \quad \int_0^x \frac{1}{\sqrt{1-t^2}} dt = \arcsin x,$$

which is valid for $-1 < x < 1$. In the Leibniz notation for indefinite integrals we may write

$$(3.55) \quad \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C.$$

Integration by parts yields the following further integration formula:

$$\int \arcsin x \, dx = x \arcsin x - \int \frac{x \, dx}{\sqrt{1-x^2}} = x \arcsin x + \sqrt{1-x^2} + C.$$

The cosine and tangent are inverted in a similar fashion. For the cosine it is customary to choose the interval $[0, \pi]$ in which to perform the inversion. (See Figure 3.15.) The resulting inverse function, called the arc cosine, is defined as follows:

$$u = \arccos v \quad \text{means} \quad v = \cos u \quad \text{and} \quad 0 \leq u \leq \pi.$$

[†] The integral in (3.54) may be used as the starting point for a completely analytic theory of the trigonometric functions, without any reference to geometry. Briefly, the idea is to begin with the arc sine function, defining it by the integral in (3.54), just as we defined the logarithm as an integral. Next, the sine function is defined as the inverse of the arc sine, and the cosine as the derivative of the sine. Many details are required to carry out this program completely and we shall not attempt to describe them here. An alternative method for introducing the trigonometric functions analytically will be mentioned in Chapter 9.

The graph of the arc cosine function is shown in Figure 3.16.

To invert the tangent we choose the open interval $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ (see Figure 3.17) and we define the arc tangent as follows:

$$u = \arctan v \quad \text{means} \quad v = \tan u \quad \text{and} \quad -\frac{\pi}{2} < u < \frac{\pi}{2}.$$

Figure 3.18 shows a portion of the graph of the arc tangent function.

The argument used to derive (3.53) can also be applied to the arc cosine and arc tangent functions, and it yields the following differentiation formulas:

$$(3.56) \quad D \arccos x = \frac{-1}{\sqrt{1-x^2}},$$

valid for $-1 < x < 1$, and

$$(3.57) \quad D \arctan x = \frac{1}{1+x^2},$$

valid for all real x .

When (3.56) is translated into an integration formula it becomes

$$(3.58) \quad \int_0^x \frac{1}{\sqrt{1-t^2}} dt = -(\arccos x - \arccos 0) = \frac{\pi}{2} - \arccos x$$

if $-1 < x < 1$. By comparing (3.58) with (3.54), we deduce the relation $\frac{1}{2}\pi - \arccos x = \arcsin x$. (This may also be deduced from the familiar identity $\sin(\frac{1}{2}\pi - y) = \cos y$ if we write $y = \arccos x$.) In the Leibniz notation for indefinite integrals we may write (3.58) as follows:

$$(3.59) \quad \int \frac{dx}{\sqrt{1-x^2}} = -\arccos x + C.$$

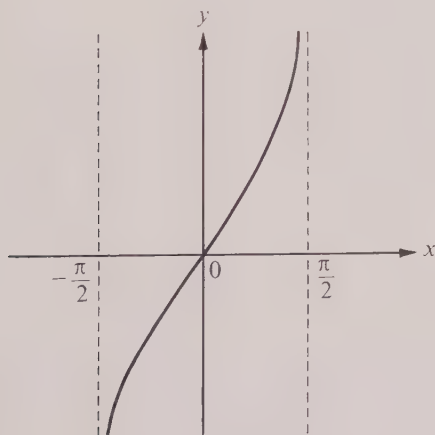


FIGURE 3.17 $y = \tan x$.

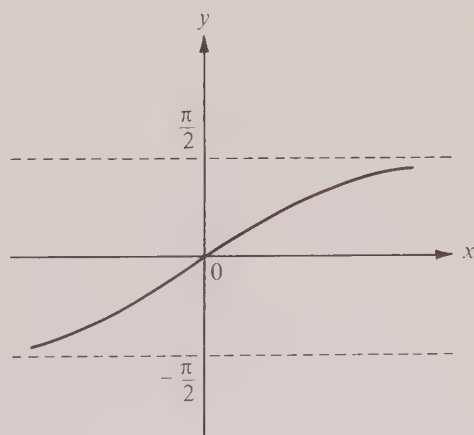


FIGURE 3.18 $y = \arctan x$.

Similarly, from (3.57) we obtain

$$(3.60) \quad \int_0^x \frac{dt}{1+t^2} = \arctan x \quad \text{or} \quad \int \frac{dx}{1+x^2} = \arctan x + C.$$

Using integration by parts in conjunction with (3.59) and (3.60), we can derive the following further integration formulas:

$$\int \arccos x \, dx = x \arccos x + \int \frac{x \, dx}{\sqrt{1-x^2}} = x \arccos x - \sqrt{1-x^2} + C,$$

$$\int \arctan x \, dx = x \arctan x - \int \frac{x \, dx}{1+x^2} = x \arctan x - \frac{1}{2} \log(1+x^2) + C.$$

The inverses of the cotangent, secant, and cosecant can be defined by means of the following formulas:

$$\operatorname{arccot} x = \frac{\pi}{2} - \arctan x \quad \text{for all real } x,$$

$$\operatorname{arcsec} x = \arccos \frac{1}{x} \quad \text{when } |x| \geq 1,$$

$$\operatorname{arccsc} x = \arcsin \frac{1}{x} \quad \text{when } |x| \geq 1.$$

Differentiation and integration formulas for these functions are listed in the exercises of the following section.

3.17 Exercises

Derive the differentiation formulas in Exercises 1 through 5.

$$1. \, D \arccos x = \frac{-1}{\sqrt{1-x^2}} \quad \text{if } -1 < x < 1,$$

$$2. \, D \arctan x = \frac{1}{1+x^2} \quad \text{for all real } x,$$

$$3. \, D \operatorname{arccot} x = \frac{-1}{1+x^2} \quad \text{for all real } x,$$

$$4. \, D \operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}} \quad \text{if } |x| > 1,$$

$$5. \, D \operatorname{arccsc} x = \frac{-1}{|x|\sqrt{x^2-1}} \quad \text{if } |x| > 1.$$

Derive the integration formulas in Exercises 6 through 10.

$$6. \, \int \operatorname{arccot} x \, dx = x \operatorname{arccot} x + \frac{1}{2} \log(1+x^2) + C.$$

$$7. \, \int \operatorname{arcsec} x \, dx = x \operatorname{arcsec} x - \frac{x}{|x|} \log \left| x + \sqrt{x^2-1} \right| + C.$$

$$8. \int \operatorname{arc} \csc x \, dx = x \operatorname{arc} \csc x + \frac{x}{|x|} \log \left| x + \sqrt{x^2 - 1} \right| + C.$$

$$9. \int (\operatorname{arc} \sin x)^2 \, dx = x(\operatorname{arc} \sin x)^2 - 2x + 2\sqrt{1 - x^2} \operatorname{arc} \sin x + C.$$

$$10. \int \frac{\operatorname{arc} \sin x}{x^2} \, dx = \log \left| \frac{1 - \sqrt{1 - x^2}}{x} \right| - \frac{\operatorname{arc} \sin x}{x} + C.$$

$$11. (a) \text{ Show that } D \left(\operatorname{arc} \cot x - \operatorname{arc} \tan \frac{1}{x} \right) = 0 \text{ for all } x \neq 0.$$

(b) Prove that there is no constant C such that $\operatorname{arc} \cot x - \operatorname{arc} \tan (1/x) = C$ for all $x \neq 0$. Explain why this does not contradict Theorem 2-5 in Section 2.15.

12. Show that

$$\operatorname{arc} \tan x + \operatorname{arc} \tan \frac{1}{x} = \begin{cases} \frac{\pi}{2} & \text{if } x > 0, \\ -\frac{\pi}{2} & \text{if } x < 0. \end{cases}$$

$$\left[\text{Hint. Compute } D \left(\operatorname{arc} \tan x + \operatorname{arc} \tan \frac{1}{x} \right). \right]$$

13. If $xy \neq 1$, show that

$$\operatorname{arc} \tan x + \operatorname{arc} \tan y = \operatorname{arc} \tan \frac{x + y}{1 - xy} + \pi \alpha(x, y),$$

where

$$\alpha(x, y) = \begin{cases} 0 & \text{if } xy < 1, \\ 1 & \text{if } xy > 1 \text{ and } x > 0, \\ -1 & \text{if } xy > 1 \text{ and } x < 0. \end{cases}$$

$$\left[\text{Hint. Show that } \frac{\partial}{\partial x} \left(\operatorname{arc} \tan x - \operatorname{arc} \tan \frac{x + y}{1 - xy} \right) = 0. \right]$$

14. If $|x| \leq 1$ and $|y| \leq 1$, show that

$$\operatorname{arc} \sin x + \operatorname{arc} \sin y = (-1)^\alpha \operatorname{arc} \sin (x\sqrt{1 - y^2} + y\sqrt{1 - x^2}) + \pi \alpha,$$

where

$$\alpha = \begin{cases} 0 & \text{if } xy \leq 0 \text{ or } x^2 + y^2 \leq 1, \\ 1 & \text{if } xy > 0, x > 0, \text{ and } x^2 + y^2 > 1, \\ -1 & \text{if } xy > 0, x < 0, \text{ and } x^2 + y^2 > 1. \end{cases}$$

Draw a figure in the xy -plane and indicate the regions where α has a fixed value.

In Exercises 15 through 28, find the derivative $f'(x)$. In each case the function f is assumed to be defined for all real x for which the given formula for $f(x)$ is meaningful.

$$15. f(x) = \operatorname{arc} \sin \frac{x}{2}.$$

$$22. f(x) = \operatorname{arc} \tan (\tan^2 x).$$

$$16. f(x) = \operatorname{arc} \cos \frac{1 - x}{\sqrt{2}}.$$

$$23. f(x) = \operatorname{arc} \tan (x + \sqrt{1 + x^2}).$$

$$17. f(x) = \operatorname{arc} \cos \frac{1}{x}.$$

$$24. f(x) = \operatorname{arc} \sin (\sin x - \cos x).$$

$$18. f(x) = \operatorname{arc} \sin (\sin x).$$

$$25. f(x) = \operatorname{arc} \cos \sqrt{1 - x^2}.$$

$$19. f(x) = \sqrt{x} - \operatorname{arc} \tan \sqrt{x}.$$

$$26. f(x) = \operatorname{arc} \tan \frac{1 + x}{1 - x}.$$

$$20. f(x) = \operatorname{arc} \tan x + \frac{1}{2} \operatorname{arc} \tan (x^3).$$

$$27. f(x) = [\operatorname{arc} \cos (x^2)]^{-2}.$$

$$21. f(x) = \operatorname{arc} \sin \frac{1 - x^2}{1 + x^2}.$$

$$28. f(x) = \log \left(\operatorname{arc} \cos \frac{1}{\sqrt{x}} \right).$$

29. Show that $dy/dx = (x + y)/(x - y)$ if $\arctan(y/x) = \log \sqrt{x^2 + y^2}$.
30. Compute d^2y/dx^2 if $y = (\arcsin x)/\sqrt{1 - x^2}$ for $|x| < 1$.
31. (a) If $f(x) = \arctan x$, show that for every integer $n \geq 0$ we have $f^{(2n)}(0) = 0$ and $f^{(2n+1)}(0) = (-1)^n(2n)!$.
- (b) Find similar formulas for $f^{(n)}(0)$ when $f(x) = \arcsin x$.

In Exercises 32 through 50, evaluate the indefinite integrals.

- | | |
|--|--|
| 32. $\int \frac{dx}{\sqrt{a^2 - x^2}}, \quad a \neq 0.$ | 41. $\int \frac{\arctan \sqrt{x}}{\sqrt{x}(1+x)} dx.$ |
| 33. $\int \frac{dx}{\sqrt{1-2x-x^2}}.$ | 42. $\int \sqrt{1-x^2} dx.$ |
| 34. $\int \frac{dx}{a^2 + x^2}, \quad a \neq 0.$ | 43. $\int \frac{x e^{\arctan x}}{(1+x^2)^{3/2}} dx.$ |
| 35. $\int \frac{dx}{a + bx^2} \quad (ab \neq 0).$ | 44. $\int \frac{e^{\arctan x}}{(1+x^2)^{3/2}} dx.$ |
| 36. $\int \frac{dx}{x^2 - x + 2}.$ | 45. $\int \frac{x^2}{(1+x^2)^2} dx.$ |
| 37. $\int x \arctan x dx.$ | 46. $\int \frac{e^x}{1 + e^{2x}} dx.$ |
| 38. $\int x^2 \arccos x dx.$ | 47. $\int \frac{\arccot e^x}{e^x} dx.$ |
| 39. $\int x(\arctan x)^2 dx.$ | 48. $\int \left(\frac{a+x}{a-x} \right)^{1/2} dx, \quad a > 0.$ |
| 40. $\int \arctan \sqrt{x} dx.$ | |
| 49. $\int \frac{dx}{\sqrt{(x-a)(b-x)}}, \quad b \neq a. \text{ [Hint. } x-a = (b-a)\sin^2 u.]$ | |
| 50. $\int \sqrt{(x-a)(b-x)} dx, \quad b \neq a.$ | |

3.18 Integration by partial fractions

We recall that a quotient of two polynomials is called a rational function. Differentiation of a rational function leads to a new rational function which may be obtained by the quotient rule for derivatives. On the other hand, integration of a rational function may lead to functions that are not rational. For example, we have

$$\int \frac{dx}{x} = \log |x| + C \quad \text{and} \quad \int \frac{dx}{1+x^2} = \arctan x + C.$$

We shall describe a method for computing the integral of any rational function and we shall find that the result can always be expressed in terms of polynomials, rational functions, inverse tangents, and logarithms.

The basic idea of the method is to decompose a given rational function into a sum of simpler fractions (called partial fractions) that can be integrated by the techniques dis-

cussed earlier. We shall describe the general procedure by means of a number of simple examples that illustrate all the essential features of the method.

Example 1. In this example we begin with two simple fractions, $1/(x - 1)$ and $1/(x + 3)$, which we know how to integrate, and see what happens when we form a linear combination of these fractions. For example, if we take twice the first fraction plus three times the second, we obtain

$$\frac{2}{x - 1} + \frac{3}{x + 3} = \frac{2(x + 3) + 3(x - 1)}{(x - 1)(x + 3)} = \frac{5x + 3}{x^2 + 2x - 3}.$$

If, now, we read this formula from right to left, it tells us that the rational function r given by $r(x) = (5x + 3)/(x^2 + 2x - 3)$ has been expressed as a linear combination of $1/(x - 1)$ and $1/(x + 3)$. Therefore we may evaluate the integral of r by writing

$$\int \frac{5x + 3}{x^2 + 2x - 3} dx = 2 \int \frac{dx}{x - 1} + 3 \int \frac{dx}{x + 3} = 2 \log |x - 1| + 3 \log |x + 3| + C.$$

Example 2. The foregoing example suggests a procedure for dealing with integrals of the form $\int (ax + b)/(x^2 + 2x - 3) dx$. For example, to evaluate $\int (2x + 5)/(x^2 + 2x - 3) dx$ we try to express the integral as a linear combination of $1/(x - 1)$ and $1/(x + 3)$ by writing

$$(3.61) \quad \frac{2x + 5}{x^2 + 2x - 3} = \frac{A}{x - 1} + \frac{B}{x + 3}$$

with constants A and B to be determined. If we can choose A and B so that Equation (3.61) is an identity, then the integral of the fraction on the left is equal to the sum of the integrals of the simpler fractions on the right. To find A and B we multiply both sides of (3.61) by $(x - 1)(x + 3)$ to remove the fractions. This gives us

$$(3.62) \quad A(x + 3) + B(x - 1) = 2x + 5.$$

At this stage there are two methods commonly used to find A and B . One method is to equate coefficients of like powers of x in (3.62). This leads to the equations $A + B = 2$ and $3A - B = 5$. Solving this pair of simultaneous equations, we obtain $A = 7/4$ and $B = 1/4$. The other method involves the substitution of two values of x in (3.62), and leads to another pair of equations for A and B . In this particular case, the presence of the factors $x - 1$ and $x + 3$ suggests that we use the values $x = 1$ and $x = -3$. When we put $x = 1$ in (3.62) the coefficient of B vanishes and we find $4A = 7$, or $A = 7/4$. Similarly, we can make the coefficient of A vanish by putting $x = -3$. This gives us $-4B = -1$, or $B = 1/4$. In any event, we have found values of A and B to satisfy (3.61), so we have

$$\int \frac{2x + 5}{x^2 + 2x - 3} dx = \frac{7}{4} \int \frac{dx}{x - 1} + \frac{1}{4} \int \frac{dx}{x + 3} = \frac{7}{4} \log |x - 1| + \frac{1}{4} \log |x + 3| + C.$$

It is clear that the method described in Example 2 also applies to integrals of the form $\int f(x)/g(x) dx$ in which f is a linear polynomial and g is a quadratic polynomial that can be factored into distinct linear factors with real coefficients, say $g(x) = (x - x_1)(x - x_2)$. In this case the quotient $f(x)/g(x)$ can be expressed as a linear combination of $1/(x - x_1)$ and $1/(x - x_2)$, and integration of $f(x)/g(x)$ leads to a corresponding combination of the logarithmic terms $\log |x - x_1|$ and $\log |x - x_2|$.

The foregoing examples involve rational functions f/g in which the degree of the numerator is less than that of the denominator. A rational function with this property is said to be a *proper* rational function. If f/g is *improper*, that is, if the degree of f is not

less than that of g , then we can express f/g as the sum of a polynomial and a proper rational function. In fact, we simply divide f by g to obtain

$$\frac{f(x)}{g(x)} = Q(x) + \frac{R(x)}{g(x)},$$

where Q and R are polynomials (called the *quotient* and *remainder*, respectively) such that the remainder has degree less than that of g . For example,

$$\frac{x^3 + 3x}{x^2 - 2x - 3} = x + 2 + \frac{10x + 6}{x^2 - 2x - 3}.$$

Therefore, in the study of integration technique, there is no loss in generality if we restrict ourselves to *proper* rational functions, and from now on we consider $\int f(x)/g(x) dx$, where f has degree less than that of g .

A general theorem in algebra states that every proper rational function can be expressed as a finite sum of fractions of the forms

$$\frac{A}{(x + a)^k} \quad \text{and} \quad \frac{Bx + C}{(x^2 + bx + c)^m},$$

where k and m are positive integers and A, B, C, a, b, c are constants with $b^2 - 4c < 0$. The condition $b^2 - 4c < 0$ means that the quadratic polynomial $x^2 + bx + c$ cannot be factored into linear factors with real coefficients or, what amounts to the same thing, the quadratic equation $x^2 + bx + c = 0$ has no real roots. Such a quadratic factor is said to be *irreducible*. When a rational function has been so expressed, we say that it has been decomposed into *partial fractions*. Therefore the problem of integrating this rational function reduces to that of integrating its partial fractions. These may be easily dealt with by the techniques described in the examples which follow.

We shall not bother to prove that partial-fraction decompositions always exist. Instead, we shall show (by means of examples) how to obtain the partial fractions in specific problems. In each case that arises the partial-fraction decomposition can be verified directly.

It is convenient to separate the discussion into cases depending on the way in which the denominator of the quotient $f(x)/g(x)$ can be factored.

CASE 1. The denominator is a product of distinct linear factors. Suppose $g(x)$ splits into n distinct linear factors, say

$$g(x) = (x - x_1)(x - x_2) \cdots (x - x_n).$$

Now notice that a linear combination of the form

$$\frac{A_1}{x - x_1} + \cdots + \frac{A_n}{x - x_n}$$

may be expressed as a single fraction with the common denominator $g(x)$, and the numerator of this fraction will be a polynomial of degree $< n$ involving the A 's. There-

fore, if we can find A 's to make this numerator equal to $f(x)$ we shall have the decomposition

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - x_1} + \cdots + \frac{A_n}{x - x_n},$$

and the integral of $f(x)/g(x)$ will be equal to $\sum_{i=1}^n A_i \log |x - x_i|$. In the next example we work out a case with $n = 3$.

Example 3. Integrate $\int \frac{2x^2 + 5x - 1}{x^3 + x^2 - 2x} dx$.

Solution. Since $x^3 + x^2 - 2x = x(x - 1)(x + 2)$, the denominator is a product of distinct linear factors and we try to find A_1 , A_2 , and A_3 such that

$$\frac{2x^2 + 5x - 1}{x^3 + x^2 - 2x} = \frac{A_1}{x} + \frac{A_2}{x - 1} + \frac{A_3}{x + 2}.$$

Clearing the fractions, we obtain

$$2x^2 + 5x - 1 = A_1(x - 1)(x + 2) + A_2x(x + 2) + A_3x(x - 1).$$

When $x = 0$ we find $-2A_1 = -1$, so $A_1 = \frac{1}{2}$. When $x = 1$ we obtain $3A_2 = 6$, $A_2 = 2$, and when $x = -2$ we find $6A_3 = -3$, or $A_3 = -\frac{1}{2}$. Therefore we have

$$\begin{aligned} \int \frac{2x^2 + 5x - 1}{x^3 + x^2 - 2x} dx &= \frac{1}{2} \int \frac{dx}{x} + 2 \int \frac{dx}{x - 1} - \frac{1}{2} \int \frac{dx}{x + 2} \\ &= \frac{1}{2} \log |x| + 2 \log |x - 1| - \frac{1}{2} \log |x + 2| + C. \end{aligned}$$

CASE 2. The denominator is a product of linear factors, some of which are repeated. We illustrate this case with an example.

Example 4. Integrate $\int \frac{x^2 + 2x + 3}{(x - 1)(x + 1)^2} dx$.

Solution. Here we try to find A_1 , A_2 , A_3 so that

$$(3.63) \quad \frac{x^2 + 2x + 3}{(x - 1)(x + 1)^2} = \frac{A_1}{x - 1} + \frac{A_2}{x + 1} + \frac{A_3}{(x + 1)^2}.$$

We need both $A_2/(x + 1)$ and $A_3/(x + 1)^2$ as well as $A_1/(x - 1)$ in order to get a polynomial of degree two in the numerator and to have as many constants as equations when we try to determine the A 's. Clearing the fractions, we obtain

$$(3.64) \quad x^2 + 2x + 3 = A_1(x + 1)^2 + A_2(x - 1)(x + 1) + A_3(x - 1).$$

Substituting $x = 1$, we find $4A_1 = 6$, so $A_1 = \frac{3}{2}$. When $x = -1$ we obtain $-2A_3 = 2$ and $A_3 = -1$. We need one more equation to determine A_2 . Since there are no other choices of x that will make any factor vanish we choose a convenient x that will help to simplify the calculations. For example, the choice $x = 0$ leads to the equation $3 = A_1 - A_2 - A_3$ from which we find $A_2 = -\frac{1}{2}$. An alternative method is to differentiate both sides of (3.64) and then substitute a convenient x . Differentiation of (3.64) leads to the equation

$$2x + 2 = 2A_1(x + 1) + A_2(x - 1) + A_2(x + 1) + A_3,$$

and if we put $x = -1$ we find $0 = -2A_2 + A_3$, so $A_2 = \frac{1}{2}A_3 = -\frac{1}{2}$, as before. Therefore we have found A 's to satisfy (3.63), so we have

$$\begin{aligned}\int \frac{x^2 + 2x + 3}{(x-1)(x+1)^2} dx &= \frac{3}{2} \int \frac{dx}{x-1} - \frac{1}{2} \int \frac{dx}{x+1} - \int \frac{dx}{(x+1)^2} \\ &= \frac{3}{2} \log |x-1| - \frac{1}{2} \log |x+1| + \frac{1}{x+1} + C.\end{aligned}$$

If, on the left of (3.63), the factor $(x+1)^3$ had appeared instead of $(x+1)^2$, we would have added an extra term $A_4/(x+1)^3$ on the right. More generally, if a linear factor $x+a$ appears p times in the denominator, then for this factor we must allow for a sum of p terms, namely

$$(3.65) \quad \sum_{k=1}^p \frac{A_k}{(x+a)^k},$$

where the A 's are constants. A sum of this type is to be used for each repeated linear factor.

CASE 3. The denominator contains irreducible quadratic factors, none of which are repeated.

Example 5. Integrate $\int \frac{3x^2 + 2x - 2}{x^3 - 1} dx$.

Solution. The denominator can be factored as the product

$$x^3 - 1 = (x-1)(x^2 + x + 1),$$

where $x^2 + x + 1$ is irreducible, and we try a decomposition of the form

$$\frac{3x^2 + 2x - 2}{x^3 - 1} = \frac{A}{x-1} + \frac{Bx + C}{x^2 + x + 1}.$$

In the fraction with denominator $x^2 + x + 1$ we have used a linear polynomial $Bx + C$ in the numerator in order to have as many constants as equations when we solve for A , B , C . Clearing the fractions and solving for A , B , and C , we find $A = 1$, $B = 2$, and $C = 3$. Therefore we have

$$\int \frac{3x^2 + 2x - 2}{x^3 - 1} dx = \int \frac{dx}{x-1} + \int \frac{2x+3}{x^2+x+1} dx.$$

The first integral on the right is $\log |x-1|$. To evaluate the second integral we write

$$\begin{aligned}\int \frac{2x+3}{x^2+x+1} dx &= \int \frac{2x+1}{x^2+x+1} dx + \int \frac{2}{x^2+x+1} dx \\ &= \log(x^2+x+1) + 2 \int \frac{dx}{(x+\frac{1}{2})^2 + \frac{3}{4}}.\end{aligned}$$

If we let $u = x + \frac{1}{2}$ and $\alpha = \sqrt{\frac{3}{4}}$, the last integral is

$$2 \int \frac{du}{u^2 + \alpha^2} = \frac{2}{\alpha} \arctan \frac{u}{\alpha} = \frac{4}{3} \sqrt{3} \arctan \frac{2x+1}{\sqrt{3}}.$$

Therefore we have

$$\int \frac{3x^2 + 2x - 2}{x^3 - 1} dx = \log |x - 1| + \log (x^2 + x + 1) + \frac{4}{3} \sqrt{3} \arctan \frac{2x + 1}{\sqrt{3}} + C.$$

CASE 4. The denominator contains irreducible quadratic factors, some of which are repeated. Here the situation is analogous to Case 2. In the partial-fraction decomposition of $f(x)/g(x)$ we allow, first of all, a sum of the form (3.65) for each linear factor, as already described. In addition, if an irreducible quadratic factor $x^2 + bx + c$ is repeated m times, we allow a sum of m terms, namely

$$\sum_{k=1}^m \frac{B_k x + C_k}{(x^2 + bx + c)^k},$$

where each numerator is linear.

Example 6. Integrate $\int \frac{x^4 - x^3 + 2x^2 - x + 2}{(x - 1)(x^2 + 2)^2} dx$.

Solution. We write

$$\frac{x^4 - x^3 + 2x^2 - x + 2}{(x - 1)(x^2 + 2)^2} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 2} + \frac{Dx + E}{(x^2 + 2)^2}.$$

Clearing the fractions and solving for A , B , C , D , and E , we find

$$A = \frac{1}{3}, \quad B = \frac{2}{3}, \quad C = -\frac{1}{3}, \quad D = -1, \quad E = 0.$$

Therefore we have

$$\begin{aligned} \int \frac{x^4 - x^3 + 2x^2 - x + 2}{(x - 1)(x^2 + 2)^2} dx &= \frac{1}{3} \int \frac{dx}{x - 1} + \int \frac{\frac{2}{3}x - \frac{1}{3}}{x^2 + 2} dx - \int \frac{x dx}{(x^2 + 2)^2} \\ &= \frac{1}{3} \int \frac{dx}{x - 1} + \frac{1}{3} \int \frac{2x dx}{x^2 + 2} - \frac{1}{3} \int \frac{dx}{x^2 + 2} - \frac{1}{2} \int \frac{2x dx}{(x^2 + 2)^2} \\ &= \frac{1}{3} \log |x - 1| + \frac{1}{3} \log (x^2 + 2) - \frac{1}{6} \arctan \frac{x}{2} + \frac{1}{2} \frac{1}{x^2 + 2} + C. \end{aligned}$$

The foregoing examples are typical of what happens in general. The problem of integrating a proper rational function reduces to that of calculating integrals of the forms

$$\int \frac{dx}{(x + a)^n}, \quad \int \frac{x dx}{(x^2 + bx + c)^m}, \quad \text{and} \quad \int \frac{dx}{(x^2 + bx + c)^m}.$$

The first integral is $\log |x + a|$ if $n = 1$ and $(x + a)^{1-n}/(1 - n)$ if $n > 1$. To treat the other two we express the quadratic as a sum of two squares by writing

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) = u^2 + \alpha^2,$$

where $u = x + b/2$ and $\alpha = \frac{1}{2}\sqrt{4c - b^2}$. (This is possible because $4c - b^2 > 0$.) The substitution $u = x + b/2$ reduces the problem to that of computing

$$(3.66) \quad \int \frac{u \, du}{(u^2 + \alpha^2)^m} \quad \text{and} \quad \int \frac{du}{(u^2 + \alpha^2)^m}.$$

The first of these is $\frac{1}{2} \log(u^2 + \alpha^2)$ if $m = 1$, and $\frac{1}{2}(u^2 + \alpha^2)^{1-m}/(1 - m)$ if $m > 1$. When $m = 1$ the second integral in (3.66) is evaluated by the formula

$$\int \frac{du}{u^2 + \alpha^2} = \frac{1}{\alpha} \arctan \frac{u}{\alpha} + C.$$

The case $m > 1$ may be reduced to the case $m = 1$ by repeated application of the recursion formula

$$\int \frac{du}{(u^2 + \alpha^2)^m} = \frac{1}{2\alpha^2(m-1)} \frac{u}{(u^2 + \alpha^2)^{m-1}} + \frac{2m-3}{2\alpha^2(m-1)} \int \frac{du}{(u^2 + \alpha^2)^{m-1}},$$

which is obtained by integration by parts. This discussion shows that every rational function may be integrated in terms of polynomials, rational functions, inverse tangents, and logarithms.

A function of two variables defined by an equation of the form

$$P(x, y) = \sum_{m=0}^p \sum_{n=0}^q a_{m,n} x^m y^n$$

is called a *polynomial in two variables*. The quotient of two such polynomials is called a *rational function of two variables*. Integrals of the form $\int R(\sin x, \cos x) \, dx$, where R is a rational function of two variables, may be reduced by the substitution $u = \tan \frac{1}{2}x$ to integrals of the form $\int r(u) \, du$ where r is a rational function of one variable. The latter integral may be evaluated by the techniques just described. We illustrate the method with a particular example.

Example 7. Integrate $\int \frac{1}{\sin x + \cos x} \, dx$.

Solution. The substitution $u = \tan \frac{1}{2}x$ gives us

$$x = 2 \arctan u, \quad dx = \frac{2}{1+u^2} \, du,$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} = \frac{2 \tan \frac{1}{2}x}{\sec^2 \frac{1}{2}x} = \frac{2u}{1+u^2},$$

$$\cos x = 2 \cos^2 \frac{x}{2} - 1 = \frac{2}{\sec^2 \frac{1}{2}x} - 1 = \frac{2}{1+u^2} - 1 = \frac{1-u^2}{1+u^2},$$

and

$$\sin x + \cos x = \frac{2u + 1 - u^2}{1 + u^2}.$$

Therefore we have

$$\int \frac{dx}{\sin x + \cos x} = -2 \int \frac{du}{u^2 - 2u - 1} = -2 \int \frac{du}{(u-a)(u-b)},$$

where $a = 1 + \sqrt{2}$ and $b = 1 - \sqrt{2}$. The method of partial fractions leads to

$$\int \frac{du}{(u-a)(u-b)} = \frac{1}{a-b} \int \left(\frac{1}{u-a} - \frac{1}{u-b} \right) du$$

and, since $a - b = 2\sqrt{2}$, we obtain

$$(3.67) \quad \int \frac{dx}{\sin x + \cos x} = \frac{\sqrt{2}}{2} \log \left| \frac{u-b}{u-a} \right| + C = \frac{\sqrt{2}}{2} \log \left| \frac{\tan \frac{1}{2}x - 1 + \sqrt{2}}{\tan \frac{1}{2}x - 1 - \sqrt{2}} \right| + C.$$

The final answer may be simplified somewhat by using suitable trigonometric identities. First we note that $\sqrt{2} - 1 = \tan \frac{1}{8}\pi$ so the numerator of the last fraction in (3.67) is $\tan \frac{1}{2}x + \tan \frac{1}{8}\pi$. In the denominator we write

$$\left| \tan \frac{x}{2} - 1 - \sqrt{2} \right| = (\sqrt{2} + 1) \left| (\sqrt{2} - 1) \tan \frac{x}{2} - 1 \right| = (\sqrt{2} + 1) \left| 1 - \tan \frac{x}{2} \tan \frac{\pi}{8} \right|.$$

Taking logarithms as indicated in (3.67), we may combine the term $-\frac{1}{2}\sqrt{2} \log(\sqrt{2} + 1)$ with the arbitrary constant and rewrite (3.67) as follows:

$$\int \frac{dx}{\sin x + \cos x} = \frac{\sqrt{2}}{2} \log \left| \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) \right| + C.$$

3.19 Exercises

Evaluate the following integrals:

$$1. \int \frac{2x+3}{(x-2)(x+5)} dx.$$

$$2. \int \frac{x dx}{(x+1)(x+2)(x+3)}.$$

$$3. \int \frac{x dx}{x^3 - 3x + 2}.$$

$$4. \int \frac{x^4 + 2x - 6}{x^3 + x^2 - 2x} dx.$$

$$5. \int \frac{8x^3 + 7}{(x+1)(2x+1)^3} dx.$$

$$6. \int \frac{4x^2 + x + 1}{x^3 - 1} dx.$$

$$7. \int \frac{x^4 dx}{x^4 + 5x^2 + 4}.$$

$$8. \int \frac{x+2}{x^2+x} dx.$$

$$9. \int \frac{dx}{x(x^2+1)^2}.$$

$$10. \int \frac{dx}{(x+1)(x+2)^2(x+3)^3}.$$

$$11. \int \frac{x dx}{(x+1)^2}.$$

$$12. \int \frac{dx}{x^3 - x}.$$

$$13. \int \frac{x^2 dx}{x^2 + x - 6}.$$

$$14. \int \frac{(x+2) dx}{x^2 - 4x + 4}.$$

$$15. \int \frac{dx}{(x^2 - 4x + 4)(x^2 - 4x + 5)}.$$

$$16. \int \frac{(x-3) dx}{x^3 + 3x^2 + 2x}.$$

17. $\int \frac{dx}{(x^2 - 1)^2}.$
18. $\int \frac{x+1}{x^3 - 1} dx.$
19. $\int \frac{x^4 + 1}{x(x^2 + 1)^2} dx.$
20. $\int \frac{dx}{x^4 - 2x^3}.$
21. $\int \frac{1 - x^3}{x(x^2 + 1)} dx.$
22. $\int \frac{dx}{x^4 - 1}.$
23. $\int \frac{dx}{x^4 + 1}.$
24. $\int \frac{x^2 dx}{(x^2 + 2x + 2)^2}.$
25. $\int \frac{4x^5 - 1}{(x^5 + x + 1)^2} dx.$
26. $\int \frac{dx}{2 \sin x - \cos x + 5}.$
27. $\int \frac{dx}{1 + a \cos x} \quad (0 < a < 1).$
28. $\int \frac{dx}{1 + a \cos x} \quad (a > 1).$
29. $\int \frac{\sin^2 x}{1 + \sin^2 x} dx.$
30. $\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} \quad (ab \neq 0).$
31. $\int \frac{dx}{(a \sin x + b \cos x)^2} \quad (a \neq 0).$
32. $\int_0^{\pi/2} \frac{\sin x dx}{1 + \cos x + \sin x}.$

3.20 Miscellaneous review exercises

1. Let $f(x) = \int_1^x (\log t)/(t + 1) dt$ if $x > 0$. Compute $f(x) + f(1/x)$. As a check, you should obtain $f(2) + f(\frac{1}{2}) = \frac{1}{2} \log^2 2$.

2. Find a function f , continuous for all x (and not everywhere zero), such that

$$f^2(x) = \int_0^x f(t) \frac{\sin t}{2 + \cos t} dt.$$

3. Try to evaluate $\int e^x/x dx$ by using integration by parts.

4. Integrate $\int_0^{\pi/2} \log(e^{\cos x}) dx$.

5. A function f is defined by the equation

$$f(x) = \sqrt{\frac{4x + 2}{x(x + 1)(x + 2)}} \quad \text{if } x > 0.$$

(a) Find the slope of the graph of f at the point for which $x = 1$.

(b) The region under the graph and above the interval $[1, 4]$ is rotated about the x -axis, thus generating a solid of revolution. Write an integral for the volume of this solid. Compute this integral and show that its value is $\pi \log(25/8)$.

6. A function F is defined by the following indefinite integral:

$$F(x) = \int_1^x \frac{e^t}{t} dt \quad \text{if } x > 0.$$

(a) For what values of x is it true that $\log x \leq F(x)$?

(b) Prove that $\int_1^x e^t/(t + a) dt = e^{-a}[F(x + a) - F(1 + a)]$.

(c) In a similar way, express the following integrals in terms of F :

$$\int_1^x \frac{e^{at}}{t} dt, \quad \int_1^x \frac{e^t}{t^2} dt, \quad \int_1^x e^{1/t} dt.$$

7. In each case, give an example of a continuous function f satisfying the conditions stated for all real x , or else explain why there is no such function:

(a) $\int_0^x f(t) dt = e^x.$

(b) $\int_0^{x^2} f(t) dt = 1 - 2x^2.$ [$2x^2$ means $2(x^2)$.]

(c) $\int_0^x f(t) dt = f^2(x) - 1.$

8. If $f(x + y) = f(x)f(y)$ for all x and y and if $f(x) = 1 + xg(x)$, where $g(x) \rightarrow 0$ as $x \rightarrow 0$, prove that (a) $f'(x)$ exists for every x , and (b) $f(x) = e^x$.

9. Given a function g which has a derivative $g'(x)$ for every real x and which satisfies the following equations:

$$g'(0) = 2 \quad \text{and} \quad g(x + y) = e^y g(x) + e^x g(y) \quad \text{for all } x \text{ and } y.$$

(a) Show that $g(2x) = 2e^x g(x)$ and find a similar formula for $g(3x)$.

(b) Generalize (a) by finding a formula relating $g(nx)$ to $g(x)$, valid for every positive integer n . Prove your result by induction.

(c) Show that $g(0) = 0$ and find the limit of $g(h)/h$ as $h \rightarrow 0$.

(d) There is a constant C such that $g'(x) = g(x) + Ce^x$ for all x . Prove this statement and find the value of C . [Hint. Use the definition of the derivative $g'(x)$.]

10. A periodic function with period a satisfies $f(x + a) = f(x)$ for all x in its domain. What can you conclude about a function which has a derivative everywhere and satisfies an equation of the form

$$f(x + a) = bf(x)$$

for all x , where a and b are positive constants?

11. Use logarithmic differentiation to derive the formulas for differentiation of products and quotients from the corresponding formulas for sums and differences.

12. Let $A = \int_0^1 e^t/(t + 1) dt$. Express the values of the following integrals in terms of A :

(a) $\int_{a-1}^a \frac{e^{-t}}{t - a - 1} dt.$

(c) $\int_0^1 \frac{e^t}{(t + 1)^2} dt.$

(b) $\int_0^1 \frac{te^{t^2}}{t^2 + 1} dt.$

(d) $\int_0^1 e^t \log(1 + t) dt.$

13. Let $p(x) = c_0 + c_1x + c_2x^2$ and let $f(x) = e^x p(x)$.

(a) Show that $f^{(n)}(0)$, the n th derivative of f at 0, is $c_0 + nc_1 + n(n - 1)c_2$.

(b) Solve the problem when p is a polynomial of degree 3.

(c) Generalize to a polynomial of degree m .

14. Let $f(x) = x \sin ax$. Show that $f^{(2n)}(x) = (-1)^n(a^{2n}x \sin ax - 2na^{2n-1} \cos ax)$.

15. Prove that

$$\sum_{k=0}^n (-1)^k \binom{n}{k} \frac{1}{k + m + 1} = \sum_{k=0}^m (-1)^k \binom{m}{k} \frac{1}{k + n + 1}.$$

[Hint. $1/(k + m + 1) = \int_0^1 t^{k+m} dt$.]

16. Let $F(x) = \int_0^x f(t) dt$. Determine a formula (or formulas) for computing $F(x)$ for all real x if f is defined as follows:

- (a) $f(t) = (t + |t|)^2$. (c) $f(t) = e^{-|t|}$.
 (b) $f(t) = \begin{cases} 1 - t^2 & \text{if } |t| \leq 1, \\ 1 - |t| & \text{if } |t| > 1. \end{cases}$ (d) $f(t) = \text{the maximum of } 1 \text{ and } t^2$.

17. A solid of revolution is generated by rotating the graph of $y = f(x)$, $0 \leq x \leq a$, about the x -axis. If, for every $a > 0$, the volume is $a^2 + a$, find the function f .

18. If a, b, a_1, b_1 are given, with $ab \neq 0$, show that there exist constants A, B, C such that

$$\int \frac{a_1 \sin x + b_1 \cos x}{a \sin x + b \cos x} dx = Ax + B \log |a \sin x + b \cos x| + C.$$

[Hint. Show that A and B exist such that

$$a_1 \sin x + b_1 \cos x = A(a \sin x + b \cos x) + B(a \cos x - b \sin x).]$$

19. In each case, find a function f satisfying the given conditions.

- (a) $f'(x^2) = 1/x$ for $x > 0$, $f(1) = 1$.
 (b) $f'(\sin^2 x) = \cos^2 x$ for all x , $f(1) = 1$.
 (c) $f'(\sin x) = \cos^2 x$ for all x , $f(1) = 1$.
 (d) $f'(\log x) = \begin{cases} 1 & \text{for } 0 < x \leq 1, \\ x & \text{for } x > 1, \end{cases} \quad f(0) = 0.$

20. A function, called the *integral logarithm* and denoted by Li , is defined as follows:

$$\text{Li}(x) = \int_2^x \frac{dt}{\log t} \quad \text{if } x \geq 2.$$

This function occurs in analytic number theory where it is proved that $\text{Li}(x)$ is a very good approximation to the number of primes $\leq x$. Derive the following properties of $\text{Li}(x)$:

- (a) $\text{Li}(x) = \frac{x}{\log x} + \int_2^x \frac{dt}{\log^2 t} - \frac{2}{\log 2}$.
 (b) $\text{Li}(x) = \frac{x}{\log x} + \sum_{k=1}^{n-1} \frac{k! x}{\log^{k+1} x} + n! \int_2^x \frac{dt}{\log^{n+1} t} + C_n$,

where C_n is a constant (depending on n). Find this constant.

(c) Show that there is a constant b such that $\int_b^{\log x} e^t/t dt = \text{Li}(x)$ and find the value of b .

(d) Express the integral $\int_c^x e^{2t}/(t-1) dt$ in terms of the integral logarithm, where $c = 1 + \log \sqrt{2}$.

(e) Let $f(x) = e^4 \text{Li}(e^{2x-4}) - e^2 \text{Li}(e^{2x-2})$ if $x > 3$. Show that

$$f'(x) = \frac{e^{2x}}{x^2 - 3x + 2}.$$

INTRODUCTION TO DIFFERENTIAL EQUATIONS

4.1 Introduction

A large variety of scientific problems arise in which one tries to determine something from its rate of change. For example, we could try to compute the position of a moving particle from a knowledge of its velocity or acceleration. Or a radioactive substance may be disintegrating at a known rate and we may be required to determine the amount of material present after a given time. In examples like these, we are trying to determine an *unknown function* from prescribed information expressed in the form of an equation involving at least one of the derivatives of the unknown function. These equations are called *differential equations* and their study forms one of the most challenging branches of mathematics.

Differential equations are classified under two main headings: *ordinary* and *partial*, depending on whether the unknown is a function of just *one* variable or of *two or more* variables. A simple example of an ordinary differential equation is the relation

$$(4.1) \qquad f'(x) = f(x)$$

which is satisfied, in particular, by the exponential function: $f(x) = e^x$. We shall see presently that every solution of (4.1) must be of the form $f(x) = Ce^x$, where C may be any constant.

On the other hand, an equation like

$$\frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2} = 0$$

is an example of a partial differential equation. This particular one, called *Laplace's equation*, appears in the theory of electricity and magnetism, fluid mechanics, and elsewhere. It has many different kinds of solutions, among which are $f(x, y) = x + 2y$, $f(x, y) = e^x \cos y$, and $f(x, y) = \log(x^2 + y^2)$.

A systematic study of differential equations began at the time of Newton and Leibniz and has continued to the present day. There is an enormous amount of literature on the subject and all we can hope to do here is consider some of the simpler types of differential equations that arise in practice. In some of the later chapters (Volume II) we shall have occasion to solve a few partial differential equations, but for the present we shall restrict

ourselves entirely to a few rather special types of ordinary differential equations that can be solved by very special devices. Some of the physical problems that lead to such equations will also be discussed, our object being to show the reader how such problems are translated into the language of differential equations.

When we work with a differential equation such as (4.1) it is customary to write y in place of $f(x)$ and y' in place of $f'(x)$, the higher derivatives being denoted by y'' , y''' , etc. Of course, other letters such as u , v , z , etc. are also used instead of y . By the *order* of an equation is meant the order of the highest derivative which appears. For example, (4.1) is a first-order equation which may be written as $y' = y$. The differential equation $y' = x^3y + \sin(xy'')$ is one of second order.

In this chapter we shall be interested primarily in first-order equations which can be solved for y' and written as follows:

$$(4.2) \quad y' = f(x, y),$$

where the expression $f(x, y)$ on the right has various special forms. The simplest case occurs when $f(x, y)$ is independent of y . In this case (4.2) becomes

$$(4.3) \quad y' = Q(x),$$

where Q is a known function. Solving the differential equation in (4.3) amounts to finding a primitive of Q . The only cases of interest for us are those for which Q is continuous on some interval. We know, from the first and second fundamental theorems of calculus, that every indefinite integral of Q is a primitive and that any primitive can be obtained from a given one by adding an appropriate constant. Therefore, the differential equation in (4.3) has an infinite number of solutions, all of which can be included in one comprehensive formula that is written as follows:

$$(4.4) \quad y = \int Q(x) dx + C.$$

The constant C (usually called an arbitrary constant of integration) may be specialized to yield the various solutions.

Example. Suppose a particle moves along a straight line in such a way that its velocity at time t is $2 \sin t$. If $f(t)$ denotes its position at time t , measured from some starting point, then we have $f'(t) = 2 \sin t$ and we conclude that $f(t) = -2 \cos t + C$. This is all we can deduce from a knowledge of the velocity alone; some other piece of information is needed to fix the position function. If we know the value of f at some particular instant, then we can determine C . For example, if $f(0) = 0$, then $C = 2$ and the position function is $f(t) = 2 - 2 \cos t$. But if $f(0) = 2$, then $C = 4$ and the position function is $f(t) = 4 - 2 \cos t$.

If it is not possible to evaluate the integral in (4.4) in terms of familiar functions, such as polynomials, rational functions, trigonometric and inverse trigonometric functions, logarithms, and exponentials, then we consider the differential equation as having been solved if the solution can be expressed in terms of integrals of known functions. In actual practice there are various methods for obtaining approximate evaluations of integrals which lead to useful information about the solution. Automatic high-speed computing machines are often designed with this kind of problem in mind.

In some respects the example just solved is typical of what happens in general. Somewhere in the process of solving a first-order differential equation, an integration is required to remove the derivative y' and in this step an arbitrary constant C appears. The way in which the arbitrary constant C enters into the solution will depend on the nature of the given differential equation. It may appear as an additive constant, as in Equation (4.4), but it is more likely to appear in some other way. For example, when we solve the equation $y' = y$ the solution is of the form $y = Ce^x$.

A solution of the form $y = G(x, C)$, in which the expression on the right involves known functions of x (or integrals of known functions of x), is called an *explicit solution* because y is given explicitly in terms of x . Actually, the equation $y = G(x, C)$ represents a whole *family* of solutions, one for each admissible value of C . Even though there may exist a function G such that $y = G(x, C)$ satisfies the differential equation, it may not be possible, in actual practice, to determine $G(x, C)$ explicitly in terms of familiar functions. What happens in many cases is that when the derivative y' is eliminated we arrive at a relation of the form

$$(4.5) \quad F(x, y, C) = 0.$$

The resulting solutions y are said to be defined *implicitly* in this case because the properties of each y are implied by Equation (4.5).

For example, in a later section we shall learn how to solve the differential equation

$$y' = \frac{y - x}{y + x}$$

and we shall find a family of solutions defined implicitly by the equation

$$(4.6) \quad \frac{1}{2} \log(x^2 + y^2) + \arctan \frac{y}{x} + C = 0.$$

It would be hopeless to try to solve (4.6) for y in terms of x . In a situation of this sort we say that Equation (4.6) represents a *family of solutions*. Each special value of C gives rise to a *particular solution* of the differential equation.

4.2 Exercises

In Exercises 1 through 9, verify that the differential equation has the two particular solutions indicated and try to guess further solutions. (Exercise 2 is worked out below as a sample.)

1. $y'' = 4y$. $y = 5e^{2x} + 7e^{-2x}$; $y = -\frac{1}{2}e^{2x} + 23e^{-2x}$.
2. $xyy' = (x+1)(y+1)$. $x(y+1) = 2e^{y-x}$; $x(y+1) = 5e^{y-x}$.
3. $y^3 + x^3y' = 0$. $x^2 + y^2 - 5x^2y^2 = 0$; $x^2 + y^2 = 13x^2y^2$.
4. $xy' - x^2 - y^2 = y$. $y = x \tan(x+1)$; $y = x \tan(x-1)$.
5. $xy' - xy^2 = y$. $y(x^2 - 2) = -2x$; $x^2y + 2x = y$.
6. $\partial f / \partial x = 2xy$. $f(x, y) = x^2y + \sin y$; $f(x, y) = y(x^2 - e^{2y})$.
7. $\partial^2 f / \partial x^2 = 2y$. $f(x, y) = y(x + \sin y)^2$; $f(x, y) = x^2y + e^y$.
8. $\partial^2 f / \partial x^2 + \partial^2 f / \partial y^2 = 0$. $f(x, y) = e^x \cos y$; $f(x, y) = \log(x^2 + y^2)$.
9. $\partial f / \partial x + \partial f / \partial y = f$. $f(x, y) = e^x \cos(x - y)$; $f(x, y) = e^y(x - y)^3$.

Solution to Exercise 2. To test $x(y + 1) = 2e^{y-x}$ we differentiate both sides with respect to x (remembering that y is a function of x), and we get $y + xy' + 1 = 2e^{y-x}(y' - 1)$. Replacing $2e^{y-x}$ by $xy + x$ and rearranging terms, we find that $xyy' = xy + x + y + 1$, which is the same as the given differential equation. The second relation is verified in the same way and these two solutions suggest that further solutions may be obtained by writing $x(y + 1) = Ce^{y-x}$, where C is any constant. Again, this is easily verified to be a solution.

4.3 Integral curves and direction fields

Suppose a differential equation of the first order, say

$$(4.7) \quad y' = f(x, y),$$

has a family of solutions described by a relation of the form

$$(4.8) \quad F(x, y, C) = 0.$$

If we introduce a rectangular coordinate system and plot all the points (x, y) whose coordinates satisfy (4.8) for a particular C , we obtain a curve called an *integral curve* of the differential equation. Different values of C usually give different integral curves, but all of them share a common geometric property described by the differential equation (4.7) which relates the slope y' at each point (x, y) of the curve to the coordinates x and y . As C takes on all its values the collection of integral curves obtained is called a *one-parameter family* of curves.

For example, when the differential equation is $y' = 3$, integration gives us $y = 3x + C$ and the integral curves form a family of straight lines, all having slope 3. The arbitrary constant C represents the y -intercept of these lines.

If the differential equation is $y' = x$, integration yields $y = \frac{1}{2}x^2 + C$ and the integral curves form a family of parabolas as shown in Figure 4.1. Again, the constant C tells us where the various curves cross the y -axis. Figure 4.2 illustrates the family of exponential curves, $y = Ce^x$, which are integral curves of the differential equation $y' = y$. Once more, C represents the y -intercept. In this case, C is also equal to the slope of the curve at the point where it crosses the y -axis.

A family of nonparallel straight lines is shown in Figure 4.3. These happen to be integral curves of the differential equation

$$(4.9) \quad y = x \frac{dy}{dx} - \frac{1}{4} \left(\frac{dy}{dx} \right)^2,$$

and a one-parameter family of solutions is given by

$$(4.10) \quad y = Cx - \frac{1}{4}C^2.$$

This family is one which possesses an *envelope*, that is, a curve having the property that at each of its points it is tangent to one of the members of the family.† The envelope here is $y = x^2$ and its graph is indicated by the dotted curve in Figure 4.3. The envelope of a family of integral curves is itself an integral curve because the slope and coordinates at

† And conversely, each member of the family is tangent to the envelope.

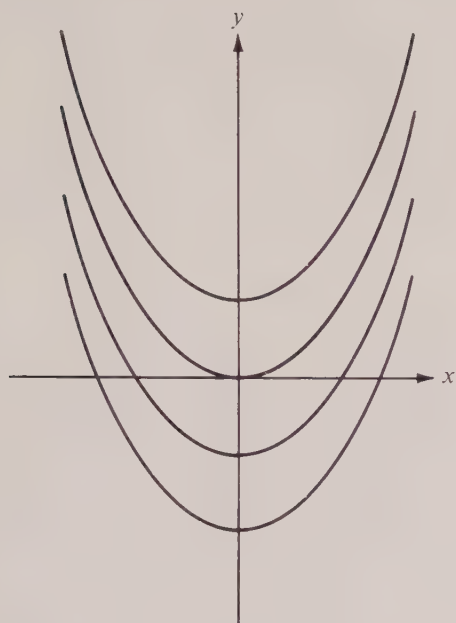


FIGURE 4.1 Integral curves of the differential equation $y' = x$.

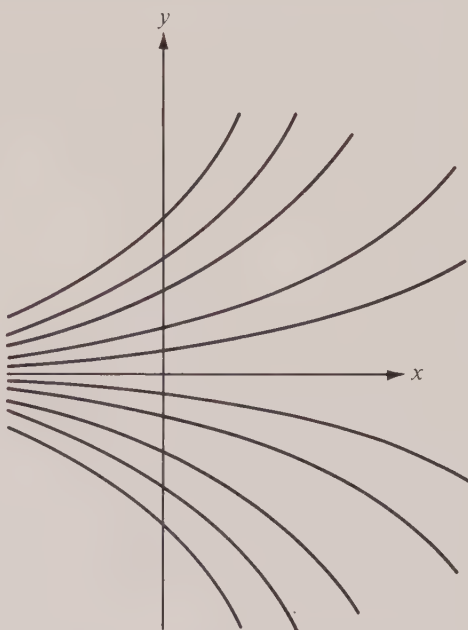


FIGURE 4.2 Integral curves of the differential equation $y' = y$.

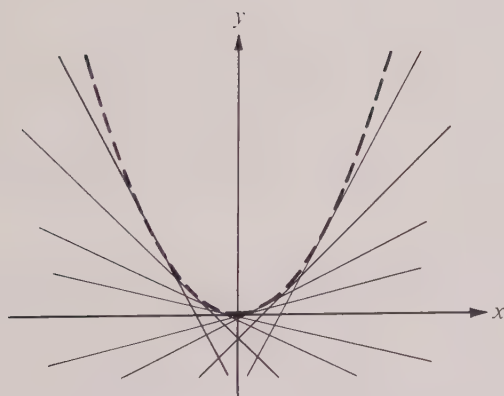


FIGURE 4.3 Integral curves of the differential equation $y = x \frac{dy}{dx} - \frac{1}{4} \left(\frac{dy}{dx} \right)^2$.

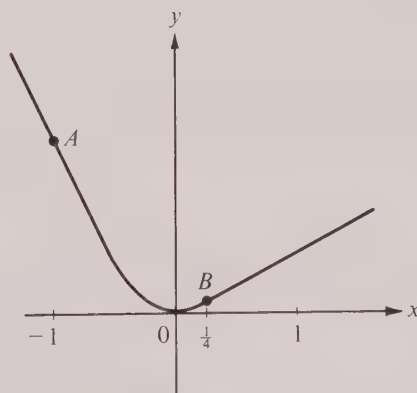


FIGURE 4.4 A solution of (4.9) that is not a member of the family in (4.10).

a point of the envelope are the same as those of one of the integral curves of the family. In this example it is easy to verify directly that $y = x^2$ is a solution of (4.9). Note that this particular solution is not a member of the family in (4.10). Further solutions, not members of the family, may be obtained by piecing together members of the family with portions of the envelope. An example is shown in Figure 4.4. The tangent line at A comes from taking $C = -2$ in (4.10) and the tangent at B comes from $C = \frac{1}{2}$. The resulting solution, $y = f(x)$, is given as follows:

$$f(x) = \begin{cases} -2x - 1 & \text{if } x \leq -1, \\ x^2 & \text{if } -1 \leq x \leq \frac{1}{4}, \\ \frac{1}{2}x - \frac{1}{16} & \text{if } x \geq \frac{1}{4}. \end{cases}$$

This function has a derivative and satisfies the differential equation in (4.9) for every real x . It is clear that an infinite number of similar examples could be constructed in the same way. This example shows that it may not be easy to exhibit all possible solutions of a differential equation. In practice, we ordinarily are not interested in *all* solutions but only in those solutions which satisfy one or more additional conditions. For example, if a solution of (4.9) is desired with a second derivative everywhere, then a function like that in Figure 4.4 must be excluded because $f''(x)$ does not exist at the point $x = -1$. However, unless such a restriction is stated, it is to be understood that a solution of a first-order differential equation is any function f for which $f'(x)$ exists and satisfies the equation for all x under consideration.

Figure 4.5 illustrates what is called a *direction field* of a differential equation. This is simply a collection of short line segments drawn tangent to the various integral curves. The particular example shown in Figure 4.5 is a direction field of the equation $y' = y$.

A direction field can be constructed without solving the differential equation. Choose a point, say (a, b) , and compute the number $f(a, b)$ obtained by substituting in the right-hand side of (4.7). If there is an integral curve through this point, its slope there must be equal to $f(a, b)$. Therefore, if we draw a short line segment through (a, b) having this slope, it will be part of a direction field of the differential equation. By drawing several of these line segments, we can get a fair idea of the general behavior of the integral curves. Sometimes such qualitative information about the solution may be all that is needed. Notice that different points $(0, b)$ on the y -axis yield different integral curves. This gives us a geometric reason for expecting an arbitrary constant to appear when we integrate a first-order equation.

4.4 Exercises

In the construction of a direction field of a differential equation, sometimes the work may be speeded considerably if we first locate those points at which the slope y' has a constant value C . For each C these points lie on a curve called an *isocline*.

1. Plot the isoclines corresponding to the constant slopes $\frac{1}{2}$, 1, $\frac{3}{2}$, and 2 for the differential equation $y' = x^2 + y^2$. With the aid of the isoclines, construct a direction field for the equation and try to determine the shape of the integral curve passing through the origin.

2. Show that the isoclines of the differential equation $y' = x + y$ form a one-parameter family of straight lines. Plot the isoclines corresponding to the constant slopes 0, $\pm\frac{1}{2}$, ± 1 , $\pm\frac{3}{2}$, ± 2 .

With the aid of the isoclines, construct a direction field and sketch the integral curve passing through the origin. One of the integral curves is also an isocline; find this curve.

3. Plot a number of isoclines and construct a direction field for the equation

$$y = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^2.$$

If you draw the direction field carefully, you should be able to determine a one-parameter family of solutions of this equation from the appearance of the direction field.

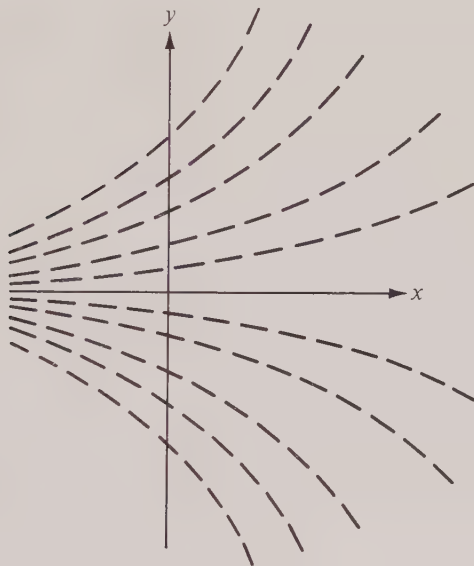


FIGURE 4.5 A direction field for the differential equation $y' = y$.

4.5 A first-order differential equation for the exponential function

The exponential function is equal to its own derivative, and the same is true of any constant multiple of the exponential. It is easy to show that these are the only functions that satisfy this property everywhere on the real axis.

To prove this, let us suppose that f is a function such that

$$(4.11) \quad f'(x) = f(x) \quad \text{for all } x.$$

First we shall find those solutions of (4.11) that are never zero. If $f(x)$ is never zero, we may divide both sides of (4.11) by $f(x)$ to obtain $f'(x)/f(x) = 1$. This equation can be rewritten in the form $G'(x) = F'(x)$, where $G(x) = \log |f(x)|$ and $F(x) = x$. Therefore we must necessarily have $G(x) = F(x) + K$, where K is a constant. This means that $\log |f(x)| = x + K$, or

$$|f(x)| = e^{x+K}.$$

Since we are assuming that $f(x)$ is never zero, we must have $f(x) > 0$ for all x or $f(x) < 0$ for all x (by continuity). In the first case we conclude that $f(x) = e^{x+K}$, and in the second that $f(x) = -e^{x+K}$. Both cases can be included in the single formula

$$(4.12) \quad f(x) = Ce^x,$$

where C has been written for $\pm e^K$. In other words, we have shown that those solutions of the differential equation $y' = y$ which are never zero are given by the formula $y = Ce^x$, where C is an arbitrary nonzero constant. When $C = 0$ this formula yields $y = 0$ which is, of course, a particular solution of the equation.

The foregoing argument does not exclude the possibility that there may exist further solutions, not included among those in (4.12). These would have to be solutions that vanish (i.e., have the value zero) at one or more points, but not everywhere. However, it is easy to show that *all* solutions of (4.11) are included in (4.12). In fact, if f is any function satisfying (4.11), we may introduce $g(x) = e^{-x}f(x)$. Then $g'(x) = e^{-x}f'(x) - e^{-x}f(x) = 0$ for all x and hence g is constant, say $g(x) = C$ for all x . This means that $C = e^{-x}f(x)$, or $f(x) = Ce^x$. Since no hypothesis other than (4.11) was made on f , it follows that *all* solutions occur in (4.12).

The method used to solve the equation $y' = y$ can be extended to more general equations as described in the next section.

4.6 Separation of variables

A first-order differential equation of the form $y' = f(x, y)$ in which the right-hand member $f(x, y)$ splits into a product of two factors, one depending on x alone and the other depending on y alone, is said to be a *separable equation*. Examples are $y' = x^3$, $y' = y$, $y' = \sin y \log x$, $y' = x/\tan y$, etc. Thus each separable equation can be expressed in the form

$$y' = Q(x)R(y),$$

where Q and R are known functions. For our purposes we assume that both Q and R are continuous wherever they are defined. When $R(y) \neq 0$ we can divide by $R(y)$ and rewrite this differential equation in the form

$$(4.13) \quad A(y)y' = Q(x),$$

where $A(y) = 1/R(y)$. If we remember that y stands for a certain unknown function, say $y = Y(x)$, then (4.13) can also be written in the form

$$(4.14) \quad A[Y(x)]Y'(x) = Q(x).$$

Integration of (4.14) gives us

$$(4.15) \quad \int A[Y(x)]Y'(x) dx = \int Q(x) dx + C.$$

If we make the substitution $y = Y(x)$ in the integral on the left, then we have $dy = Y'(x) dx$ and Equation (4.15) becomes

$$(4.16) \quad \int A(y) dy = \int Q(x) dx + C.$$

If G is any primitive of A , and H any primitive of Q , then (4.16) takes the form

$$(4.17) \quad G(y) = H(x) + C.$$

This is an equation that must be satisfied by every solution y of the differential equation $A(y)y' = Q(x)$. Conversely, if y satisfies (4.17), then $G'(y)y' = H'(x)$, which means that $A(y)y' = Q(x)$. Therefore Equation (4.17) gives us an implicit representation of a one-parameter family of solutions, expressed in terms of a primitive G of A and a primitive H of Q . Although primitives G and H always exist, it is not always possible to express them simply in terms of familiar functions.

The process by which we passed from (4.13) to (4.16) may be expressed more briefly if we use the Leibniz notation for derivatives. First we write dy/dx for y' in (4.13) and then rewrite this in the form

$$A(y) dy = Q(x) dx$$

which, of course, is equivalent to (4.14). Now we simply attach integral signs to both sides of this equation and add an arbitrary constant C to obtain (4.16). The justification of this last step is provided by the foregoing derivation of (4.16) from (4.15).

The technique just described is called the *method of separation of variables*. Although the variables can be separated only in very special equations, the method is rather important in practice because many physical problems lead to separable equations.

Example. An important special case of a separable equation is

$$(4.18) \quad \frac{dy}{dx} + P(x)y = 0,$$

where P is a known function. Separating the variables and integrating, we find

$$\int \frac{dy}{y} = - \int P(x) dx + K$$

or $\log |y| = - \int P(x) dx + K$. If we argue as in Section 4.5, we are led to the solution

$$(4.19) \quad y = Ce^{-\int P(x) dx}.$$

Note. Again, since we have divided by y (which requires $y \neq 0$) we must present a separate argument to prove that *all* solutions of (4.18) occur in (4.19). To do this, let y be any solution of (4.18) and introduce $g(x) = ye^{\int P(x) dx}$. Then the derivative of g is given by

$$g'(x) = \frac{d}{dx} (ye^{\int P(x) dx}) = y'e^{\int P(x) dx} + yP(x)e^{\int P(x) dx} = e^{\int P(x) dx} [y' + P(x)y].$$

Since y is assumed to satisfy (4.18), the expression in square brackets vanishes. Therefore $g'(x) = 0$ and hence $g(x) = C$ for some constant C . In other words, $y = Ce^{-\int P(x) dx}$.

The separate argument given in the foregoing paragraph can be reworded to provide a new method for solving the differential equation in (4.18). If we multiply the left member of (4.18) by $e^{\int P(x) dx}$, we obtain

$$e^{\int P(x) dx} [y' + P(x)y] = y'e^{\int P(x) dx} + yP(x) e^{\int P(x) dx} = \frac{d}{dx} (ye^{\int P(x) dx}).$$

Thus, multiplication of both sides of (4.18) by $e^{\int P(x) dx}$ gives us the equation

$$(4.20) \quad \frac{d}{dx} (ye^{\int P(x) dx}) = 0.$$

In other words, if y satisfies (4.18) then it also satisfies (4.20) and hence

$$(4.21) \quad ye^{\int P(x) dx} = C$$

for some constant C . Therefore *every* solution of (4.18) has the form (4.21). Conversely, if y is given by (4.21) for any particular C it is easy to check that y satisfies (4.18). Therefore Equation (4.21) provides a complete catalog of all solutions of (4.18).

The factor $e^{\int P(x) dx}$, which converts (4.18) into an equation like (4.20) which can be integrated at once, is called an *integrating factor* of (4.18). Many differential equations can be solved by finding an appropriate integrating factor although it may not be easy in practice to guess what factor should be used. Methods for discovering integrating factors for certain special equations are known and they are discussed in more detailed investigations of the theory of differential equations.

4.7 Exercises

Solve the differential equations in Exercises 1 through 12 by separating the variables.

1. $y' = x^3/y^2$.
2. $\tan x \cos y = -y' \tan y$.
3. $(x+1)y' + y^2 = 0$.
4. $y' = (y-1)(y-2)$.
5. $y\sqrt{1-x^2}y' = x$.
6. $(x-1)y' = xy$.
7. $(1-x^2)^{1/2}y' + 1 + y^2 = 0$.
8. $xy(1+x^2)y' - (1+y^2) = 0$.
9. $(x^2-4)y' = y$.
10. $xyy' = 1 + x^2 + y^2 + x^2y^2$.
11. $yy' = e^{x+2y} \sin x$.
12. $x dx + y dy = xy(x dy - y dx)$.

In Exercises 13 through 16, find functions f , continuous on the whole real axis, which satisfy the conditions given. When it is easy to enumerate all of them, do so; in any case, find as many as you can.

13. $f(x) = 2 + \int_1^x f(t) dt$.
14. $f(x)f'(x) = 5x$, $f(0) = 1$.
15. $f'(x) + 2xe^{f(x)} = 0$, $f(0) = 0$.
16. $f^2(x) + [f'(x)]^2 = 1$. Note, $f(x) = -1$ is one solution.

17. A nonnegative function f , continuous on the whole real axis, has the property that its ordinate set over an arbitrary interval has an area proportional to the length of the interval. Find f .

18. Solve Exercise 17 if the area is proportional to the difference of the function values at the endpoints of the interval.

19. Solve Exercise 18 when "difference" is replaced by "sum."
20. Solve Exercise 18 when "difference" is replaced by "product."

4.8 First-order linear equations

There are no methods known for solving all first-order differential equations of the form $y' = f(x, y)$ but there are many special tricks that work in certain cases. For example, separation of variables can always be used when the equation is separable. In this section we deal with another type of equation that can always be solved. It has the form

$$(4.22) \quad y' + P(x)y = Q(x),$$

where P and Q are known functions, and it is called a first-order *linear* equation because all the terms involving the unknown function y and its derivative y' can be written as a linear combination of y and y' . The functions P and Q are assumed to be continuous on some open interval. We seek a solution y defined on this interval.

Equation (4.18), treated at the end of Section 4.6, is a special case of a linear equation in which $Q(x) = 0$. We solved (4.18) by separation of variables and also by use of an integrating factor. If we examine the second solution, we see that the integrating factor used there also works here. Multiplying both sides of (4.22) by $e^{\int P(x) dx}$, we obtain

$$(4.23) \quad \frac{d}{dx} (ye^{\int P(x) dx}) = Q(x)e^{\int P(x) dx}.$$

Integration of (4.23) yields

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} + C,$$

from which we obtain

$$(4.24) \quad y = e^{-\int P(x) dx} \left[\int Q(x)e^{\int P(x) dx} + C \right].$$

It is easy to check that every such y actually satisfies (4.22), and hence *all* solutions of (4.22) are given by (4.24).

Here we have a favorable case in which the solution y has been expressed explicitly in terms of x . It is fortunate that this equation, which can be solved so easily, arises in a great number of important applications. Some of these are discussed in Section 4.10.

Example. Solve the first-order linear equation $xy' + (1 - x)y = e^{2x}$.

Solution. To transform the equation to the form shown in (4.22) we divide through by x (assuming $x \neq 0$) to obtain

$$(4.25) \quad y' + \left(\frac{1}{x} - 1 \right) y = \frac{e^{2x}}{x}.$$

This is now a special case of (4.22) with $P(x) = 1/x - 1$ and $Q(x) = e^{2x}/x$. To solve it we may use (4.24). First we compute

$$\int P(x) dx = \int \left(\frac{1}{x} - 1 \right) dx = \log |x| - x.$$

Then we have

$$e^{\int P(x) dx} = e^{\log |x| - x} = |x|e^{-x}$$

and (4.24) gives us

$$y = \frac{e^x}{|x|} \left(\int \frac{e^{2x}}{x} |x|e^{-x} dx + C \right).$$

If $x > 0$ we have $|x| = x$ and the solution is

$$(4.26) \quad y = \frac{e^x}{x} \left(\int e^x dx + C \right) = \frac{e^{2x}}{x} + C \frac{e^x}{x}.$$

If $x < 0$ we have $|x| = -x$ and the solution is

$$(4.27) \quad y = \frac{e^x}{-x} \left(\int -e^x dx + C \right) = \frac{e^{2x}}{x} - C \frac{e^x}{x}.$$

Since C is an arbitrary constant, we may transform (4.27) into (4.26) by replacing C by $-C$. Hence all solutions of (4.25), valid for all $x \neq 0$, are given by the formula

$$y = \frac{e^{2x}}{x} + C \frac{e^x}{x}.$$

We shall show now that exactly one of these solutions tends to a limit as $x \rightarrow 0$. In fact, when $C = -1$ the foregoing formula becomes

$$y = e^x \left(\frac{e^x - 1}{x} \right).$$

Since the factor in parentheses is the difference quotient for the derivative of e^x at $x = 0$, we see that $y \rightarrow 1$ as $x \rightarrow 0$. For a general C the solution can be written in the form

$$y = e^x \left(\frac{e^x - 1}{x} \right) + (1 + C) \frac{e^x}{x}.$$

As $x \rightarrow 0$ the first term tends to 1, and if $C \neq -1$ the second term is unbounded. Therefore the solution tends to a limit as $x \rightarrow 0$ if and only if $C = -1$.

4.9 Exercises

Solve the differential equations in Exercises 1 through 10.

1. $y' - 3y = e^{2x}$.

6. $xy' - 2y = x^5$.

2. $y' \sin x + y \cos x = 1$.

7. $x(x+1)y' + y = x(x+1)^2e^{-x^2}$.

3. $xy' - y = x \sin x$.

8. $y' + y \cot x = 2 \cos x$.

4. $dx/dt + x = e^{2t}$.

9. $(x-2)(x-3)y' + 2y = (x-1)(x-2)$.

5. $y' + y \tan x = \sin 2x$.

10. $y' + xy = x^3$, given that $y = 0$ when $x = 0$.

11. An equation of the form $y' + P(x)y = Q(x)y^n$ ($n \neq 1$) is called a *Bernoulli equation*. Show that this may be transformed into a linear first-order equation for a new unknown function v , where $v = y^{1-n}$. Obtain a formula generalizing formula (4.24) of Section 4.8.

Solve the equations in Exercises 12 through 15 by the method suggested in Exercise 11.

12. $y' - 4y = 2e^x y^{1/2}$.

14. $xy' - 2y = 4x^3 y^{1/2}$.

13. $y' - y + y^2(x^2 + x + 1) = 0$.

15. $xy' + y = y^2 x^2 \log x$.

The equations in Exercises 16 and 17 may be transformed to linear first-order equations by a suitable change of variable. Solve them by this method.

16. $y' + x \tan y = x^2 \sec y$. *Note.* The solution involves the integral $\int x^2 e^{x^2/2} dx$. Do not attempt to evaluate this integral.

17. $2xyy' + (1 + x)y^2 = e^x$.

18. An equation of the form $y' + P(x)y + Q(x)y^2 = R(x)$ is called a *Riccati equation*. (There is no known method for solving the general Riccati equation.) (a) Prove that if u is a known solution of this equation, then there are further solutions of the form $y = u + 1/v$, where v satisfies a first-order linear equation.

(b) The equation $y' + y + y^2 = 2$ has two constant solutions. Start with each of these constant solutions and use part (a) to find a corresponding family of solutions. Are the two families identical?

19. Prove that there is exactly one function f , continuous on the positive real axis, such that

$$f(x) = 1 + \frac{1}{x} \int_1^x f(t) dt$$

for all $x > 0$ and find this function.

20. The function f defined by the equation

$$f(x) = xe^{(1-x^2)/2} - xe^{-x^2/2} \int_1^x t^{-2} e^{t^2/2} dt$$

for $x > 0$ has the properties that (i) it is continuous on the positive real axis, and (ii) it satisfies the equation

$$f(x) = 1 - x \int_1^x f(t) dt$$

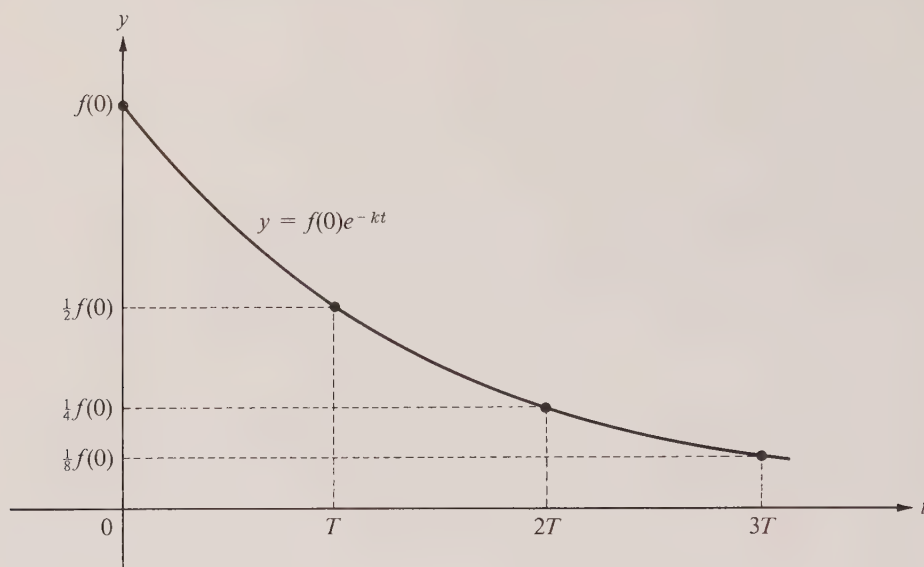
for all $x > 0$. Find all functions with these two properties.

4.10 Some physical problems leading to first-order differential equations

One of the most important aspects of the study of differential equations for the student of science or engineering is concerned with “setting up” differential equations that may help to solve certain physical problems. In this section we discuss a few relatively simple problems that lead to equations of the first order. In each case the differential equation occurs as a translation of some physical law, such as Newton’s second law of motion, a “conservation” law, etc.† We present these examples here in the hope that they may help the reader learn to set up differential equations by himself when he meets similar situations in practice. Further applications of differential equations to geometric problems will be discussed later.

Example 1: Radioactive decay. Although various radioactive elements show marked differences in their rates of decay, they all seem to share a common property—the rate at which a given substance decomposes at any instant is proportional to the amount present at that instant. If we de-

† Our purpose here is not to justify these laws, but rather to deduce logical consequences from the assumption that these laws are true. Actually, the physical laws we use here are only approximations to reality and their motivation properly belongs to the sciences from which the various problems emanate. If intuition or experimental evidence agrees with the results deduced mathematically, then one feels that the law is a good description of reality. If not, one tries to find a more suitable law to work from.

FIGURE 4.6 Radioactive decay with half-life T .

note by $y = f(t)$ the amount present at time t , the derivative $y' = f'(t)$ represents the rate of change of y at time t and the “law of decay” states that

$$y' = -ky,$$

where k is a positive constant (called the *decay constant*) whose actual value depends on the particular element that is decomposing. The minus sign comes in because y decreases as t increases, and hence y' is always negative. The general solution of this equation is

$$y = Ce^{-kt}.$$

The integration constant C represents the amount of substance present when $t = 0$. Therefore we can write the solution in terms of $f(t)$ as follows:

$$(4.28) \quad f(t) = f(0)e^{-kt}.$$

To determine the decay constant k it is necessary to make a measurement of the amount present for some value of $t > 0$.

It is interesting to see what information can be deduced from (4.28) without knowing the exact value of $f(0)$ or of k . First we observe that there is no finite time t at which $f(t)$ will be zero because the exponential e^{-kt} never vanishes. Therefore it is not useful to study the “total lifetime” of a radioactive substance. However, it is possible to determine the time required for any particular *fraction* of a sample to decay. The fraction $\frac{1}{2}$ is usually chosen for convenience and the time T at which $f(T)/f(0) = \frac{1}{2}$ is called the *half-life* of the substance. This can be determined by solving the equation $e^{-kT} = \frac{1}{2}$ for T . Taking logarithms, we get $-kT = -\log 2$ or $T = (\log 2)/k$. This equation relates the half-life to the decay constant. Since we have

$$\frac{f(t+T)}{f(t)} = \frac{f(0)e^{-k(t+T)}}{f(0)e^{-kt}} = e^{-kT} = \frac{1}{2}$$

we see that the half-life is the same for every sample of a given material. Figure 4.6 illustrates the general shape of a radioactive decay curve.

Example 2: Falling body in a resisting medium. A body of mass m is dropped from rest from a great height in the earth's atmosphere. Assume that it falls in a straight line and that the only forces acting on it are the earth's gravitational attraction (mg , where g is the acceleration due to gravity, assumed to be constant) and a resisting force (due to air resistance) which is proportional to its velocity. It is required to discuss the resulting motion.

Let $s = f(t)$ denote the distance the body has fallen at time t and let $v = s' = f'(t)$ denote its velocity. The assumption that it falls from rest means that $f'(0) = 0$.

There are two forces acting on the body, a downward force mg (due to its weight) and an upward force $-kv$ (due to air resistance), where k is some positive constant. Newton's second law states that the net sum of the forces acting on the body at any instant is equal to the product of its mass m and its acceleration. If we denote the acceleration at time t by a , then $a = v' = s''$ and Newton's law gives us the equation

$$ma = mg - kv.$$

This can be considered as a second-order differential equation for the displacement s or as a first-order equation for the velocity v . As a first-order equation for v it can be dealt with as a separable equation or as a linear equation and it leads to the general solution

$$v = \frac{mg}{k} + Ce^{-kt/m}.$$

Since $v = 0$ when $t = 0$ the integration constant C is equal to $-mg/k$ and the formula for velocity becomes

$$(4.29) \quad v = \frac{mg}{k}(1 - e^{-kt/m}).$$

Notice that as t increases without bound the exponential term tends to zero and the velocity tends to a limiting value mg/k . If we differentiate (4.29), we find that the acceleration at every instant is

$$a = ge^{-kt/m}.$$

This tends to the limiting value 0 as t increases indefinitely, which means that the air resistance tends to balance out the force of gravity.

Since $v = s'$, Equation (4.29) is itself a differential equation for the displacement s , and it may be integrated directly to give

$$s = \frac{mg}{k}t + \frac{gm^2}{k^2}e^{-kt/m} + C.$$

Since $s = 0$ when $t = 0$ we find $C = -gm^2/k^2$ and the equation of motion becomes

$$s = \frac{mg}{k}t + \frac{gm^2}{k^2}(e^{-kt/m} - 1).$$

If the initial velocity is v_0 when $t = 0$, formula (4.29) for the velocity at time t must be replaced by

$$v = \frac{mg}{k}(1 - e^{-kt/m}) + v_0e^{-kt/m}.$$

It is interesting to note that for *every* initial velocity (positive, negative, or zero) the limiting velocity, as t increases without bound, is mg/k , a number independent of v_0 . The reader should convince himself, on physical grounds, that this seems reasonable.

Example 3: A temperature problem. The rate at which a body changes temperature is proportional to the difference between its temperature and that of the surrounding medium. (This is called *Newton's law of cooling*.) If $y = f(t)$ is the (unknown) temperature of the body at time t and if $A(t)$ denotes the (known) temperature of the surrounding medium, Newton's law leads to the differential equation

$$(4.30) \quad y' = -k(y - A(t)),$$

where k is a positive constant. Thus, y satisfies a linear first-order equation.

Consider now a specific problem in which a body cools from 200° to 100° in 40 minutes while immersed in a medium whose temperature is kept constant, say $A(t) = 10^\circ$, and let us use this information to compute k . Since $A(t)$ is constant we can separate the variables in (4.30) to get

$$\int_{200}^{100} \frac{dy}{y - 10} = -k \int_0^{40} dt,$$

where we have inserted specific limits of integration instead of using indefinite integrals with an arbitrary constant. If we carry out the integration and evaluate k , we find

$$\log 90 - \log 190 = -40k, \quad k = \frac{1}{40} \log \frac{19}{9} = \frac{\log 19 - \log 9}{40}.$$

Next, let us compute the time required for this same material to cool from 200° to 100° if the temperature of the medium is kept at 5° . Denoting the unknown time by T , we get

$$\int_{200}^{100} \frac{dy}{y - 5} = -k \int_0^T dt,$$

so that $\log(95/195) = -kT$. This gives us

$$T = \frac{1}{k} \log \frac{39}{19} = 40 \frac{\log 39 - \log 19}{\log 19 - \log 9} = 40 \frac{0.719}{0.747} = 38.5 \text{ minutes}.$$

The differential equation in (4.30) tells us that the rate of cooling decreases considerably as the temperature of the body begins to approach the temperature of the medium. To illustrate, let us find the time required to cool the same substance from 100° to 10° with the medium kept at 5° . The calculation leads to $\log(5/95) = -kT$, or

$$T = \frac{1}{k} \log 19 = 40 \frac{\log 19}{\log 19 - \log 9} = \frac{40(2.944)}{0.747} = 158 \text{ minutes}.$$

Note that the temperature drop from 100° to 10° takes more than four times as long as the change from 200° to 100° .

Example 4: A dilution problem. A tank contains 100 gallons of brine whose concentration is 2.5 pounds of salt per gallon. Brine containing 2 pounds of salt per gallon runs into the tank at a rate of 5 gallons per minute and the mixture (kept uniform by stirring) runs out at the same rate. Find the amount of salt in the tank at every instant.

Let $y = f(t)$ denote the number of pounds of salt in the tank at time t minutes after mixing begins. There are two factors which cause y to change, the incoming brine which brings salt in at a rate of 10 pounds per minute and the outgoing mixture which removes salt at a rate of $5(y/100)$ pounds per minute. (The fraction $y/100$ represents the concentration at time t .) Hence the differential equation is

$$y' = 10 - \frac{y}{20} = \frac{1}{20} (200 - y).$$

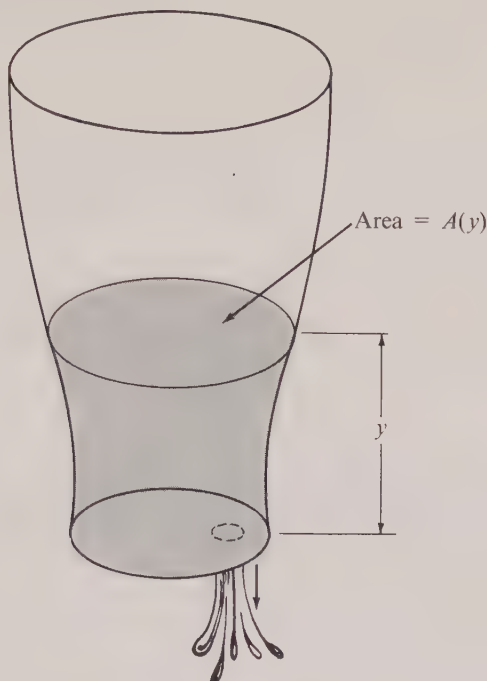


FIGURE 4.7 Flow of fluid through an orifice.

We separate the variables and integrate, and then use the fact that $y = 250$ when $t = 0$ to obtain

$$(4.31) \quad y = 200 + 50e^{-t/20}.$$

This equation shows that $y > 200$ for all t and that $y \rightarrow 200$ as t increases without bound. Hence the minimum salt content is 200 pounds. (This could also have been guessed from the statement of the problem.) Equation (4.31) can be solved for t in terms of y to yield

$$t = 20 \log \left(\frac{50}{y - 200} \right).$$

This enables us to find the time at which the salt content will be a given amount y , provided $200 < y < 250$.

Example 5: Flow of fluid through an orifice. Suppose we are given a tank (not necessarily cylindrical) containing a fluid. The fluid flows from the tank through a sharp-edged orifice. If there were no friction (and hence no loss of energy) the speed of the jet would be equal to $\sqrt{2gy}$ feet per second, where y denotes the height (in feet) of the surface above the orifice.† (See Figure 4.7.) If A_0 denotes the area (in square feet) of the orifice, then $A_0\sqrt{2gy}$ represents the number of cubic feet per second of fluid flowing from the orifice. Because of friction, the jet stream contracts somewhat and the actual rate of discharge is more nearly $cA_0\sqrt{2gy}$, where c is an experimentally determined number called the *discharge coefficient*. For ordinary sharp-edged orifices the approximate value of c is 0.60.

† If a particle of mass m falls freely through a distance y and reaches a speed v , its kinetic energy $\frac{1}{2}mv^2$ must be equal to the potential energy mgy (the work done in lifting it up a distance y). Solving for v , we get $v = \sqrt{2gy}$.

Using this and taking $g = 32$, we find that the speed of the jet is $4.8\sqrt{y}$ feet per second, and therefore the rate of discharge of volume is $4.8A_0\sqrt{y}$ cubic feet per second.

Let $V(y)$ denote the volume of the fluid in the tank when the height of the fluid is y . If the cross-sectional area of the tank at the height u is $A(u)$, then we have $V(y) = \int_0^y A(u) du$, from which we obtain $dV/dy = A(y)$. The argument in the foregoing paragraph tells us that the rate of change of volume with respect to time is $dV/dt = -4.8A_0\sqrt{y}$ cubic feet per second, the minus sign coming in because the volume is decreasing. By the chain rule we have

$$\frac{dV}{dt} = \frac{dV}{dy} \frac{dy}{dt} = A(y) \frac{dy}{dt}.$$

Combining this with the equation $dV/dt = -4.8A_0\sqrt{y}$, we obtain the differential equation

$$(4.32) \quad A(y) \frac{dy}{dt} = -4.8A_0\sqrt{y},$$

a first-order equation in which the variables are separable.

Consider now a specific case in which $A(y)$ is constant, say $A(y) = A$ for all y , and suppose the level of the fluid is lowered from 10 feet to 9 feet in 10 minutes (600 seconds). These data can be combined with the differential equation (4.32) as follows:

$$-\int_{10}^9 \frac{dy}{\sqrt{y}} = k \int_0^{600} dt,$$

where $k = 4.8A_0/A$. Using this, we can determine k and we find

$$\frac{\sqrt{10} - \sqrt{9}}{1/2} = 600k \quad \text{or} \quad k = \frac{\sqrt{10} - 3}{300}.$$

Now we can compute the time required for the level to fall from one given value to any other. For example, if at time t_1 the level is 7 feet and at time t_2 it is 1 foot (t_1, t_2 measured in minutes, say), then we must have

$$-\int_7^1 \frac{dy}{\sqrt{y}} = k \int_{60t_1}^{60t_2} dt,$$

which yields

$$\begin{aligned} t_2 - t_1 &= \frac{2(\sqrt{7} - 1)}{60k} = 10 \frac{\sqrt{7} - 1}{\sqrt{10} - 3} = \frac{10(\sqrt{7} - 1)(\sqrt{10} + 3)}{10 - 9} = (10)(1.645)(6.162) \\ &= 101.3 \text{ minutes.} \end{aligned}$$

4.11 Exercises

1. The half-life for radium is approximately 1600 years. Find what percentage of a given quantity of radium disintegrates in 100 years.

2. A man wearing a parachute jumps from a great height. The combined weight of man and parachute is 192 pounds. Let $v(t)$ denote his speed (in feet per second) at time t seconds after falling. During the first 10 seconds, before the parachute opens, assume the air resistance is $\frac{3}{4}v(t)$ pounds. Thereafter, while the parachute is open, assume the resistance is $12v(t)$ pounds. Assume the acceleration of gravity is 32 ft/sec^2 and find explicit formulas for the speed $v(t)$ at time t . (You may use the approximation $e^{-5/4} = 37/128$ in your calculations.)

3. Carry out the details of the computations in Example 2 of Section 4.10 and verify the results given there.

4. Refer to Example 2 of Section 4.10 and show that the differential equation can be written as follows:

$$v \frac{dv}{ds} = \frac{k}{m} \left(\frac{gm}{k} - v \right).$$

Solve this equation and obtain an equation relating v and s . Check your result with the formulas for v and s derived in the example.

5. Modify Example 2 of Section 4.10 by assuming the air resistance is proportional to v^2 .

(a) Solve the resulting differential equation for v . Express v both in terms of the distance s and in terms of the time t . Show that

$$v^2 = \frac{mg}{k} (1 - e^{-2ks/m}); \quad v = \sqrt{\frac{mg}{k} \frac{e^{ct} - e^{-ct}}{e^{ct} + e^{-ct}}} = \sqrt{\frac{mg}{k}} \tanh ct,$$

where $c = \sqrt{kg/m}$.

(b) From the results of (a), find the limiting value of v as t increases indefinitely.

6. A body in a room at 60° cools from 200° to 120° in half an hour.

(a) Show that its temperature after t minutes is $60 + 140e^{-kt}$, where $k = (\log 7 - \log 3)/30$.

(b) Show that the time t required to reach a temperature of T degrees is given by the formula $t = [\log 140 - \log (T - 60)]/k$, where $60 < T \leq 200$.

(c) Find the time at which the temperature is 90° .

(d) Find a formula for the temperature of the body at time t if the room temperature is not kept constant but falls at a rate of 1° each ten minutes. Assume the room temperature is 60° when the body temperature is 200° .

7. A thermometer has been stored in a room whose temperature is 75° . Five minutes after being taken outdoors it reads 65° . After another five minutes it reads 60° . Compute the outdoor temperature.

8. In a tank are 100 gallons of brine containing 50 pounds of dissolved salt. Water runs into the tank at the rate of 3 gallons per minute, and the concentration is kept uniform by stirring. How much salt is in the tank at the end of one hour if the mixture runs out at a rate of 2 gallons per minute?

9. Refer to Exercise 8. Suppose the bottom of the tank is covered with a mixture of salt and insoluble material. Assume that the salt dissolves at a rate proportional to the difference between the concentration of the solution and that of a saturated solution (3 pounds of salt per gallon), and that if the water were fresh 1 pound of salt would dissolve per minute. How much salt will be in solution at the end of one hour?

10. A tank with vertical sides has a square cross-section of area 4 square feet. Water is leaving the tank through an orifice of area $5/3$ square inches. If the water level is initially 2 feet above the orifice, find the time required for the level to drop 1 foot.

11. Refer to the preceding problem. If water also flows into the tank at the rate of 100 cubic inches per second, show that the water level approaches the value $(25/24)^2$ feet above the orifice, regardless of the initial water level.

12. A tank has the shape of a right circular cone with its vertex up. Find the time required to empty a liquid from the tank through an orifice in its base. Express your result in terms of the dimensions of the cone and the area A_0 of the orifice.

Population growth. In a study of the growth of a population (whether human, animal, or bacterial), the function which counts the number x of individuals present at time t is necessarily a *step function* taking on only integer values. Therefore the true *rate of growth* dx/dt is zero (when t lies in an open interval where x is constant), or else the derivative dx/dt does not exist (when x jumps from one integer to another). Nevertheless, useful information can often be obtained if we assume

that the population x is a continuous function of t with a continuous derivative dx/dt at each instant. We then postulate various "laws of growth" for the population, depending on the factors in the environment which may stimulate or hinder growth.

For example, if environment has little or no effect, it seems reasonable to assume that the rate of growth is proportional to the amount present. The simplest kind of growth law takes the form

$$(4.33) \quad \frac{dx}{dt} = kx,$$

where k is a constant that depends on the particular kind of population. Conditions may develop which cause the factor k to change with time, and the growth law (4.33) can be generalized as follows:

$$(4.34) \quad \frac{dx}{dt} = k(t)x.$$

If, for some reason, the population cannot exceed a certain maximum M (for example, because the food supply may be exhausted), we may reasonably suppose that the rate of growth is jointly proportional to both x and $M - x$. Thus we have a second type of growth law:

$$(4.35) \quad \frac{dx}{dt} = kx(M - x),$$

where, as in (4.33), k may be constant or, more generally, k may change with time. Technological improvements may tend to increase or decrease the value of M slowly, and hence we can generalize (4.35) even further by allowing M to change with time.

13. Express x as a function of t for each of the "growth laws" in (4.33) and (4.35) (with k and M both constant). Show that the result for (4.35) can be expressed as follows:

$$(4.36) \quad x = \frac{M}{1 + e^{-\alpha(t-t_0)}},$$

where α is a constant and t_0 is the time at which $x = M/2$.

14. Assume the growth law in formula (4.36) of Exercise 13, and suppose a census is taken at three equally spaced time intervals t_1, t_2, t_3 , the resulting numbers being x_1, x_2, x_3 . Show that this suffices to determine M and that, in fact, we have

$$(4.37) \quad M = x_2 \frac{x_3(x_2 - x_1) - x_1(x_3 - x_2)}{x_2^2 - x_1x_3}.$$

15. Derive a formula that generalizes (4.36) of Exercise 13 for the growth law (4.35) when k is not necessarily constant. Express the result in terms of the time t_0 for which $x = M/2$.

16. The Census Bureau reported the following population figures (in millions) for the United States at ten-year intervals from 1790 to 1950: 3.9, 5.3, 7.2, 9.6, 12.9, 17, 23, 31, 39, 50, 63, 76, 92, 108, 122, 135, 150.

(a) Use Equation (4.37) to determine a value of M on the basis of the census figures for 1790, 1850, and 1910.

(b) Same as (a) for the years 1910, 1930, 1950.

(c) On the basis of your calculations in (a) and (b) would you be inclined to accept or reject the growth law (4.35) for the population of the United States?

17. (a) Plot a graph of $\log x$ as a function of t , where x denotes the population figures quoted in Exercise 16. Use this graph to show that the growth law (4.33) was very nearly satisfied from 1790 to 1910. Determine a reasonable average value of k for this period.

(b) Determine a reasonable average value of k for the period from 1920 to 1950, assume that

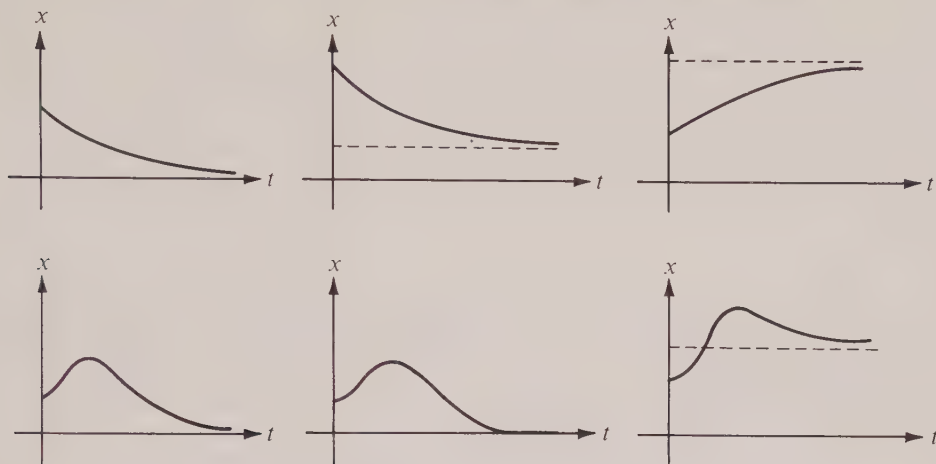


FIGURE 4.8 Exercise 18.

the growth law (4.33) will hold for this k , and predict the United States population for the years 2000 and 2050.

18. The presence of toxins in a certain medium destroys a strain of bacteria at a rate jointly proportional to the number of bacteria present and to the amount of toxin. If there were no toxins present, the bacteria would grow at a rate proportional to the amount present. Let x denote the number of living bacteria present at time t . Assume that the amount of toxin is increasing at a constant rate and that the production of toxin begins at time $t = 0$. Set up a differential equation for x . Solve the differential equation. One of the curves shown in Figure 4.8 best represents the general behavior of x as a function of t . State your choice and explain your reasoning.

4.12 A second-order differential equation for the sine and cosine

The importance of the exponential function in both pure and applied mathematics is due to the fact that it enables us to solve many of the simple first-order differential equations. It is equally important in the study of second- and higher-order equations but here it shares honors with the sine and cosine functions.

It is easy to find a simple differential equation satisfied by the sine and cosine. If we write $u = \sin x$ and $v = \cos x$, then we have $u' = \cos x = v$ and $v' = -\sin x = -u$. Differentiating once more, we find $u'' = -u$ and $v'' = -v$. In other words, both the sine and cosine satisfy the second-order differential equation

$$(4.38) \quad y'' + y = 0.$$

This equation and some of its generalizations arise in a variety of problems concerned with vibrating mechanisms, electrical oscillations, planetary motion, and other phenomena where periodicity is present.

A systematic study of second-order equations related to vibrations will not be undertaken here. In this section we shall deal only with a simple generalization of (4.38) which

describes (at least approximately) the motion of a point on a vibrating mechanism, such as a plucked string or a vibrating tuning fork. More precisely, we assume that a particle is constrained to move in a straight line with its acceleration directed toward a fixed point of the line and proportional to the displacement from that point. If we take the origin as the fixed point and let y be the displacement at time x , then the acceleration y'' must be negative when y is positive and positive when y is negative. Therefore we can write $y'' = -k^2y$, or

$$(4.39) \quad y'' + k^2y = 0,$$

where k^2 is a positive constant. This is called the differential equation of *simple harmonic motion*. (The same equation arises in electric-circuit theory where it is called the equation of the harmonic oscillator.)

It is easy to check that

$$y = \sin kx \quad \text{and} \quad y = \cos kx$$

are two particular solutions of (4.39). Further solutions may be obtained by forming linear combinations of the type

$$(4.40) \quad y = A \sin kx + B \cos kx,$$

where A and B are arbitrary constants.

It is not difficult to prove that *all* solutions of (4.39) are to be found in (4.40). There is a special technique for doing this that applies also to more general equations. The idea is to write y as a product of two new functions, say

$$(4.41) \quad y = uv.$$

Differentiating, we get $y' = uv' + u'v$, $y'' = uv'' + 2u'v' + u''v$, and Equation (4.39) becomes

$$(4.42) \quad uv'' + 2u'v' + u''v + k^2uv = 0.$$

The two terms involving u are $uv'' + k^2uv = (v'' + k^2v)u$. These terms will drop out if v is any particular solution of (4.39). For example, if we choose $v = \cos kx$, Equation (4.42) becomes

$$u'' \cos kx - 2ku' \sin kx = 0.$$

This equation has an integrating factor $\cos kx$ which transforms it into

$$(4.43) \quad \frac{d}{dx} (u' \cos^2 kx) = 0.$$

This implies $u' = C/\cos^2 kx$, where C is an arbitrary constant. Integrating again, we obtain $u = (C/k) \tan kx + B$, where B is another arbitrary constant. Substituting back in (4.41), we find $y = (C/k) \sin kx + B \cos kx$. Replacing C/k by A , we obtain (4.40).

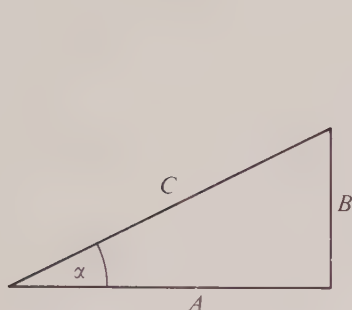


FIGURE 4.9

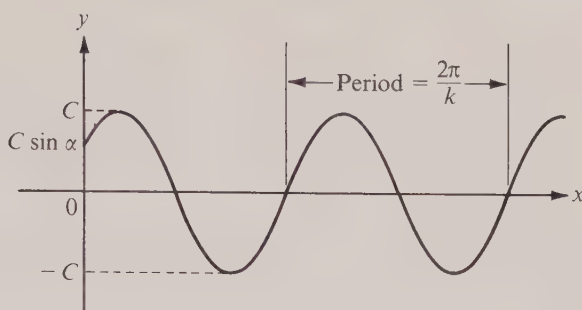


FIGURE 4.10 Simple harmonic motion.

By juggling the arbitrary constants, we can express the solution in (4.40) in terms of the sine or cosine alone. For example, if we introduce new constants C and α , where

$$C = \sqrt{A^2 + B^2} \quad \text{and} \quad \alpha = \arctan \frac{B}{A},$$

then we have (see Figure 4.9) $A = C \cos \alpha$, $B = C \sin \alpha$, and Equation (4.40) becomes

$$y = C \cos \alpha \sin kx + C \sin \alpha \cos kx = C \sin(kx + \alpha).$$

When the solution is written in this way the constants C and α have a simple geometric interpretation (see Figure 4.10). The extreme values of y , which occur when $\sin(kx + \alpha) = \pm 1$, are $\pm C$. When $x = 0$ the initial displacement is $C \sin \alpha$. As x increases, the particle oscillates between the extreme values $+C$ and $-C$ with period $2\pi/k$. The angle $kx + \alpha$ is called the *phase angle* and α itself is called the initial value of the phase angle.

4.13 Exercises

In Exercises 1 through 5, a particle is assumed to be moving in simple harmonic motion, according to the equation $y = C \sin(kx + \alpha)$. The *velocity* of the particle is defined to be the derivative y' . The *frequency* of the motion is the reciprocal of the period. (Period = $2\pi/k$; frequency = $k/2\pi$.) The frequency represents the number of cycles completed in unit time, provided $k > 0$.

1. Find the amplitude C if the frequency is $1/\pi$ and if the initial values of y and y' (when $x = 0$) are 2 and 4, respectively.
2. Find the velocity when the displacement y is zero, given that the amplitude is 7 and the frequency is 10.
3. Show that the equation of motion can also be written as follows:

$$y = A \cos(mx + \beta).$$

Find equations that relate the constants A , m , β , and C , k , α .

4. Find the equation of motion given that $y = 3$ and $y' = 0$ when $x = 0$, and that the period is $\frac{1}{2}$.

5. Find the amplitude of the motion given that the period is 2π and the velocity is $\pm v_0$ when $y = y_0$.

6. (a) Let (a_1, b_1) and (a_2, b_2) be two points in the plane such that $a_1 - a_2 \neq n\pi$, where n is an integer. Prove that there is exactly one solution of the differential equation $y'' + y = 0$ whose graph passes through these two points.

(b) Is the statement in part (a) ever true if $a_1 - a_2$ is a multiple of π ?

(c) Generalize the result in part (a) for the equation $y'' + k^2y = 0$. Discuss also the case $k = 0$.

7. If (a, b) is a given point in the plane and if m is a given real number, prove that the differential equation $y'' + k^2y = 0$ has exactly one solution whose graph passes through (a, b) and has the slope m there. Discuss also the case $k = 0$.

8. (a) Show that the differential equation $y'' - k^2y = 0$ has the solutions $y = e^{kx}$ and $y = e^{-kx}$.

(b) If $k \neq 0$, show that every solution of $y'' - k^2y = 0$ has the form

$$y = Ae^{kx} + Be^{-kx},$$

where A and B are constants. Discuss also the case $k = 0$.

(c) If $k \neq 0$, show that every solution of $y'' - k^2y = 0$ has the form

$$y = a \sinh kx + b \cosh kx,$$

where a and b are constants. How are a and b related to the constants A and B in part (b)?

9. Solve Exercise 7 when the equation $y'' + k^2y = 0$ is replaced by the equation $y'' - k^2y = 0$.

10. If a particle undergoing simple harmonic motion is suddenly subjected to a resisting force proportional to its velocity, the new motion satisfies the following differential equation:

$$(4.44) \quad y'' + 2cy' + k^2y = 0,$$

where c and k^2 are constants, $k > 0$. When $c \neq 0$ the equation is said to represent a *damped vibration*. The nature of the vibration is determined by the relative sizes of c and k and is classified as follows:

(i) *Critical damping*: $c = k$.

(ii) *Overcritical damping*: $c > k$.

(iii) *Undercritical damping*: $0 < c < k$.

(iv) *No damping* (simple harmonic motion): $c = 0$.

(a) Verify that in the first three cases the following functions are solutions of (4.44):

(i) $y = (A + Bx)e^{-kx}$.

(ii) $y = Ae^{ax} + Be^{bx}$, where $a = -c + \sqrt{c^2 - k^2}$, $b = -c - \sqrt{c^2 - k^2}$.

(iii) $y = Ae^{-cx} \sin(\beta x + \alpha)$, where $\beta = \sqrt{k^2 - c^2}$.

(b) Determine values of k and c corresponding to the following special cases and sketch the general shape of the graphs:

(i) $y = (1 + x)e^{-x}$. (ii) $y = e^{-x} + 2e^{-2x}$. (iii) $y = 2e^{-x} \sin 2x$.

4.14 Homogeneous first-order equations

We conclude this introductory chapter on differential equations with a discussion of a rather special kind of first-order equation,

$$(4.45) \quad y' = f(x, y),$$

in which the right-hand side has a special property known as *homogeneity*. This means that

$$(4.46) \quad f(tx, ty) = f(x, y)$$

for all x, y , and all $t \neq 0$. In other words, replacement of x by tx and y by ty has no effect on the value of $f(x, y)$. Equations of the form (4.45) which have this property are called *homogeneous* (sometimes called *homogeneous of degree zero*). Examples are

$$y' = \frac{y-x}{y+x}, \quad y' = \left(\frac{x^2 + y^2}{xy} \right)^3, \quad y' = \frac{x}{y} \sin \frac{x^2 + y^2}{x^2 - y^2}, \quad y' = \log x - \log y.$$

If we use (4.46) with $t = 1/x$, the differential equation in (4.45) becomes

$$(4.47) \quad y' = f\left(1, \frac{y}{x}\right).$$

The appearance of the quotient y/x on the right suggests that we introduce a new unknown function v where $v = y/x$. Then $y = vx$, $y' = v'x + v$, and this substitution transforms (4.47) into

$$v'x + v = f(1, v) \quad \text{or} \quad x \frac{dv}{dx} = f(1, v) - v.$$

This last equation can be solved for v by separating the variables. Knowing v , we obtain the solution of (4.45) by replacing v by y/x .

Example. Solve the differential equation $y' = (y - x)/(y + x)$.

Solution. We rewrite the equation as follows:

$$y' = \frac{y/x - 1}{y/x + 1}.$$

The substitution $v = y/x$ transforms this into

$$x \frac{dv}{dx} = \frac{v - 1}{v + 1} - v = -\frac{1 + v^2}{v + 1}.$$

Separating the variables, we get

$$\int \frac{v}{1 + v^2} dv + \int \frac{1}{1 + v^2} dv = -\int \frac{dx}{x}.$$

Integration yields

$$\frac{1}{2} \log(1 + v^2) + \arctan v = -\log|x| + C.$$

Replacing v by y/x , we have

$$\frac{1}{2} \log(x^2 + y^2) - \frac{1}{2} \log x^2 + \arctan \frac{y}{x} = -\log|x| + C,$$

and since $\log x^2 = 2 \log|x|$ this simplifies to

$$\frac{1}{2} \log(x^2 + y^2) + \arctan \frac{y}{x} = C.$$

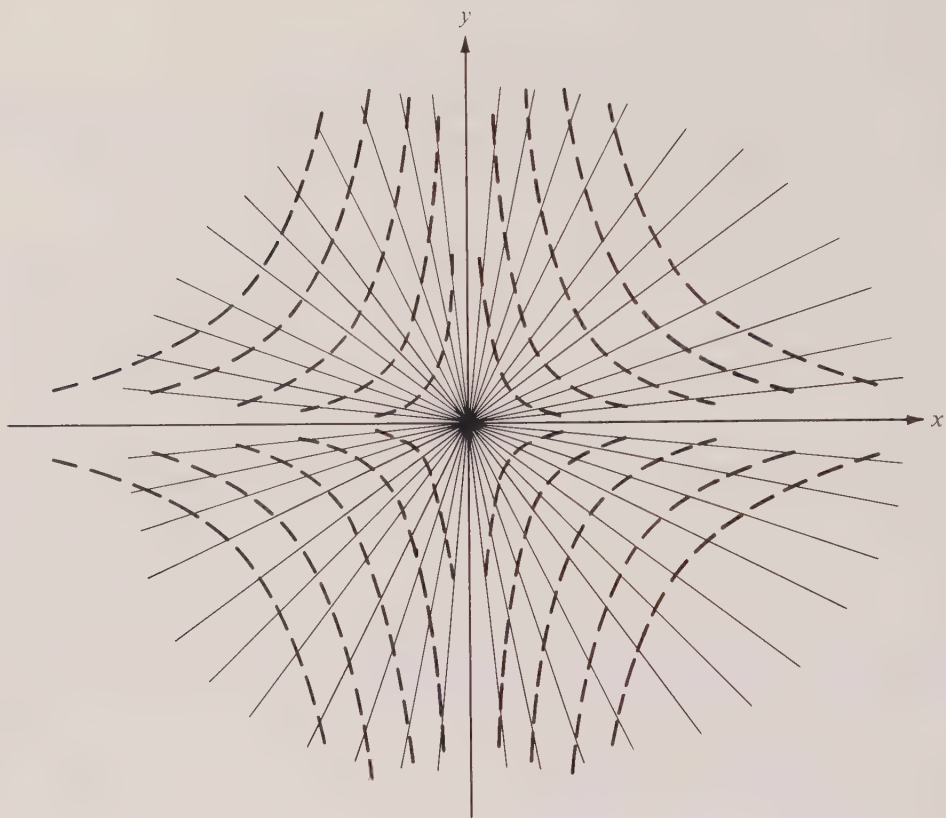


FIGURE 4.11 A direction field for the differential equation $y' = -2y/x$. The isoclines are straight lines through the origin.

There are some interesting geometric properties possessed by the solutions of a homogeneous equation $y' = f(x, y)$. First of all, it is easy to show that straight lines through the origin are isoclines[†] of the equation. This property is illustrated in Figure 4.11 which shows a direction field of the differential equation $y' = -2y/x$. The isocline corresponding to slope c has the equation $-2y/x = c$, or $y = -\frac{1}{2}cx$, and is therefore a line of slope $-\frac{1}{2}c$ through the origin. To prove the property in general, consider a line of slope m through the origin. Then $y = mx$ for all (x, y) on this line; in particular, the point $(1, m)$ is on the line. Suppose now, for the sake of simplicity, that there is an integral curve through each point of the line $y = mx$. The slope of the integral curve through a point (a, b) on this line is $f(a, b) = f(a, ma)$. If $a \neq 0$ we may use the homogeneity property in (4.46) to write $f(a, ma) = f(1, m)$. In other words, if $(a, b) \neq (0, 0)$, the integral curve through (a, b) has the same slope as the integral curve through $(1, m)$. Therefore the line $y = mx$ is an isocline, as asserted. (It can also be shown that these are the only isoclines of a homogeneous equation.)

[†] We recall that an isocline of $y' = f(x, y)$ is a curve along which the slope y' is constant.

This property of the isoclines suggests a property of the integral curves known as *invariance under similarity transformations*. We recall that a similarity transformation carries a set S into a new set kS obtained by multiplying the coordinates of each point of S by a constant factor $k > 0$. Every line through the origin remains fixed under a similarity transformation. Therefore, the isoclines of a homogeneous equation do not change under a similarity transformation; hence the appearance of the direction field does not change either. This suggests that similarity transformations carry integral curves into integral curves. To prove this analytically, let us assume that S is an integral curve described by an explicit formula of the form

$$(4.48) \quad y = F(x).$$

To say that S is an integral curve of $y' = f(x, y)$ means that we have

$$(4.49) \quad F'(x) = f(x, F(x))$$

for all x under consideration. Now choose any point (x, y) on kS . Then the point $(x/k, y/k)$ lies on S and hence its coordinates satisfy (4.48), so we have $y/k = F(x/k)$ or $y = kF(x/k)$. In other words, the curve kS is described by the equation $y = G(x)$, where $G(x) = kF(x/k)$. Note that the derivative of G is given by

$$G'(x) = kF'\left(\frac{x}{k}\right) \cdot \frac{1}{k} = F'\left(\frac{x}{k}\right).$$

To prove that kS is an integral curve of $y' = f(x, y)$ it will suffice to show that $G'(x) = f(x, G(x))$ or, what is the same thing, that

$$(4.50) \quad F'\left(\frac{x}{k}\right) = f\left(x, kF\left(\frac{x}{k}\right)\right).$$

But if we replace x by x/k in Equation (4.49) and then use the homogeneity property with $t = k$, we obtain

$$F'\left(\frac{x}{k}\right) = f\left(\frac{x}{k}, F\left(\frac{x}{k}\right)\right) = f\left(x, kF\left(\frac{x}{k}\right)\right),$$

and this proves (4.50). In other words, we have shown that kS is an integral curve whenever S is. A simple example in which this geometric property is quite obvious is the homogeneous equation $y' = -x/y$ whose integral curves form a one-parameter family of concentric circles given by the equation $x^2 + y^2 = C$.

It can also be shown that, if the integral curves of a first-order equation $y' = f(x, y)$ are invariant under similarity transformations, then the differential equation is necessarily homogeneous.

4.15 Miscellaneous exercises on differential equations

1. Show that the substitution $y = x/v$ transforms a homogeneous equation $y' = f(x, y)$ into a first-order equation in v in which the variables are separable. (Sometimes this substitution leads to

integrals that are easier to evaluate than those obtained by the substitution $y = xv$ discussed in the text.)

Solve the differential equations in Exercises 2 through 11.

$$2. y' = \frac{-x}{y}.$$

$$7. x^2y' + xy + 2y^2 = 0.$$

$$3. y' = 1 + \frac{y}{x}.$$

$$8. y^2 dx + (x^2 - xy + y^2) dy = 0.$$

$$4. y' = \frac{x^2 + 2y^2}{xy}.$$

$$9. y' = \frac{y(x^2 + xy + y^2)}{x(x^2 + 3xy + y^2)}.$$

$$5. (2y^2 - x^2) dy + 3xy dx = 0.$$

$$10. y' = \frac{y}{x} + x \sin \frac{y}{x}.$$

$$6. xy' = y - \sqrt{x^2 + y^2}.$$

$$11. x(y + 4x)y' + y(x + 4y) = 0.$$

12. The equation $xy'' - y' + (1 - x)y = 0$ possesses a solution of the form $y = e^{mx}$, where m is constant. Determine this solution explicitly.

13. Solve the differential equation $(x + y^3) + 6xy^2y' = 0$ by making a suitable change of variable which converts it into a linear equation.

14. Solve the differential equation $(1 + y^2e^{2x})y' + y = 0$ by introducing a change of variable of the form $y = ue^{mx}$, where m is constant and u is a new unknown function.

15. (a) Given a function f which satisfies the relations

$$2f'(x) = f\left(\frac{1}{x}\right) \quad \text{if } x > 0, \quad f(1) = 2,$$

let $y = f(x)$ and show that y satisfies a differential equation of the form

$$x^2y'' + axy' + by = 0,$$

where a and b are constants. Determine a and b .

(b) Find a solution of the form $f(x) = Cx^n$.

16. (a) Let u be a nonzero solution of the second-order equation

$$y'' + P(x)y' + Q(x)y = 0.$$

Show that the substitution $y = uv$ converts the equation

$$y'' + P(x)y' + Q(x)y = R(x)$$

into a first-order linear equation for v .

(b) Obtain a nonzero solution of the equation $y'' - 4y' + x^2(y' - 4y) = 0$ by inspection, and use the method of part (a) to find a solution of

$$y'' - 4y' + x^2(y' - 4y) = 2xe^{-x^3/3}$$

such that $y = 0$ and $y' = 4$ when $x = 0$.

17. If k is a nonzero constant, show that the differential equation $y'' + k^2y = R(x)$ has the particular solution

$$y = \frac{1}{k} \int_0^x R(t) \sin k(x - t) dt.$$

18. Scientists at the Ajax Atomics Works isolated one gram of a new radioactive element called Deteriorum. It was found to decay at a rate proportional to the *square* of the amount present. After one year, $\frac{1}{2}$ gram remained.

(a) Set up and solve the differential equation for the mass of Deteriorum remaining at time t .

(b) Evaluate the decay constant in units of $\text{gm}^{-1} \text{yr}^{-1}$.

19. In the preceding problem suppose the word *square* were replaced by *square root*, the other data remaining the same. Show that in this case the substance would decay entirely within a finite time, and find this time.

20. At the beginning of the Gold Rush, the population of Coyote Gulch, Arizona was 365. From then on, the population would have grown by a factor of e each year, except for the high rate of “accidental” death, amounting to one victim per day among every 100 citizens. By solving an appropriate differential equation determine, as functions of time, (a) the actual population of Coyote Gulch t years from the day the Gold Rush began, and (b) the cumulative number of fatalities.

21. With what speed should a rocket be fired upward so that it never returns to earth? (Neglect all forces except the earth’s gravitational attraction.)

22. If a strain of bacteria grows at a rate proportional to the amount present and if the population doubles in one hour, by how much will it increase at the end of two hours?